

Epigenetic Control of Temperature Response in *Ostreococcus tauri* (ECTOR)

Christina Arvanitidou (carvanitidou@us.es) and Francisco J. Romero-Campero (fran@us.es)

2024-03-15

H3K27me3 genome wide distribution analysis

We load the genome of *Ostreococcus tauri* and compute the length of its chromosomes. We determine the length of the largest chromosome and for technical reasons we reverse the vector of lengths. This will be used to represent graphically the genome wide distribution of H3K27me3.

```
library(seqinr)

ostta.genome <- read.fasta(file = "ostta_genome/ostreococcus_tauri.fa",
                           seqtype = "DNA")
chr.lens <- sapply(X = getSequence(ostta.genome), FUN = length)

number.chrs <- length(getSequence(ostta.genome))
max.chr.len <- max(chr.lens)
chr.lens <- rev(chr.lens)
```

We load the H3K27me3 peaks identified in cultures grown under constant light at 20°C. We represent each chromosome using a rectangle. Each H3K27me3 peak is represented in its precise genome location using a blue rectangle.

```
ll.20 <- read.table(file="h3k27me3/h3k27me3_ll_20_peaks.bed",
                    header = F)

ll.20 <- ll.20[,1:3]

nrow(ll.20)
```

```
## [1] 948
```

```
#jpeg(filename = "images/h3k27me3_genome_wide_distribution_ll_20.jpg", res = 300, width = 900, height = 900)
#par(mar=c(0,0,0,0))
plot(x=c(0,max.chr.len), y=c(0,4*number.chrs), col = "white",
     xlab = "", ylab = "", axes=F)

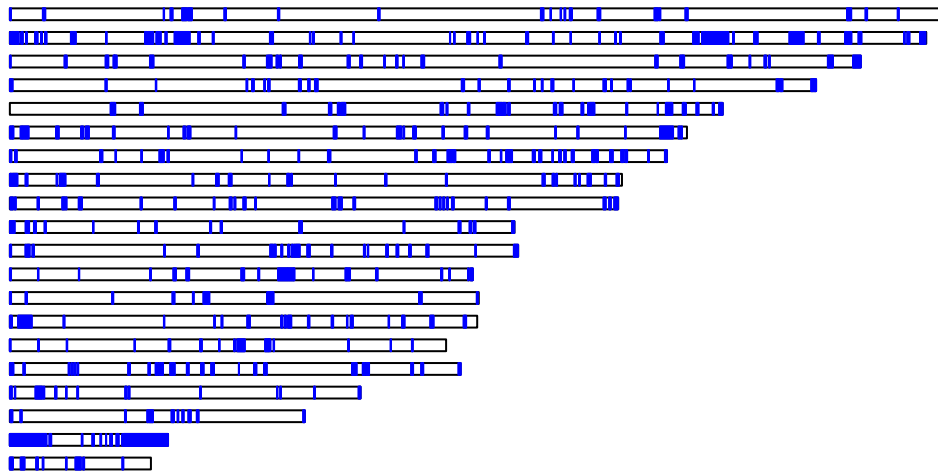
for(i in 1:number.chrs)
{
  polygon(x = c(0, chr.lens[i], chr.lens[i], 0),
         y = c(4*i+1, 4*i+1, 4*i-1, 4*i-1),
         col = "white")
}

for(i in 1:nrow(ll.20))
```

```
{
  current.chr <- ll.20[i,1]
  current.start <- ll.20[i,2]
  current.end <- ll.20[i,3]

  current.line <- (20:1)[current.chr]

  polygon(x = c(current.start,
                current.end,
                current.end,
                current.start),
        y = c(4*current.line+1,
              4*current.line+1,
              4*current.line-1,
              4*current.line-1),
        col = "blue",border="blue")
}
```



```
#dev.off()
```

We identified 948 genomic regions significantly occupied by H3K27me3 in the three different independent biological replicates. We compute the percentage of the whole genome occupied by H3K27me3 and the percentage in each chromosome. A barplot is used to represent the percentage of H3K27me3 in each chromosome. Z-scores and p-values are computed for each chromosome. A gradient from white to blue is used to represent significance levels.

```
h3k27me3.peaks.len <- ll.20$V3 - ll.20$V2
sum(h3k27me3.peaks.len)/sum(chr.lens)
```

```
## [1] 0.1521059
```

```
ll.20 <- cbind(ll.20,h3k27me3.peaks.len)
colnames(ll.20) <- c("chr","start","end","length")
head(ll.20)
```

```
##   chr  start    end length
## 1   1    177   1228   1051
## 2   1    178   1228   1050
## 3   1 38779 41436   2657
## 4   1 38779 41436   2657
```

```

## 5    1 180449 180706    257
## 6    1 180449 180775    326

number.of.h3k27.peaks.chr <- vector(mode = "numeric",length = 20)
peaks.len.chr <- vector(mode = "numeric",length = 20)

for(i in 1:20)
{
  number.of.h3k27.peaks.chr[i] <- nrow(subset(ll.20, chr == i))
  peaks.len.chr[i] <- sum(subset(ll.20, chr == i)[,4])
}

percentage.chr.k27 <- peaks.len.chr/rev(chr.lens)

z.scores <- (percentage.chr.k27 - mean(percentage.chr.k27))/sd(percentage.chr.k27)

which(z.scores > 2.5)

## [1] 19

p.vals <- pnorm((percentage.chr.k27 - mean(percentage.chr.k27[-19]))/sd(percentage.chr.k27[-19]),lower.tail=FALSE)

p.vals

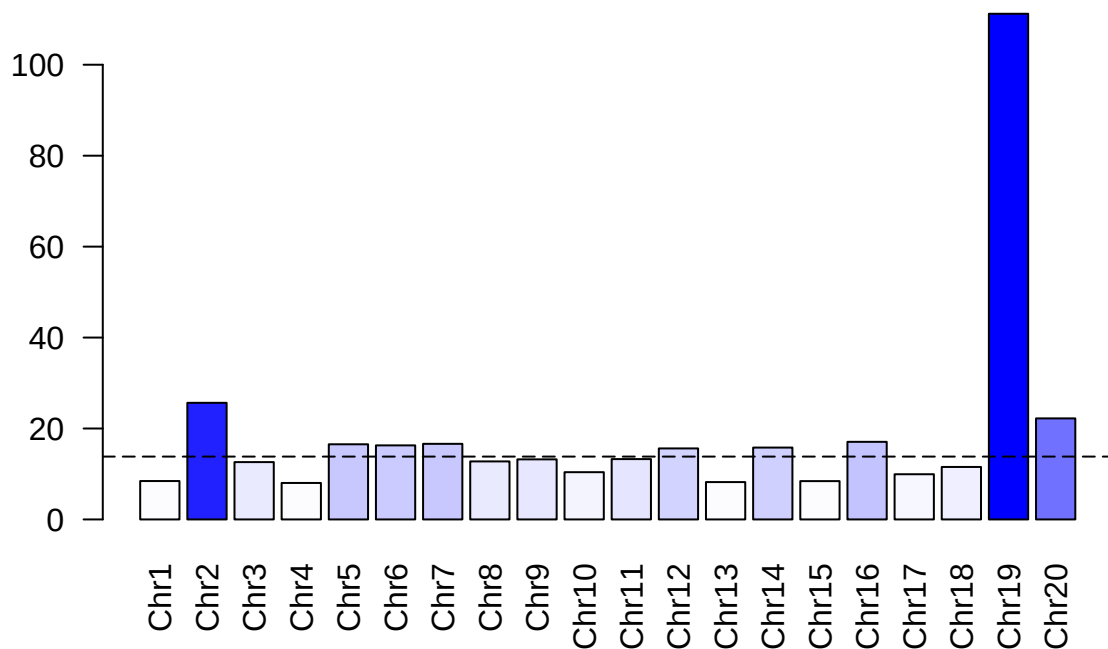
## [1] 8.708847e-01 6.495816e-03 6.007867e-01 8.884634e-01 2.849912e-01
## [6] 3.024475e-01 2.780904e-01 5.883321e-01 5.511531e-01 7.645109e-01
## [11] 5.459650e-01 3.528228e-01 8.806263e-01 3.382840e-01 8.715538e-01
## [16] 2.477236e-01 7.927559e-01 6.840704e-01 2.242258e-93 3.877563e-02

values.for.colors <- round(-log10(p.vals)*40)
values.for.colors[19] <- 100

white.blue.palette <- colorRampPalette(c('white','blue'))

#jpeg(filename = "images/h3k27me3_chr_barplot_ll_20.jpg",res = 300,width = 1800,height = 900)
barplot(100*percentage.chr.k27,border="black",names.arg = paste0("Chr",1:20),las=2,col=white.blue.palette)
abline(h = 100*mean(percentage.chr.k27[-19]),lty=5)

```



```
#dev.off()
```

Approximately 15.2% of the *Ostreococcus* genome is occupied by H3K27me3. H3K27me3 is mostly uniformly distributed over the different chromosome although it significantly accumulates (p-value < 0.01) in Chr2 (sexual chromosome) and Chr19 (viral resistance chromosome) with p-values 6.5×10^{-3} and 2.24×10^{-93} respectively.

We determine the gene features occupied by each H3K27me3 mark and the H3K27me3 marked genes.

```
library(TxDb.Otauri.JGI)
```

```
## Loading required package: GenomicFeatures
## Loading required package: BiocGenerics
##
## Attaching package: 'BiocGenerics'
## The following objects are masked from 'package:stats':
##
##   IQR, mad, sd, var, xtabs
## The following objects are masked from 'package:base':
##
##   anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##   colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##   get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##   match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##   Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
##   table, tapply, union, unique, unsplit, which.max, which.min
## Loading required package: S4Vectors
## Loading required package: stats4
##
## Attaching package: 'S4Vectors'
## The following object is masked from 'package:utils':
##
```

```

##      findMatches
## The following objects are masked from 'package:base':
##
##      expand.grid, I, unname
## Loading required package: IRanges
## Loading required package: GenomeInfoDb
## Loading required package: GenomicRanges
## Loading required package: AnnotationDbi
## Loading required package: Biobase
## Welcome to Bioconductor
##
##      Vignettes contain introductory material; view with
##      'browseVignettes()'. To cite Bioconductor, see
##      'citation("Biobase)", and for packages 'citation("pkgname)".

txdb <- TxDb.Otari.JGI

ostta.genes <- as.data.frame(genes(txdb))
head(ostta.genes)

##           seqnames start   end width strand   gene_id
## ostta01g00010      1   244  1932  1689      - ostta01g00010
## ostta01g00020      1  2077  4029  1953      + ostta01g00020
## ostta01g00030      1  4157  5299  1143      + ostta01g00030
## ostta01g00040      1  5392  7041  1650      - ostta01g00040
## ostta01g00050      1  7386  9605  2220      - ostta01g00050
## ostta01g00060      1  9780 10412   633      - ostta01g00060

target.genes <- vector(mode = "character",length = nrow(11.20))
peak.annotation <- vector(mode = "character",length = nrow(11.20))

for(i in 1:nrow(11.20))
{
  current.chr <- 11.20[i,1]
  current.start <- 11.20[i,2]
  current.end <- 11.20[i,3]

  complete.gene.body <- subset(ostta.genes,
                              seqnames == current.chr &
                              start >= current.start &
                              end <= current.end)

  overlap.tss.pos.strand <- subset(ostta.genes,
                                  seqnames == current.chr &
                                  start >= current.start & start <= current.end &
                                  strand == "+")

  overlap.tss.neg.strand <- subset(ostta.genes,
                                  seqnames == current.chr &
                                  end >= current.start & end <= current.end &
                                  strand == "-")

```

```

overlap.tes.pos.strand <- subset(ostta.genes,
                                seqnames == current.chr &
                                end >= current.start & end <= current.end &
                                strand == "+")

overlap.tes.neg.strand <- subset(ostta.genes,
                                seqnames == current.chr &
                                start >= current.start & start <= current.end &
                                strand == "-")

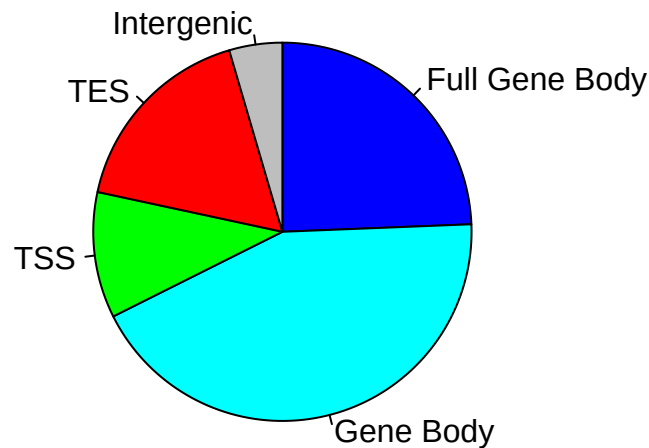
inside.gene.body <- subset(ostta.genes, seqnames == current.chr &
                           start <= current.start & end >= current.end)

if(nrow(complete.gene.body) > 0)
{
  target.genes[i] <- paste(unlist(complete.gene.body$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "gene.body"
} else if(nrow(overlap.tss.pos.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tss.pos.strand$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "TSS"
} else if(nrow(overlap.tss.neg.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tss.neg.strand$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "TSS"
} else if(nrow(overlap.tes.pos.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.pos.strand$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(overlap.tes.neg.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.neg.strand$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(inside.gene.body) > 0)
{
  target.genes[i] <- paste(unlist(inside.gene.body$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "internal.gene.body"
} else
{
  peak.annotation[i] <- "intergenic"
}
}

freq.peak.annotation <- table(peak.annotation)
#jpeg(filename = "images/h3k27me3_pie_chart_ll_20.jpg",res = 300,width = 900,height = 900)
#par(mar=c(0,0,0,0))

```

```
pie(freq.peak.annotation[c("gene.body",
                           "internal.gene.body",
                           "TSS",
                           "TES",
                           "intergenic")],
    labels = c("Full Gene Body",
               "Gene Body",
               "TSS",
               "TES",
               "Intergenic"),
    clockwise = T, border = "black", col=c("blue","cyan","green","red","grey"))
```



```
#dev.off()
```

```
percentage.freq.peak.annotation <-
  100*freq.peak.annotation/sum(freq.peak.annotation)
```

```
percentage.freq.peak.annotation["gene.body"] +
percentage.freq.peak.annotation["internal.gene.body"]
```

```
## gene.body
## 67.61603
```

```
percentage.freq.peak.annotation
```

```
## peak.annotation
##      gene.body      intergenic internal.gene.body      TES
##      24.367089       4.535865      43.248945      17.088608
##      TSS
##      10.759494
```

```
ll.20.peak.annotation <- cbind(ll.20[,1:3],
                                data.frame(peak.annotation,target.genes))
```

```
head(ll.20.peak.annotation)
```

```
##   chr  start    end peak.annotation  target.genes
## 1   1    177   1228          TES ostta01g00010
## 2   1    178   1228          TES ostta01g00010
## 3   1   38779  41436      gene.body ostta01g00260
```

```
## 4 1 38779 41436 gene.body ostta01g00260
## 5 1 180449 180706 intergenic
## 6 1 180449 180775 intergenic

write.table(x = ll.20.peak.annotation,
            file = "tables/H3K27me3_LL_20C_peak_annotation.tsv",
            quote = F, sep = "\t", row.names = F)

all.k27.genes <- unique(unlist(sapply(X = ll.20.peak.annotation$target.genes,
                                     FUN = function(x) {strsplit(x, ",")[[1]]})))
names(all.k27.genes) <- NULL
length(all.k27.genes)
```

```
## [1] 475
```

```
length(all.k27.genes)/7668
```

```
## [1] 0.06194575
```

We found 475 genes associated to H3K27me3 marks corresponding to 6.2% of the entire *Ostreococcus* genome. H3K27me3 is mostly located occupying either the entire gene body (24.4%) or internal regions of gene body (43.2%). Although some regions occupied by H3K27me3 has been found overlapping only with the TSS (10.8%) or the TES (17.1%). H3K27me3 was found in intergenic regions only in 4.5% of the cases.

Next we use metageneplots to determine the shape of the H3K27me3 over its occupied genes. We consider three different sets of marked genes, genes with their entire gene body marked, genes with H3K27 inside their gene body with no overlap in the TSS or TES and genes with H3K27me3 overlapping their TSS or TES.

```
## Function to compute ChIP signal over different tiles
ntile.signal <- function(signal.vector, number.tiles)
{
  step.len <- floor(length(signal.vector)/number.tiles)
  number.plus.1 <- length(signal.vector) %>% number.tiles
  pos.plus.1 <- sort(sample(1:100, number.plus.1, replace = F))

  acum <- c(0)
  for(i in 1:number.tiles)
  {
    if(i %in% pos.plus.1)
    {
      acum <- c(acum, acum[i]+step.len+1)
    } else
    {
      acum <- c(acum, acum[i]+step.len)
    }
  }

  acum[1] <- 1

  n.tile.signal.vector <- vector(mode = "numeric", length = number.tiles)
  for(i in 2:length(acum))
  {
    n.tile.signal.vector[(i-1)] <- mean(signal.vector[acum[i-1]:acum[i]])
  }

  return(n.tile.signal.vector)
}
```

```

## Function to generate the data for a metageneplot
metageneplot <- function(gene.names,genes,bw.file.name,ntile.txs,ntile.gene.body)
{
  res <- matrix(nrow=length(gene.names),ncol=2*ntile.txs+ntile.gene.body)
  rownames(res) <- gene.names

  bw.file <- import.bw(con = bw.file.name)
  bw.file <- as.data.frame(bw.file)

  for(i in 1:length(gene.names))
  {
    #print(i)
    current.gene.chr <- genes[gene.names[i],"seqnames"]
    current.gene.start <- genes[gene.names[i],"start"]
    current.gene.end <- genes[gene.names[i],"end"]
    current.gene.strand <- genes[gene.names[i],"strand"]

    gene.body.signal <- subset(bw.file, seqnames == current.gene.chr &
                             start >= current.gene.start &
                             end <= current.gene.end)[["score"]]

    upstream.start.signal <- subset(bw.file, seqnames == current.gene.chr &
                                   start >= (current.gene.start - 2000) &
                                   end <= current.gene.start)[["score"]]

    downstream.end.signal <- subset(bw.file, seqnames == current.gene.chr &
                                   start >= current.gene.end &
                                   end <= (current.gene.end + 2000))[["score"]]

    ntile.signal.complete <-
      c(ntile.signal(signal.vector = upstream.start.signal,number.tiles = ntile.txs),
        ntile.signal(signal.vector = gene.body.signal,number.tiles = ntile.gene.body),
        ntile.signal(signal.vector = downstream.end.signal,number.tiles = ntile.txs))

    if(current.gene.strand == "+")
    {
      res[i,] <- ntile.signal.complete
    } else
    {
      res[i,] <- rev(ntile.signal.complete)
    }
  }

  return(res)
}

## Genes with their entire gene body marked
k27.gene.body <- unique(unlist(sapply(
  X = subset(ll.20.peak.annotation,
             peak.annotation == "gene.body")[["target.genes"]],
  FUN = function(x) {strsplit(x,",")[1]})))
names(k27.gene.body) <- NULL
length(k27.gene.body)

```

```
## [1] 167
```

```
## Genes with H3K27 inside their gene body with no overlap  
## in the TSS or TES.
```

```
k27.internal.gene.body <- unique(unlist(sapply(  
  X = subset(11.20.peak.annotation,  
    peak.annotation == "internal.gene.body")["target.genes"],  
  FUN = function(x) {strsplit(x, ",")[[1]]}))  
names(k27.internal.gene.body) <- NULL  
length(k27.internal.gene.body)
```

```
## [1] 182
```

```
## Genes with H3K27me3 overlapping their TSS
```

```
k27.tss <- unique(unlist(sapply(  
  X = subset(11.20.peak.annotation,  
    peak.annotation == "TSS")["target.genes"],  
  FUN = function(x) {strsplit(x, ",")[[1]]}))  
names(k27.tss) <- NULL  
length(k27.tss)
```

```
## [1] 53
```

```
## Genes with H3K27me3 overlapping their TES
```

```
k27.tes <- unique(unlist(sapply(  
  X = subset(11.20.peak.annotation,  
    peak.annotation == "TES")["target.genes"],  
  FUN = function(x) {strsplit(x, ",")[[1]]}))  
names(k27.tes) <- NULL  
length(k27.tes)
```

```
## [1] 82
```

Next we generate the metagenepLOTS for genes where the H3K27me3 occupies their entire gene body including TSS and TES.

```
library(rtracklayer)
```

```
metagenepLOT.data.k27.gene.body.1 <-  
  metagenepLOT(gene.names = k27.gene.body,  
    genes = ostta.genes,  
    bw.file.name =  
      "h3k27me3/h3k27me3_11_20_chip_1.bw",  
    ntile.txs = 10,  
    ntile.gene.body = 20)  
median.metagenepLOT.data.k27.gene.body.1 <-  
  apply(X = metagenepLOT.data.k27.gene.body.1,  
    MARGIN = 2,  
    FUN = median, na.rm = T)
```

```
metagenepLOT.data.k27.gene.body.2 <-  
  metagenepLOT(gene.names = k27.gene.body,  
    genes = ostta.genes,  
    bw.file.name =  
      "h3k27me3/h3k27me3_11_20_chip_2.bw",  
    ntile.txs = 10,  
    ntile.gene.body = 20)
```

```

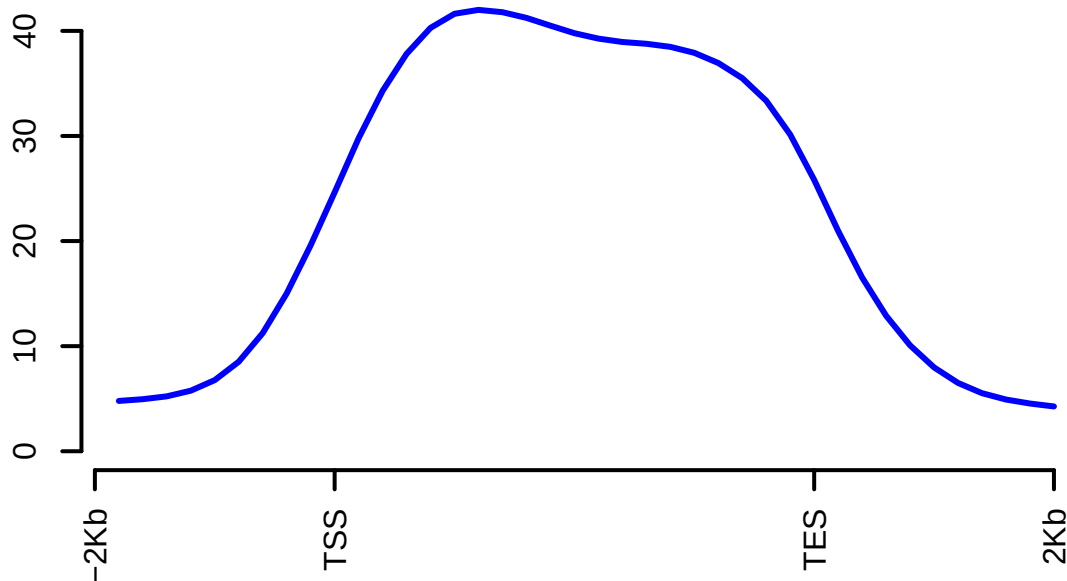
median.metagenesplot.data.k27.gene.body.2 <-
  apply(X = metagenesplot.data.k27.gene.body.2,
        MARGIN = 2,
        FUN = median, na.rm = T)

metagenesplot.data.k27.gene.body.3 <-
  metagenesplot(gene.names = k27.gene.body,
                genes = ostta.genes,
                bw.file.name =
                  "h3k27me3/h3k27me3_11_20_chip_3.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
median.metagenesplot.data.k27.gene.body.3 <-
  apply(X = metagenesplot.data.k27.gene.body.3,
        MARGIN = 2,
        FUN = median, na.rm = T)

median.metagenesplot.data.k27.gene.body <-
  (median.metagenesplot.data.k27.gene.body.1 +
   median.metagenesplot.data.k27.gene.body.2 +
   median.metagenesplot.data.k27.gene.body.3)/3

plot(smooth.spline(x=1:40,y=median.metagenesplot.data.k27.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,45),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),
     las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metagenesplots for genes where the H3K27me3 occupies part of the gene body without overlapping with the TSS and TES.

```

metagenesplot.data.k27.internal.gene.body.1 <-
  metagenesplot(gene.names = k27.internal.gene.body,
                genes = ostta.genes,
                bw.file.name =
                  "h3k27me3/h3k27me3_11_20_chip_1.bw",

```

```

        ntile.txs = 10,
        ntile.gene.body = 20)
median.metagenepLOT.data.k27.internal.gene.body.1 <-
  apply(X = metagenepLOT.data.k27.internal.gene.body.1,
        MARGIN = 2,
        FUN = median,na.rm = T)

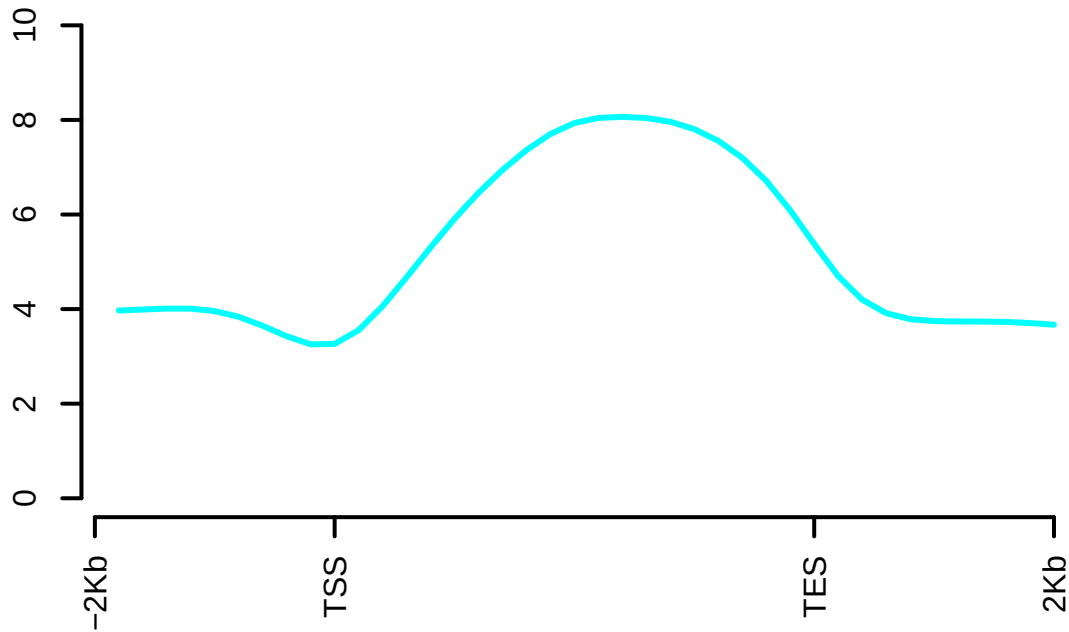
metagenepLOT.data.k27.internal.gene.body.2 <-
  metagenepLOT(gene.names = k27.internal.gene.body,
               genes = ostta.genes,
               bw.file.name =
                 "h3k27me3/h3k27me3_11_20_chip_2.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k27.internal.gene.body.2 <-
  apply(X = metagenepLOT.data.k27.internal.gene.body.2,
        MARGIN = 2,
        FUN = median,na.rm = T)

metagenepLOT.data.k27.internal.gene.body.3 <-
  metagenepLOT(gene.names = k27.internal.gene.body,
               genes = ostta.genes,
               bw.file.name =
                 "h3k27me3/h3k27me3_11_20_chip_3.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k27.internal.gene.body.3 <-
  apply(X = metagenepLOT.data.k27.internal.gene.body.3,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenepLOT.data.k27.internal.gene.body <-
  (median.metagenepLOT.data.k27.internal.gene.body.1 +
   median.metagenepLOT.data.k27.internal.gene.body.2 +
   median.metagenepLOT.data.k27.internal.gene.body.3)/3

plot(smooth.spline(x=1:40,y=median.metagenepLOT.data.k27.internal.gene.body,
                  spar = 0.5),
     type="l",col="cyan",lwd=3,ylim=c(0,10),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),
     las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metagenepLOTS for genes where the H3K27me3 occupies only the TSS and the beginning of the gene.

```
metagenepLOT.data.k27.tss.1 <-
  metagenepLOT(gene.names = k27.tss,
    genes = ostta.genes,
    bw.file.name =
      "h3k27me3/h3k27me3_11_20_chip_1.bw",
    ntile.txs = 10,
    ntile.gene.body = 20)
median.metagenepLOT.data.k27.tss.1 <-
  apply(X = metagenepLOT.data.k27.tss.1,
    MARGIN = 2,
    FUN = median, na.rm = T)

metagenepLOT.data.k27.tss.2 <-
  metagenepLOT(gene.names = k27.tss,
    genes = ostta.genes,
    bw.file.name =
      "h3k27me3/h3k27me3_11_20_chip_2.bw",
    ntile.txs = 10,
    ntile.gene.body = 20)
median.metagenepLOT.data.k27.tss.2 <-
  apply(X = metagenepLOT.data.k27.tss.2,
    MARGIN = 2,
    FUN = median, na.rm = T)

metagenepLOT.data.k27.tss.3 <-
  metagenepLOT(gene.names = k27.tss,
    genes = ostta.genes,
    bw.file.name =
      "h3k27me3/h3k27me3_11_20_chip_3.bw",
    ntile.txs = 10,
    ntile.gene.body = 20)
```

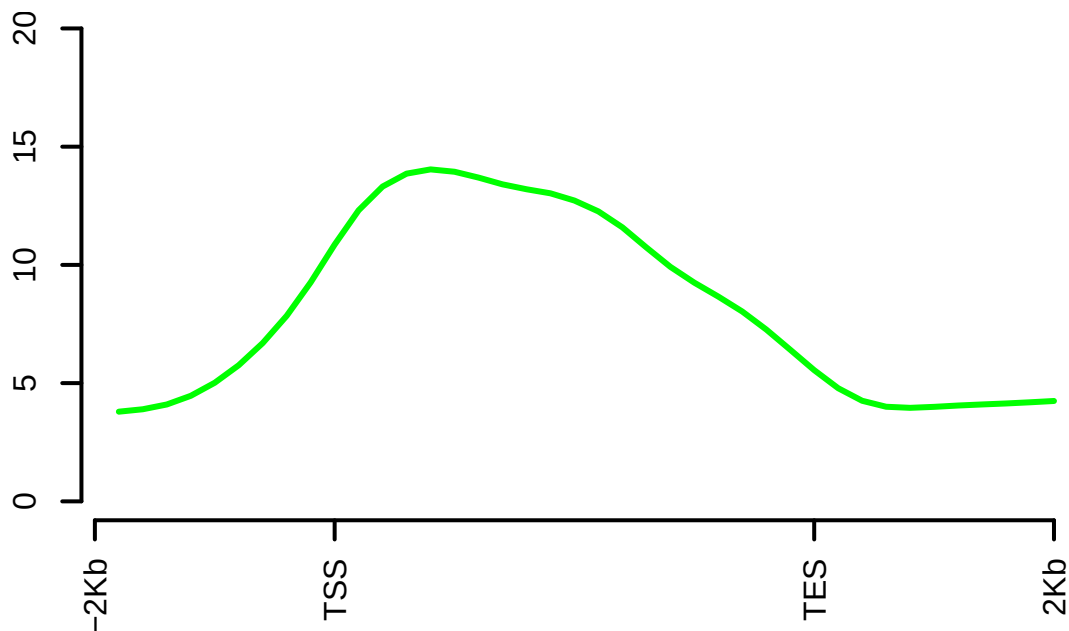
```

median.metagenesplot.data.k27.tss.3 <-
  apply(X = metagenesplot.data.k27.tss.3,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenesplot.data.k27.tss <-
  (median.metagenesplot.data.k27.tss.1 +
   median.metagenesplot.data.k27.tss.2 +
   median.metagenesplot.data.k27.tss.3)

plot(smooth.spline(x=1:40,y=median.metagenesplot.data.k27.tss,spar = 0.5),
     type="l",col="green",lwd=3,ylim=c(0,20),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),
     las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metagenesplots for genes where the H3K27me3 occupies only the TES and the end of the gene.

```

metagenesplot.data.k27.tes.1 <-
  metagenesplot(gene.names = k27.tes,
                genes = ostta.genes,
                bw.file.name =
                  "h3k27me3/h3k27me3_11_20_chip_1.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
median.metagenesplot.data.k27.tes.1 <-
  apply(X = metagenesplot.data.k27.tes.1,
        MARGIN = 2,
        FUN = median,na.rm = T)

metagenesplot.data.k27.tes.2 <-
  metagenesplot(gene.names = k27.tes,
                genes = ostta.genes,

```

```

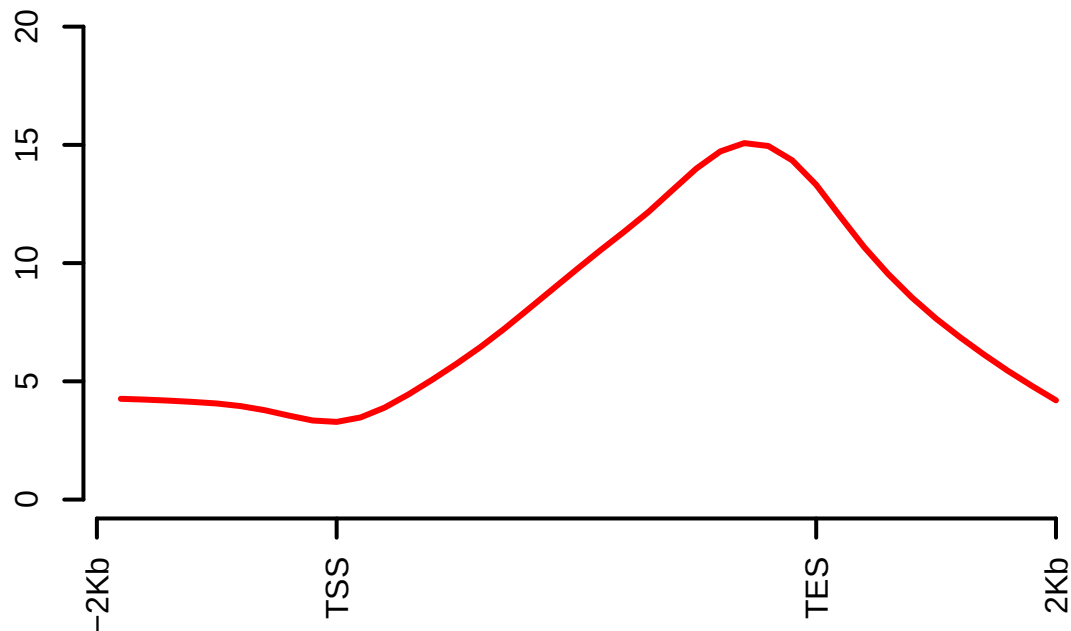
        bw.file.name =
            "h3k27me3/h3k27me3_11_20_chip_2.bw",
        ntile.txs = 10,
        ntile.gene.body = 20)
median.metagenepLOT.data.k27.tes.2 <-
    apply(X = metagenepLOT.data.k27.tes.2,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenepLOT.data.k27.tes.3 <-
    metagenepLOT(gene.names = k27.tes,
        genes = ostta.genes,
        bw.file.name =
            "h3k27me3/h3k27me3_11_20_chip_3.bw",
        ntile.txs = 10,
        ntile.gene.body = 20)
median.metagenepLOT.data.k27.tes.3 <-
    apply(X = metagenepLOT.data.k27.tes.3,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenepLOT.data.k27.tes <-
    (median.metagenepLOT.data.k27.tes.1 +
    median.metagenepLOT.data.k27.tes.2 +
    median.metagenepLOT.data.k27.tes.3)

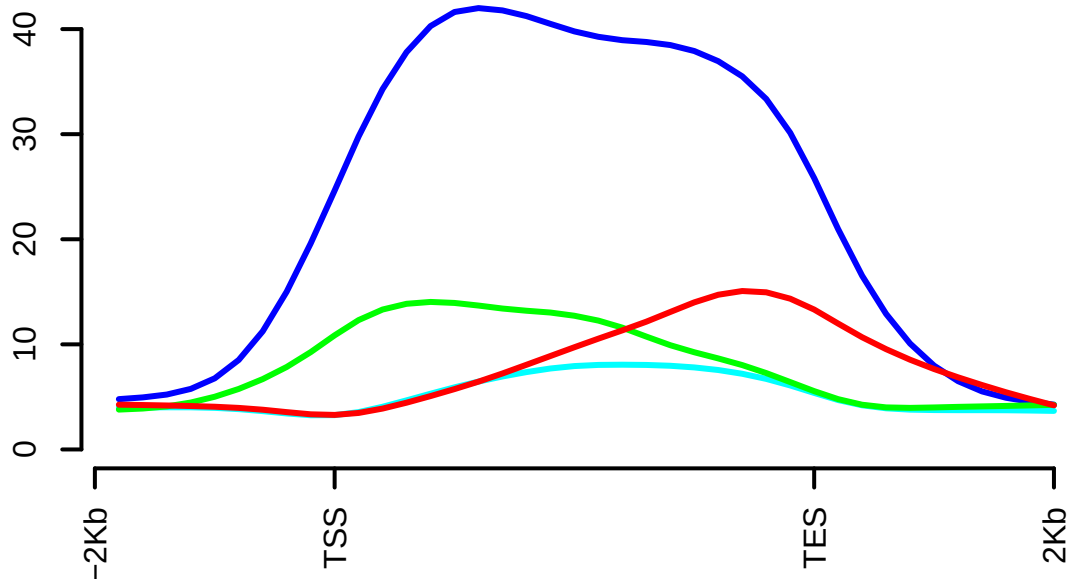
plot(smooth.spline(x=1:40,y=median.metagenepLOT.data.k27.tes,spar = 0.5),
    type="l",col="red",lwd=3,ylim=c(0,20),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),
    las=2,lwd=2)
axis(side = 2,lwd=2)

```



In order to compare the different metagenepLOTS are represented in the same plot.

```
#jpeg(filename = "images/h3k27me3_metageneplots.jpg",res = 250,width = 900,height = 900)
plot(smooth.spline(x=1:40,y=median.metageneplot.data.k27.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,45),axes=F,xlab="",ylab="")
lines(smooth.spline(x=1:40,y=median.metageneplot.data.k27.internal.gene.body,
                    spar = 0.5),
      type="l",col="cyan",lwd=3)
lines(smooth.spline(x=1:40,y=median.metageneplot.data.k27.tss,spar = 0.5),
      type="l",col="green",lwd=3)
lines(smooth.spline(x=1:40,y=median.metageneplot.data.k27.tes,spar = 0.5),
      type="l",col="red",lwd=3)
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),
     las=2,lwd=2)
axis(side = 2,lwd=2)
```



```
#dev.off()
```

It can be observed that the level of H3K27me3 mark varies depending on the part of gene that it occupies. Next we check if these differences are significant.

```
max.metageneplot.data.k27.gene.body.1 <-
  apply(X = metageneplot.data.k27.gene.body.1,
        MARGIN = 1,
        FUN = max,na.rm = T)
max.metageneplot.data.k27.gene.body.2 <-
  apply(X = metageneplot.data.k27.gene.body.2,
        MARGIN = 1,
        FUN = max,na.rm = T)
max.metageneplot.data.k27.gene.body.3 <-
  apply(X = metageneplot.data.k27.gene.body.3,
        MARGIN = 1,
        FUN = max,na.rm = T)

max.metageneplot.data.k27.gene.body <- (max.metageneplot.data.k27.gene.body.1 +
                                         max.metageneplot.data.k27.gene.body.2 +
                                         max.metageneplot.data.k27.gene.body.3)/3
```

```

max.metagenepLOT.data.k27.internal.gene.body.1 <-
  apply(X = metagenepLOT.data.k27.internal.gene.body.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepLOT.data.k27.internal.gene.body.2 <-
  apply(X = metagenepLOT.data.k27.internal.gene.body.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepLOT.data.k27.internal.gene.body.3 <-
  apply(X = metagenepLOT.data.k27.internal.gene.body.3,
        MARGIN = 1,
        FUN = max, na.rm = T)

max.metagenepLOT.data.k27.internal.gene.body <-
  (max.metagenepLOT.data.k27.internal.gene.body.1 +
   max.metagenepLOT.data.k27.internal.gene.body.2 +
   max.metagenepLOT.data.k27.internal.gene.body.3)/3

max.metagenepLOT.data.k27.tss.1 <-
  apply(X = metagenepLOT.data.k27.tss.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepLOT.data.k27.tss.2 <-
  apply(X = metagenepLOT.data.k27.tss.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepLOT.data.k27.tss.3 <-
  apply(X = metagenepLOT.data.k27.tss.3,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepLOT.data.k27.tss <- (max.metagenepLOT.data.k27.tss.1 +
                                   max.metagenepLOT.data.k27.tss.2 +
                                   max.metagenepLOT.data.k27.tss.3)/3

max.metagenepLOT.data.k27.tes.1 <-
  apply(X = metagenepLOT.data.k27.tes.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepLOT.data.k27.tes.2 <-
  apply(X = metagenepLOT.data.k27.tes.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepLOT.data.k27.tes.3 <-
  apply(X = metagenepLOT.data.k27.tes.3,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepLOT.data.k27.tes <- (max.metagenepLOT.data.k27.tes.1 +
                                   max.metagenepLOT.data.k27.tes.2 +
                                   max.metagenepLOT.data.k27.tes.3)/3

shapiro.test(max.metagenepLOT.data.k27.gene.body)

```

```
##
```

```

## Shapiro-Wilk normality test
##
## data: max.metagenepLOT.data.k27.gene.body
## W = 0.86429, p-value = 3.986e-11
shapiro.test(max.metagenepLOT.data.k27.internal.gene.body)

##
## Shapiro-Wilk normality test
##
## data: max.metagenepLOT.data.k27.internal.gene.body
## W = 0.54989, p-value < 2.2e-16
shapiro.test(max.metagenepLOT.data.k27.tss)

##
## Shapiro-Wilk normality test
##
## data: max.metagenepLOT.data.k27.tss
## W = 0.90815, p-value = 0.0006201
shapiro.test(max.metagenepLOT.data.k27.tes)

##
## Shapiro-Wilk normality test
##
## data: max.metagenepLOT.data.k27.tes
## W = 0.9206, p-value = 8.327e-05
wilcox.test(x = max.metagenepLOT.data.k27.gene.body,
            y = max.metagenepLOT.data.k27.internal.gene.body,
            alternative = "greater")$p.value

## [1] 9.081301e-41
wilcox.test(x = max.metagenepLOT.data.k27.gene.body,
            y = max.metagenepLOT.data.k27.tss,
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenepLOT.data.k27.gene.body and max.metagenepLOT.data.k27.tss
## W = 6938, p-value = 2.457e-10
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = max.metagenepLOT.data.k27.gene.body,
            y = max.metagenepLOT.data.k27.tes,
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenepLOT.data.k27.gene.body and max.metagenepLOT.data.k27.tes
## W = 11253, p-value < 2.2e-16
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = max.metagenepLOT.data.k27.tss,
            y = max.metagenepLOT.data.k27.tes)

```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenepilot.data.k27.tss and max.metagenepilot.data.k27.tes
## W = 2336, p-value = 0.464
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(x = max.metagenepilot.data.k27.tss,
            y = max.metagenepilot.data.k27.tes)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenepilot.data.k27.tss and max.metagenepilot.data.k27.tes
## W = 2336, p-value = 0.464
## alternative hypothesis: true location shift is not equal to 0
```

Indeed the differences between the metagenepilots are highly significant except the ones corresponding to the marks overlapping with the TSS or TES.

Now we analyse the gene expression level of marked genes.

```
temp.gene.expression <-
  read.table(file="rnaseq_data/ostta_temperature_gene_expression.tsv",
            header = T, sep = "\t")
```

```
gene.expression.20 <-
  (temp.gene.expression$T20_1 +
   temp.gene.expression$T20_2 +
   temp.gene.expression$T20_3)/3
```

```
names(gene.expression.20) <- rownames(temp.gene.expression)
```

```
non.k27.genes <-
  setdiff(names(gene.expression.20),
          c(k27.gene.body, k27.tss, k27.tes, k27.internal.gene.body))
```

```
summary(gene.expression.20[k27.gene.body])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.00425  0.19024   0.90820  11.71827  4.71063  178.30990
```

```
summary(gene.expression.20[k27.internal.gene.body])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.0042    7.0391   15.7631   45.3239   38.2289  1012.0876
```

```
summary(gene.expression.20[k27.tss])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.0042    0.7497    2.7891   25.2631   14.8434   555.2781
```

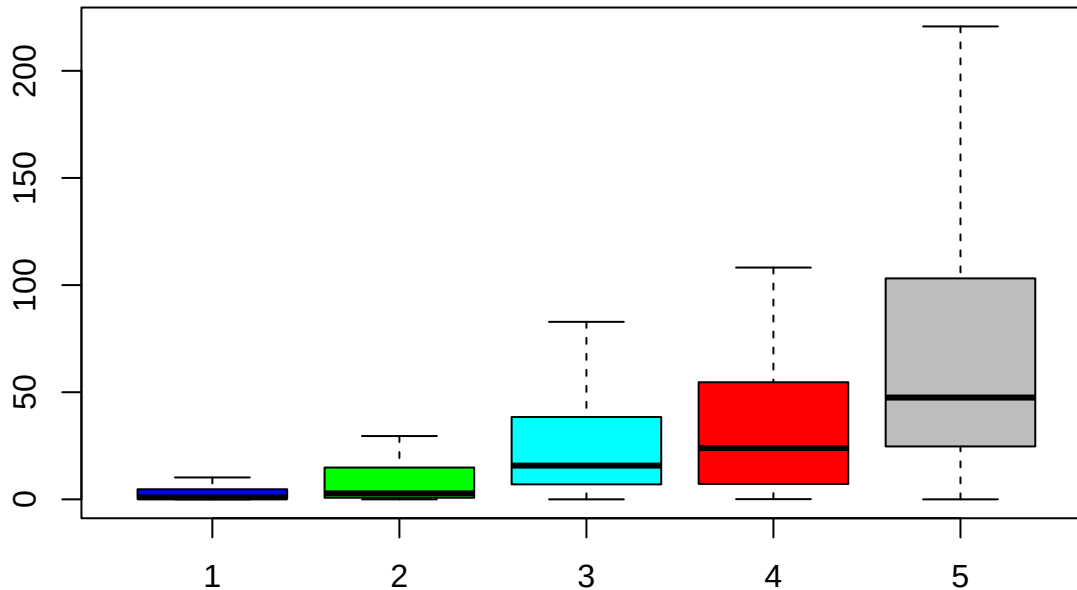
```
summary(gene.expression.20[k27.tes])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.1009    7.2239   23.9375   52.3546   53.8551   456.6316
```

```
summary(gene.expression.20[non.k27.genes])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
##    0.004   24.694   47.506   124.745  103.126 13604.318
```

```
#jpeg(filename = "images/h3k27me3_expression_barplot_different_genes.jpg",res = 250,width = 600,height = 400)
boxplot(gene.expression.20[k27.gene.body],
        gene.expression.20[k27.tss],
        gene.expression.20[k27.internal.gene.body],
        gene.expression.20[k27.tes],
        gene.expression.20[non.k27.genes],
        col=c("blue","green","cyan","red","grey"),
        outline=F)
```



```
#dev.off()

wilcox.test(x = gene.expression.20[k27.gene.body],
            y = gene.expression.20[k27.tss],
            alternative = "less")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.gene.body] and gene.expression.20[k27.tss]
## W = 3302.5, p-value = 0.002715
## alternative hypothesis: true location shift is less than 0
```

```
wilcox.test(x = gene.expression.20[k27.tss],
            y = gene.expression.20[k27.internal.gene.body],
            alternative = "less")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.tss] and gene.expression.20[k27.internal.gene.body]
## W = 2605.5, p-value = 1.789e-07
## alternative hypothesis: true location shift is less than 0
```

```
wilcox.test(x = gene.expression.20[k27.internal.gene.body],
            y = gene.expression.20[k27.tes],
            alternative = "less")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.internal.gene.body] and gene.expression.20[k27.tes]
## W = 6782.5, p-value = 0.1185
## alternative hypothesis: true location shift is less than 0
```

```
wilcox.test(x = gene.expression.20[k27.tes],
            y = gene.expression.20[non.k27.genes],
            alternative = "less")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.tes] and gene.expression.20[non.k27.genes]
## W = 189958, p-value = 1.429e-08
## alternative hypothesis: true location shift is less than 0
```

We can see all H3K27me3 genes present levels of expression significantly lower than non-marked genes. Specifically, genes whose entire gene body or TSS is occupied by H3K27me3 are completely repressed. Whereas, genes where only the TES or gene body is occupied by H3K27me3 are expressed although at a significantly lower level than non-marked genes.

In what follows we perform a functional analysis over the H3K27me3 marked genes. First, we analyze the relationship between H3K27me3 peaks and transposable elements.

```
dna.transposons <-
  read.table(file="te_analysis/dna_transposons_unsorted.bed",
             sep="\t",header=F)
nrow(dna.transposons)
```

```
## [1] 35
```

```
ltr.transposons <-
  read.table(file="te_analysis/ltr_transposons_unsorted.bed",
             sep="\t", header=F)
nrow(ltr.transposons)
```

```
## [1] 69
```

```
unclassified.transposons <-
  read.table(file="te_analysis/unknown_transposons_unsorted.bed",
             sep="\t",header = F)
nrow(unclassified.transposons)
```

```
## [1] 988
```

```
overlapping.peaks <- function(target.peak, peaks)
{
  subset(peaks, V1 == target.peak[1] & (
    (V2 >= target.peak[2] & V2 <= target.peak[3]) |
    (V3 >= target.peak[2] & V3 <= target.peak[3]) |
    (V2 <= target.peak[2] & V3 >= target.peak[3])))
}
```

```

peak.te <- vector(mode = "character",length = nrow(ll.20.peak.annotation))

for(i in 1:nrow(ll.20.peak.annotation))
{
  current.peak <- unlist(ll.20.peak.annotation[i,1:3])

  if(nrow(overlapping.peaks(current.peak,dna.transposons)) > 0)
  {
    peak.te[i] <- "DNA_transposon"
  } else if (nrow(overlapping.peaks(current.peak,ltr.transposons)) > 0)
  {
    peak.te[i] <- "LTR_transposon"
  } else if (nrow(overlapping.peaks(current.peak,unclassified.transposons)) > 0)
  {
    peak.te[i] <- "Unclassified_transposon"
  } else
  {
    peak.te[i] <- "No_transposon"
  }
}

100*table(peak.te)/sum(table(peak.te))

```

```

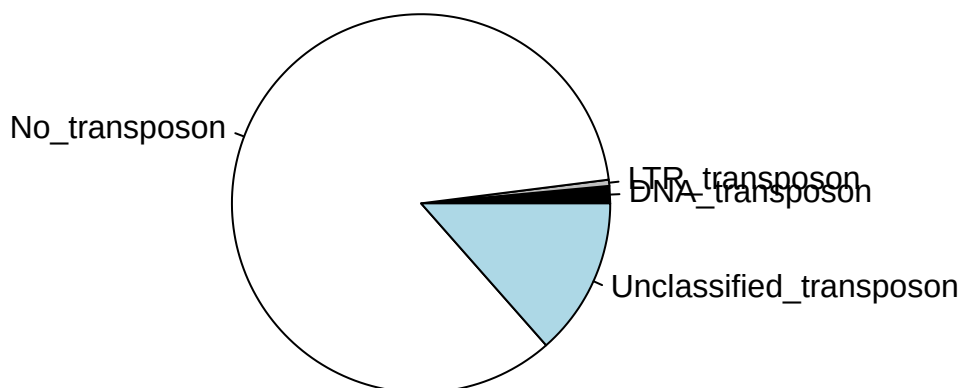
## peak.te
##          DNA_transposon          LTR_transposon          No_transposon
##          1.4767932          0.5274262          84.4936709
## Unclassified_transposon
##          13.5021097

```

```

pie(table(peak.te), col=c("black","grey","white","lightblue"))

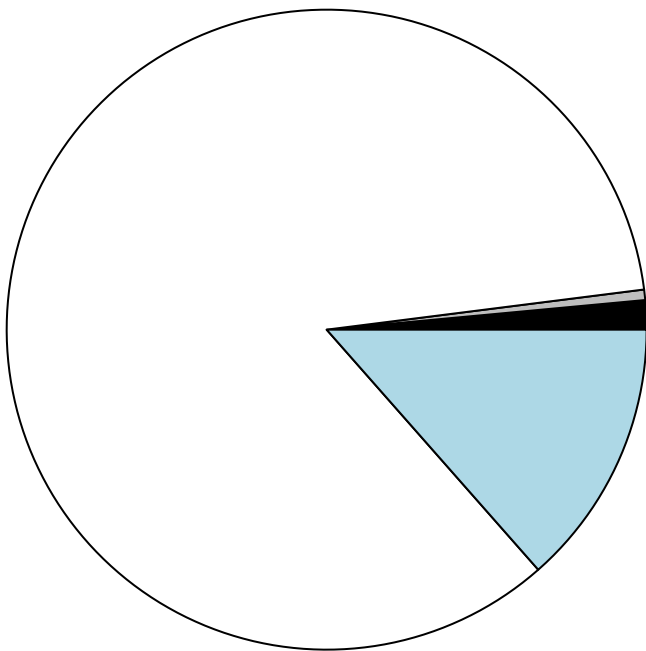
```



```

#jpeg(filename = "images/h3k27me3_transposons.jpg",res = 300,width = 900,height = 900)
par(mar=c(0,0,0,0))
pie(table(peak.te), col=c("black","grey","white","lightblue"),labels = "")

```



```
#dev.off()

nrow(dna.transposons)

## [1] 35
100*table(peak.te)["DNA_transposon"]/nrow(dna.transposons)

## DNA_transposon
##           40
nrow(ltr.transposons)

## [1] 69
100*table(peak.te)["LTR_transposon"]/nrow(ltr.transposons)

## LTR_transposon
##      7.246377
nrow(unclassified.transposons)

## [1] 988
100*table(peak.te)["Unclassified_transposon"]/nrow(unclassified.transposons)

## Unclassified_transposon
##      12.95547
table(l1.20.peak.annotation$peak.annotation[peak.te == "DNA_transposon"])

##
##      gene.body      intergenic internal.gene.body
##           4           6           4
table(l1.20.peak.annotation$peak.annotation[peak.te == "LTR_transposon"])

##
```

```
## gene.body intergenic
##      2      3
table(ll.20.peak.annotation$peak.annotation[peak.te == "Unclassified_transposon"])
```

	gene.body	intergenic	internal.gene.body	TES
	46	23	20	26
TSS				
	13			

```
table(ll.20.peak.annotation$peak.annotation)
```

	gene.body	intergenic	internal.gene.body	TES
	231	43	410	162
TSS				
	102			

```
(table(ll.20.peak.annotation$peak.annotation[peak.te == "DNA_transposon"])["intergenic"] +
table(ll.20.peak.annotation$peak.annotation[peak.te == "LTR_transposon"])["intergenic"] +
table(ll.20.peak.annotation$peak.annotation[peak.te == "Unclassified_transposon"])["intergenic"]) / tab

## intergenic
##      0.744186

ll.20.peak.annotation <- cbind(ll.20.peak.annotation,peak.te)
write.table(x = ll.20.peak.annotation,file = "tables/H3K27me3_LL_20C_peak_annotation.tsv",quote = F,sep
```

H3K27me3 is only partially associated with Transposable Elements. More than 84% of the peaks did not overlap with any TE. Only 1.5%, 0.5% and 13.5% of the H3K27me3 peaks were found overlapping with DNA, LTR and unclassified transposons respectively. Most TE were neither associated with H3K27me3 peaks, specifically 14 out of 35 (40%) DNA transposons, 5 out of 69 (7.25%) LTR transposons and 128 out of 988 (13%) unclassified transposons were H3K27me3 marked. Nonetheless, most H3K27me3 marks located at intergenic regions (74.42%) were associated with transposons.

```
library(clusterProfiler)

##
## clusterProfiler v4.14.4 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##
## Please cite:
##
## G Yu. Thirteen years of clusterProfiler. The Innovation. 2024,
## 5(6):100722
##
## Attaching package: 'clusterProfiler'
## The following object is masked from 'package:AnnotationDbi':
##
##      select
## The following object is masked from 'package:IRanges':
##
##      slice
## The following object is masked from 'package:S4Vectors':
##
```

```

##      rename
## The following object is masked from 'package:stats':
##
##      filter
library(enrichplot)

## enrichplot v1.26.5 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##
## Please cite:
##
## Guangchuang Yu, Fei Li, Yide Qin, Xiaochen Bo, Yibo Wu and Shengqi
## Wang. GOSemSim: an R package for measuring semantic similarity among GO
## terms and gene products. Bioinformatics. 2010, 26(7):976-978
library(org.Otauriv5.eg.db)

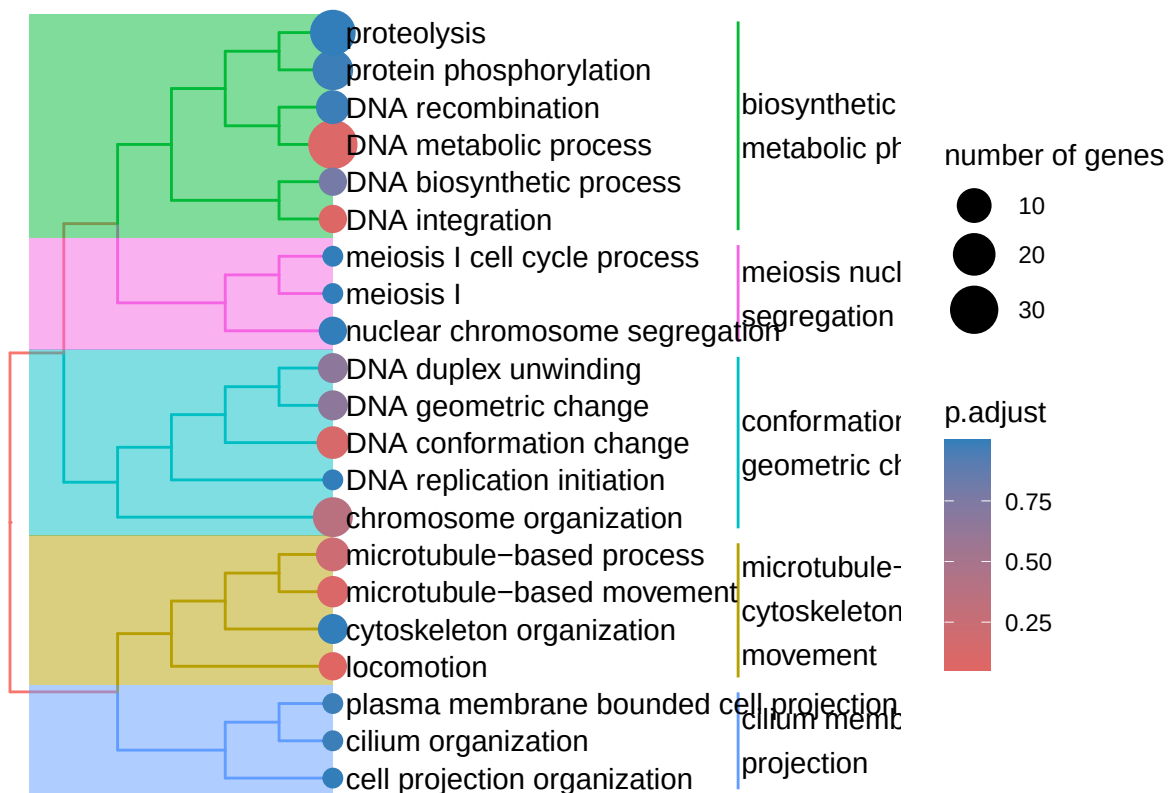
##
k27.go.terms <- enrichGO(gene = all.k27.genes, pvalueCutoff = 1,
                        qvalueCutoff = 1,
                        OrgDb = org.Otauriv5.eg.db,
                        ont = "BP",
                        keyType = "GID")
k27.go.terms.filtered <- k27.go.terms
k27.go.terms.filtered@result <- subset(k27.go.terms@result, pvalue < 0.05)

k27.go.terms.df <- as.data.frame(k27.go.terms.filtered)

write.table(x = k27.go.terms.df, file = "go_enrichment_k27.tsv",
           quote = F, sep = "\t", row.names = F)

treeplot(x = pairwise_termsim(k27.go.terms.filtered))

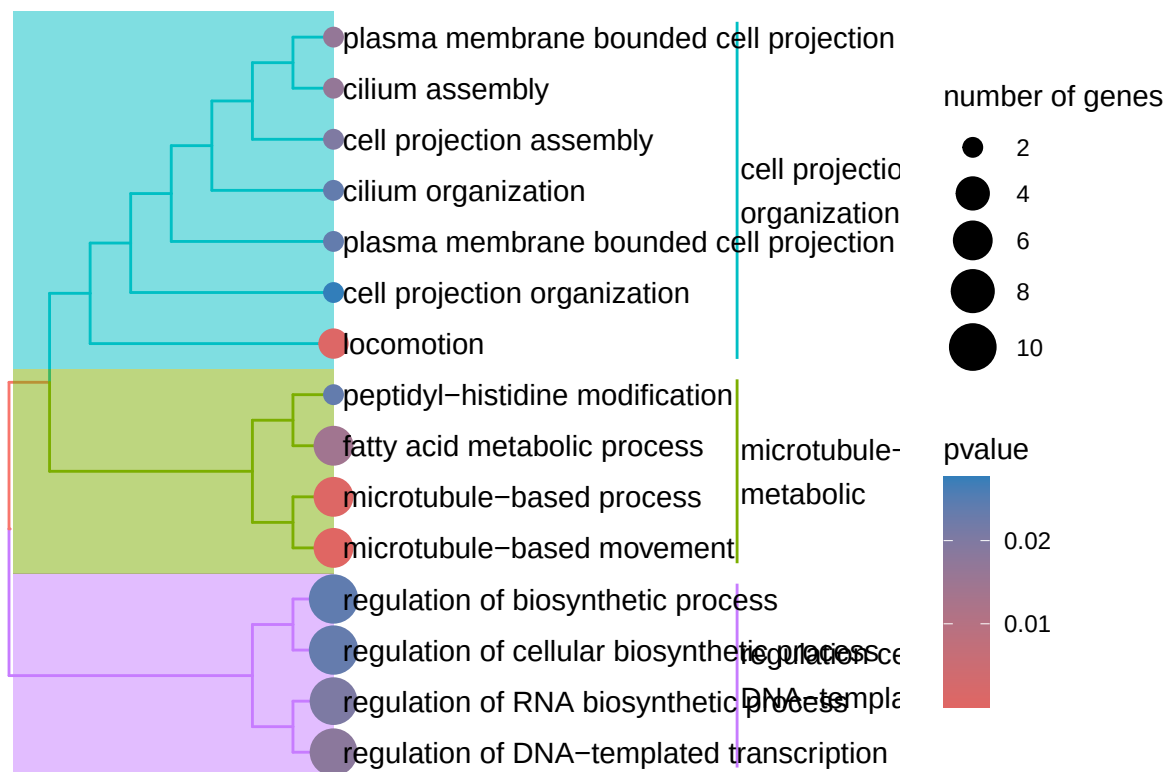
```



```
k27.go.terms <- enrichGO(gene = c(k27.gene.body,k27.tss), pvalueCutoff = 1,
  qvalueCutoff = 1,
  OrgDb = org.Otauriv5.eg.db,
  ont = "BP",
  keyType = "GID")
k27.go.terms.filtered <- k27.go.terms
k27.go.terms.filtered@result <- subset(k27.go.terms@result, pvalue < 0.05)

k27.go.terms.df <- as.data.frame(k27.go.terms.filtered)

#jpeg(filename = "images/h3k27me3_treeplot_gene_body.jpg",res = 400,width = 3500,height = 1300)
treeplot(x = pairwise_termsim(k27.go.terms.filtered),
  cluster.params = list(n = 3),color ="pvalue", showCategory = 15)
```



```
#dev.off()

write.table(x = k27.go.terms.df, file = "a_go_enrichment_full_gene_body_tss_k27.tsv",
            quote = F, sep = "\t", row.names = F)
```

Our GO enrichment analysis suggests that H3K27me3 is repressing the assembly of bounded cellular projection, microtubule-based movement and transcription regulation. Other processes are also significantly affected such as fatty acid metabolism and cell differentiation.

For example, the transcription factors ostta14g02090 (Arabidopsis ortholog APETALA2 AT4G36920), ostta17g00450 (Arabidopsis ortholog BELL1 AT5G41410), ostta11g02490, ostta07g04340 (Arabidopsis orthologs WRKY AT2G04880) and ostta02g03030 (Arabidopsis ortholog CSP, cold shock protein, AT2G17870).

```
library(rtracklayer)
library(ChIPpeakAnno)

genes.data.df <- as.data.frame(genes(txdb))
cds.data <- as.data.frame(cds(txdb))
exons.data <- as.data.frame(exons(txdb))

gene.profile <- function(gene.name,
                          promoter_length=1000,
                          selected.bigwig.files,
                          selected.bed.files,
                          line.colors = "blue",
                          area.colors = "lightblue")
{
  target.gene.body <- genes.data.df[gene.name,]
  target.gene.chr <- as.character(target.gene.body$seqnames)
  target.gene.start <- target.gene.body$start
```

```

target.gene.end <- target.gene.body$end

target.gene.strand <- as.character(target.gene.body$strand)

## Extract cds annotation
cds.data.target.gene <- subset(cds.data,
                               seqnames == target.gene.chr &
                               (start >= target.gene.start &
                                end <= target.gene.end))

## Extract exons annotation
exons.data.target.gene <- subset(exons.data,
                                 seqnames == target.gene.chr &
                                 (start >= target.gene.start &
                                  end <= target.gene.end))

## Determine the genome range to plot including promoter, gene body and 5' UTR
## This depends on whether the gene is on the forward or reverse strand
range.to.plot <- target.gene.body

if(target.gene.strand == "+")
{
  range.to.plot$start <- range.to.plot$start - promoter_length
  range.to.plot$end <- range.to.plot$end + promoter_length
} else if (target.gene.strand == "-")
{
  range.to.plot$end <- range.to.plot$end + promoter_length
  range.to.plot$start <- range.to.plot$start - promoter_length
}

## Compute the length of the genome range to represent
current.length <- range.to.plot$end - range.to.plot$start

## Compute profile in gene
## Since ChIPpeakAnno needs more than one region to plot our region
## is duplicated
regions.plot <- GRanges(rbind(range.to.plot,range.to.plot))

## Import signal from the bigwig files
cvglists <- sapply(selected.bigwig.files, import,
                   format="BigWig",
                   which=regions.plot,
                   as="RleList")

## Compute signal in the region to plot
chip.signal <- featureAlignedSignal(cvglists, regions.plot,
                                   upstream=ceiling(current.length/2),
                                   downstream=ceiling(current.length/2),
                                   n.tile=current.length)

## Compute mean signal
if(target.gene.strand == "+")
{

```

```

    chip.signal.mean <- colMeans(chip.signal[[1]],na.rm = TRUE)
  } else if (target.gene.strand == "-")
  {
    chip.signal.mean <- rev(colMeans(chip.signal[[1]],na.rm = TRUE))
  }

  ## Normalization
  #chip.signal.mean <- 20 * chip.signal.mean / max(chip.signal.mean)
  upper.lim <- 1.1*max(chip.signal.mean)

  gene.height <- -25
  cord.x <- 1:current.length

  ## Exon width to plot
  exon.width <- 2#1#2

  ## Cds width to plot
  cds.width <- 3#1.5#3

  ## Width of the rectangle representing the peak reagon
  peak.width <- 1#0.5#1

  ## Extract exons for target gene
  exons.data.target.gene <- subset(exons.data, seqnames == target.gene.chr & (start >= target.gene.start & end <= target.gene.end))

  ## Transform exon coordinates to current range
  min.pos <- min(exons.data.target.gene$start)

  exons.data.target.gene$start <- exons.data.target.gene$start - min.pos + promoter_length
  exons.data.target.gene$end <- exons.data.target.gene$end - min.pos + promoter_length

  ## Extract cds for target gene
  cds.data.target.gene <- subset(cds.data, seqnames == target.gene.chr & (start >= target.gene.start & end <= target.gene.end))

  cds.data.target.gene$start <- cds.data.target.gene$start - min.pos + promoter_length
  cds.data.target.gene$end <- cds.data.target.gene$end - min.pos + promoter_length

  plot(cord.x, rep(gene.height,length(cord.x)),type="l",col="black",lwd=3,ylab="",
        cex.lab=2,axes=FALSE,xlab="",main="",cex.main=2,
        ylim=c(-35,upper.lim),xlim=c(0,max(cord.x)))

  ## Represent exons
  for(i in 1:nrow(exons.data.target.gene))
  {
    # Determine start/end for each exon
    current.exon.start <- exons.data.target.gene$start[i]
    current.exon.end <- exons.data.target.gene$end[i]

    ## Determine coordinates for each exon polygon and represent it
    exon.x <- c(current.exon.start,current.exon.end,current.exon.end,current.exon.start)
    exon.y <- c(gene.height + exon.width, gene.height + exon.width, gene.height - exon.width, gene.height - exon.width)

    polygon(x = exon.x, y = exon.y, col = "blue",border = "blue")
  }

```

```

}

if(nrow(cds.data.target.gene) > 0)
{
  for(i in 1:nrow(cds.data.target.gene))
  {
    # Determine current cds start/end
    current.cds.start <- cds.data.target.gene$start[i]
    current.cds.end <- cds.data.target.gene$end[i]

    # Determine current cds coordinates for the polygon and represent it
    cds.x <- c(current.cds.start,current.cds.end,current.cds.end,current.cds.start)
    cds.y <- c(gene.height + cds.width, gene.height + cds.width, gene.height - cds.width, gene.height - cds.width)
    polygon(x = cds.x, y = cds.y, col = "blue",border = "blue")
  }
}

## Draw arrow to represent transcription direction
if(target.gene.strand == "+")
{
  lines(c(promoter_length,promoter_length,promoter_length+100),
        y=c(gene.height,gene.height+5,gene.height+5),lwd=3)
  lines(c(promoter_length+50,promoter_length+100),y=c(gene.height+6,gene.height+5),lwd=3)
  lines(c(promoter_length+50,promoter_length+100),y=c(gene.height+4,gene.height+5),lwd=3)
} else if (target.gene.strand == "-")
{
  lines(c(current.length - promoter_length, current.length - promoter_length, current.length - promoter_length - 100),
        y=c(gene.height,gene.height+5,gene.height+5),lwd=3)
  lines(c(current.length - promoter_length-50, current.length - promoter_length - 100),y=c(gene.height+6,gene.height+5),lwd=3)
  lines(c(current.length - promoter_length-50, current.length - promoter_length - 100),y=c(gene.height+4,gene.height+5),lwd=3)
}

## Draw promoter range
if(target.gene.strand == "+")
{
  axis(side = 1,labels = c(- promoter_length, - promoter_length / 2,"TSS"),at = c(1,promoter_length/2))
} else if(target.gene.strand == "-")
{
  axis(side = 1,labels = c("TSS",- promoter_length / 2,- promoter_length),at = c(current.length-promoter_length,
current.length - promoter_length / 2, - promoter_length))
}

## Draw gene name
text(x = current.length / 2, y = -33 ,
     labels = bquote(italic(. (gene.name))),cex = 1.7,font = 3)

## Extract bed file name 1 and read it
current.peaks <- read.table(file=selected.bed.files,header = F, as.is = T)
current.peaks <- current.peaks[,1:3]
colnames(current.peaks) <- c("seqnames","start","end")
peak.coordinates <- subset(current.peaks, seqnames == as.character(range.to.plot$seqnames) & start >=
current.peaks.to.plot <- peak.coordinates[,2:3]

## Transform coordinates
current.peaks.to.plot <- current.peaks.to.plot - range.to.plot$start

```

```

## Check if there are peaks for the target gene
if(nrow(current.peaks.to.plot) > 0)
{
  #motifs.in.peaks <- vector(mode="list", length=nrow(current.peaks.to.plot))
  for(j in 1:nrow(current.peaks.to.plot))
  {
    ## Extract start and end point of each peak region
    current.peak.start <- current.peaks.to.plot[j,1]
    current.peak.end <- current.peaks.to.plot[j,2]

    ## Computer coordinates for polygon and draw it
    peak.x <- c(current.peak.start,current.peak.end,
                current.peak.end,current.peak.start)
    peak.y <- c(peak.width - 12, peak.width - 12,
                - peak.width - 12, - peak.width - 12)

    polygon(x = peak.x, y = peak.y, col = "blue", border = "blue",lwd=2)
  }
}

## Draw profiles if bw file is provided
## Compute base line for current TF
current.base.line <- - 10

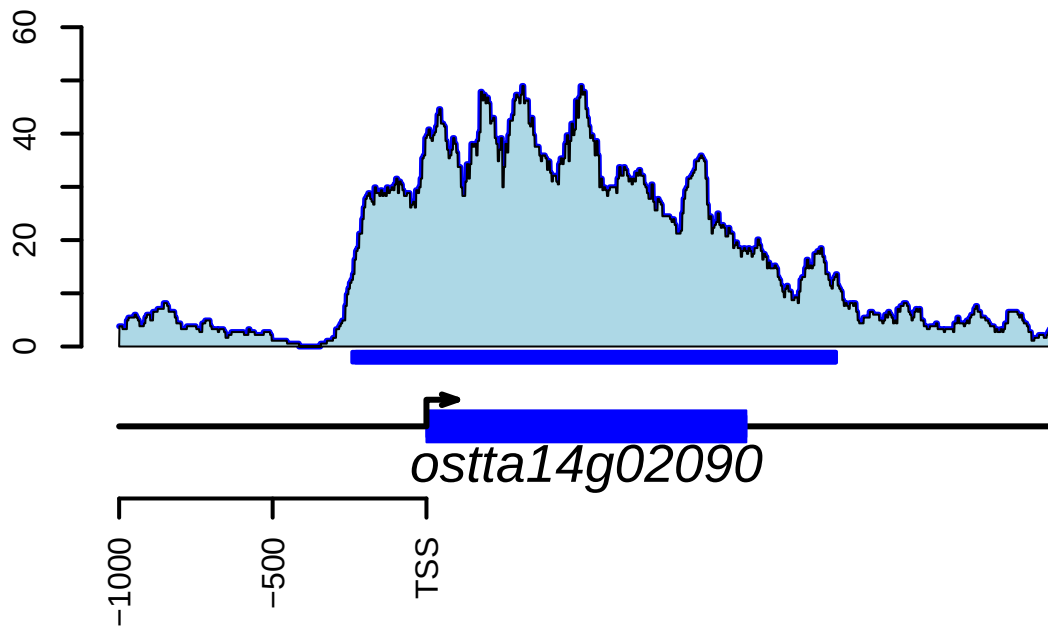
## Represent signal from the current TF
lines(chip.signal.mean+current.base.line,type="l",col=line.colors,lwd=3)
## Determine polygon coordinates and represent it
cord.y <- c(current.base.line,chip.signal.mean+current.base.line,current.base.line)
cord.x <- 1:length(cord.y)
polygon(cord.x,cord.y,col=area.colors)

axis(side = 2,lwd = 2,
      at = seq(from=0,to=ceiling(upper.lim/10)*10,by=10)+current.base.line,
      labels = seq(from=0,to=ceiling(upper.lim/10)*10,by=10))
}

#jpeg(filename = "images/example_ostta14g02090.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta14g02090",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")

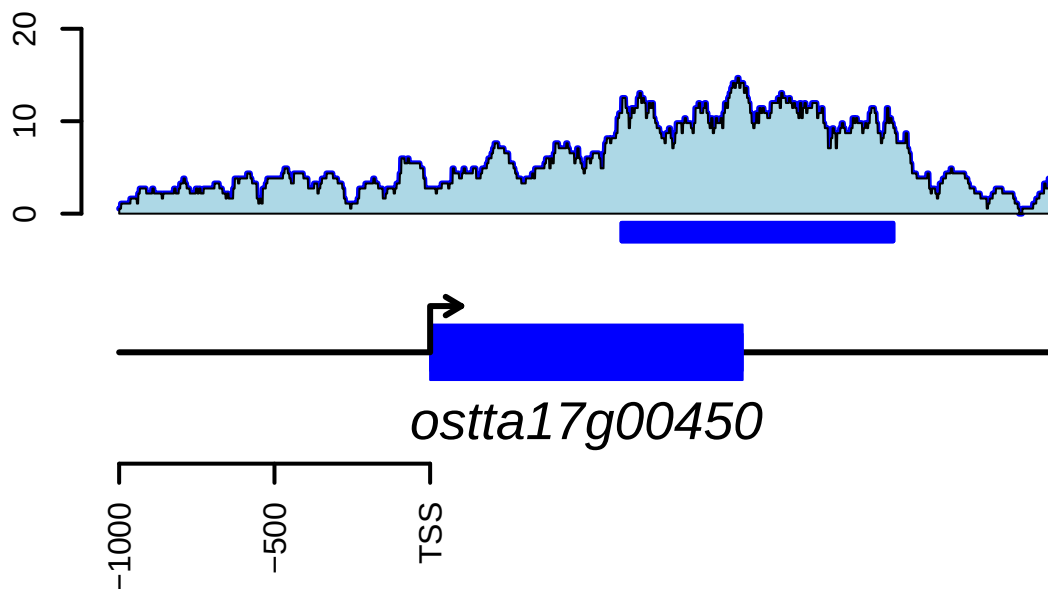
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

```



```
#dev.off()
#jpeg(filename = "images/example_ostta17g00450.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta17g00450",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

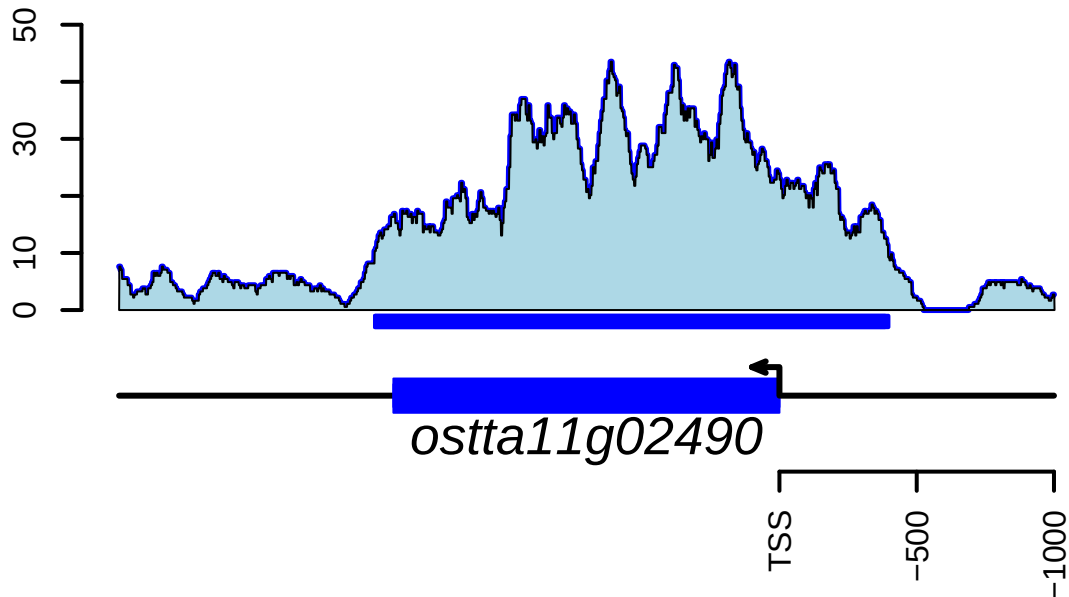
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
#jpeg(filename = "images/example_ostta11g02490.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta11g02490",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
```

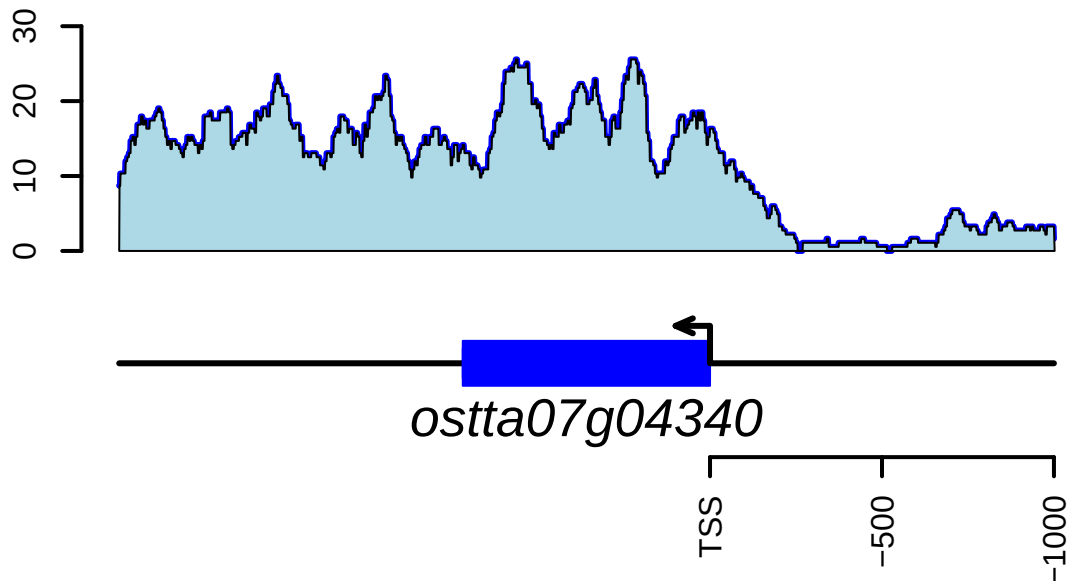
```
selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =  
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



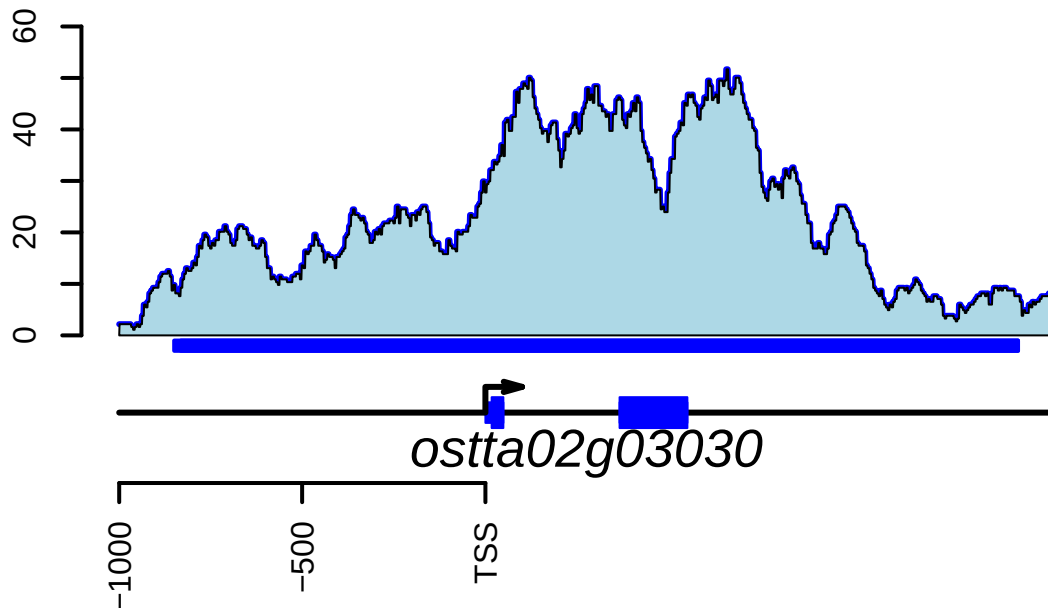
```
#dev.off()  
#jpeg(filename = "images/example_ostta07g04340.jpg",res = 250,width = 3800,height = 1500)  
gene.profile(gene.name="ostta07g04340",  
             promoter_length=1000,  
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",  
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =  
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
#jpeg(filename = "images/example_ostta02g03030.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta02g03030",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
jpeg(filename = "images/example_ostta11g01720.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta11g01720",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

```
dev.off()
```

```
## pdf
## 2
```

```
jpeg(filename = "images/example_ostta05g02370.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta05g02370",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

```
dev.off()
```

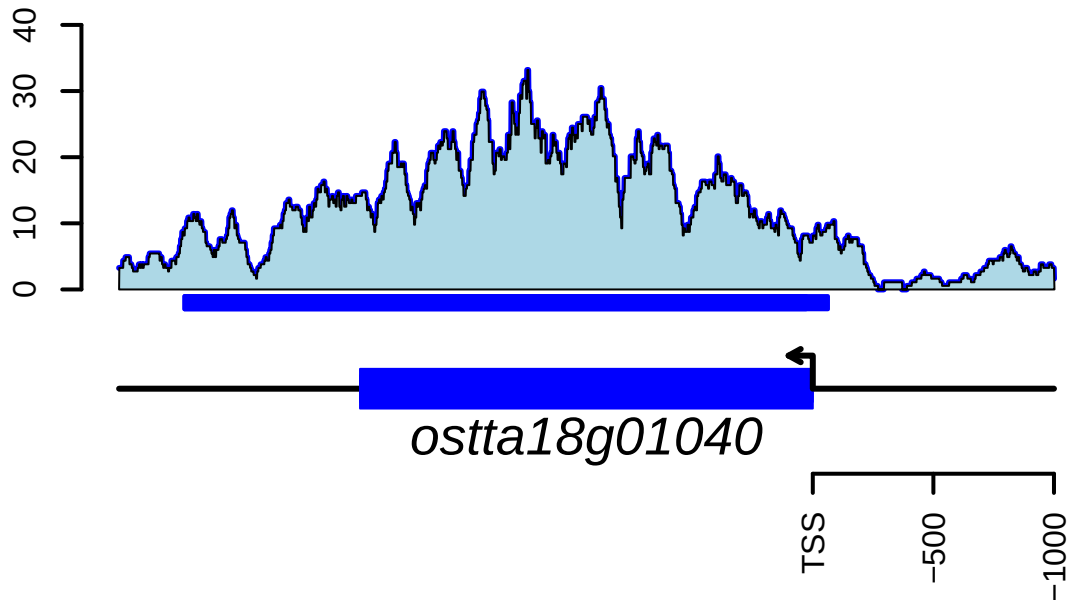
```
## pdf
```

```
## 2
```

Examples of genes involved in microtubule-based movement are ostta18g01040 (kinesin), ostta08g03710 (tubulin alpha), ostta06g04460 and ostta14g00070 (dyenin).

```
#jpeg(filename = "images/example_ostta18g01040.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta18g01040",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

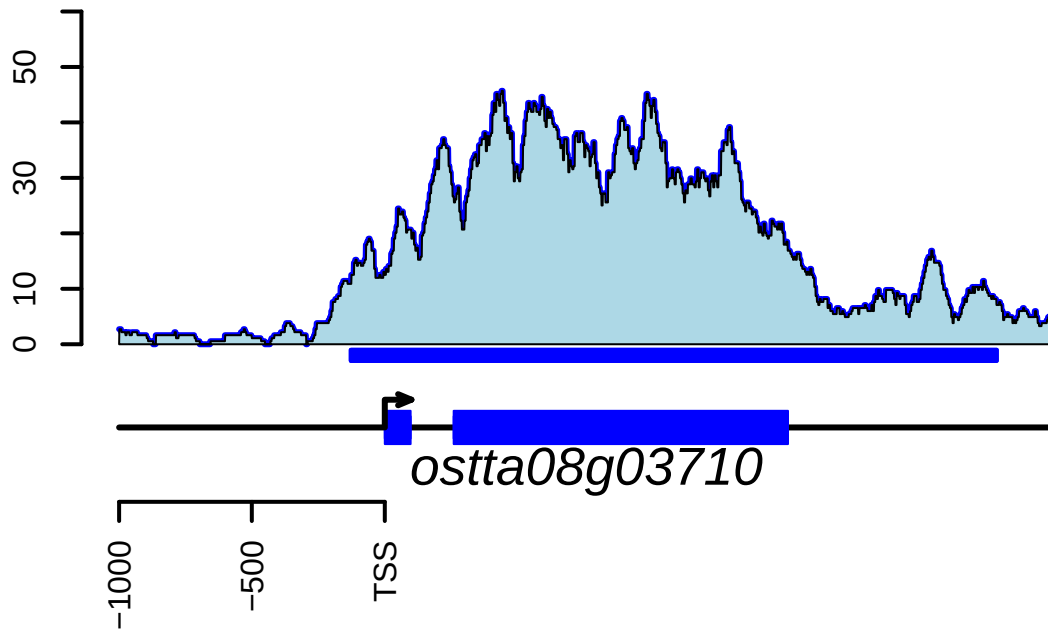
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/example_ostta08g03710.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta08g03710",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

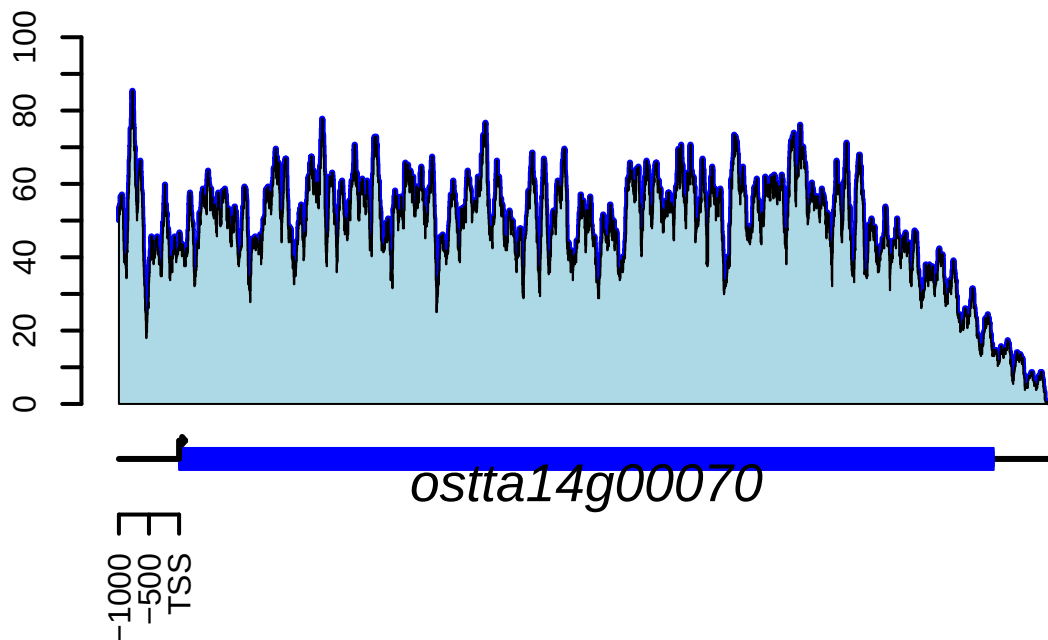
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/example_ostta14g00070.jpg", res = 250, width = 3800, height = 1500)
gene.profile(gene.name="ostta14g00070",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

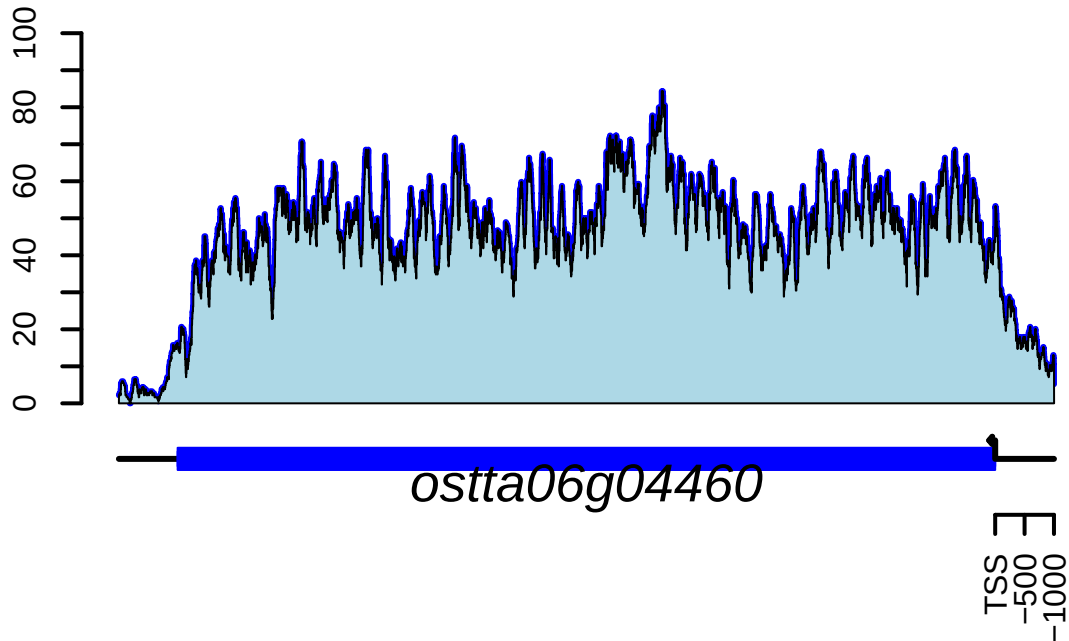
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

```
#jpeg(filename = "images/example_ostta06g04460.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta06g04460",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

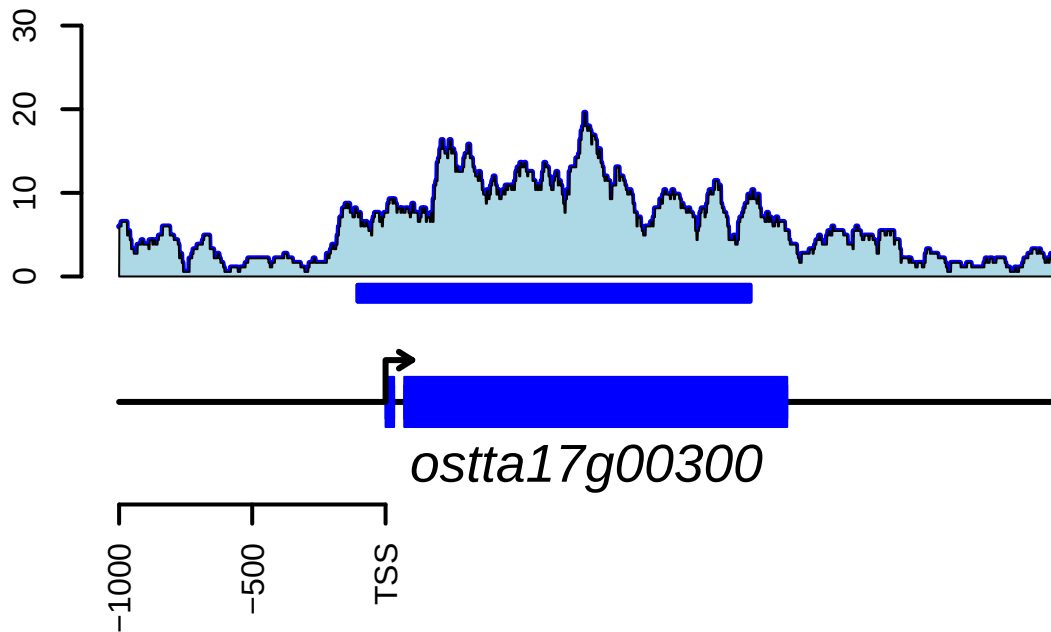


```
#dev.off()
```

Examples of genes involved in fatty acid metabolism are ostta17g00300, enoyl-ACP reductase, a component of the fatty acid synthase complex and ostta03g02570, acetyl-CoA synthetase, and ostta02g00710, ostta02g00720, ostta02g00780, Polyketide synthases.

```
#jpeg(filename = "images/example_ostta17g00300.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta17g00300",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

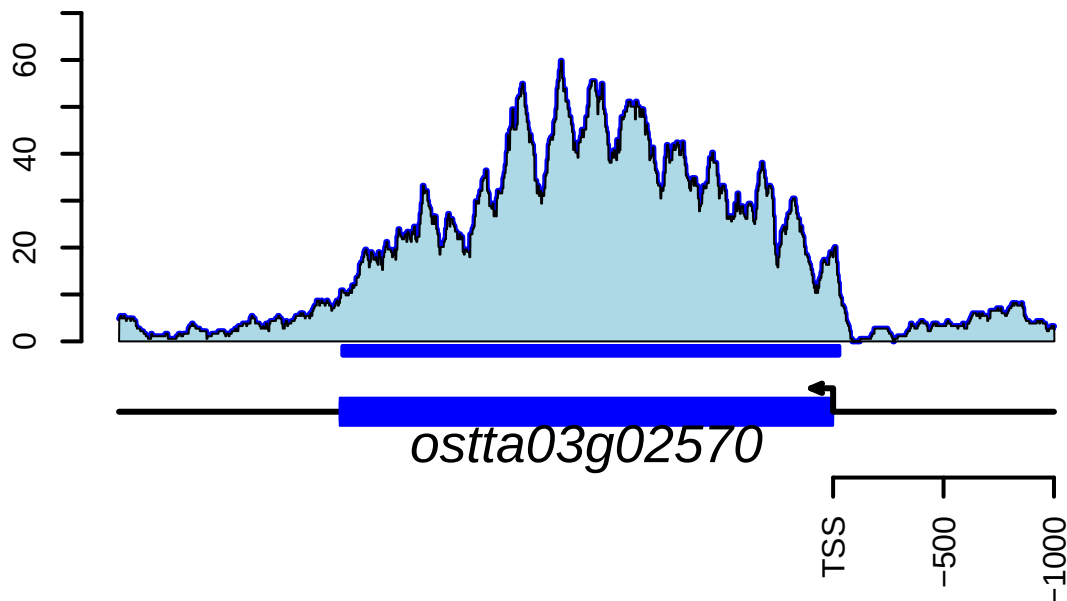
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/example_ostta03g02570.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta03g02570",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/example_ostta02g00710.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta02g00710",
```

```

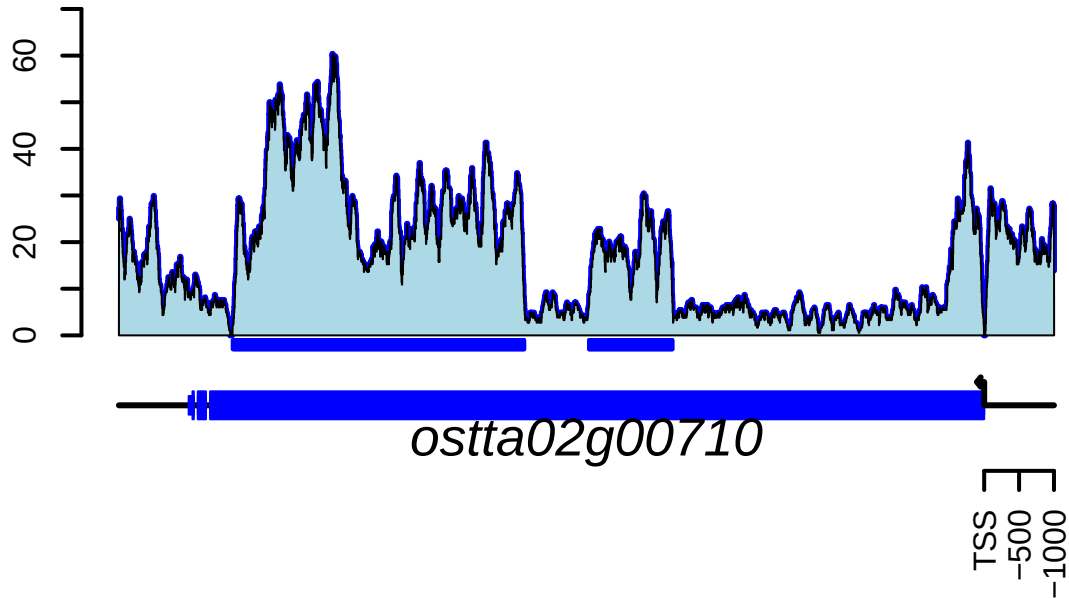
promoter_length=1000,
selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")

```

```

## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

```



```

#dev.off()

```

```

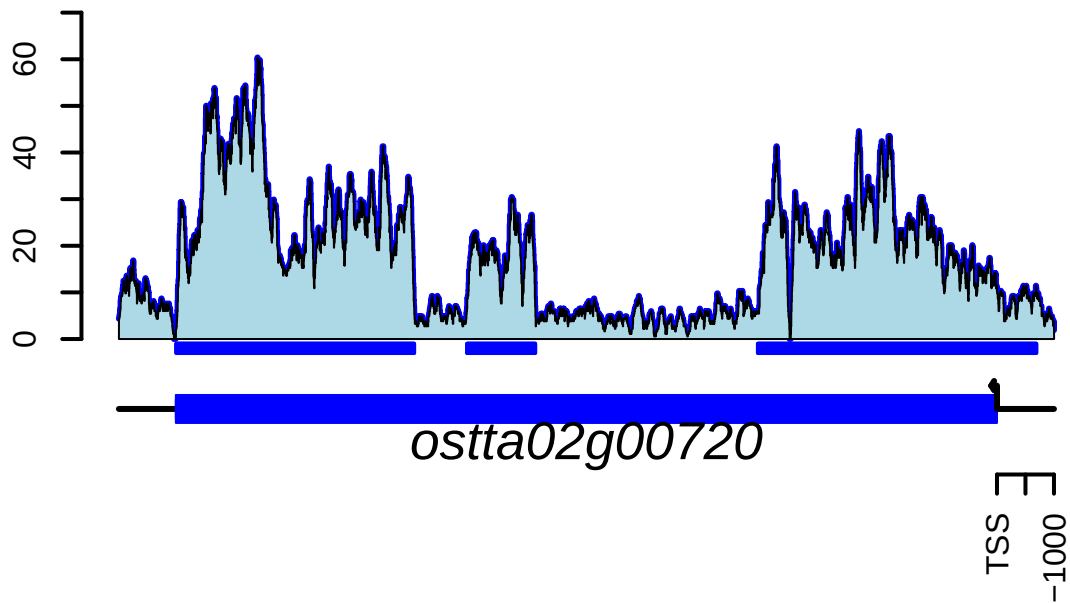
#jpeg(filename = "images/example_ostta02g00720.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta02g00720",
promoter_length=1000,
selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")

```

```

## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

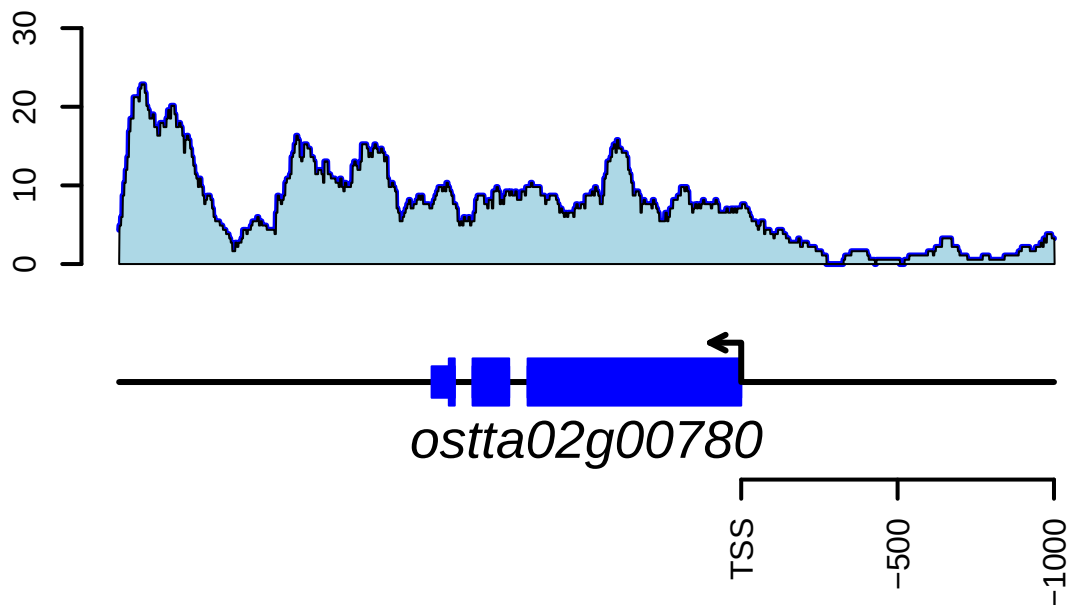
```



```
#dev.off()

#jpeg(filename = "images/example_ostta02g00780.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta02g00780",
             promoter_length=1000,
             selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
             selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

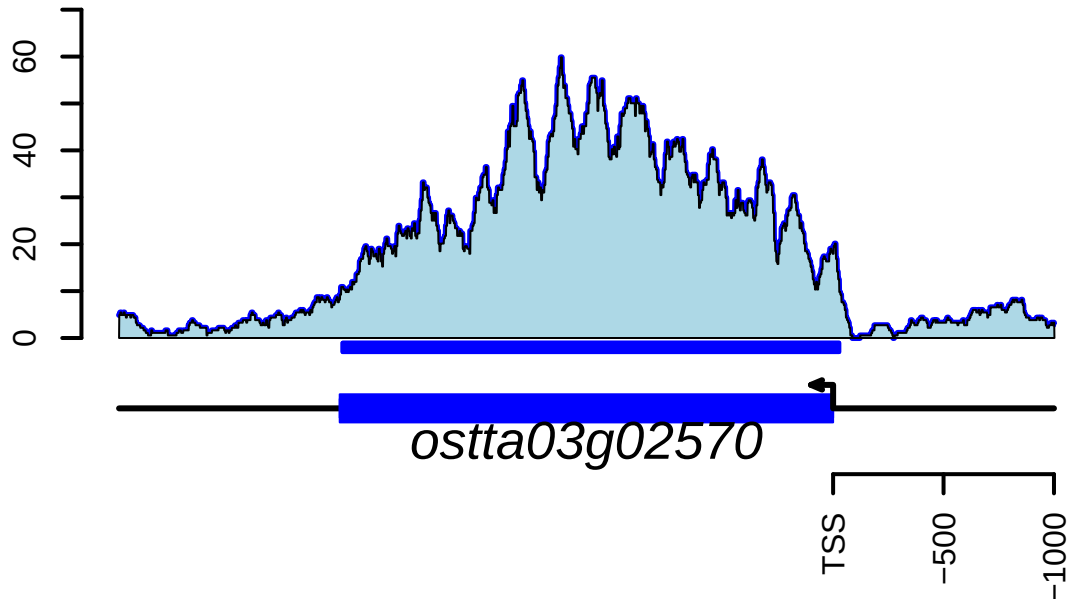


```
#dev.off()

#jpeg(filename = "images/example_ostta03g02570.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta03g02570",
             promoter_length=1000,
```

```
selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

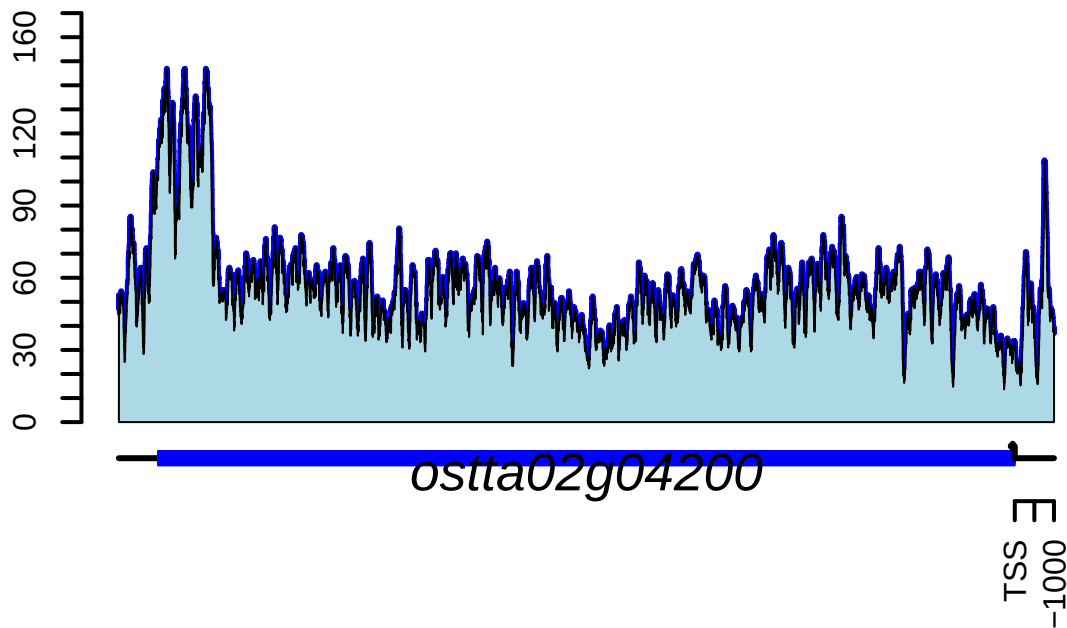
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

```
#jpeg(filename = "images/example_ostta02g04200.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta02g04200",
promoter_length=1000,
selected.bigwig.files="h3k27me3/h3k27me3_ll_20_chip_1.bw",
selected.bed.files="h3k27me3/h3k27me3_ll_20_peaks.bed")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

We also perform KEGG pathway enrichment analysis.

```
gene.ids <- ostta.genes$gene_id
k27.enrich.kegg <- enrichKEGG(gene = paste0("OT_",c(k27.gene.body,k27.tss)),#all.k27.genes),
                             universe = paste0("OT_",gene.ids),
                             organism = "ota",
                             pAdjustMethod = "BH",
                             pvalueCutoff = 0.05)
```

```
## Reading KEGG annotation online: "https://rest.kegg.jp/link/ota/pathway"...
```

```
## Reading KEGG annotation online: "https://rest.kegg.jp/list/pathway/ota"...
```

```
df.activated.enrich.kegg <- as.data.frame(k27.enrich.kegg)
head(df.activated.enrich.kegg)
```

```
##           category subcategory      ID
## ota04814 Cellular Processes Cell motility ota04814
##           Description GeneRatio BgRatio RichFactor
## ota04814 Motor proteins - Ostreococcus tauri      4/21 17/1748 0.2352941
##           FoldEnrichment zScore      pvalue      p.adjust      qvalue
## ota04814      19.58543 8.48904 3.318484e-05 0.0007300665 0.0007300665
##                                           geneID
## ota04814 OT_ostta14g00070/OT_ostta18g01040/OT_ostta06g04460/OT_ostta08g03710
##           Count
## ota04814      4
```

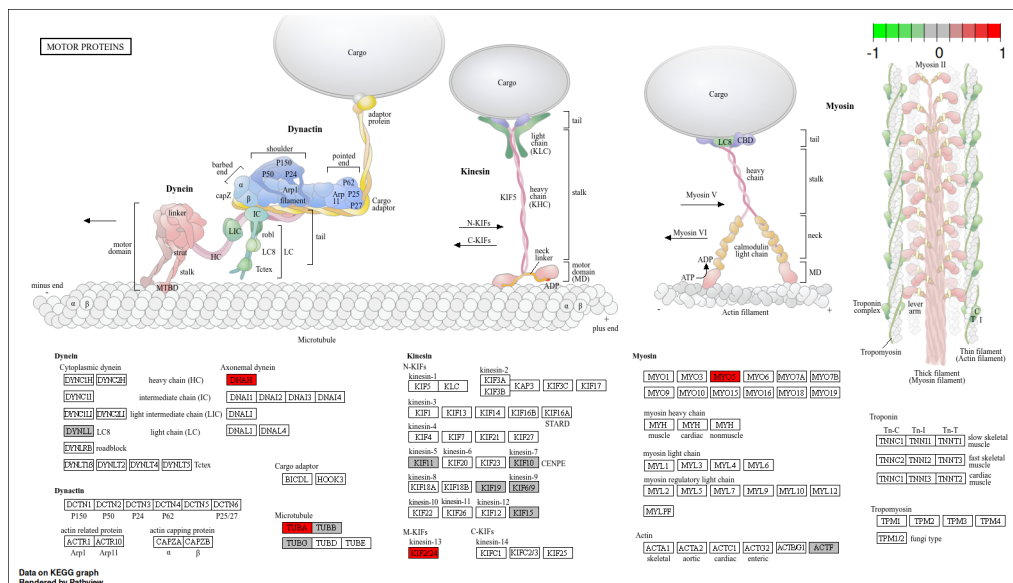
```
library(pathview)
```

```
##
```

```
## #####
## Pathview is an open source software package distributed under GNU General
## Public License version 3 (GPLv3). Details of GPLv3 is available at
## http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
```

```
## The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
## license agreement (details at http://www.kegg.jp/kegg/legal.html).
## #####
```

Consistently we found the cellular process “Cell motility” significantly represented in the H3K27me3 marked genes.



We next explore the association between ChIP signal and gene expression. The Reads Per Kilobase of peak and Million of mapped reads (RPKM) is computed. Marked genes with different levels of ChIP signal according to the quantiles of the distribution are separated into different sets and their expression level represented.

43

```

      # "../chipseq_data/bed_data/h3k27me3/temp20c/k27_20_ll_chip_1_counts.bed",
      header = F, sep = "\t")
count.2 <-
  read.table(
    file=
      "h3k27me3/h3k27me3_ll_20_chip_2_counts.bed",
      # "../chipseq_data/bed_data/h3k27me3/temp20c/k27_20_ll_chip_2n_counts.bed",
      header = F, sep = "\t")
count.3 <-
  read.table(
    file=
      "h3k27me3/h3k27me3_ll_20_chip_3_counts.bed",
      # "../chipseq_data/bed_data/h3k27me3/temp20c/k27_20_ll_chip_3n_counts.bed",
      header = F, sep = "\t")
#count.1 <- count.1[,c(1:3,10)]
#count.2 <- count.2[,c(1:3,10)]
#count.3 <- count.3[,c(1:3,10)]

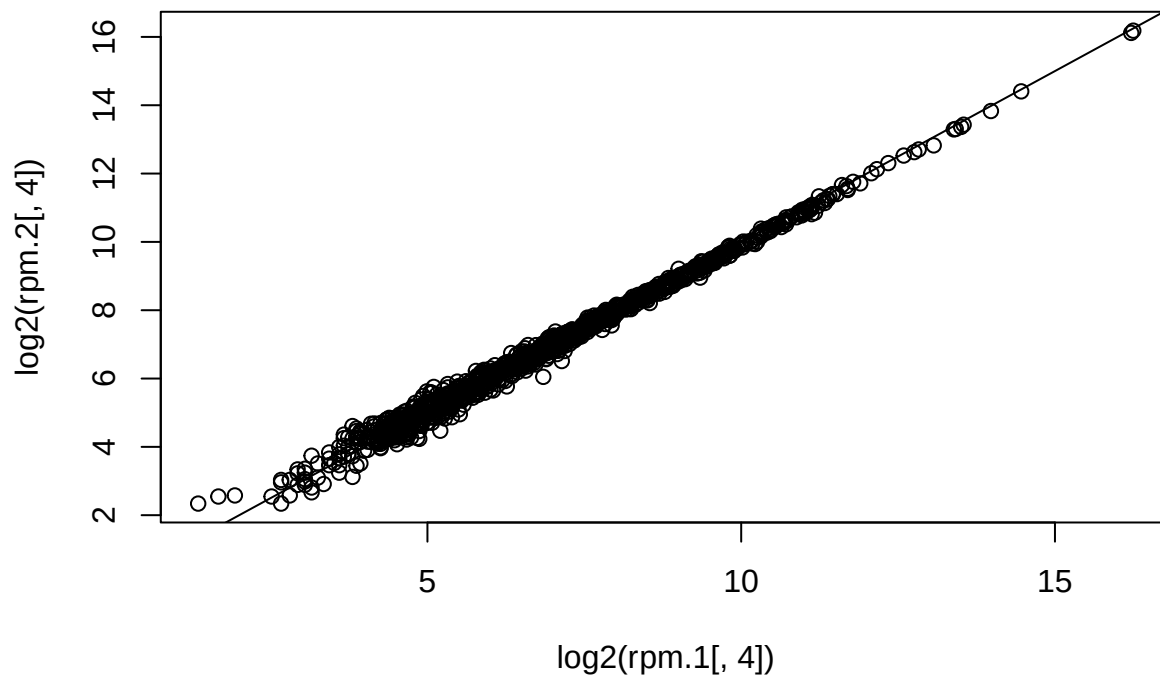
mapped.reads.k27.20c.1 <- 3146662/1E6
mapped.reads.k27.20c.2 <- 7702115/1E6
mapped.reads.k27.20c.3 <- 7102648/1E6

rpm.1 <- count.1
rpm.2 <- count.2
rpm.3 <- count.3

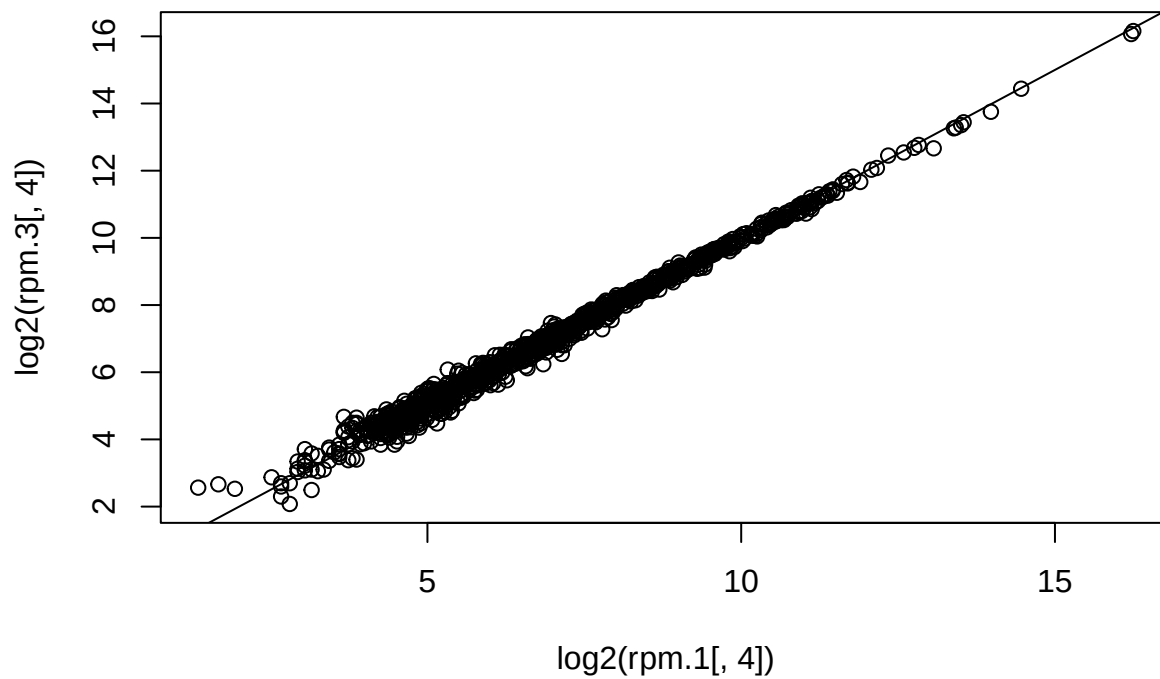
rpm.1[,4] <- rpm.1[,4]/mapped.reads.k27.20c.1
rpm.2[,4] <- rpm.2[,4]/mapped.reads.k27.20c.2
rpm.3[,4] <- rpm.3[,4]/mapped.reads.k27.20c.3

plot(log2(rpm.1[,4]), log2(rpm.2[,4]))
lines(x=c(0,100), y=c(0,100))

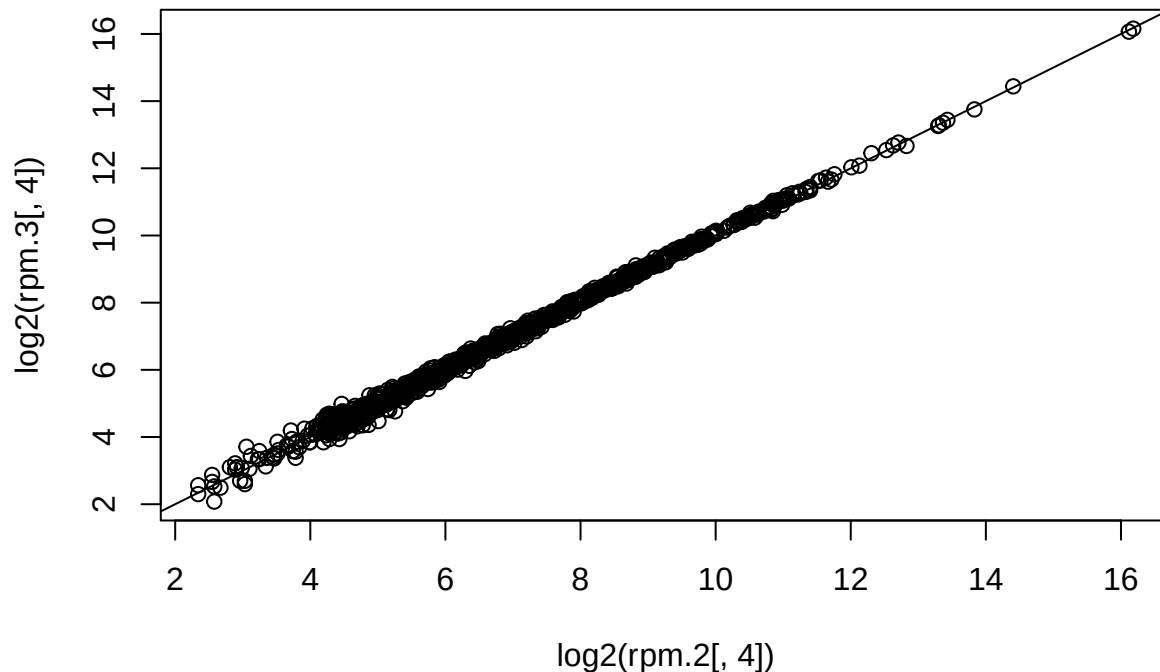
```



```
plot(log2(rpm.1[,4]),log2(rpm.3[,4]))
lines(x=c(0,100),y=c(0,100))
```



```
plot(log2(rpm.2[,4]),log2(rpm.3[,4]))
lines(x=c(0,100),y=c(0,100))
```



```
rpm <- rpm.1
rpm[,4] <- (rpm.1[,4] + rpm.2[,4] + rpm.3[,4])/3

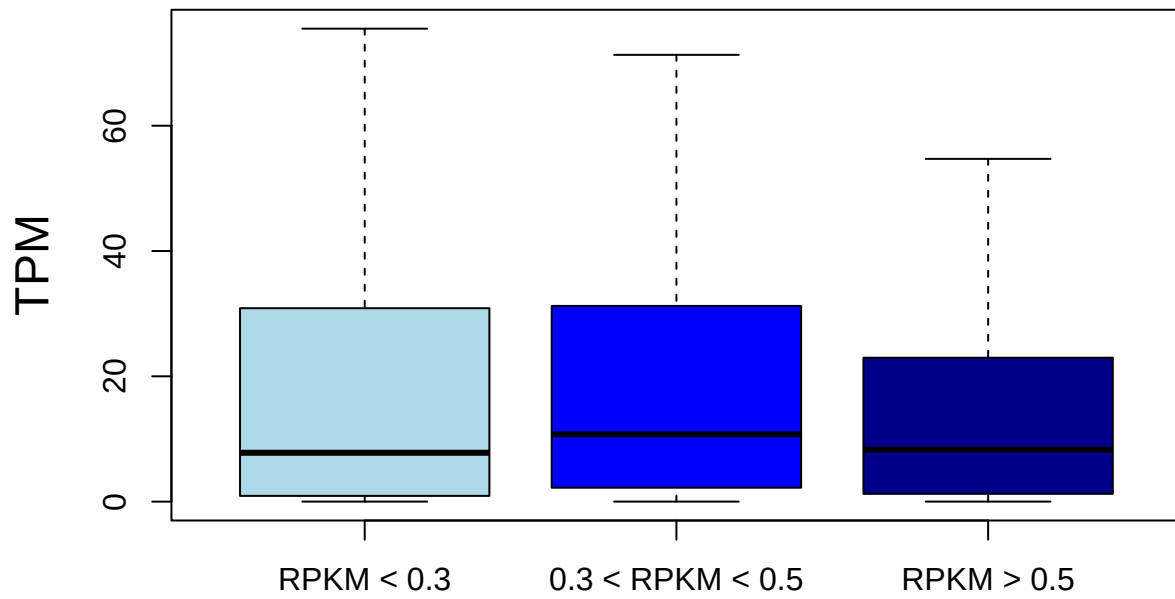
rpkm <- rpm
rpkm[,4] <- (rpm[,4]/(rpm[,3]-rpm[,2]))

genes.2nd.quant <-
  (ll.20.peak.annotation$target.genes)[which(rpkm[,4] < quantile(rpkm[,4])[3])]
genes.2nd.quant <- genes.2nd.quant[genes.2nd.quant != ""]
genes.2nd.quant <- unlist(strsplit(x = genes.2nd.quant, split = ","))

genes.3rd.quant <-
  ll.20.peak.annotation$target.genes[which(rpkm[,4] >= quantile(rpkm[,4])[3] &
                                           rpkm[,4] < quantile(rpkm[,4])[4])]
genes.3rd.quant <- genes.3rd.quant[genes.3rd.quant != ""]
genes.3rd.quant <- unlist(strsplit(x = genes.3rd.quant, split = ","))

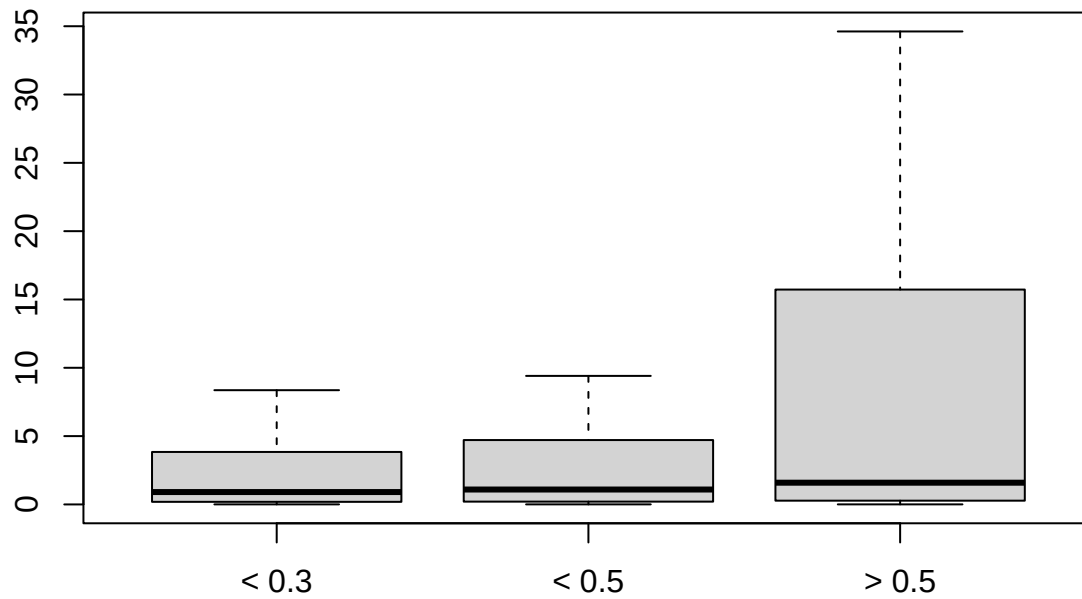
genes.4th.quant <-
  ll.20.peak.annotation$target.genes[which(rpkm[,4] >= quantile(rpkm[,4])[4])]
genes.4th.quant <- genes.4th.quant[genes.4th.quant != ""]
genes.4th.quant <- unlist(strsplit(x = genes.4th.quant, split = ","))

##jpeg(filename = "../images/boxplot_chip_signal_gene_expression.jpg",width = 400,height = 600)
boxplot(gene.expression.20[genes.2nd.quant],
        gene.expression.20[genes.3rd.quant],
        gene.expression.20[genes.4th.quant],
        names=c("RPKM < 0.3", "0.3 < RPKM < 0.5", "RPKM > 0.5"),
        ylab="TPM", cex.lab=1.5, outline=F, col=c("lightblue","blue","darkblue"))
```

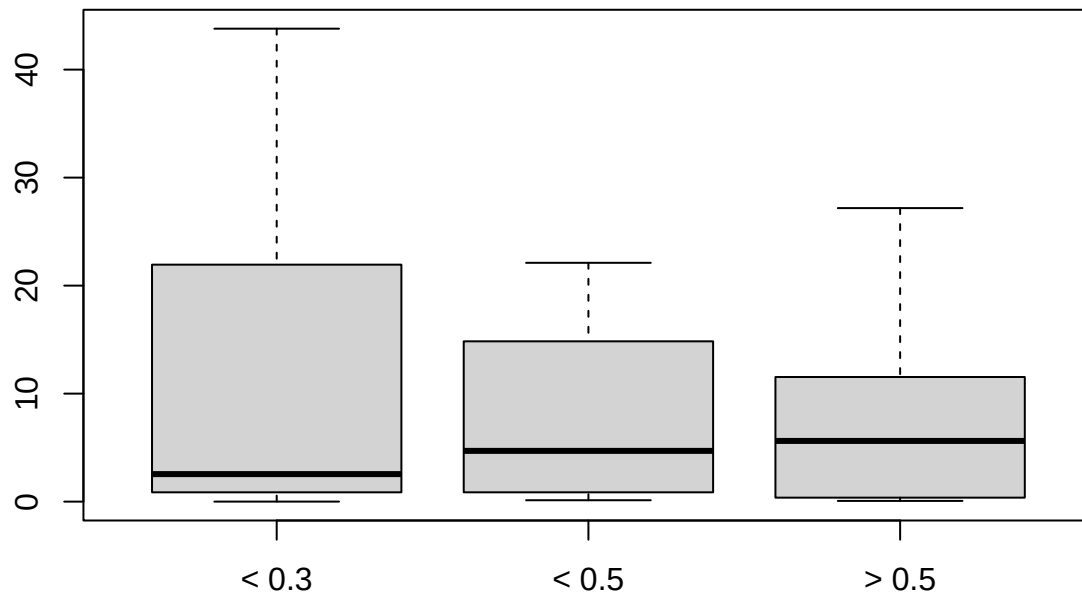


```
##dev.off()
```

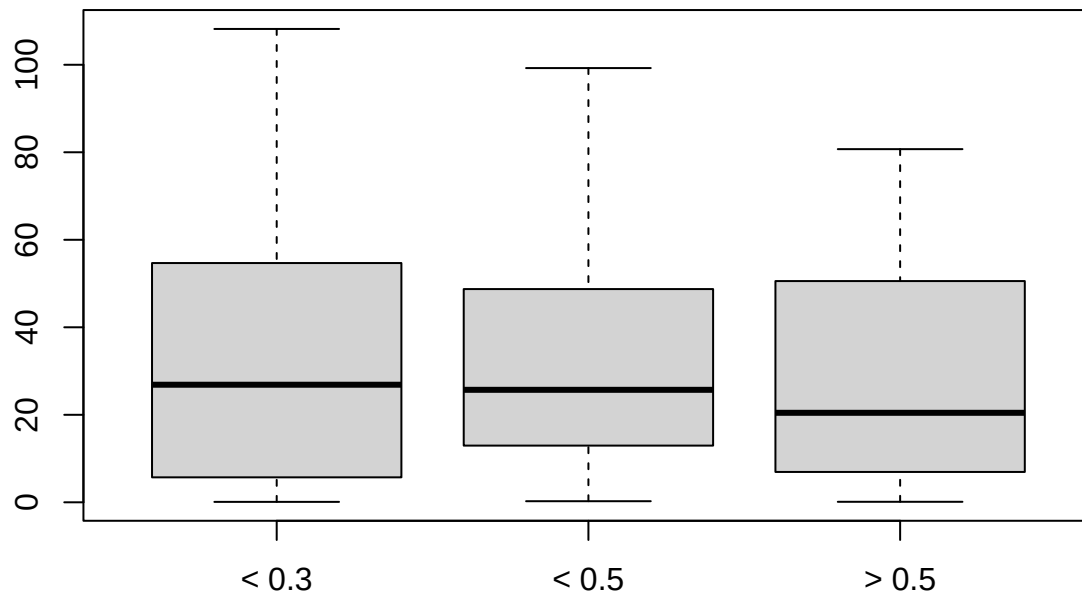
```
boxplot(gene.expression.20[intersect(k27.gene.body,genes.2nd.quant)],
gene.expression.20[intersect(k27.gene.body,genes.3rd.quant)],
gene.expression.20[intersect(k27.gene.body,genes.4th.quant)],
names=c("< 0.3", "< 0.5", "> 0.5"),outline=F)
```



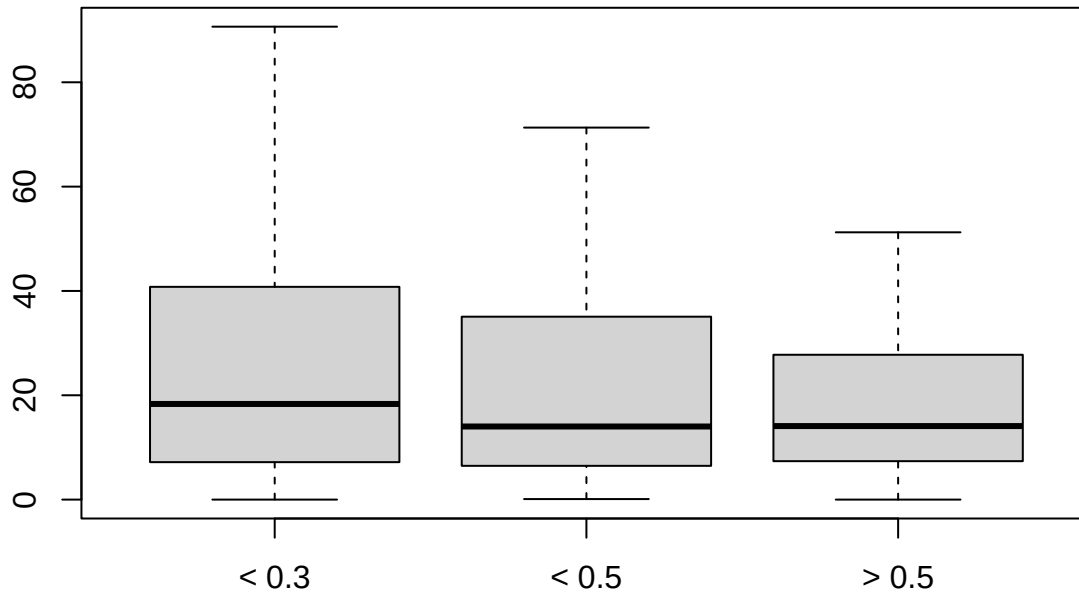
```
boxplot(gene.expression.20[intersect(k27.tss,genes.2nd.quant)],
gene.expression.20[intersect(k27.tss,genes.3rd.quant)],
gene.expression.20[intersect(k27.tss,genes.4th.quant)],
names=c("< 0.3", "< 0.5", "> 0.5"),outline=F)
```



```
boxplot(gene.expression.20[intersect(k27.tes,genes.2nd.quant)],
gene.expression.20[intersect(k27.tes,genes.3rd.quant)],
gene.expression.20[intersect(k27.tes,genes.4th.quant)],
names=c("< 0.3", "< 0.5", "> 0.5"),outline=F)
```



```
boxplot(gene.expression.20[intersect(k27.internal.gene.body,genes.2nd.quant)],
gene.expression.20[intersect(k27.internal.gene.body,genes.3rd.quant)],
gene.expression.20[intersect(k27.internal.gene.body,genes.4th.quant)],
names=c("< 0.3", "< 0.5", "> 0.5"),outline=F)
```



H3K4me3 genome wide distribution analysis

We load the H3K4me3 peaks identified in cultures grown under constant light at 20°C. We represent each chromosome using a rectangle. Each H3K4me3 peak is represented in its precise genome location using a dark orange rectangle.

```
k4.ll.20 <- read.table(file="h3k4me3/h3k4me3_ll_20_peaks.bed",
                      header = F)

k4.ll.20 <- k4.ll.20[,1:3]

nrow(k4.ll.20)

## [1] 1887

#jpeg(filename = "images/h3k27me3_genome_wide_distribution_ll_20.jpg",res = 300,width = 900,height = 900)
#par(mar=c(0,0,0,0))
plot(x=c(0,max.chr.len), y=c(0,4*number.chrs), col = "white", xlab = "", ylab = "", axes=F)

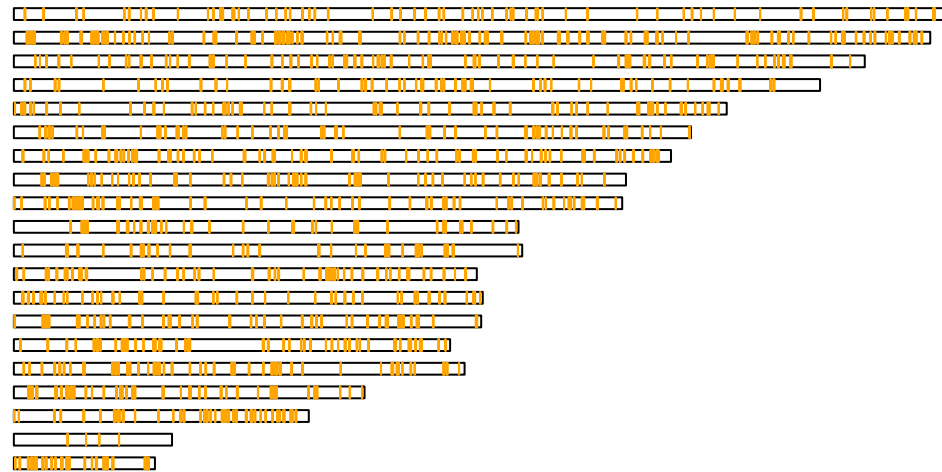
for(i in 1:number.chrs)
{
  polygon(x = c(0, chr.lens[i], chr.lens[i], 0),
         y = c(4*i+1, 4*i+1, 4*i-1, 4*i-1),
         col = "white")
}

for(i in 1:nrow(k4.ll.20))
{
  current.chr <- k4.ll.20[i,1]
  current.start <- k4.ll.20[i,2]
  current.end <- k4.ll.20[i,3]

  current.line <- (20:1)[current.chr]

  polygon(x = c(current.start, current.end, current.end, current.start),
         y = c(4*current.line+1, 4*current.line+1, 4*current.line-1, 4*current.line-1),
```

```
col = "orange",border="orange")
}
```



```
#dev.off()
```

We identified 1887 genomic regions significantly occupied by H3K4me3 in the three different independent biological replicates. We compute the percentage of the whole genome occupied by H3K4me3 and the percentage in each chromosome. A barplot is used to represent the percentage of H3K4me3 in each chromosome. Z-scores and p-values are computed for each chromosome. A gradient from white to dark orange is used to represent significance levels.

```
h3k4me3.peaks.len <- k4.ll.20$V3 - k4.ll.20$V2
sum(h3k4me3.peaks.len)/sum(chr.len)
```

```
## [1] 0.09848293
```

```
k4.ll.20 <- cbind(k4.ll.20,h3k4me3.peaks.len)
colnames(k4.ll.20) <- c("chr","start","end","length")
head(k4.ll.20)
```

```
##   chr start  end length
## 1    1 12592 13416    824
## 2    1 12592 13425    833
## 3    1 34347 35199    852
## 4    1 34338 35218    880
## 5    1 73004 73889    885
## 6    1 73004 73847    843
```

```
number.of.h3k4.peaks.chr <- vector(mode = "numeric",length = 20)
k4.peaks.len.chr <- vector(mode = "numeric",length = 20)
```

```
for(i in 1:20)
{
  number.of.h3k4.peaks.chr[i] <- nrow(subset(k4.ll.20, chr == i))
  k4.peaks.len.chr[i] <- sum(subset(k4.ll.20, chr == i)[,4])
}
```

```
percentage.chr.k4 <- k4.peaks.len.chr/rev(chr.len)
```

```
k4.z.scores <- (percentage.chr.k4 - mean(percentage.chr.k4))/sd(percentage.chr.k4)
```

```

which(k4.z.scores > 2.5)

## [1] 20
which(k4.z.scores < -2.5)

## integer(0)
p.vals <- pnorm((percentage.chr.k4 - mean(percentage.chr.k4))/sd(percentage.chr.k4),lower.tail = F)
p.vals

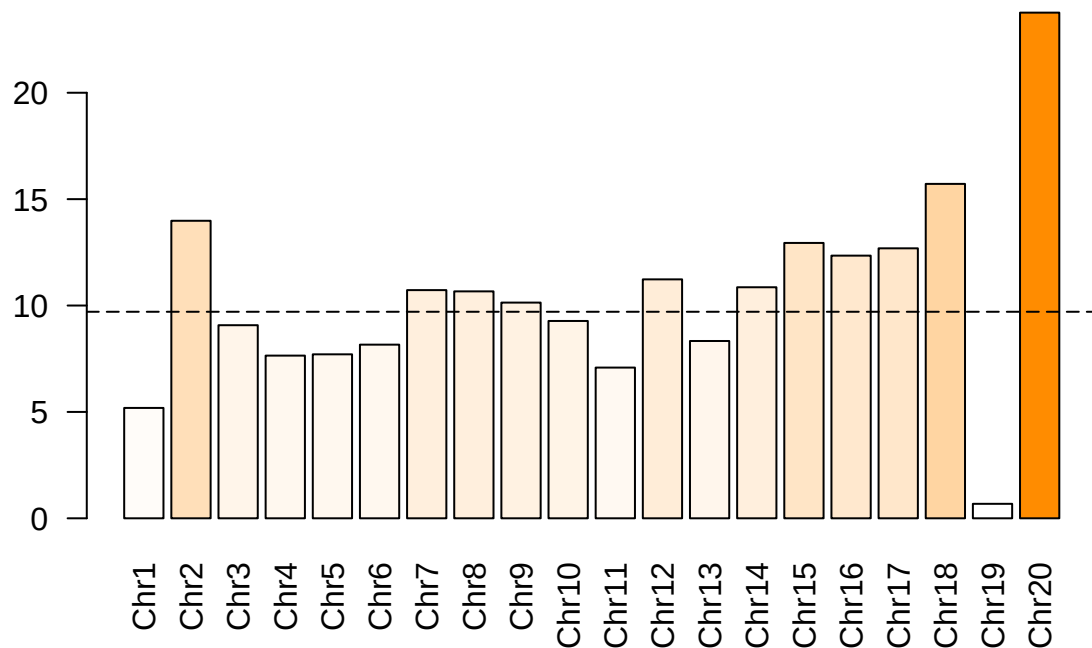
## [1] 0.873663502 0.216966583 0.614994480 0.727536970 0.723105760 0.688829001
## [7] 0.472527598 0.477687215 0.523775546 0.598238166 0.766844241 0.428813799
## [13] 0.675368814 0.460899975 0.289660719 0.335886597 0.308903007 0.122621329
## [19] 0.983449925 0.001723036

values.for.colors <- round(-log10(p.vals)*40+1)
values.for.colors[20] <- 100

white.darkorange.palette <- colorRampPalette(c('white','darkorange'))

#jpeg(filename = "images/h3k27me3_chr_barplot_ll_20.jpg",res = 300,width = 1800,height = 900)
barplot(100*percentage.chr.k4,border="black",names.arg = paste0("Chr",1:20),las=2,col=white.darkorange.palette)
)
abline(h = 100*mean(percentage.chr.k4[-20]),lty=5)

```



```
#dev.off()
```

Approximately 9.8% of the *Ostreococcus* genome is occupied by H3K4me3. H3K4me3 is mostly uniformly distributed over the different chromosome although it significantly accumulates (p-value < 0.01) in Chr20 with p-value 1.7×10^{-3} .

We determine the gene features occupied by each H3K4me3 mark and the H3K4me3 marked genes.

```

library(TxDb.Otauri.JGI)
txdb <- TxDb.Otauri.JGI

```

```

ostta.genes <- as.data.frame(genes(txdb))
head(ostta.genes)

##           seqnames start   end width strand   gene_id
## ostta01g00010      1   244  1932  1689      - ostta01g00010
## ostta01g00020      1  2077  4029  1953      + ostta01g00020
## ostta01g00030      1  4157  5299  1143      + ostta01g00030
## ostta01g00040      1  5392  7041  1650      - ostta01g00040
## ostta01g00050      1  7386  9605  2220      - ostta01g00050
## ostta01g00060      1  9780 10412   633      - ostta01g00060

target.genes <- vector(mode = "character",length = nrow(k4.ll.20))
peak.annotation <- vector(mode = "character",length = nrow(k4.ll.20))

for(i in 1:nrow(k4.ll.20))
{
  current.chr <- k4.ll.20[i,1]
  current.start <- k4.ll.20[i,2]
  current.end <- k4.ll.20[i,3]

  complete.gene.body <- subset(ostta.genes,
                              seqnames == current.chr &
                              start >= current.start &
                              end <= current.end)

  overlap.tss.pos.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   start >= current.start & start <= current.end &
                                   strand == "+")

  overlap.tss.neg.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   end >= current.start & end <= current.end &
                                   strand == "-")

  overlap.tes.pos.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   end >= current.start & end <= current.end &
                                   strand == "+")

  overlap.tes.neg.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   start >= current.start & start <= current.end &
                                   strand == "-")

  inside.gene.body <- subset(ostta.genes, seqnames == current.chr &
                             start <= current.start & end >= current.end)

  if(nrow(complete.gene.body) > 0)
  {
    target.genes[i] <- paste(unlist(complete.gene.body$gene_id),collapse = ",")
    peak.annotation[i] <- "gene.body"
  } else if((nrow(overlap.tss.pos.strand) > 0) | (nrow(overlap.tss.neg.strand) > 0))

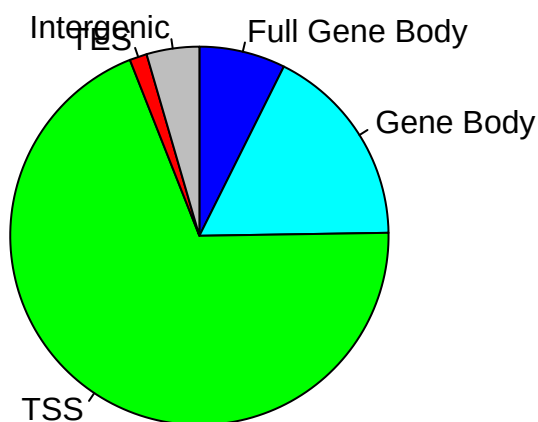
```

```

{
  target.genes[i] <- paste(c(unlist(overlap.tss.pos.strand$gene_id),
                             unlist(overlap.tss.neg.strand$gene_id)),collapse = ",")
  peak.annotation[i] <- "TSS"
} else if(nrow(overlap.tes.pos.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.pos.strand$gene_id),collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(overlap.tes.neg.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.neg.strand$gene_id),collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(inside.gene.body) > 0)
{
  target.genes[i] <- paste(unlist(inside.gene.body$gene_id),collapse = ",")
  peak.annotation[i] <- "internal.gene.body"
} else
{
  peak.annotation[i] <- "intergenic"
}
}

freq.peak.annotation <- table(peak.annotation)
#jpeg(filename = "images/h3k4me3_pie_chart_ll_20.jpg",res = 300,width = 900,height = 900)
#par(mar=c(0,0,0,0))
pie(freq.peak.annotation[c("gene.body",
                           "internal.gene.body",
                           "TSS",
                           "TES",
                           "intergenic")],
    labels = c("Full Gene Body",
               "Gene Body",
               "TSS",
               "TES",
               "Intergenic"),
    clockwise = T, border = "black", col=c("blue", "cyan", "green", "red", "grey"))

```



```

#dev.off()

percentage.freq.peak.annotation <- 100*freq.peak.annotation/sum(freq.peak.annotation)

```

```
percentage.freq.peak.annotation
```

```
## peak.annotation
##           gene.body      intergenic internal.gene.body      TES
##           7.366190      4.504505      17.382088      1.483837
##           TSS
##           69.263381

k4.ll.20.peak.annotation <- cbind(k4.ll.20[,1:3],data.frame(peak.annotation,target.genes))

head(k4.ll.20.peak.annotation)
```

```
##   chr start  end peak.annotation      target.genes
## 1   1 12592 13416      TSS ostta01g00090,ostta01g00080
## 2   1 12592 13425      TSS ostta01g00090,ostta01g00080
## 3   1 34347 35199      TSS ostta01g00220,ostta01g00210
## 4   1 34338 35218      TSS ostta01g00220,ostta01g00210
## 5   1 73004 73889      TSS      ostta01g00485
## 6   1 73004 73847      TSS      ostta01g00485
```

```
write.table(x = k4.ll.20.peak.annotation,file = "tables/H3K4me3_LL_20C_peak_annotation.tsv",quote = F,s
```

```
all.k4.genes <- unique(unlist(sapply(X =k4.ll.20.peak.annotation$target.genes,
                                   FUN = function(x) {strsplit(x,",")[[1]]})))
names(all.k4.genes) <- NULL
length(all.k4.genes)
```

```
## [1] 1247
```

```
length(all.k4.genes)/7668
```

```
## [1] 0.1626239
```

We found 1247 genes associated to H3K4me3 marks corresponding to 16.3% of the entire *Ostreococcus* genome. H3K4me3 is mostly located occupying the TSS (69.3%) with marginal percentages for internal regions of gene body (17.4%), intergenic regions (4.5%), entire gene bodies (7.3%) and TES (1.5%).

Next we use metageneplots to determine the shape of the H3K4me3 over its occupied genes. We consider three different sets of marked genes, genes with their entire gene body marked, genes with H3K4me3 inside their gene body with no overlap in the TSS or TES and genes with H3K4me3 overlapping their TSS or TES

Next we generate the metageneplots for genes where the H3K4me3 occupies their entire gene body including TSS and TES.

```
library(rtracklayer)
## Genes with their entire gene body marked
k4.gene.body <- unique(unlist(sapply(X = subset(k4.ll.20.peak.annotation,
                                              peak.annotation == "gene.body")["target.genes"],
                                  FUN = function(x) {strsplit(x,",")[[1]]})))
names(k4.gene.body) <- NULL
length(k4.gene.body)
```

```
## [1] 86
```

```
## Genes with H3K27 inside their gene body with no overlap
## in the TSS or TES.
```

```
k4.internal.gene.body <- unique(unlist(sapply(X = subset(k4.ll.20.peak.annotation,
                                                         peak.annotation == "internal.gene.body")["target.genes"],
```

```

FUN = function(x) {strsplit(x, ",")[[1]]})

names(k4.internal.gene.body) <- NULL
length(k4.internal.gene.body)

## [1] 244

## Genes with H3K27me3 overlapping their TSS
k4.tss <- unique(unlist(sapply(X = subset(k4.ll.20.peak.annotation,
                                          peak.annotation == "TSS")["target.genes"],
                              FUN = function(x) {strsplit(x, ",")[[1]]})))

names(k4.tss) <- NULL
length(k4.tss)

## [1] 936

## Genes with H3K27me3 overlapping their TES
k4.tes <- unique(unlist(sapply(X = subset(k4.ll.20.peak.annotation,
                                          peak.annotation == "TES")["target.genes"],
                              FUN = function(x) {strsplit(x, ",")[[1]]})))

names(k4.tes) <- NULL
length(k4.tes)

## [1] 22

metagenesplot.data.k4.gene.body.1 <-
  metagenesplot(gene.names = k4.gene.body,
               genes = ostta.genes,
               bw.file.name = "h3k4me3/h3k4me3_ll_20_chip_1.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenesplot.data.k4.gene.body.1 <-
  apply(X = metagenesplot.data.k4.gene.body.1,
        MARGIN = 2,
        FUN = median, na.rm = T)

metagenesplot.data.k4.gene.body.2 <-
  metagenesplot(gene.names = k4.gene.body,
               genes = ostta.genes,
               bw.file.name = "h3k4me3/h3k4me3_ll_20_chip_2.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenesplot.data.k4.gene.body.2 <-
  apply(X = metagenesplot.data.k4.gene.body.2,
        MARGIN = 2,
        FUN = median, na.rm = T)

metagenesplot.data.k4.gene.body.3 <-
  metagenesplot(gene.names = k4.gene.body,
               genes = ostta.genes,
               bw.file.name = "h3k4me3/h3k4me3_ll_20_chip_3.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenesplot.data.k4.gene.body.3 <-
  apply(X = metagenesplot.data.k4.gene.body.3,
        MARGIN = 2,
        FUN = median, na.rm = T)

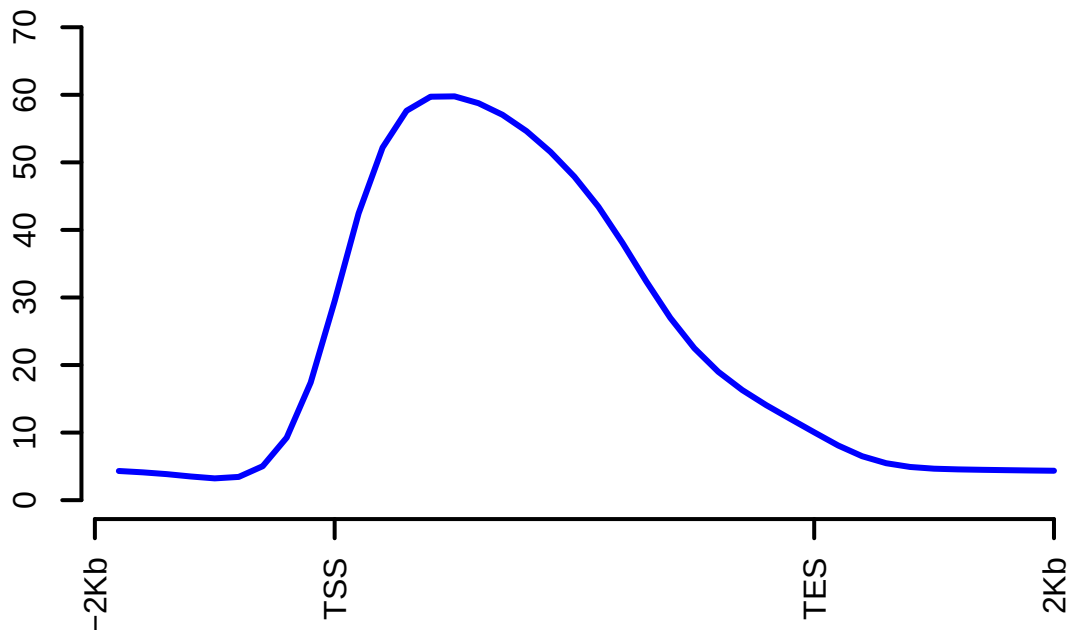
```

```

median.metagenesplot.data.k4.gene.body <- (median.metagenesplot.data.k4.gene.body.1 +
      median.metagenesplot.data.k4.gene.body.2 +
      median.metagenesplot.data.k4.gene.body.3)/3

plot(smooth.spline(x=1:40,y=median.metagenesplot.data.k4.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,70),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb", "TSS", "TES", "2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metagenesplots for genes where the H3K4me3 occupies part of the gene body without overlapping with the TSS and TES.

```

metagenesplot.data.k4.internal.gene.body.1 <-
  metagenesplot(gene.names = k4.internal.gene.body,
    genes = ostta.genes,
    bw.file.name = "h3k4me3/h3k4me3_11_20_chip_1.bw",
    ntile.txs = 10,
    ntile.gene.body = 20)
median.metagenesplot.data.k4.internal.gene.body.1 <-
  apply(X = metagenesplot.data.k4.internal.gene.body.1,
    MARGIN = 2,
    FUN = median,na.rm = T)

metagenesplot.data.k4.internal.gene.body.2 <-
  metagenesplot(gene.names = k4.internal.gene.body,
    genes = ostta.genes,
    bw.file.name = "h3k4me3/h3k4me3_11_20_chip_2.bw",
    ntile.txs = 10,
    ntile.gene.body = 20)
median.metagenesplot.data.k4.internal.gene.body.2 <-
  apply(X = metagenesplot.data.k4.internal.gene.body.2,
    MARGIN = 2,
    FUN = median,na.rm = T)

```

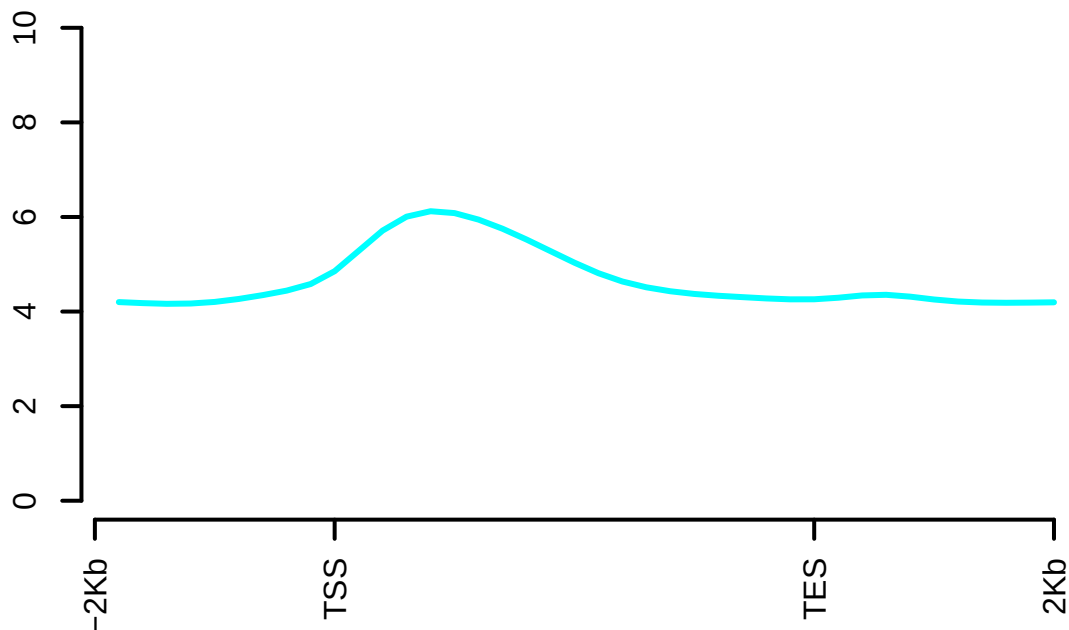
```

metagenesplot.data.k4.internal.gene.body.3 <-
  metagenesplot(gene.names = k4.internal.gene.body,
    genes = ostta.genes,
    bw.file.name = "h3k4me3/h3k4me3_11_20_chip_3.bw",
    ntile.txs = 10,
    ntile.gene.body = 20)
median.metagenesplot.data.k4.internal.gene.body.3 <-
  apply(X = metagenesplot.data.k4.internal.gene.body.3,
    MARGIN = 2,
    FUN = median, na.rm = T)

median.metagenesplot.data.k4.internal.gene.body <- (median.metagenesplot.data.k4.internal.gene.body.1 +
  median.metagenesplot.data.k4.internal.gene.body.2 +
  median.metagenesplot.data.k4.internal.gene.body.3)/3

plot(smooth.spline(x=1:40,y=median.metagenesplot.data.k4.internal.gene.body,spar = 0.5),
  type="l",col="cyan",lwd=3,ylim=c(0,10),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metagenesplots for genes where the H3K4me3 occupies only the TSS and the beginning of the gene.

```

metagenesplot.data.k4.tss.1 <-
  metagenesplot(gene.names = k4.tss,
    genes = ostta.genes,
    bw.file.name = "h3k4me3/h3k4me3_11_20_chip_1.bw",
    ntile.txs = 10,
    ntile.gene.body = 20)
median.metagenesplot.data.k4.tss.1 <-
  apply(X = metagenesplot.data.k4.tss.1,
    MARGIN = 2,
    FUN = median, na.rm = T)

```

```

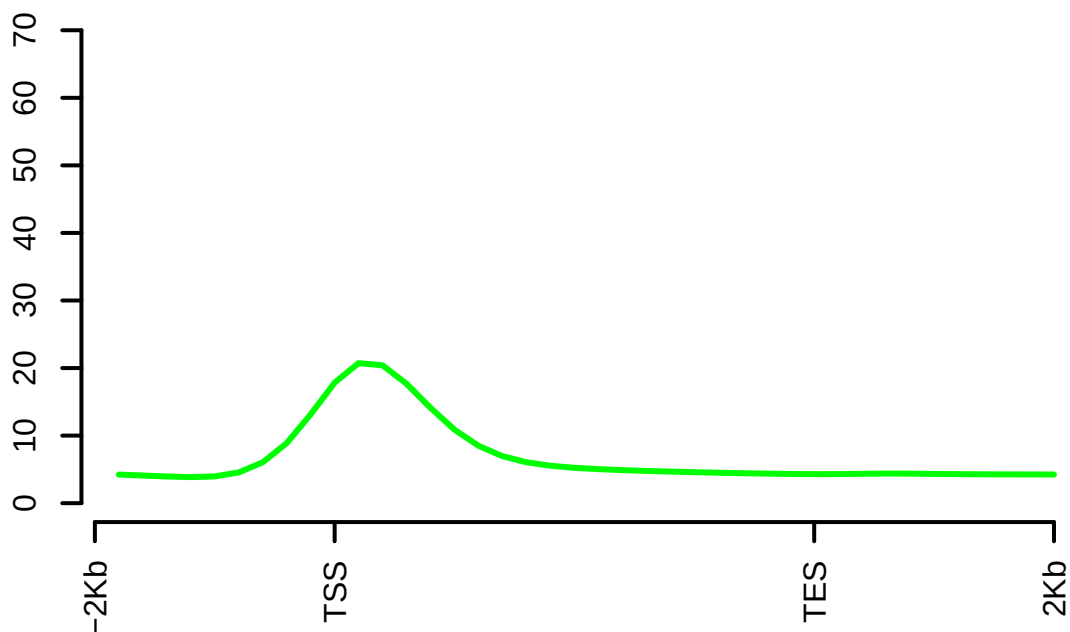
metageneplot.data.k4.tss.2 <-
  metageneplot(gene.names = k4.tss,
               genes = ostta.genes,
               bw.file.name = "h3k4me3/h3k4me3_1l_20_chip_2.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k4.tss.2 <-
  apply(X = metageneplot.data.k4.tss.2,
        MARGIN = 2,
        FUN = median, na.rm = T)

metageneplot.data.k4.tss.3 <-
  metageneplot(gene.names = k4.tss,
               genes = ostta.genes,
               bw.file.name = "h3k4me3/h3k4me3_1l_20_chip_3.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k4.tss.3 <-
  apply(X = metageneplot.data.k4.tss.3,
        MARGIN = 2,
        FUN = median, na.rm = T)

median.metageneplot.data.k4.tss <-
  (median.metageneplot.data.k4.tss.1 +
   median.metageneplot.data.k4.tss.2 +
   median.metageneplot.data.k4.tss.3)/3

plot(smooth.spline(x=1:40,y=median.metageneplot.data.k4.tss,spar = 0.5),
     type="l",col="green",lwd=3,ylim=c(0,70),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metageneplots for genes where the H3K4me3 occupies only the TES and the end of the gene.

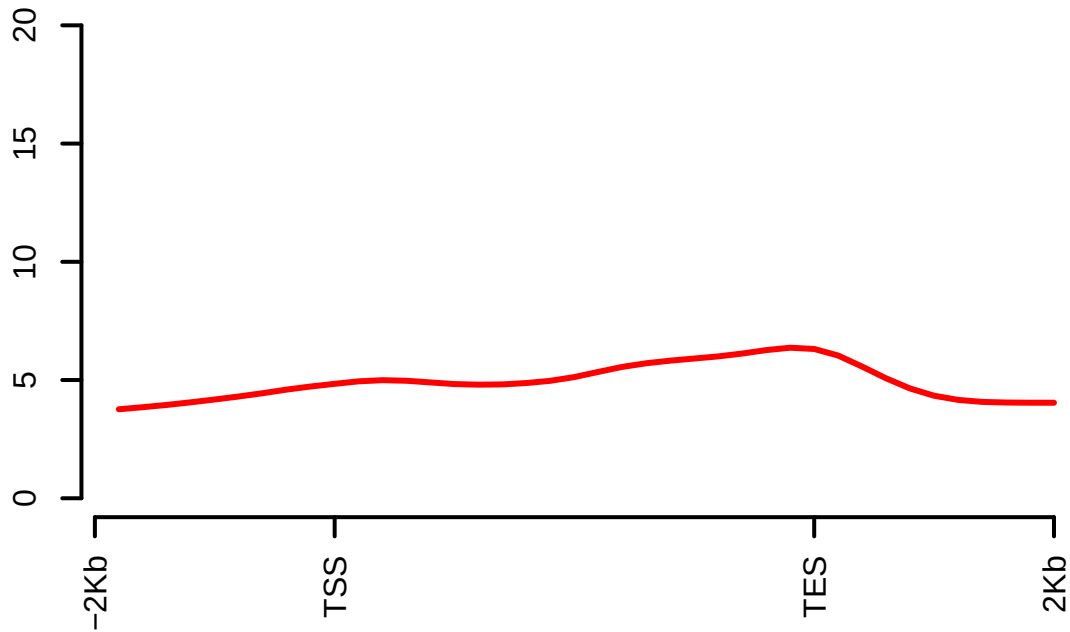
```
metageneplot.data.k4.tes.1 <-
  metageneplot(gene.names = k4.tes,
               genes = ostta.genes,
               bw.file.name = "h3k4me3/h3k4me3_1l_20_chip_1.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k4.tes.1 <-
  apply(X = metageneplot.data.k4.tes.1,
        MARGIN = 2,
        FUN = median, na.rm = T)

metageneplot.data.k4.tes.2 <-
  metageneplot(gene.names = k4.tes,
               genes = ostta.genes,
               bw.file.name = "h3k4me3/h3k4me3_1l_20_chip_2.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k4.tes.2 <-
  apply(X = metageneplot.data.k4.tes.2,
        MARGIN = 2,
        FUN = median, na.rm = T)

metageneplot.data.k4.tes.3 <-
  metageneplot(gene.names = k4.tes,
               genes = ostta.genes,
               bw.file.name = "h3k4me3/h3k4me3_1l_20_chip_3.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k4.tes.3 <-
  apply(X = metageneplot.data.k4.tes.3,
        MARGIN = 2,
        FUN = median, na.rm = T)

median.metageneplot.data.k4.tes <-
  (median.metageneplot.data.k4.tes.1 +
   median.metageneplot.data.k4.tes.2 +
   median.metageneplot.data.k4.tes.3)/3

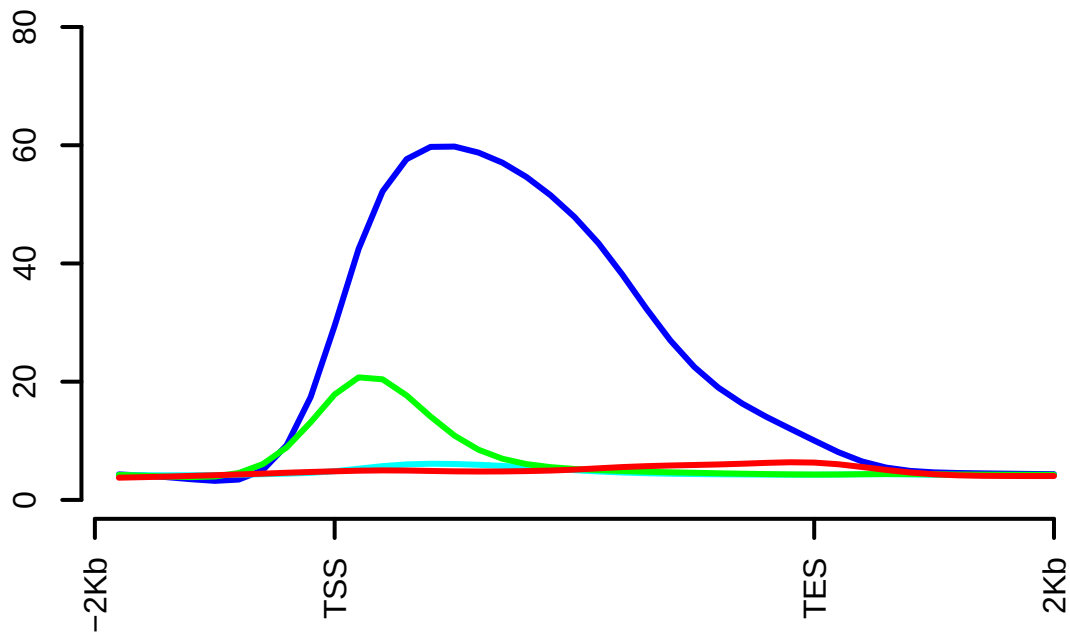
plot(smooth.spline(x=1:40,y=median.metageneplot.data.k4.tes,spar = 0.5),
     type="l",col="red",lwd=3,ylim=c(0,20),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)
```



In order to com-

pare the different metagenepoints are represented in the same plot.

```
#jpeg(filename = "images/h3k4me3_metagenepoints_2.jpg",res = 250,width = 900,height = 900)
plot(smooth.spline(x=1:40,y=median.metagenepoint.data.k4.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,80),axes=F,xlab="",ylab="")
lines(smooth.spline(x=1:40,y=median.metagenepoint.data.k4.internal.gene.body,spar = 0.5),
      type="l",col="cyan",lwd=3)
lines(smooth.spline(x=1:40,y=median.metagenepoint.data.k4.tss,spar = 0.5),
      type="l",col="green",lwd=3)
lines(smooth.spline(x=1:40,y=median.metagenepoint.data.k4.tes,spar = 0.5),
      type="l",col="red",lwd=3)
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb", "TSS", "TES", "2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)
```



```
#dev.off()
```

It can be observed that the level of H3K4me3 mark varies depending on the part of gene that it occupies. Next we check if these differences are significant.

```
max.metagenesplot.data.k4.gene.body.1 <-  
  apply(X = metagenesplot.data.k4.gene.body.1,  
        MARGIN = 1,  
        FUN = max, na.rm = T)  
max.metagenesplot.data.k4.gene.body.2 <-  
  apply(X = metagenesplot.data.k4.gene.body.2,  
        MARGIN = 1,  
        FUN = max, na.rm = T)  
max.metagenesplot.data.k4.gene.body.3 <-  
  apply(X = metagenesplot.data.k4.gene.body.3,  
        MARGIN = 1,  
        FUN = max, na.rm = T)  
  
max.metagenesplot.data.k4.gene.body <- (max.metagenesplot.data.k4.gene.body.1 +  
                                         max.metagenesplot.data.k4.gene.body.2 +  
                                         max.metagenesplot.data.k4.gene.body.3)/3  
  
max.metagenesplot.data.k4.internal.gene.body.1 <-  
  apply(X = metagenesplot.data.k4.internal.gene.body.1,  
        MARGIN = 1,  
        FUN = max, na.rm = T)  
max.metagenesplot.data.k4.internal.gene.body.2 <-  
  apply(X = metagenesplot.data.k4.internal.gene.body.2,  
        MARGIN = 1,  
        FUN = max, na.rm = T)  
max.metagenesplot.data.k4.internal.gene.body.3 <-  
  apply(X = metagenesplot.data.k4.internal.gene.body.3,  
        MARGIN = 1,  
        FUN = max, na.rm = T)  
  
max.metagenesplot.data.k4.internal.gene.body <- (max.metagenesplot.data.k4.internal.gene.body.1 +  
                                                  max.metagenesplot.data.k4.internal.gene.body.2 +  
                                                  max.metagenesplot.data.k4.internal.gene.body.3)/3  
  
max.metagenesplot.data.k4.tss.1 <-  
  apply(X = metagenesplot.data.k4.tss.1,  
        MARGIN = 1,  
        FUN = max, na.rm = T)  
max.metagenesplot.data.k4.tss.2 <-  
  apply(X = metagenesplot.data.k4.tss.2,  
        MARGIN = 1,  
        FUN = max, na.rm = T)  
max.metagenesplot.data.k4.tss.3 <-  
  apply(X = metagenesplot.data.k4.tss.3,  
        MARGIN = 1,  
        FUN = max, na.rm = T)  
max.metagenesplot.data.k4.tss <- (max.metagenesplot.data.k4.tss.1 +  
                                   max.metagenesplot.data.k4.tss.2 +
```

```

max.metagenepplot.data.k4.tss.3)/3

max.metagenepplot.data.k4.tes.1 <-
  apply(X = metagenepplot.data.k4.tes.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.tes.2 <-
  apply(X = metagenepplot.data.k4.tes.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.tes.3 <-
  apply(X = metagenepplot.data.k4.tes.3,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k4.tes <- (max.metagenepplot.data.k4.tes.1 +
  max.metagenepplot.data.k4.tes.2 +
  max.metagenepplot.data.k4.tes.3)/3

shapiro.test(max.metagenepplot.data.k4.gene.body)

##
## Shapiro-Wilk normality test
##
## data: max.metagenepplot.data.k4.gene.body
## W = 0.70701, p-value = 8.273e-12
shapiro.test(max.metagenepplot.data.k4.internal.gene.body)

##
## Shapiro-Wilk normality test
##
## data: max.metagenepplot.data.k4.internal.gene.body
## W = 0.6156, p-value < 2.2e-16
shapiro.test(max.metagenepplot.data.k4.tss)

##
## Shapiro-Wilk normality test
##
## data: max.metagenepplot.data.k4.tss
## W = 0.795, p-value < 2.2e-16
shapiro.test(max.metagenepplot.data.k4.tes)

##
## Shapiro-Wilk normality test
##
## data: max.metagenepplot.data.k4.tes
## W = 0.80079, p-value = 0.0005148
wilcox.test(x = max.metagenepplot.data.k4.gene.body,
            y = max.metagenepplot.data.k4.internal.gene.body,
            alternative = "greater")$p.value

## [1] 1.875932e-28

```

```
wilcox.test(x = max.metagenesplot.data.k4.gene.body,
            y = max.metagenesplot.data.k4.tss,
            alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k4.gene.body and max.metagenesplot.data.k4.tss
## W = 53648, p-value = 1.569e-07
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = max.metagenesplot.data.k4.gene.body,
            y = max.metagenesplot.data.k4.tes,
            alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k4.gene.body and max.metagenesplot.data.k4.tes
## W = 1629, p-value = 9.638e-08
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = max.metagenesplot.data.k4.tss,
            y = max.metagenesplot.data.k4.tes)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k4.tss and max.metagenesplot.data.k4.tes
## W = 15122, p-value = 0.0001688
## alternative hypothesis: true location shift is not equal to 0
```

Indeed the differences between all metagenesplots are highly significant.

Now we analyse the gene expression level of H3K4me3 marked genes.

```
non.k4.genes <- setdiff(names(gene.expression.20),c(k4.gene.body,k4.tss,k4.tes,k4.internal.gene.body))
```

```
summary(gene.expression.20[k4.gene.body])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##  0.753    46.257   149.039   621.958   726.707  13604.318
```

```
summary(gene.expression.20[k4.internal.gene.body])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##  4.072    27.564    49.727   164.370    99.501  10865.045
```

```
summary(gene.expression.20[k4.tss])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##  0.004    32.072    84.633   316.656   299.432  9906.184
```

```
summary(gene.expression.20[k4.tes])
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##  14.23    34.07    57.47    86.53   103.25   555.28
```

```
summary(gene.expression.20[non.k4.genes])
```

```

##      Min.   1st Qu.   Median     Mean  3rd Qu.     Max.
##      0.004   21.574   41.660   83.567   86.601  6961.033

jpeg(filename = "images/h3k24me3_expression_barplot_different_genes2.jpg",res = 250,width = 600,height = 400)
boxplot(gene.expression.20[k4.gene.body],
        gene.expression.20[k4.tss],
        gene.expression.20[k4.internal.gene.body],
        gene.expression.20[k4.tes],
        gene.expression.20[non.k4.genes],
        col=c("blue","green","cyan","red","grey"),
        outline=F)
dev.off()

## pdf
##      2

wilcox.test(x = gene.expression.20[k4.gene.body],
            y = gene.expression.20[k4.tss],
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data:  gene.expression.20[k4.gene.body] and gene.expression.20[k4.tss]
## W = 46868, p-value = 0.005753
## alternative hypothesis: true location shift is greater than 0

wilcox.test(x = gene.expression.20[k4.tss],
            y = gene.expression.20[k4.internal.gene.body],
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data:  gene.expression.20[k4.tss] and gene.expression.20[k4.internal.gene.body]
## W = 137830, p-value = 3.087e-07
## alternative hypothesis: true location shift is greater than 0

wilcox.test(x = gene.expression.20[k4.internal.gene.body],
            y = gene.expression.20[k4.tes],
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data:  gene.expression.20[k4.internal.gene.body] and gene.expression.20[k4.tes]
## W = 2541.5, p-value = 0.6605
## alternative hypothesis: true location shift is greater than 0

wilcox.test(x = gene.expression.20[k4.tes],
            y = gene.expression.20[non.k4.genes],
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data:  gene.expression.20[k4.tes] and gene.expression.20[non.k4.genes]
## W = 83054, p-value = 0.07689

```

```
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = gene.expression.20[k4.gene.body],
            y = gene.expression.20[non.k4.genes],
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k4.gene.body] and gene.expression.20[non.k4.genes]
## W = 395364, p-value = 2.761e-12
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = gene.expression.20[k4.internal.gene.body],
            y = gene.expression.20[non.k4.genes],
            alternative = "greater")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k4.internal.gene.body] and gene.expression.20[non.k4.genes]
## W = 883412, p-value = 0.0003477
## alternative hypothesis: true location shift is greater than 0
wilcox.test(x = gene.expression.20[k4.tss],
            y = gene.expression.20[non.k4.genes],
            alternative = "greater")$p.value

## [1] 1.47008e-51
wilcox.test(x = gene.expression.20[k4.tes],
            y = gene.expression.20[non.k4.genes],
            alternative = "greater")$p.value

## [1] 0.07689168
```

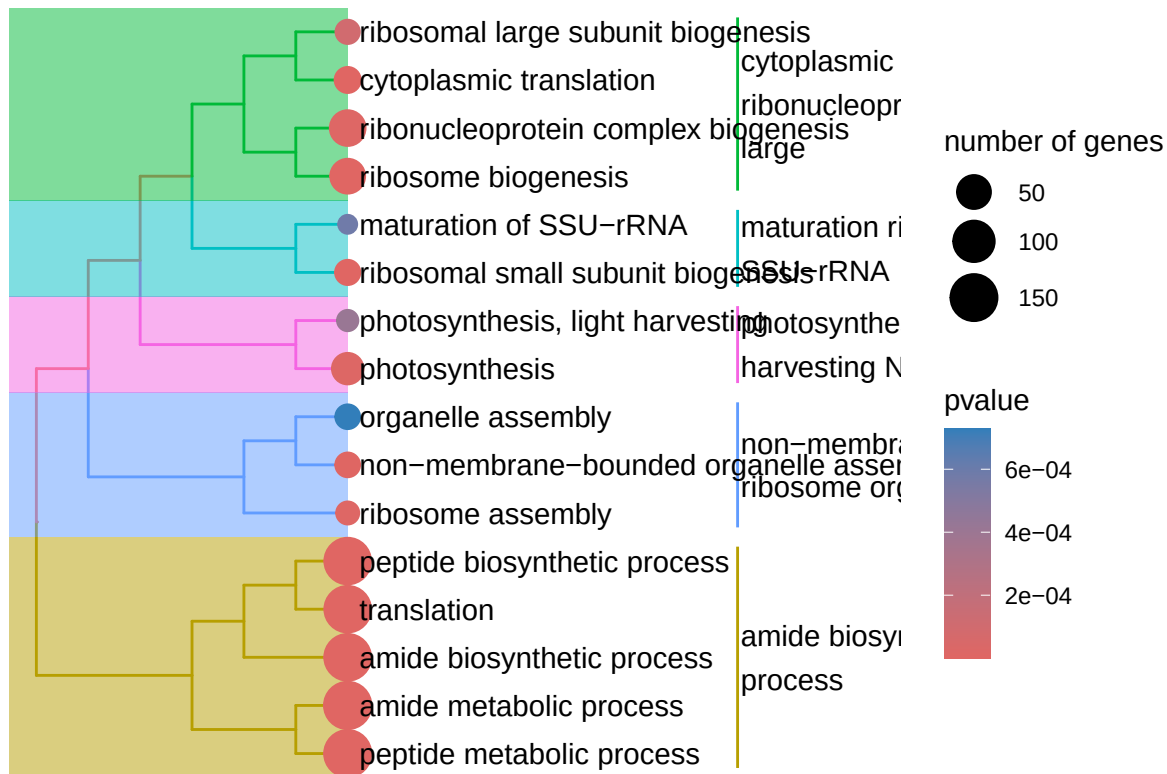
We can see H3K4me3 genes marked in the TSS or the entire gene present levels of expression significantly higher than non-marked genes. Whereas, genes where the mark is internal to the gene body or localized only to the TES are not significantly different from the non-marked genes.

In what follows we perform a functional analysis over the H3K4me3 marked genes. First we study the enrichment in GO terms.

```
library(clusterProfiler)
library(enrichplot)
library(org.Otauriv5.eg.db)
k4.enrichment <- enrichGO(gene = c(k4.tss,k4.gene.body), pvalueCutoff = 0.05,qvalueCutoff = 0.05,
                        OrgDb = org.Otauriv5.eg.db,
                        ont = "BP",
                        keyType = "GID")
k4.enrichment.df <- as.data.frame(k4.enrichment)

write.table(x = k4.enrichment.df, file = "go_enrichment_k4.tsv",quote = F,sep = "\t",row.names = F)

#jpeg(filename = "images/h3k4me3_treeplot.jpg",res = 400,width = 4500,height = 1500)
treeplot(x = pairwise_termsim(k4.enrichment),color = "pvalue")
```



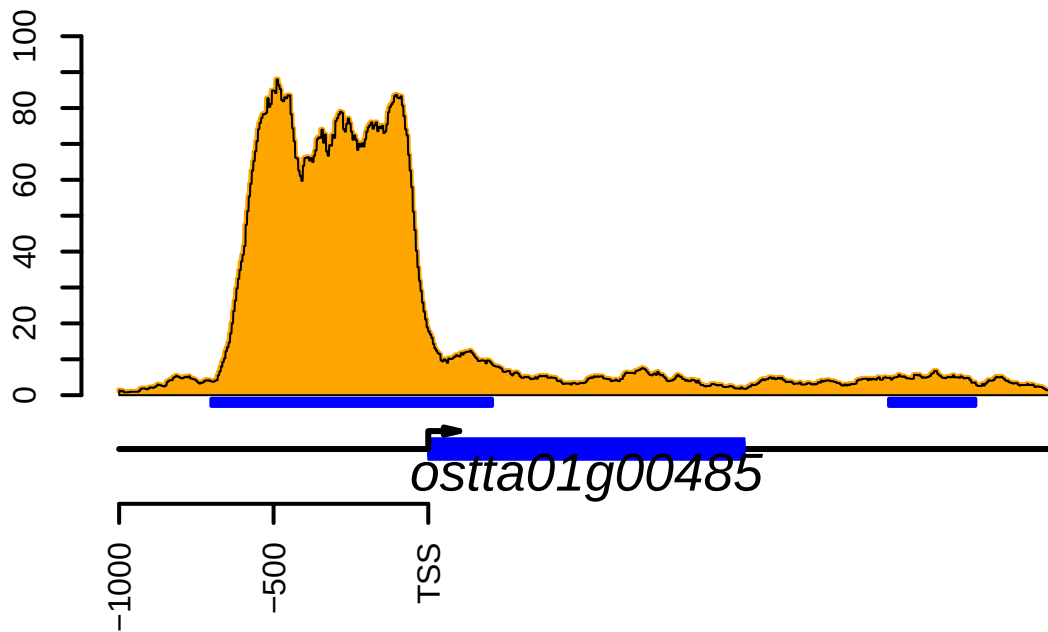
```
#dev.off()
```

Our GO enrichment analysis suggests that H3K4me3 is associated with gene activation in the processes biological processes ribosome biogenesis and assembly, translation, photosynthesis and positive regulation.

For example, large subunit ribosomal protein L3e, ostta01g00485, and small subunit ribosomal protein S8e, ostta14g02310, are H3K4me3 marked genes.

```
#jpeg(filename = "images/example_ostta01g00485.jpg", res = 250, width = 3800, height = 1500)
gene.profile(gene.name="ostta01g00485",
  promoter_length=1000,
  selected.bigwig.files="h3k4me3/h3k4me3_11_20_chip_1.bw",
  selected.bed.files="h3k4me3/h3k4me3_peaks.bed",
  line.colors = "orange", area.colors = "orange")
```

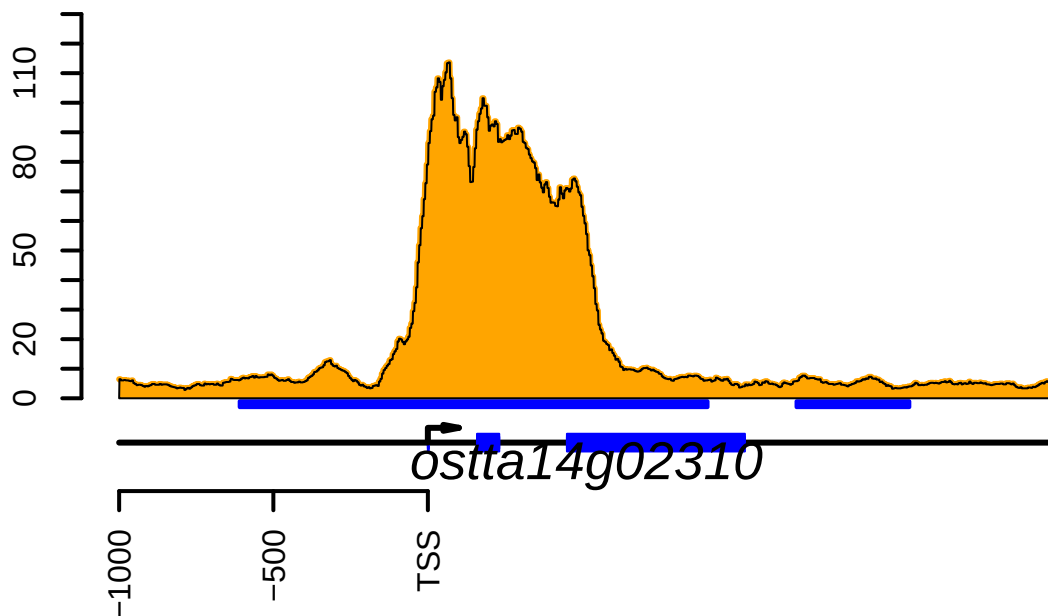
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/example_ostta14g02310.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta14g02310",
             promoter_length=1000,
             selected.bigwig.files="h3k4me3/h3k4me3_11_20_chip_1.bw",
             selected.bed.files="h3k4me3/h3k4me3_peaks.bed",
             line.colors = "orange",area.colors = "orange")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



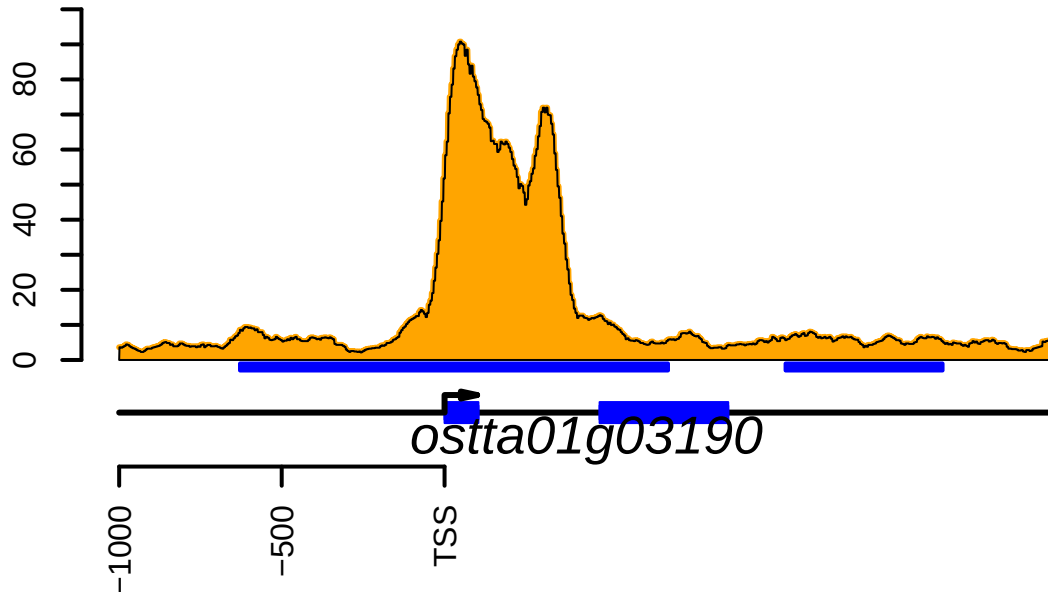
```
#dev.off()
```

Genes encoding translation initiation factor 5A, ostta01g03190, and 3B ostta15g01830 are also H3K4me3

marked.

```
#jpeg(filename = "images/example_ostta01g03190.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta01g03190",
             promoter_length=1000,
             selected.bigwig.files="h3k4me3/h3k4me3_l1_20_chip_1.bw",
             selected.bed.files="h3k4me3/h3k4me3_peaks.bed",
             line.colors = "orange",area.colors = "orange")
```

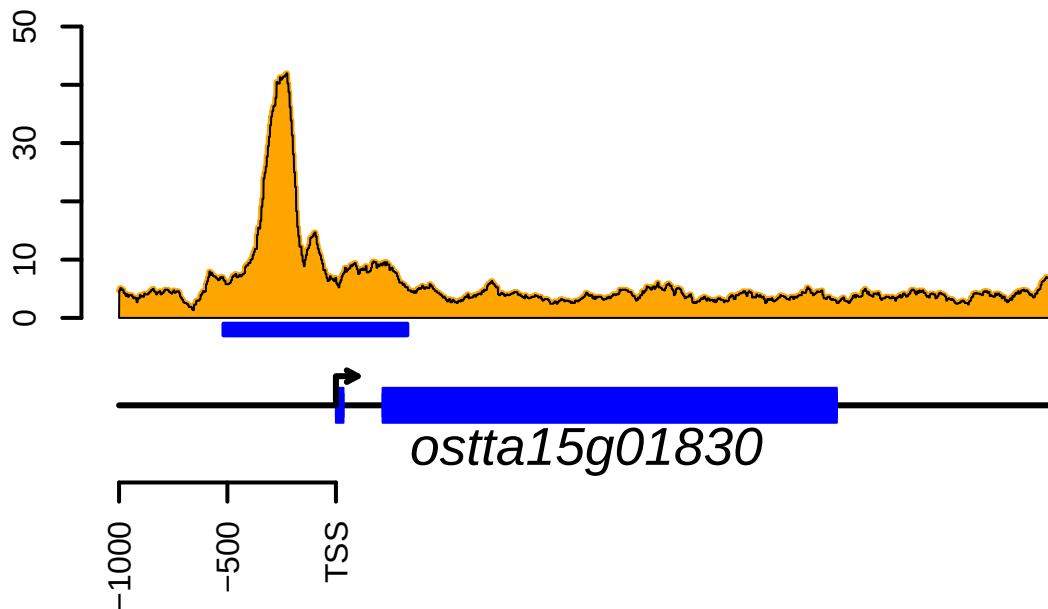
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/example_ostta15g01830.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta15g01830",
             promoter_length=1000,
             selected.bigwig.files="h3k4me3/h3k4me3_l1_20_chip_1.bw",
             selected.bed.files="h3k4me3/h3k4me3_peaks.bed",
             line.colors = "orange",area.colors = "orange")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

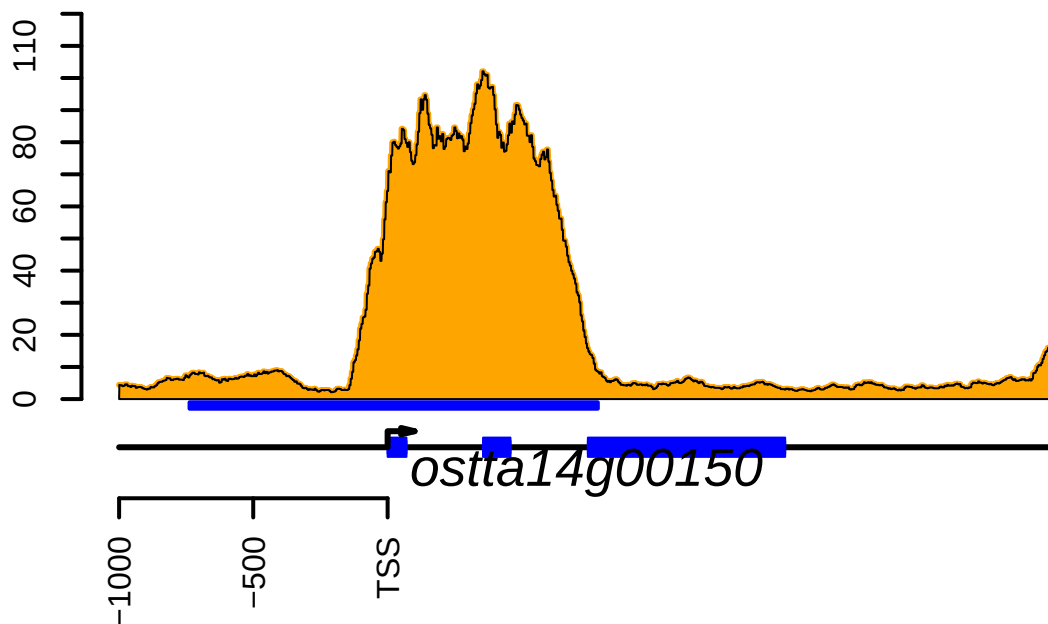


```
#dev.off()
```

Genes encoding photosystem II oxygen-evolving enhancer protein 1, 2 and 3 PsbO, ostta14g00150, PsbP, ostta14g02630, and PsbQ, ostta16g01620.

```
#jpeg(filename = "images/example_ostta14g00150.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta14g00150",
  promoter_length=1000,
  selected.bigwig.files="h3k4me3/h3k4me3_l1_20_chip_1.bw",
  selected.bed.files="h3k4me3/h3k4me3_peaks.bed",
  line.colors = "orange",area.colors = "orange")
```

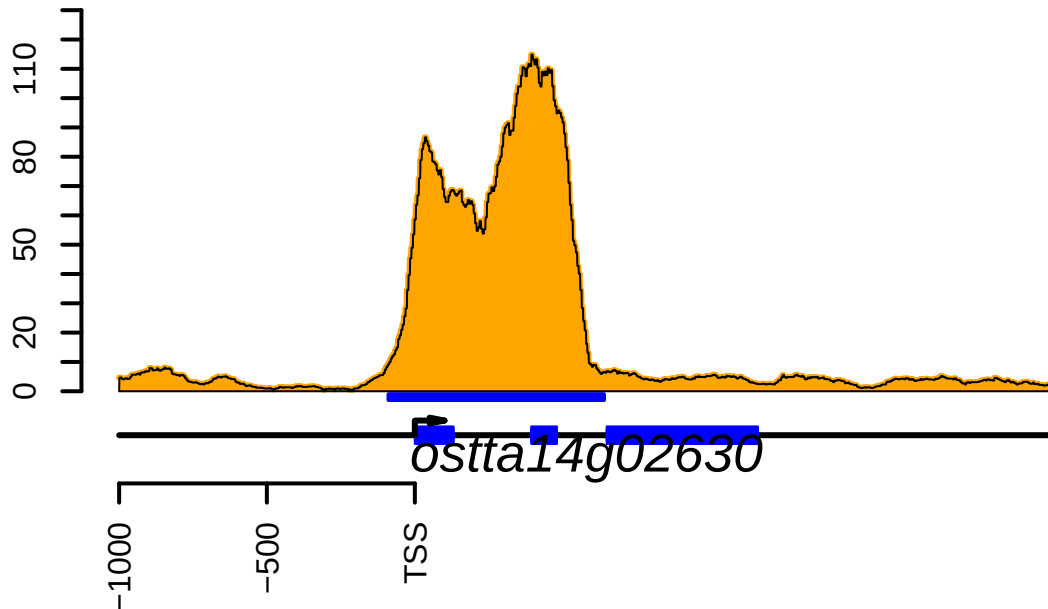
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/example_ostta14g02630.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta14g02630",
  promoter_length=1000,
  selected.bigwig.files="h3k4me3/h3k4me3_l1_20_chip_1.bw",
  selected.bed.files="h3k4me3/h3k4me3_peaks.bed",
  line.colors = "orange",area.colors = "orange")
```

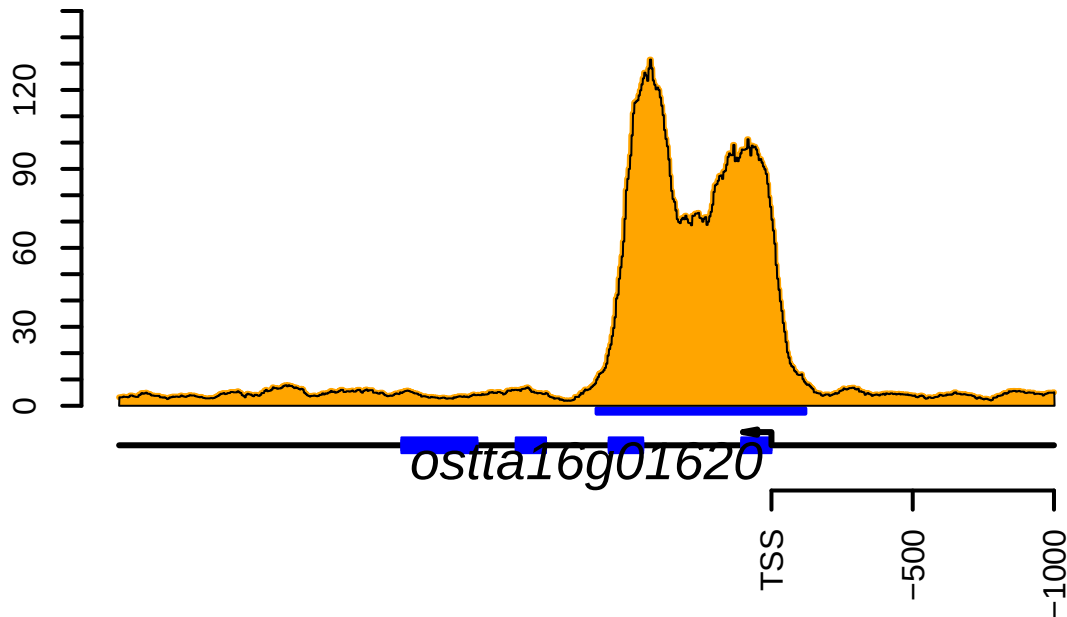
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/example_ostta16g01620.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta16g01620",
  promoter_length=1000,
  selected.bigwig.files="h3k4me3/h3k4me3_l1_20_chip_1.bw",
  selected.bed.files="h3k4me3/h3k4me3_peaks.bed",
  line.colors = "orange",area.colors = "orange")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

H3K4me3 was found in more than 25% of the annotated transcription factors in *Ostreococcus* more than the 7% detected with H3K27me3 suggesting a role in controlling master regulators in gene expression.

```
ostta.tfs <- read.table(file = "transcription_factors_list.tsv",header = T,as.is=T)
```

```
length(intersect(ostta.tfs$gene_id,all.k27.genes))
```

```
## [1] 7
```

```
length(intersect(ostta.tfs$gene_id,all.k27.genes))/nrow(ostta.tfs)
```

```
## [1] 0.07070707
```

```
length(intersect(ostta.tfs$gene_id,all.k4.genes))
```

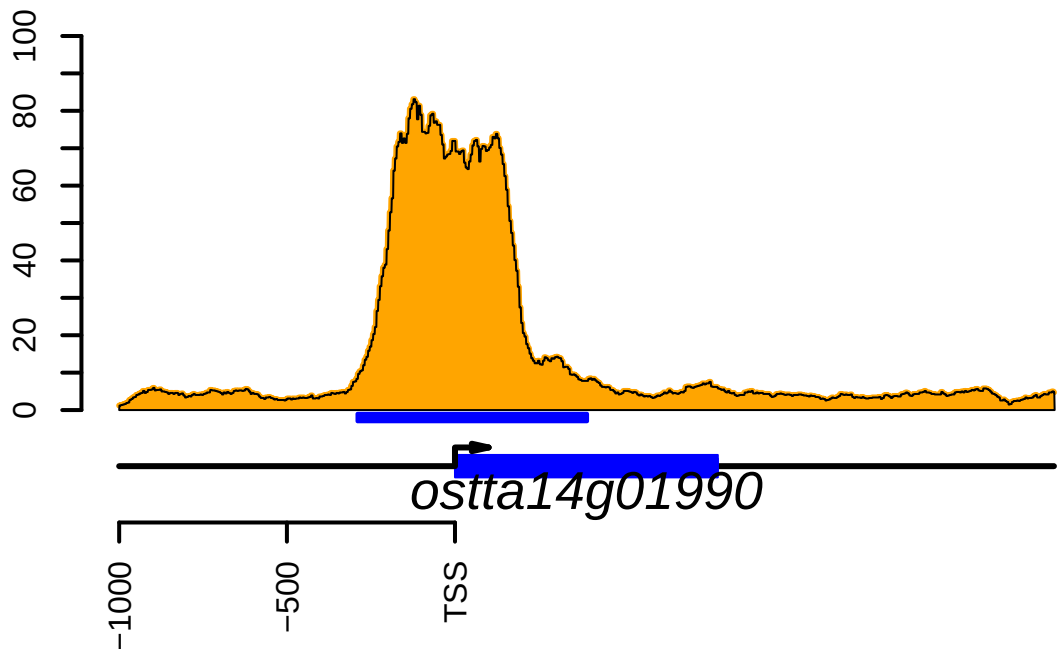
```
## [1] 25
```

```
length(intersect(ostta.tfs$gene_id,all.k4.genes))/nrow(ostta.tfs)
```

```
## [1] 0.2525253
```

```
#jpeg(filename = "images/example_ostta14g01990.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta14g01990",
  promoter_length=1000,
  selected.bigwig.files="h3k4me3/h3k4me3_11_20_chip_1.bw",
  selected.bed.files="h3k4me3/h3k4me3_peaks.bed",
  line.colors = "orange",area.colors = "orange")
```

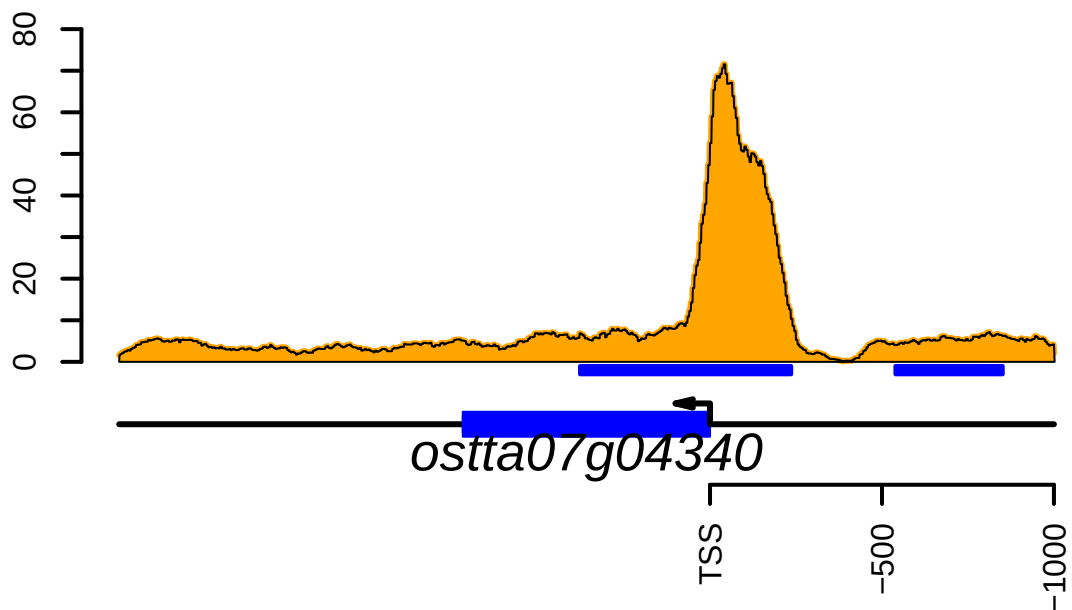
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/example_ostta07g04340.jpg",res = 250,width = 3800,height = 1500)
gene.profile(gene.name="ostta07g04340",
             promoter_length=1000,
             selected.bigwig.files="h3k4me3/h3k4me3_l1_20_chip_1.bw",
             selected.bed.files="h3k4me3/h3k4me3_peaks.bed",
             line.colors = "orange",area.colors = "orange")
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

H3K27me3 and H3K4me3 common marked genes

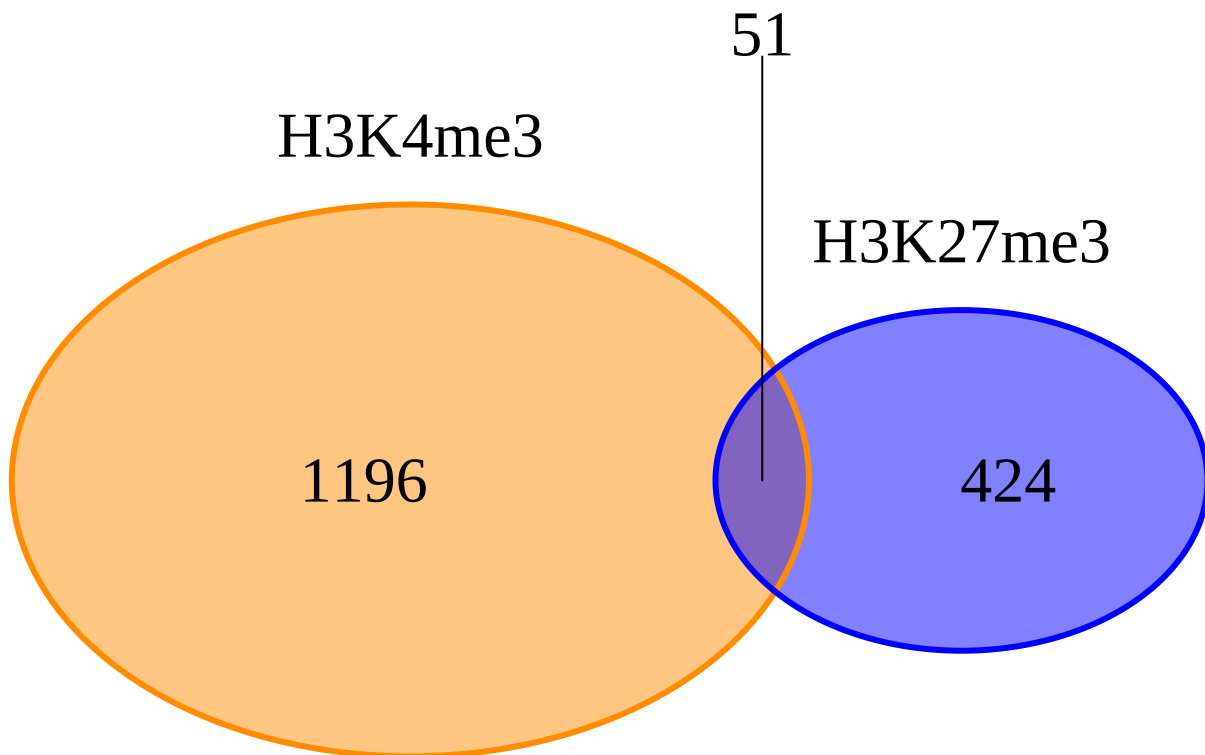
The set of H3K27me3 and H3K4me3 genes only overlapped marginally.

```
library(VennDiagram)
```

```
## Loading required package: grid
```

```
## Loading required package: futile.logger
```

```
#jpeg(filename = "images/h3k27me3_h3k4me3_venndiagram2.jpg",res = 250,width = 1500,height = 1500)
grid.newpage()
draw.pairwise.venn(area1 = length(all.k27.genes),cat.pos = c(180,180),cat.dist = 0.05,
  area2 = length(all.k4.genes),
  cross.area = length(intersect(all.k27.genes,all.k4.genes)),
  lwd = 3,category = c("H3K27me3","H3K4me3"),euler.d = T,
  col = c("blue","darkorange"),
  fill = c("blue","darkorange"),alpha = 0.5,
  cex = 2,
  cat.cex = 2)
```



```
## (polygon[GRID.polygon.201], polygon[GRID.polygon.202], polygon[GRID.polygon.203], polygon[GRID.polygon.204])
#dev.off()
```

Only 51 genes were identified marked with both H3K27me3 and H3K4me3. Next we explore the positioning of these two marks in these genes and their level of expression.

```
k4.k27.genes <- unique(intersect(all.k4.genes, all.k27.genes))
k4.k27.gene.annotation <- subset(11.20.peak.annotation, target.genes %in% k4.k27.genes)
```

```
100*table(k4.k27.gene.annotation[!duplicated(k4.k27.gene.annotation$target.genes),][["peak.annotation"]])
```

```
##
##      gene.body internal.gene.body      TES      TSS
##      6.250000      64.583333      25.000000      4.166667
```

In most genes with both marks H3K27me3 is located in inner regions of gene bodies or the TES (89.6%).

```
non.k4.k27.genes <- intersect(non.k4.genes,non.k27.genes)
summary(gene.expression.20[all.k4.genes])
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.004  30.075   75.506   301.709 257.607 13604.318
```

```
summary(gene.expression.20[all.k27.genes])
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0042  0.9851   8.1853   32.6813 28.8598 1012.0876
```

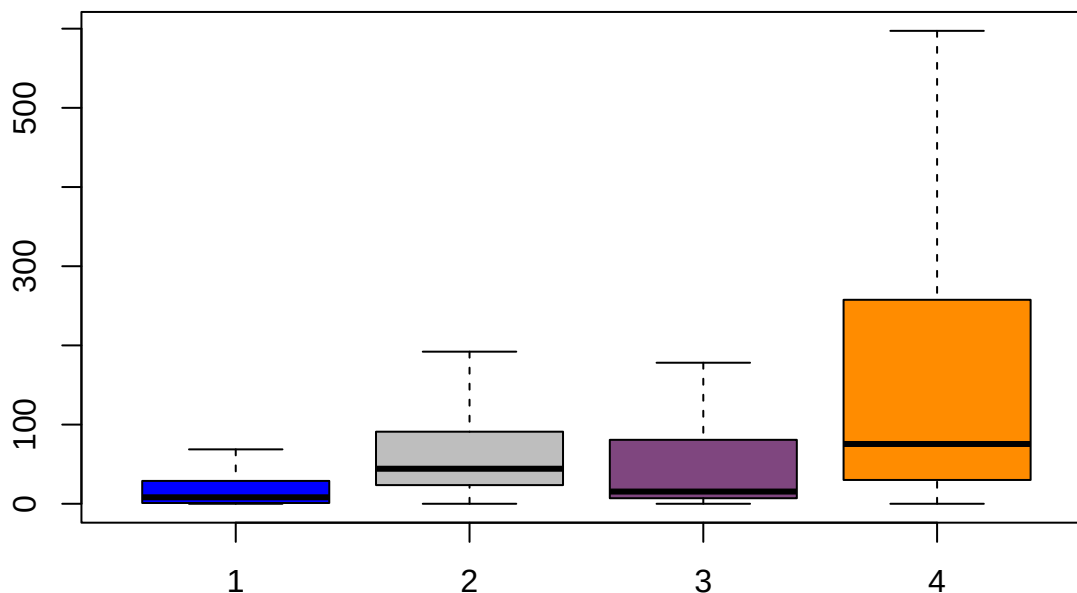
```
summary(gene.expression.20[k4.k27.genes])
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0042  7.0014 15.4122  88.9229 80.7583 730.2898
```

```
summary(gene.expression.20[non.k4.k27.genes])
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.004  23.563  44.240  87.643  91.023 6961.033
```

```
#jpeg(filename = "images/bivalent_expression_barplot2.jpg",res = 250,width = 600,height = 1000)
boxplot(gene.expression.20[all.k27.genes],
        gene.expression.20[non.k4.k27.genes],
        gene.expression.20[k4.k27.genes],
        gene.expression.20[all.k4.genes],
        col=c("blue","grey",colorRampPalette(c('blue','darkorange'))(3)[2],"darkorange"),outline=F)
```



```
#dev.off()
```

```
wilcox.test(x = gene.expression.20[all.k27.genes], y = gene.expression.20[non.k4.k27.genes],alternative
```

```
## [1] 1.561584e-102
wilcox.test(x = gene.expression.20[k4.k27.genes], y = gene.expression.20[non.k4.k27.genes], alternative = "g")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k4.k27.genes] and gene.expression.20[non.k4.k27.genes]
## W = 110522, p-value = 0.0003191
## alternative hypothesis: true location shift is less than 0
wilcox.test(x = gene.expression.20[all.k4.genes], y = gene.expression.20[non.k4.k27.genes], alternative = "g")

## [1] 2.084754e-43
wilcox.test(x = gene.expression.20[all.k4.genes], y = gene.expression.20[k4.k27.genes], alternative = "g")

## [1] 1.450592e-07
```

H3K27me3 consensus peaks covering different temperatures

We determined the genome wide distribution of H3K27me3 under different temperatures, 10°C, 14°C, 20°C and 26°C. We found differences as a response to temperature. In order to compare H3K27me3 marks under different temperature we merge all the peaks detected consistently in the three replicates for each temperature to generate a set of consensus peaks. Next we analyze the genome wide distribution of the consensus H3K27me3 peaks.

```
k27.consensus <- read.table(file="h3k27me3/h3k27me3_peaks.bed",
                           header = F)

nrow(k27.consensus)

## [1] 1056

#jpeg(filename = "images/h3k27me3_genome_wide_distribution_ll_20.jpg", res = 300, width = 900, height = 900)
#par(mar=c(0,0,0,0))
plot(x=c(0,max.chr.len), y=c(0,4*number.chrs), col = "white",
     xlab = "", ylab = "", axes=F)

for(i in 1:number.chrs)
{
  polygon(x = c(0, chr.lens[i], chr.lens[i], 0),
         y = c(4*i+1, 4*i+1, 4*i-1, 4*i-1),
         col = "white")
}

for(i in 1:nrow(k27.consensus))
{
  current.chr <- k27.consensus[i,1]
  current.start <- k27.consensus[i,2]
  current.end <- k27.consensus[i,3]

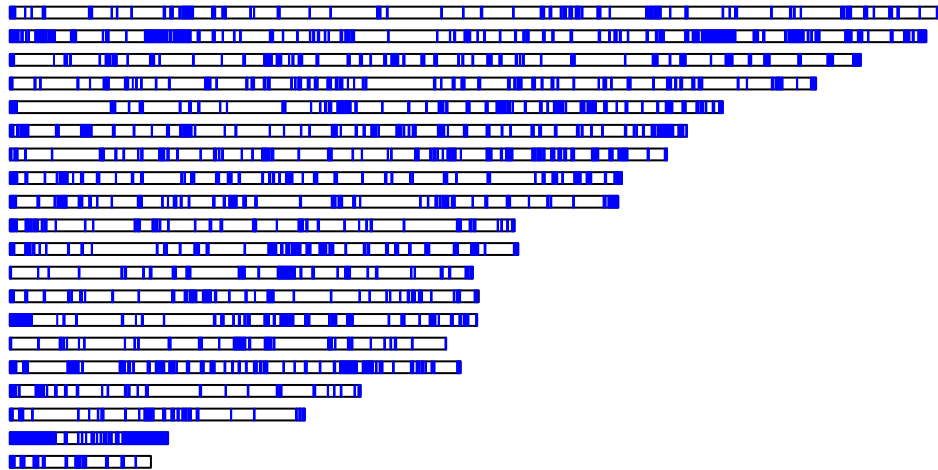
  current.line <- (20:1)[current.chr]

  polygon(x = c(current.start,
                current.end,
                current.end,
                current.start),
```

```

    y = c(4*current.line+1,
          4*current.line+1,
          4*current.line-1,
          4*current.line-1),
    col = "blue",border="blue")
}

```



```
#dev.off()
```

We identified 1056 genomic regions significantly occupied by H3K27me3 in the three different biological replicates at 10°C, 14°C, 20°C or 26°C. We compute the percentage of the whole genome occupied by H3K27me3 and the percentage in each chromosome. A barplot is used to represent the percentage of H3K27me3 in each chromosome. Z-scores and p-independent values are computed for each chromosome. A gradient from white to blue is used to represent significance levels.

```

h3k27me3.peaks.len <- k27.consensus$V3 - k27.consensus$V2
sum(h3k27me3.peaks.len)/sum(chr.lens)

```

```
## [1] 0.1445867
```

```

k27.consensus <- cbind(k27.consensus,h3k27me3.peaks.len)
colnames(k27.consensus) <- c("chr","start","end","length")
head(k27.consensus)

```

```

##   chr start   end length
## 1   1     0  1529   1529
## 2   1  2910  6526   3616
## 3   1 17915 18081    166
## 4   1 25485 25705    220
## 5   1 38421 41511   3090
## 6   1 92611 93775   1164

```

```

number.of.h3k27.peaks.chr <- vector(mode = "numeric",length = 20)
peaks.len.chr <- vector(mode = "numeric",length = 20)

```

```

for(i in 1:20)
{
  number.of.h3k27.peaks.chr[i] <- nrow(subset(k27.consensus, chr == i))
  peaks.len.chr[i] <- sum(subset(k27.consensus, chr == i)[,4])
}

```

```

percentage.chr.k27 <- peaks.len.chr/rev(chr.lens)

z.scores <- (percentage.chr.k27 - mean(percentage.chr.k27))/sd(percentage.chr.k27)

which(z.scores > 2.5)

## [1] 19

p.vals <- pnorm((percentage.chr.k27 - mean(percentage.chr.k27[-19]))/sd(percentage.chr.k27[-19]),lower.tail=FALSE)

p.vals

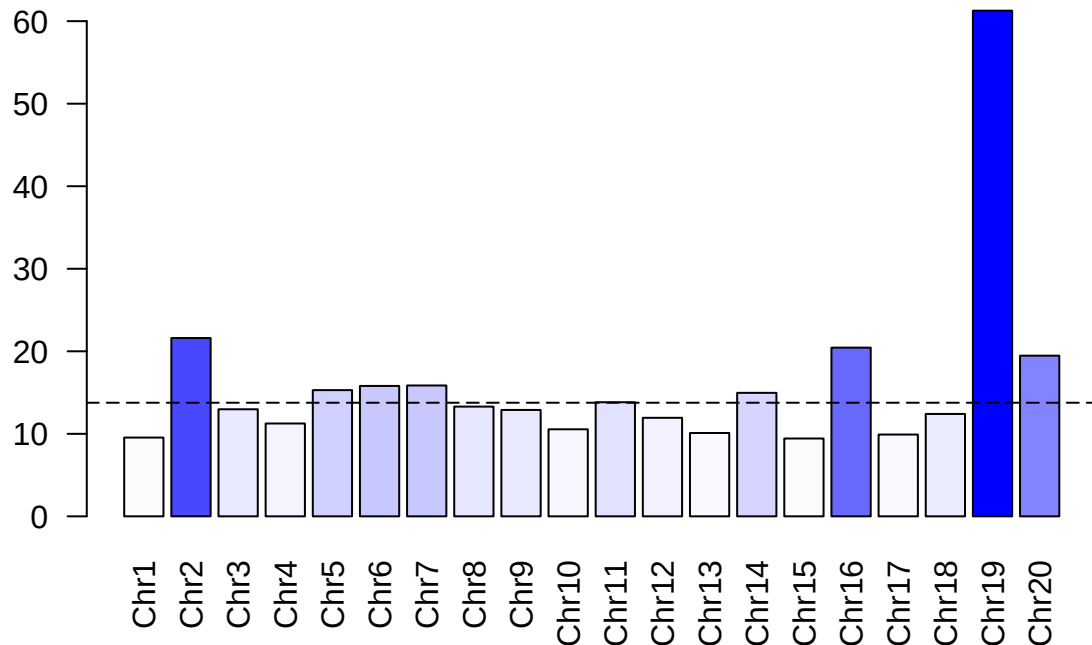
## [1] 8.765293e-01 1.575844e-02 5.859998e-01 7.547631e-01 3.374230e-01
## [6] 2.884019e-01 2.831669e-01 5.510880e-01 5.946692e-01 8.114598e-01
## [11] 4.934411e-01 6.916986e-01 8.428383e-01 3.712700e-01 8.827484e-01
## [16] 3.355930e-02 8.549079e-01 6.452332e-01 4.048493e-39 5.885202e-02

values.for.colors <- round(-log10(p.vals)*40)
values.for.colors[19] <- 100

white.blue.palette <- colorRampPalette(c('white','blue'))

#jpeg(filename = "images/h3k27me3_chr_barplot_ll_20.jpg",res = 300,width = 1800,height = 900)
barplot(100*percentage.chr.k27,border="black",names.arg = paste0("Chr",1:20),las=2,col=white.blue.palette[values.for.colors])
abline(h = 100*mean(percentage.chr.k27[-19]),lty=5)

```



```
#dev.off()
```

Approximately 14.5% of the *Ostreococcus* genome is occupied by H3K27me3. H3K27me3 is mostly uniformly distributed over the different chromosome although it significantly accumulates (p-value < 0.01) in Chr2 (sexual chromosome), Chr16 and Chr19 (viral resistance chromosome) with p-values 1.58×10^{-2} , 3.36×10^{-2} and 4.05×10^{-39} respectively.

We determine the gene features occupied by each H3K27me3 mark and the H3K27me3 marked genes.

```

library(TxDb.Otari.JGI)
txdb <- TxDb.Otari.JGI

ostta.genes <- as.data.frame(genes(txdb))
head(ostta.genes)

##           seqnames start   end width strand      gene_id
## ostta01g00010      1   244  1932  1689      - ostta01g00010
## ostta01g00020      1  2077  4029  1953      + ostta01g00020
## ostta01g00030      1  4157  5299  1143      + ostta01g00030
## ostta01g00040      1  5392  7041  1650      - ostta01g00040
## ostta01g00050      1  7386  9605  2220      - ostta01g00050
## ostta01g00060      1  9780 10412   633      - ostta01g00060

target.genes <- vector(mode = "character",length = nrow(k27.consensus))
peak.annotation <- vector(mode = "character",length = nrow(k27.consensus))

for(i in 1:nrow(k27.consensus))
{
  current.chr <- k27.consensus[i,1]
  current.start <- k27.consensus[i,2]
  current.end <- k27.consensus[i,3]

  complete.gene.body <- subset(ostta.genes,
                              seqnames == current.chr &
                              start >= current.start &
                              end <= current.end)

  overlap.tss.pos.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   start >= current.start & start <= current.end &
                                   strand == "+")

  overlap.tss.neg.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   end >= current.start & end <= current.end &
                                   strand == "-")

  overlap.tes.pos.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   end >= current.start & end <= current.end &
                                   strand == "+")

  overlap.tes.neg.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   start >= current.start & start <= current.end &
                                   strand == "-")

  inside.gene.body <- subset(ostta.genes, seqnames == current.chr &
                             start <= current.start & end >= current.end)

  if(nrow(complete.gene.body) > 0)
  {

```

```

target.genes[i] <- paste(unlist(complete.gene.body$gene_id),
                        collapse = ",")
peak.annotation[i] <- "gene.body"
} else if(nrow(overlap.tss.pos.strand) > 0)
{
target.genes[i] <- paste(unlist(overlap.tss.pos.strand$gene_id),
                        collapse = ",")

peak.annotation[i] <- "TSS"
} else if(nrow(overlap.tss.neg.strand) > 0)
{
target.genes[i] <- paste(unlist(overlap.tss.neg.strand$gene_id),
                        collapse = ",")

peak.annotation[i] <- "TSS"
} else if(nrow(overlap.tes.pos.strand) > 0)
{
target.genes[i] <- paste(unlist(overlap.tes.pos.strand$gene_id),
                        collapse = ",")

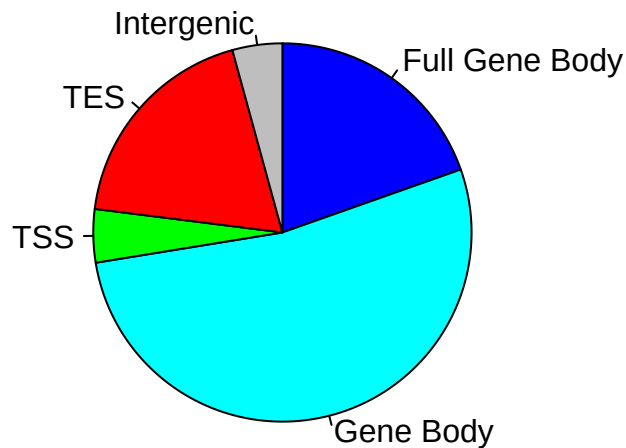
peak.annotation[i] <- "TES"
} else if(nrow(overlap.tes.neg.strand) > 0)
{
target.genes[i] <- paste(unlist(overlap.tes.neg.strand$gene_id),
                        collapse = ",")

peak.annotation[i] <- "TES"
} else if(nrow(inside.gene.body) > 0)
{
target.genes[i] <- paste(unlist(inside.gene.body$gene_id),
                        collapse = ",")

peak.annotation[i] <- "internal.gene.body"
} else
{
peak.annotation[i] <- "intergenic"
}
}

freq.peak.annotation <- table(peak.annotation)
#jpeg(filename = "images/h3k27me3_pie_chart_ll_20.jpg",res = 300,width = 900,height = 900)
#par(mar=c(0,0,0,0))
pie(freq.peak.annotation[c("gene.body",
                          "internal.gene.body",
                          "TSS",
                          "TES",
                          "intergenic")],
    labels = c("Full Gene Body",
               "Gene Body",
               "TSS",
               "TES",
               "Intergenic"),
    clockwise = T, border = "black", col=c("blue","cyan","green","red","grey"))

```



```
#dev.off()

percentage.freq.peak.annotation <-
  100*freq.peak.annotation/sum(freq.peak.annotation)

percentage.freq.peak.annotation["gene.body"] +
percentage.freq.peak.annotation["internal.gene.body"]

## gene.body
## 72.44318

percentage.freq.peak.annotation

## peak.annotation
##      gene.body      intergenic internal.gene.body      TES
##      19.602273      4.261364      52.840909      18.750000
##      TSS
##      4.545455

k27.consensus.peak.annotation <- cbind(k27.consensus[,1:3],
                                         data.frame(peak.annotation,target.genes))

rownames(k27.consensus.peak.annotation) <- apply(X = k27.consensus.peak.annotation[,1:3],
  MARGIN = 1,
  FUN =
    function(x)
    {
      paste(x[1],paste(x[2],x[3],sep="-"),sep=":")
    })

head(k27.consensus.peak.annotation)

##      chr start  end peak.annotation target.genes
## 1:0-1529    1    0 1529          TES ostta01g00010
## 1:2910-6526  1 2910 6526        gene.body ostta01g00030
## 1:17915-18081 1 17915 18081 internal.gene.body ostta01g00100
## 1:25485-25705 1 25485 25705 internal.gene.body ostta01g00150
## 1:38421-41511 1 38421 41511        gene.body ostta01g00260
## 1:92611-93775 1 92611 93775 internal.gene.body ostta01g00620
```

```
write.table(x = k27.consensus.peak.annotation,
           file = "tables/H3K27me3_consensus_peak_annotation.tsv",
           quote = F, sep = "\t", row.names = F)

all.k27.genes <- unique(unlist(sapply(X = k27.consensus.peak.annotation$target.genes,
                                     FUN = function(x) {strsplit(x, ",")[[1]]})))
names(all.k27.genes) <- NULL
length(all.k27.genes)
```

```
## [1] 994
```

```
length(all.k27.genes)/7668
```

```
## [1] 0.1296296
```

```
## Genes with their entire gene body marked
k27.gene.body <- unique(unlist(sapply(
  X = subset(k27.consensus.peak.annotation,
             peak.annotation == "gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.gene.body) <- NULL
length(k27.gene.body)
```

```
## [1] 314
```

```
## Genes with H3K27me3 overlapping their TSS
k27.tss <- unique(unlist(sapply(
  X = subset(k27.consensus.peak.annotation,
             peak.annotation == "TSS")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.tss) <- NULL
length(k27.tss)
```

```
## [1] 49
```

```
## Genes with H3K27me3 overlapping their TES
k27.tes <- unique(unlist(sapply(
  X = subset(k27.consensus.peak.annotation,
             peak.annotation == "TES")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.tes) <- NULL
length(k27.tes)
```

```
## [1] 198
```

```
## Genes with H3K27 inside their gene body with no overlap
## in the TSS or TES.
k27.internal.gene.body <- unique(unlist(sapply(
  X = subset(k27.consensus.peak.annotation,
             peak.annotation == "internal.gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.internal.gene.body) <- NULL
k27.internal.gene.body <- setdiff(k27.internal.gene.body, c(k27.tss, k27.tes))
length(k27.internal.gene.body)
```

```
## [1] 433
```

We found 994 genes associated to H3K27me3 marks corresponding to 13% of the entire *Ostreococcus* genome.

H3K27me3 is mostly located occupying either the entire gene body (19.6%, 314 genes) or internal regions of gene body (52.8%, 433 genes). Although some regions occupied by H3K27me3 has been found overlapping only with the TSS (4.5%, 49 genes) or the TES (18.8%, 198 genes). H3K27me3 was found in intergenic regions only in 4.3% of the cases.

In what follows we perform a functional analysis over the H3K27me3 consensus marked genes. First, we analyze the relationship between H3K27me3 consensus peaks and transposable elements.

```
dna.transposons <-
  read.table(file="te_analysis/dna_transposons_unsorted.bed",
             sep="\t",header=F)
nrow(dna.transposons)

## [1] 35

ltr.transposons <-
  read.table(file="te_analysis/ltr_transposons_unsorted.bed",
             sep="\t", header=F)
nrow(ltr.transposons)

## [1] 69

unclassified.transposons <-
  read.table(file="te_analysis/unknown_transposons_unsorted.bed",
             sep="\t",header = F)
nrow(unclassified.transposons)

## [1] 988

overlapping.peaks <- function(target.peak, peaks)
{
  subset(peaks, V1 == target.peak[1] & (
    (V2 >= target.peak[2] & V2 <= target.peak[3]) |
    (V3 >= target.peak[2] & V3 <= target.peak[3]) |
    (V2 <= target.peak[2] & V3 >= target.peak[3])))
}

peak.te <- vector(mode = "character",length = nrow(k27.consensus.peak.annotation))

for(i in 1:nrow(k27.consensus.peak.annotation))
{
  current.peak <- unlist(k27.consensus.peak.annotation[i,1:3])

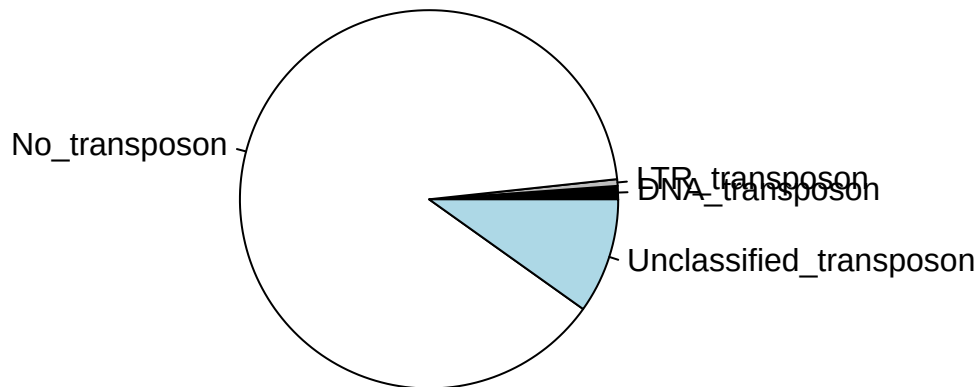
  if(nrow(overlapping.peaks(current.peak,dna.transposons)) > 0)
  {
    peak.te[i] <- "DNA_transposon"
  } else if (nrow(overlapping.peaks(current.peak,ltr.transposons)) > 0)
  {
    peak.te[i] <- "LTR_transposon"
  } else if (nrow(overlapping.peaks(current.peak,unclassified.transposons)) > 0)
  {
    peak.te[i] <- "Unclassified_transposon"
  } else
  {
    peak.te[i] <- "No_transposon"
  }
}
```

```
}
```

```
100*table(peak.te)/sum(table(peak.te))
```

```
## peak.te
##      DNA_transposon      LTR_transposon      No_transposon
##      1.1363636      0.5681818      88.4469697
## Unclassified_transposon
##      9.8484848
```

```
pie(table(peak.te), col=c("black","grey","white","lightblue"))
```



```
nrow(dna.transposons)
```

```
## [1] 35
```

```
100*table(peak.te)["DNA_transposon"]/nrow(dna.transposons)
```

```
## DNA_transposon
##      34.28571
```

```
nrow(ltr.transposons)
```

```
## [1] 69
```

```
100*table(peak.te)["LTR_transposon"]/nrow(ltr.transposons)
```

```
## LTR_transposon
##      8.695652
```

```
nrow(unclassified.transposons)
```

```
## [1] 988
```

```
100*table(peak.te)["Unclassified_transposon"]/nrow(unclassified.transposons)
```

```
## Unclassified_transposon
##      10.52632
```

```
table(k27.consensus.peak.annotation$peak.annotation[peak.te == "DNA_transposon"])
```

```
##
##      gene.body      intergenic internal.gene.body      TES
##      3          4          2          3
```

```
table(k27.consensus.peak.annotation$peak.annotation[peak.te == "LTR_transposon"])
```

```
##
## gene.body intergenic
##      3      3
table(k27.consensus.peak.annotation$peak.annotation[peak.te == "Unclassified_transposon"])

##
##      gene.body      intergenic internal.gene.body      TES
##      33      25      21      17
##      TSS
##      8
table(k27.consensus.peak.annotation$peak.annotation)

##
##      gene.body      intergenic internal.gene.body      TES
##      207      45      558      198
##      TSS
##      48
(table(k27.consensus.peak.annotation$peak.annotation[peak.te == "DNA_transposon"])["intergenic"] +
table(k27.consensus.peak.annotation$peak.annotation[peak.te == "LTR_transposon"])["intergenic"] +
table(k27.consensus.peak.annotation$peak.annotation[peak.te == "Unclassified_transposon"])["intergenic"])

## intergenic
## 0.7111111
k27.consensus.peak.annotation <- cbind(k27.consensus.peak.annotation, peak.te)
colnames(k27.consensus.peak.annotation)[1:3] <- c("chr", "start", "end")
write.table(x = k27.consensus.peak.annotation, file = "tables/H3K27me3_consensus_peak_annotation.tsv", qu
```

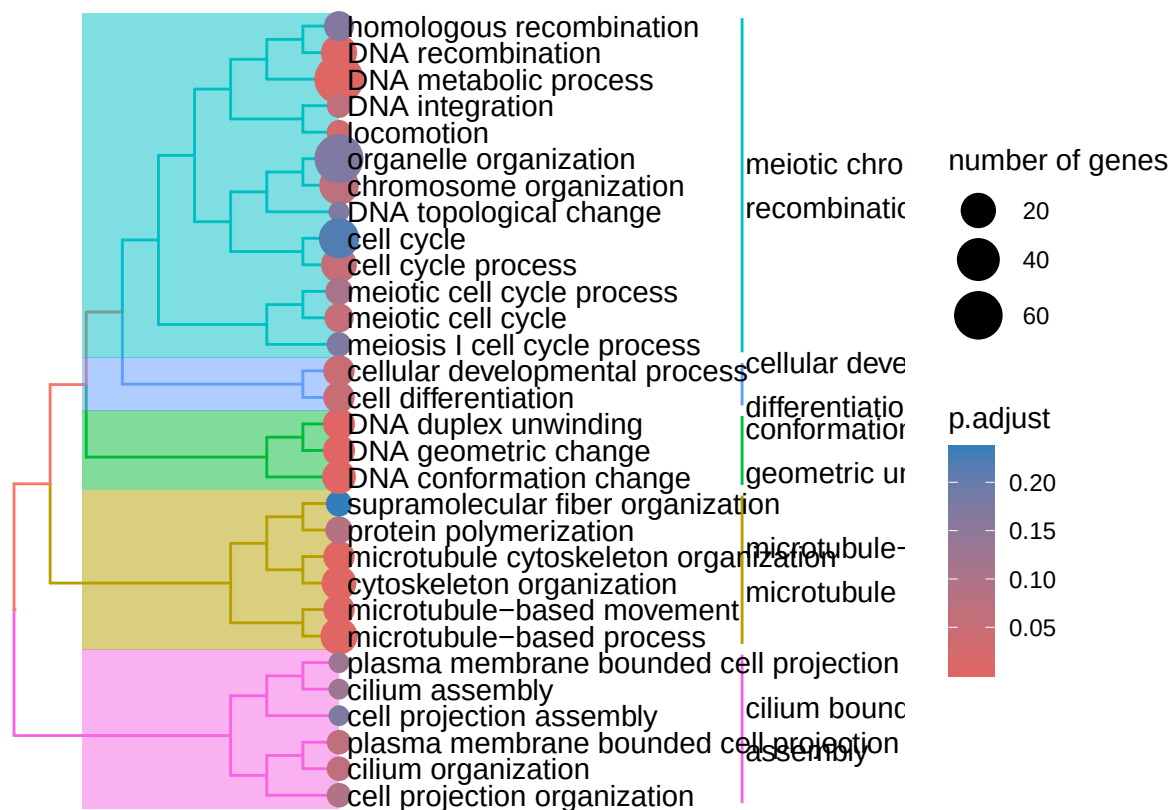
H3K27me3 is only partially associated with Transposable Elements. More than 88.4% of the peaks did not overlap with any TE. Only 1.1%, 0.6% and 9.9% of the H3K27me3 peaks were found overlapping with DNA, LTR and unclassified transposons respectively. Most TE were neither associated with H3K27me3 peaks, specifically 12 out of 35 (34.3%) DNA transposons, 6 out of 69 (8.7%) LTR transposons and 104 out of 988 (10.5%) unclassified transposons were H3K27me3 marked. Nonetheless, most H3K27me3 marks located at intergenic regions (71.1%) were associated with transposons.

```
library(clusterProfiler)
library(enrichplot)
library(org.Otauriv5.eg.db)
k27.go.terms <- enrichGO(gene = all.k27.genes, pvalueCutoff = 1,
                        qvalueCutoff = 1,
                        OrgDb = org.Otauriv5.eg.db,
                        ont = "BP",
                        keyType = "GID")
k27.go.terms.filtered <- k27.go.terms
k27.go.terms.filtered@result <- subset(k27.go.terms@result, pvalue < 0.05)

k27.go.terms.df <- as.data.frame(k27.go.terms.filtered)

write.table(x = k27.go.terms.df, file = "go_enrichment_consensus_k27.tsv",
            quote = F, sep = "\t", row.names = F)

treemap(x = pairwise_termsim(k27.go.terms.filtered))
```



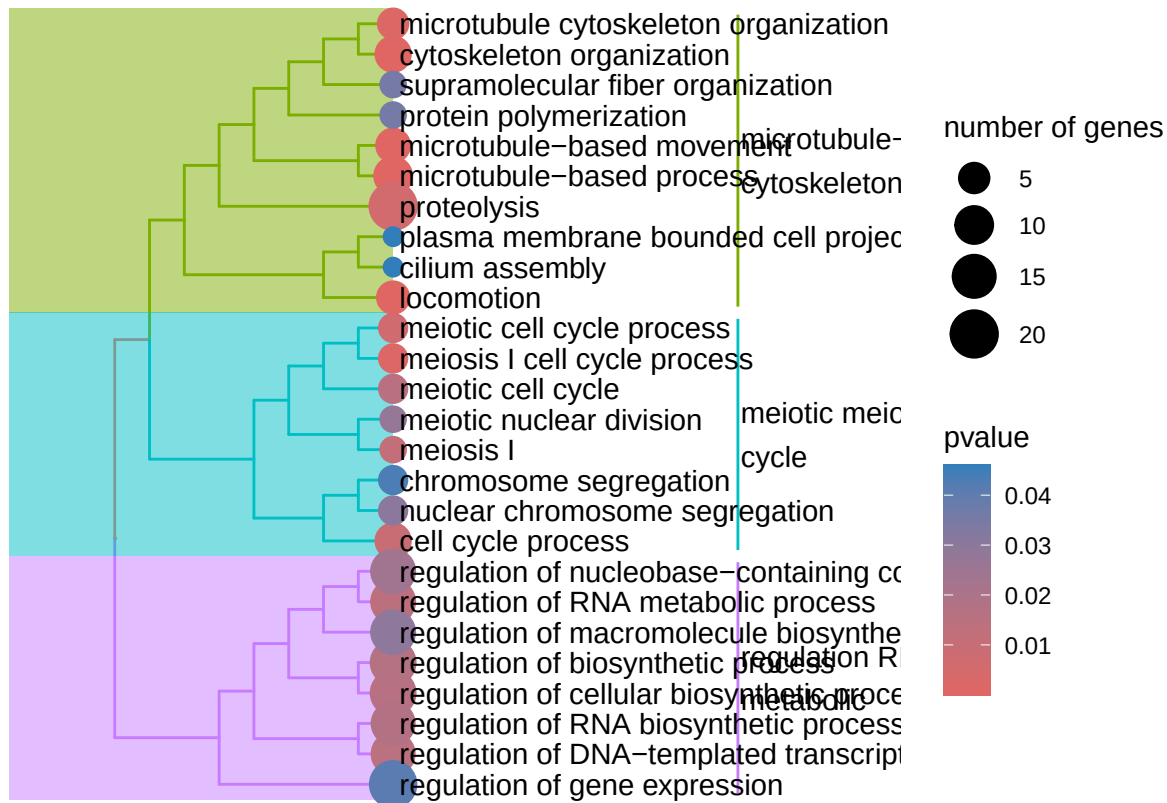
```
k27.go.terms <- enrichGO(gene = c(k27.gene.body,k27.tss), pvalueCutoff = 1,
  qvalueCutoff = 1,
  OrgDb = org.Otauriv5.eg.db,
  ont = "BP",
  keyType = "GID")
```

```
k27.go.terms.filtered <- k27.go.terms
```

```
k27.go.terms.filtered@result <- subset(k27.go.terms@result, pvalue < 0.05)
```

```
k27.go.terms.df <- as.data.frame(k27.go.terms.filtered)
```

```
#jpeg(filename = "images/consensus_h3k27me3_treeplot_gene_body.jpg",res = 250,width = 3500,height = 130)
treeplot(x = pairwise_termsim(k27.go.terms.filtered),cluster.params = list(n = 3),color = "pvalue")
```



```
#dev.off()

write.table(x = k27.go.terms.df, file = "go_enrichment_full_gene_body_tss_consensus_k27.tsv",
            quote = F, sep = "\t", row.names = F)
```

Our GO enrichment analysis suggests that H3K27me3 is repressing the assembly of bounded cellular projection, microtubule-based movement and transcription regulation. Other processes are also significantly affected such as fatty acid metabolism and cell differentiation.

H3K4me3 consensus peaks covering different temperatures

We determined the genome wide distribution of H3K4me3 under different temperatures, 10°C, 14°C, 20°C and 26°C. We found differences as a response to temperature. In order to compare H3K24me3 marks under different temperature we merge all the peaks detected consistently in the three replicates for each temperature to generate a set of consensus peaks. Next we analyze the genome wide distribution of the consensus H3K4me3 peaks.

```
k4.consensus <- read.table(file="h3k4me3/h3k4me3_peaks.bed",
                           header = F)

nrow(k4.consensus)

## [1] 4476

#jpeg(filename = "images/h3k27me3_genome_wide_distribution_ll_20.jpg", res = 300, width = 900, height = 900)
#par(mar=c(0,0,0,0))
plot(x=c(0,max.chr.len), y=c(0,4*number.chrs), col = "white",
     xlab = "", ylab = "", axes=F)

for(i in 1:number.chrs)
```

```

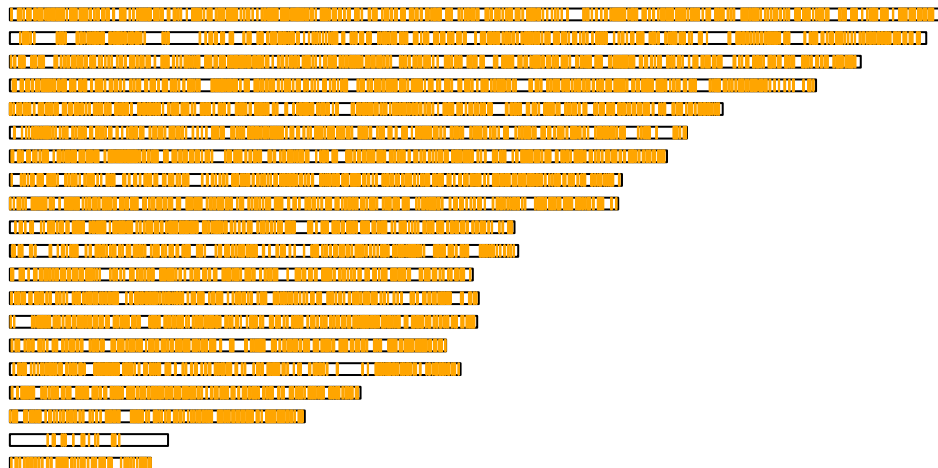
{
  polygon(x = c(0, chr.lens[i], chr.lens[i], 0),
    y = c(4*i+1, 4*i+1, 4*i-1, 4*i-1),
    col = "white")
}

for(i in 1:nrow(k4.consensus))
{
  current.chr <- k4.consensus[i,1]
  current.start <- k4.consensus[i,2]
  current.end <- k4.consensus[i,3]

  current.line <- (20:1)[current.chr]

  polygon(x = c(current.start,
    current.end,
    current.end,
    current.start),
    y = c(4*current.line+1,
      4*current.line+1,
      4*current.line-1,
      4*current.line-1),
    col = "orange",border="orange")
}

```



```
#dev.off()
```

We identified 4476 genomic regions significantly occupied by H3K4me3 in the three different biological replicates at 10°C, 14°C, 20°C or 26°C. We compute the percentage of the whole genome occupied by H3K4me3 and the percentage in each chromosome. A barplot is used to represent the percentage of H3K27me3 in each chromosome. Z-scores and p-independent values are computed for each chromosome. A gradient from white to blue is used to represent significance levels.

```

h3k4me3.peaks.len <- k4.consensus$V3 - k4.consensus$V2
sum(h3k4me3.peaks.len)/sum(chr.lens)

```

```
## [1] 0.2127566
```

```

k4.consensus <- cbind(k4.consensus,h3k4me3.peaks.len)
colnames(k4.consensus) <- c("chr","start","end","length")

```

```

head(k4.consensus)

##   chr start   end length
## 1    1  1456  2448    992
## 2    1  9308 10501   1193
## 3    1 11126 11452    326
## 4    1 12564 13452    888
## 5    1 16087 16450    363
## 6    1 20482 20649    167

number.of.h3k4.peaks.chr <- vector(mode = "numeric",length = 20)
peaks.len.chr <- vector(mode = "numeric",length = 20)

for(i in 1:20)
{
  number.of.h3k4.peaks.chr[i] <- nrow(subset(k4.consensus, chr == i))
  peaks.len.chr[i] <- sum(subset(k4.consensus, chr == i)[,4])
}

percentage.chr.k4 <- peaks.len.chr/rev(chr.lens)

z.scores <- (percentage.chr.k4 - mean(percentage.chr.k4))/sd(percentage.chr.k4)

which(z.scores > 2.5)

## integer(0)

p.vals <- pnorm((percentage.chr.k4 - mean(percentage.chr.k4))/sd(percentage.chr.k4),lower.tail = F)

p.vals

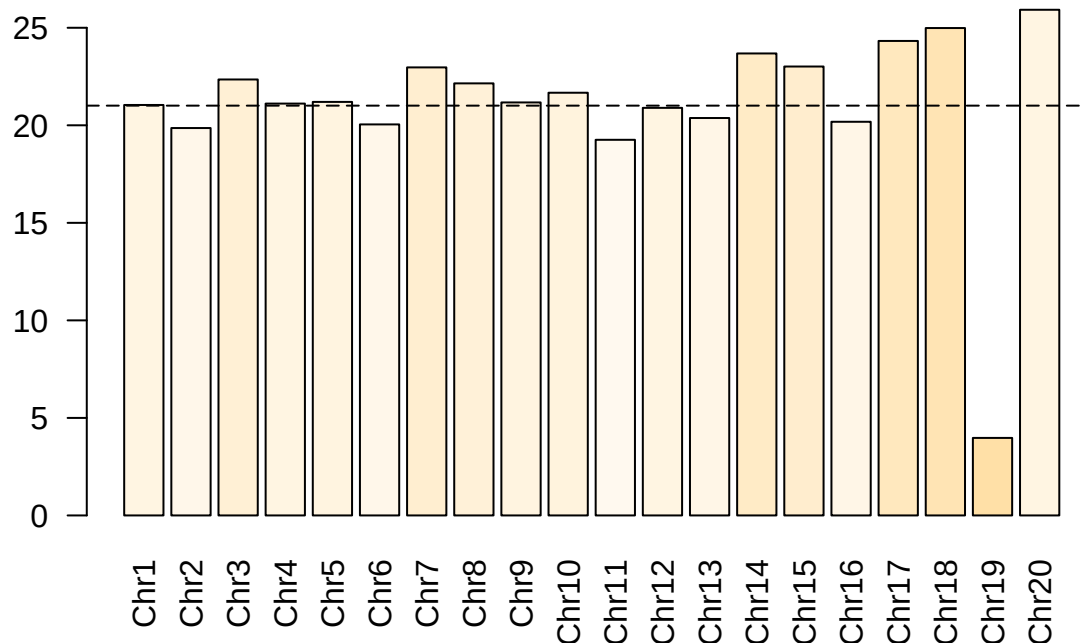
## [1] 0.4970257 0.6032652 0.3800408 0.4904729 0.4826958 0.5868156 0.3276454
## [8] 0.3975750 0.4852270 0.4402379 0.6551233 0.5103526 0.5574981 0.2711668
## [15] 0.3239413 0.5748505 0.2250500 0.1825120 0.9999478 0.1314986

values.for.colors <- round(-log10(p.vals)*40)

white.blue.palette <- colorRampPalette(c('white','orange'))

#jpeg(filename = "images/h3k27me3_chr_barplot_ll_20.jpg",res = 300,width = 1800,height = 900)
barplot(100*percentage.chr.k4,border="black",names.arg = paste0("Chr",1:20),las=2,col=white.blue.palette)
)
abline(h = 100*mean(percentage.chr.k4),lty=5)

```



```
#dev.off()
```

Approximately 14.5% of the *Ostreococcus* genome is occupied by H3K27me3. H3K27me3 is mostly uniformly distributed over the different chromosome although it significantly accumulates (p-value < 0.01) in Chr2 (sexual chromosome), Chr16 and Chr19 (viral resistance chromosome) with p-values 1.58×10^{-2} , 3.36×10^{-2} and 4.05×10^{-39} respectively.

We determine the gene features occupied by each H3K27me3 mark and the H3K27me3 marked genes.

```
library(TxDb.Otauri.JGI)
txdb <- TxDb.Otauri.JGI
```

```
ostta.genes <- as.data.frame(genes(txdb))
head(ostta.genes)
```

```
##          seqnames start   end width strand   gene_id
## ostta01g00010      1    244  1932  1689      - ostta01g00010
## ostta01g00020      1   2077  4029  1953      + ostta01g00020
## ostta01g00030      1   4157  5299  1143      + ostta01g00030
## ostta01g00040      1   5392  7041  1650      - ostta01g00040
## ostta01g00050      1   7386  9605  2220      - ostta01g00050
## ostta01g00060      1  9780 10412   633      - ostta01g00060
```

```
target.genes <- vector(mode = "character",length = nrow(k4.consensus))
peak.annotation <- vector(mode = "character",length = nrow(k4.consensus))
```

```
for(i in 1:nrow(k4.consensus))
{
  current.chr <- k4.consensus[i,1]
  current.start <- k4.consensus[i,2]
  current.end <- k4.consensus[i,3]

  complete.gene.body <- subset(ostta.genes,
                               seqnames == current.chr &
                               start >= current.start &
```

```

        end <= current.end)

overlap.tss.pos.strand <- subset(ostta.genes,
                                seqnames == current.chr &
                                start >= current.start & start <= current.end &
                                strand == "+")

overlap.tss.neg.strand <- subset(ostta.genes,
                                seqnames == current.chr &
                                end >= current.start & end <= current.end &
                                strand == "-")

overlap.tes.pos.strand <- subset(ostta.genes,
                                seqnames == current.chr &
                                end >= current.start & end <= current.end &
                                strand == "+")

overlap.tes.neg.strand <- subset(ostta.genes,
                                seqnames == current.chr &
                                start >= current.start & start <= current.end &
                                strand == "-")

inside.gene.body <- subset(ostta.genes, seqnames == current.chr &
                           start <= current.start & end >= current.end)

if(nrow(complete.gene.body) > 0)
{
  target.genes[i] <- paste(unlist(complete.gene.body$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "gene.body"
} else if(nrow(overlap.tss.pos.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tss.pos.strand$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "TSS"
} else if(nrow(overlap.tss.neg.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tss.neg.strand$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "TSS"
} else if(nrow(overlap.tes.pos.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.pos.strand$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(overlap.tes.neg.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.neg.strand$gene_id),
                           collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(inside.gene.body) > 0)
{

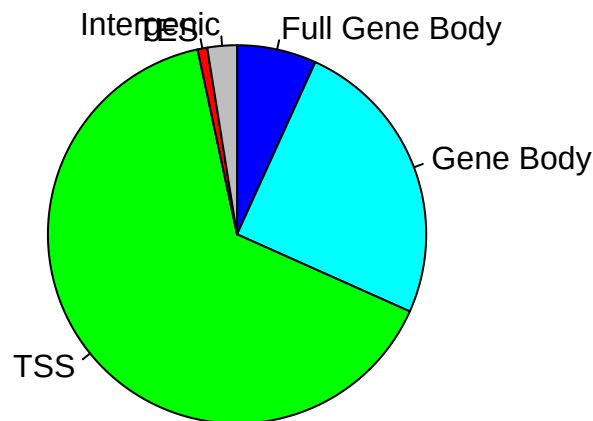
```

```

target.genes[i] <- paste(unlist(inside.gene.body$gene_id),
                        collapse = ",")
peak.annotation[i] <- "internal.gene.body"
} else
{
  peak.annotation[i] <- "intergenic"
}
}

freq.peak.annotation <- table(peak.annotation)
#jpeg(filename = "images/h3k27me3_pie_chart_ll_20.jpg",res = 300,width = 900,height = 900)
#par(mar=c(0,0,0,0))
pie(freq.peak.annotation[c("gene.body",
                          "internal.gene.body",
                          "TSS",
                          "TES",
                          "intergenic")],
    labels = c("Full Gene Body",
               "Gene Body",
               "TSS",
               "TES",
               "Intergenic"),
    clockwise = T, border = "black", col=c("blue","cyan","green","red","grey"))

```



```

#dev.off()

percentage.freq.peak.annotation <-
  100*freq.peak.annotation/sum(freq.peak.annotation)

percentage.freq.peak.annotation["gene.body"] +
percentage.freq.peak.annotation["internal.gene.body"]

```

```

## gene.body
## 31.68007

```

```
percentage.freq.peak.annotation
```

```

## peak.annotation
##      gene.body      intergenic internal.gene.body      TES
##      6.7917784      2.5245755      24.8882931      0.8266309

```

```

##           TSS
##      64.9687221

k4.consensus.peak.annotation <- cbind(k4.consensus[,1:3],
                                     data.frame(peak.annotation,target.genes))

rownames(k4.consensus.peak.annotation) <- apply(X = k4.consensus.peak.annotation[,1:3],
        MARGIN = 1,
        FUN =
        function(x)
        {
            paste(x[1],paste(x[2],x[3],sep="-"),sep=":")
        })

head(k4.consensus.peak.annotation)

##           chr start   end peak.annotation target.genes
## 1:1456-2448     1 1456 2448           TSS ostta01g00020
## 1:9308-10501    1 9308 10501      gene.body ostta01g00060
## 1:11126-11452   1 11126 11452           TSS ostta01g00070
## 1:12564-13452   1 12564 13452      gene.body ostta01g00080
## 1:16087-16450   1 16087 16450           TSS ostta01g00100
## 1:20482-20649   1 20482 20649 internal.gene.body ostta01g00110

write.table(x = k4.consensus.peak.annotation,
        file = "tables/H3K4me3_consensus_peak_annotation.tsv",
        quote = F,sep = "\t",row.names = F)

all.k4.genes <- unique(unlist(sapply(X =k4.consensus.peak.annotation$target.genes,
        FUN = function(x) {strsplit(x,",")[[1]]})))

names(all.k4.genes) <- NULL
length(all.k4.genes)

## [1] 4383

length(all.k4.genes)/7668

## [1] 0.5715962

## Genes with their entire gene body marked
k4.gene.body <- unique(unlist(sapply(
    X = subset(k4.consensus.peak.annotation,
        peak.annotation == "gene.body")[[ "target.genes"]],
    FUN = function(x) {strsplit(x,",")[[1]]})))
names(k4.gene.body) <- NULL
length(k4.gene.body)

## [1] 340

## Genes with H3K27me3 overlapping their TSS
k4.tss <- unique(unlist(sapply(
    X = subset(k4.consensus.peak.annotation,
        peak.annotation == "TSS")[[ "target.genes"]],
    FUN = function(x) {strsplit(x,",")[[1]]})))
names(k4.tss) <- NULL
length(k4.tss)

## [1] 2910

```

```
## Genes with H3K27me3 overlapping their TES
k4.tes <- unique(unlist(sapply(
  X = subset(k4.consensus.peak.annotation,
             peak.annotation == "TES")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]}))
names(k4.tes) <- NULL
length(k4.tes)

## [1] 37

## Genes with H3K27 inside their gene body with no overlap
## in the TSS or TES.
k4.internal.gene.body <- unique(unlist(sapply(
  X = subset(k4.consensus.peak.annotation,
             peak.annotation == "internal.gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]}))
names(k4.internal.gene.body) <- NULL
k4.internal.gene.body <- setdiff(k4.internal.gene.body, c(k27.tss, k27.tes))
length(k4.internal.gene.body)

## [1] 1076
```

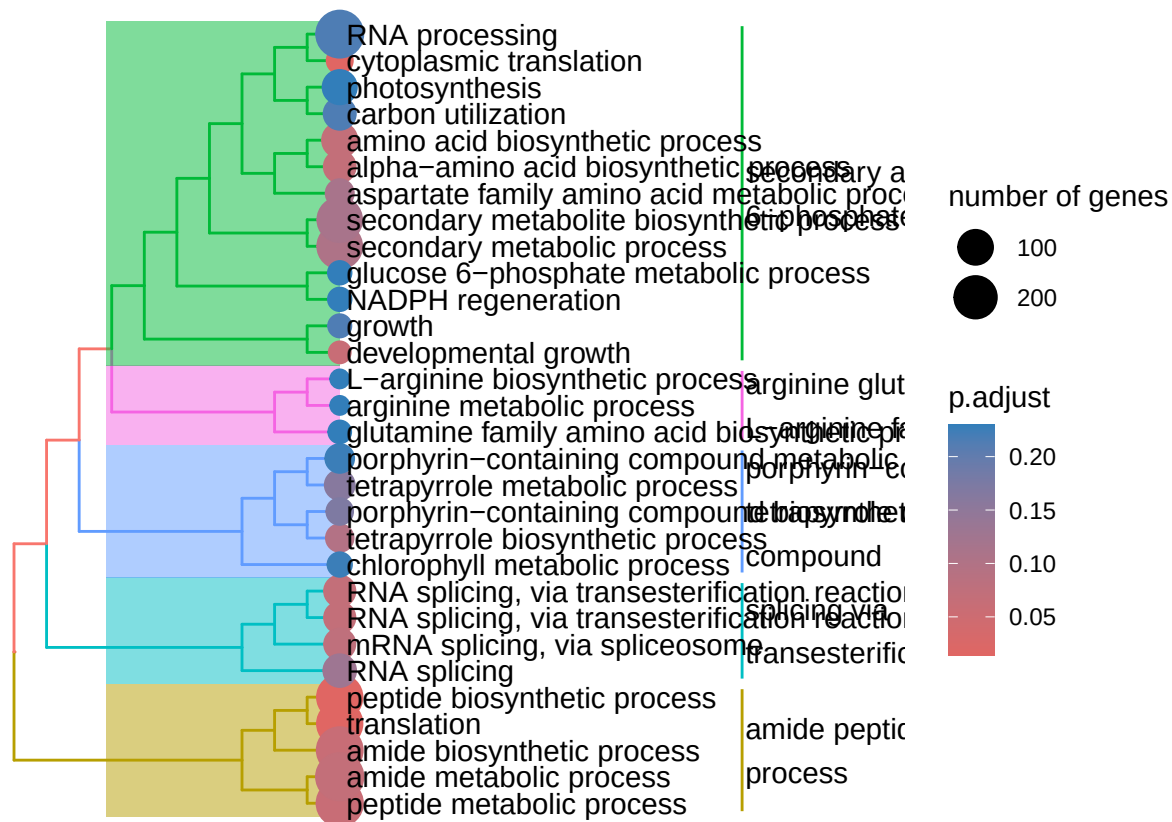
We found 994 genes associated to H3K27me3 marks corresponding to 13% of the entire *Ostreococcus* genome. H3K27me3 is mostly located occupying either the entire gene body (19.6%, 314 genes) or internal regions of gene body (52.8%, 433 genes). Although some regions occupied by H3K27me3 has been found overlapping only with the TSS (4.5%, 49 genes) or the TES (18.8%, 198 genes). H3K27me3 was found in intergenic regions only in 4.3% of the cases.

```
library(clusterProfiler)
library(enrichplot)
library(org.Otauriv5.eg.db)
k4.go.terms <- enrichGO(gene = all.k4.genes, pvalueCutoff = 1,
                        qvalueCutoff = 1,
                        OrgDb = org.Otauriv5.eg.db,
                        ont = "BP",
                        keyType = "GID")
k4.go.terms.filtered <- k4.go.terms
k4.go.terms.filtered@result <- subset(k4.go.terms@result, pvalue < 0.05)

k4.go.terms.df <- as.data.frame(k4.go.terms.filtered)

write.table(x = k27.go.terms.df, file = "go_enrichment_consensus_k27.tsv",
            quote = F, sep = "\t", row.names = F)

treeplot(x = pairwise_termsim(k4.go.terms.filtered))
```



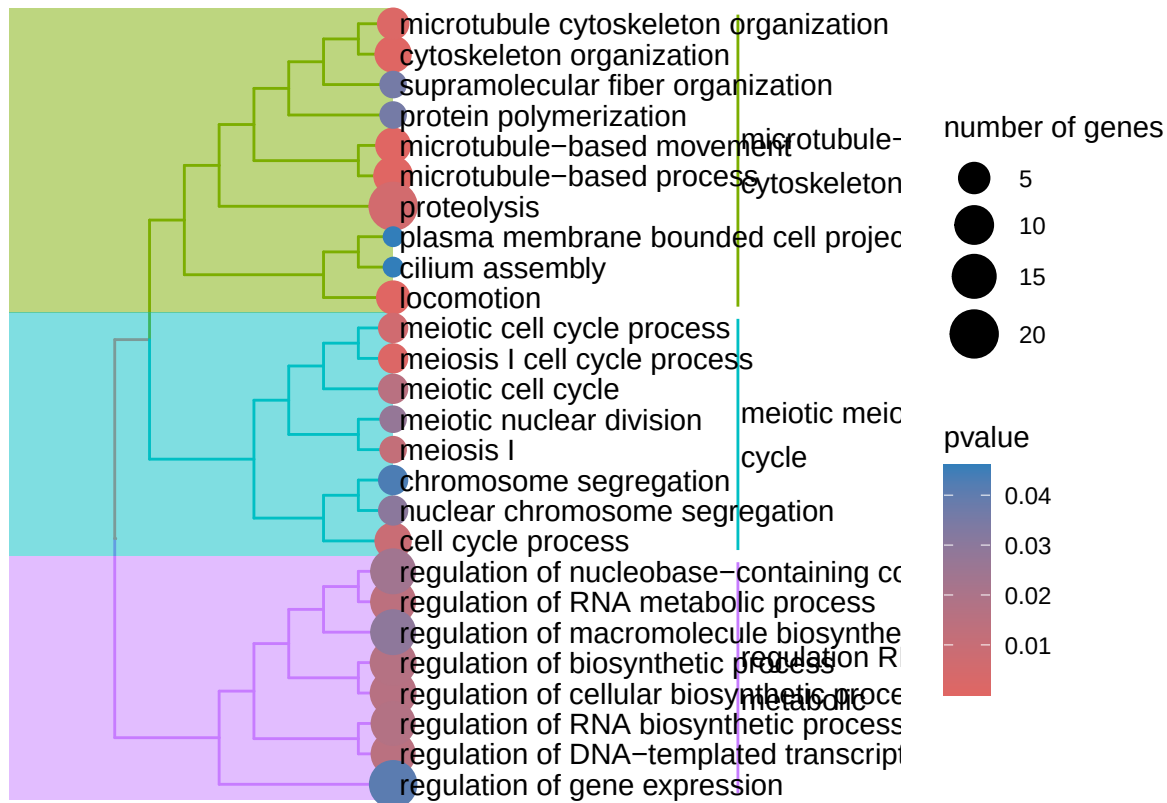
```
k27.go.terms <- enrichGO(gene = c(k27.gene.body,k27.tss), pvalueCutoff = 1,
  qvalueCutoff = 1,
  OrgDb = org.0tauriv5.eg.db,
  ont = "BP",
  keyType = "GID")
```

```
k27.go.terms.filtered <- k27.go.terms
```

```
k27.go.terms.filtered@result <- subset(k27.go.terms@result, pvalue < 0.05)
```

```
k27.go.terms.df <- as.data.frame(k27.go.terms.filtered)
```

```
#jpeg(filename = "images/consensus_h3k27me3_treeplot_gene_body.jpg",res = 250,width = 3500,height = 1300)
treeplot(x = pairwise_termsim(k27.go.terms.filtered),cluster.params = list(n = 3),color = "pvalue")
```



```
#dev.off()
```

```
write.table(x = k27.go.terms.df, file = "go_enrichment_full_gene_body_tss_consensus_k27.tsv",
           quote = F, sep = "\t", row.names = F)
```

Our GO enrichment analysis suggests that H3K27me3 is repressing the assembly of bounded cellular projection, microtubule-based movement and transcription regulation. Other processes are also significantly affected such as fatty acid metabolism and cell differentiation.

H3K27me3 response to different temperature and constant light

We study the changes in H3K27me3 as a response to temperature. We estimate ChIP signals using CPM (Counts Per Million of mapped reads).

```
## Million of mapped reads for each sample
mapped.reads.k27.20c.zt8.1 <- 2254676/1E6
mapped.reads.k27.20c.zt8.2 <- 2639017/1E6
mapped.reads.k27.20c.zt16.1 <- 3605883/1E6
mapped.reads.k27.20c.zt16.2 <- 3405826/1E6

mapped.reads.k27.10c.1 <- 6764067/1E6
mapped.reads.k27.10c.2 <- 6429853/1E6
mapped.reads.k27.10c.3 <- 6625634/1E6

mapped.reads.k27.14c.1 <- 24537863/1E6
mapped.reads.k27.14c.2 <- 6592650/1E6
mapped.reads.k27.14c.3 <- 4317544/1E6

mapped.reads.k27.20c.1 <- 3146662/1E6
```

```

mapped.reads.k27.20c.2 <- 7702115/1E6
mapped.reads.k27.20c.3 <- 7102648/1E6

mapped.reads.k27.26c.1 <- 5635098/1E6
mapped.reads.k27.26c.2 <- 4986577/1E6
mapped.reads.k27.26c.3 <- 3988645/1E6

## Load read counts and normalization by dividing by the millions
## of mapped reads
chip.signal.zt8.1 <- read.table(
  file = "h3k27me3/h3k27me3_ld_20_zt8_chip_1_counts.bed",
  header = F, sep = "\t")
chip.signal.zt8.1[,4] <- chip.signal.zt8.1[,4]/mapped.reads.k27.20c.zt8.1

chip.signal.zt8.2 <- read.table(
  file = "h3k27me3/h3k27me3_ld_20_zt8_chip_2_counts.bed",
  header = F, sep = "\t")
chip.signal.zt8.2[,4] <- chip.signal.zt8.2[,4]/mapped.reads.k27.20c.zt8.2

chip.signal.zt16.1 <- read.table(
  file = "h3k27me3/h3k27me3_ld_20_zt16_chip_1_counts.bed",
  header = F, sep = "\t")
chip.signal.zt16.1[,4] <- chip.signal.zt16.1[,4]/mapped.reads.k27.20c.zt16.1

chip.signal.zt16.2 <- read.table(
  file = "h3k27me3/h3k27me3_ld_20_zt16_chip_2_counts.bed",
  header = F, sep = "\t")
chip.signal.zt16.2[,4] <- chip.signal.zt16.2[,4]/mapped.reads.k27.20c.zt16.2

chip.signal.20.1 <- read.table(
  file = "h3k27me3/h3k27me3_ll_20_chip_1_counts.bed",
  header = F, sep = "\t")
chip.signal.20.1[,4] <- chip.signal.20.1[,4]/mapped.reads.k27.20c.1

chip.signal.20.2 <- read.table(
  file = "h3k27me3/h3k27me3_ll_20_chip_2_counts.bed",
  header = F, sep = "\t")
chip.signal.20.2[,4] <- chip.signal.20.2[,4]/mapped.reads.k27.20c.2

chip.signal.20.3 <- read.table(
  file = "h3k27me3/h3k27me3_ll_20_chip_3_counts.bed",
  header = F, sep = "\t")
chip.signal.20.3[,4] <- chip.signal.20.3[,4]/mapped.reads.k27.20c.3

chip.signal.10.1 <- read.table(
  file = "h3k27me3/h3k27me3_ll_10_chip_1_counts.bed",
  header = F, sep = "\t")
chip.signal.10.1[,4] <- chip.signal.10.1[,4]/mapped.reads.k27.10c.1

chip.signal.10.2 <- read.table(
  file = "h3k27me3/h3k27me3_ll_10_chip_2_counts.bed",
  header = F, sep = "\t")

```

```

chip.signal.10.2[,4] <- chip.signal.10.2[,4]/mapped.reads.k27.10c.2

chip.signal.10.3 <- read.table(
  file = "h3k27me3/h3k27me3_ll_10_chip_3_counts.bed",
  header = F, sep = "\t")
chip.signal.10.3[,4] <- chip.signal.10.3[,4]/mapped.reads.k27.10c.3

chip.signal.14.1 <- read.table(
  file = "h3k27me3/h3k27me3_ll_14_chip_1_counts.bed",
  header = F, sep = "\t")
chip.signal.14.1[,4] <- chip.signal.14.1[,4]/mapped.reads.k27.14c.1

chip.signal.14.2 <- read.table(
  file = "h3k27me3/h3k27me3_ll_14_chip_2_counts.bed",
  header = F, sep = "\t")
chip.signal.14.2[,4] <- chip.signal.14.2[,4]/mapped.reads.k27.14c.2

chip.signal.14.3 <- read.table(
  file = "h3k27me3/h3k27me3_ll_14_chip_3_counts.bed",
  header = F, sep = "\t")
chip.signal.14.3[,4] <- chip.signal.14.3[,4]/mapped.reads.k27.14c.3

chip.signal.26.1 <- read.table(
  file = "h3k27me3/h3k27me3_ll_26_chip_1_counts.bed",
  header = F, sep = "\t")
chip.signal.26.1[,4] <- chip.signal.26.1[,4]/mapped.reads.k27.26c.1

chip.signal.26.2 <- read.table(
  file = "h3k27me3/h3k27me3_ll_26_chip_2_counts.bed",
  header = F, sep = "\t")
chip.signal.26.2[,4] <- chip.signal.26.2[,4]/mapped.reads.k27.26c.2

chip.signal.26.3 <- read.table(
  file = "h3k27me3/h3k27me3_ll_26_chip_3_counts.bed",
  header = F, sep = "\t")
chip.signal.26.3[,4] <- chip.signal.26.3[,4]/mapped.reads.k27.26c.3

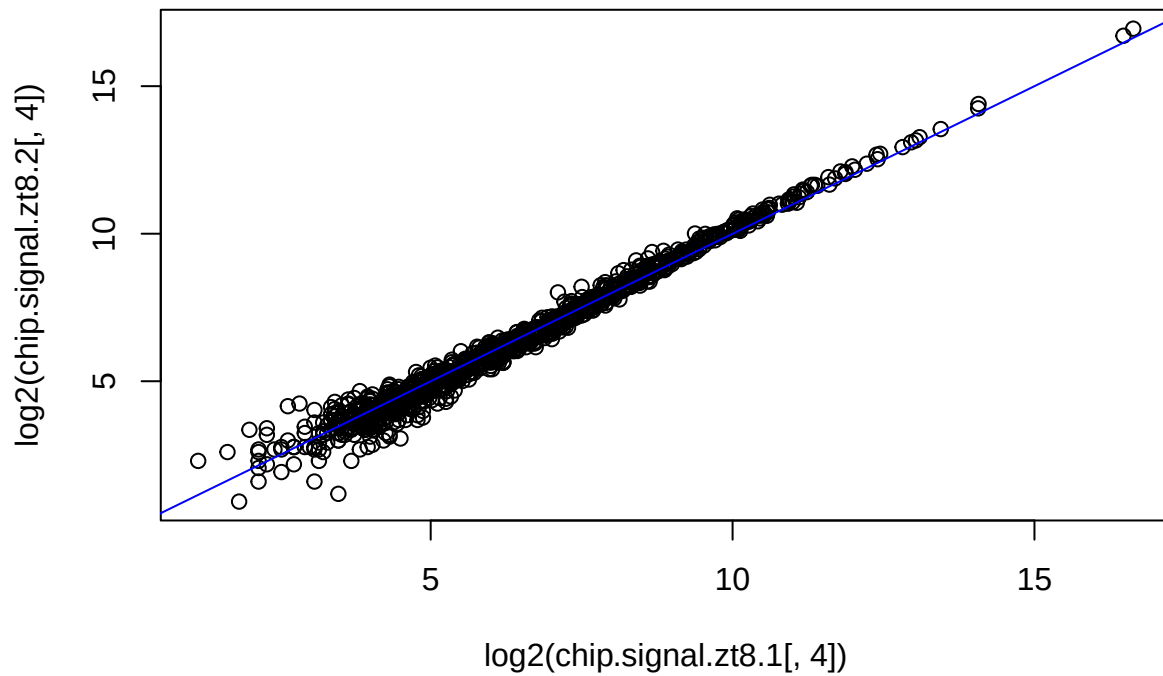
```

Scatter plots between replicates to check similarity.

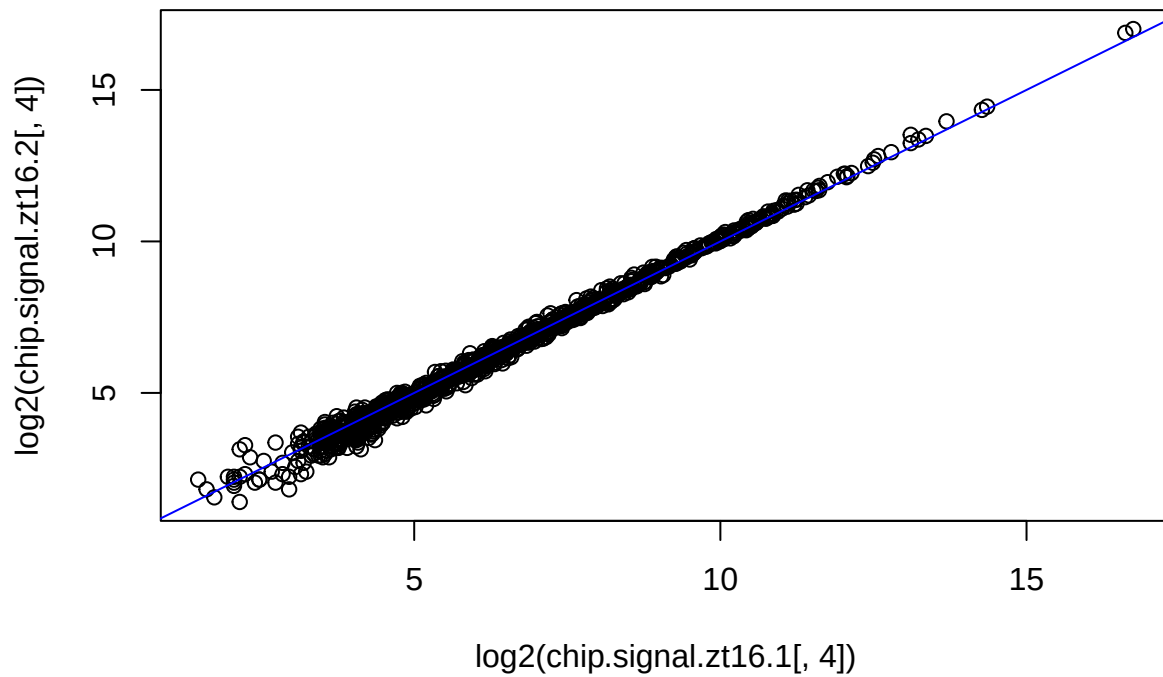
```

plot(log2(chip.signal.zt8.1[,4]), log2(chip.signal.zt8.2[,4]))
lines(x=c(0,20), y=c(0,20), col="blue")

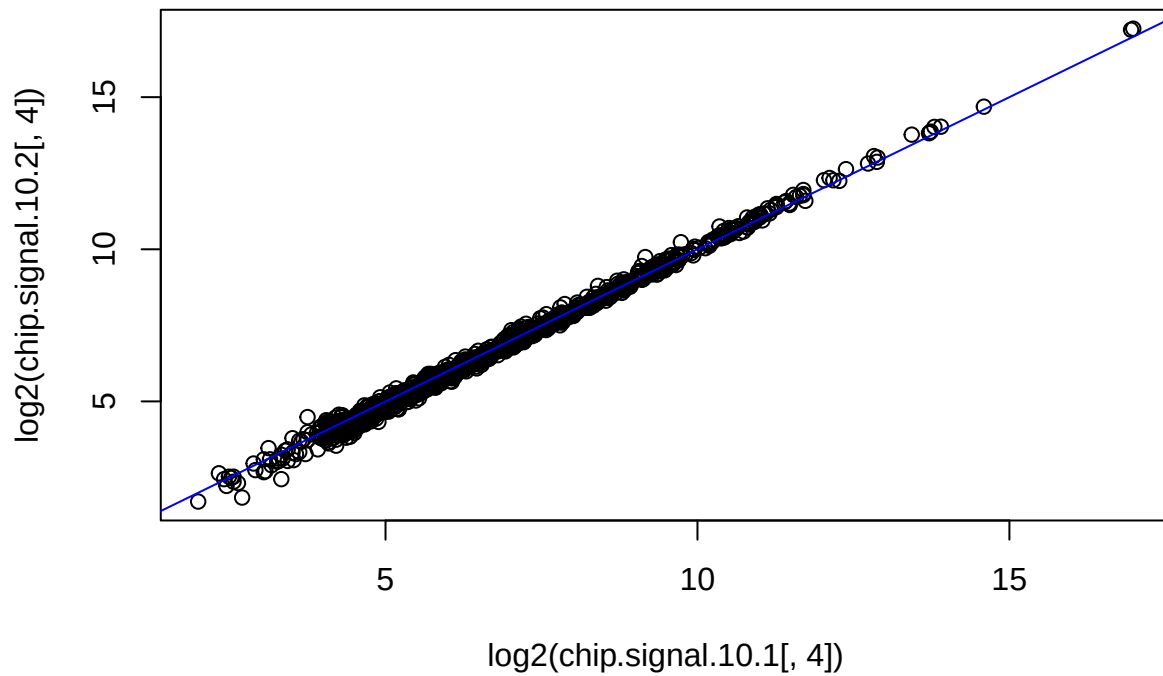
```



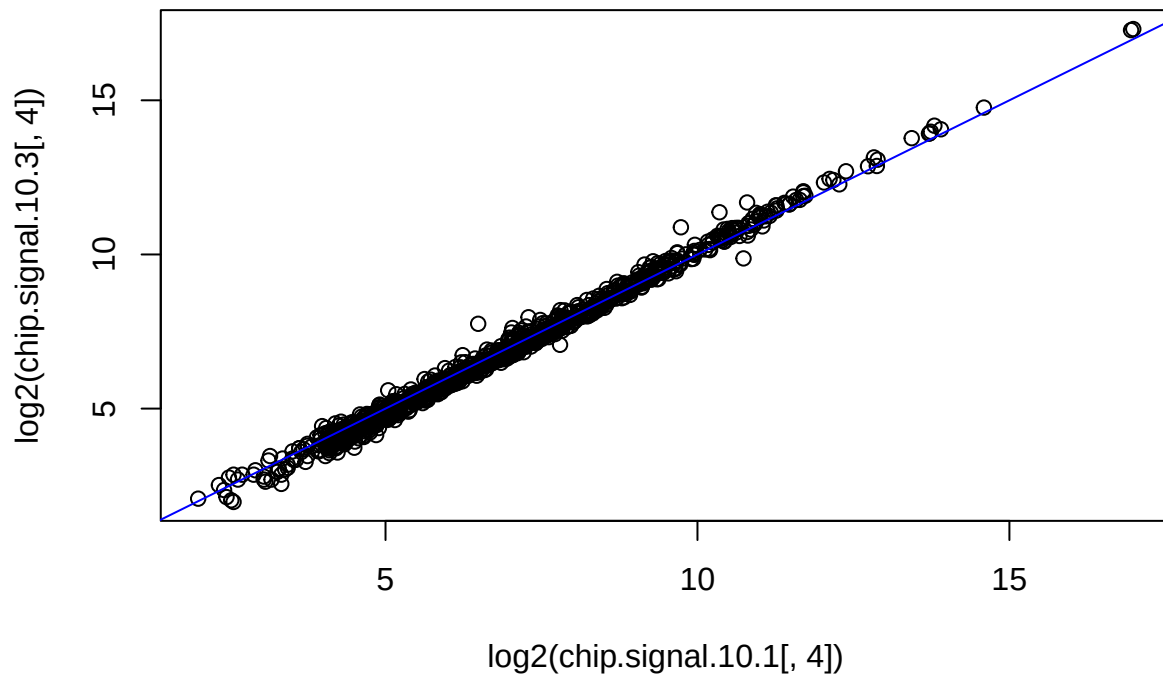
```
plot(log2(chip.signal.zt16.1[,4]),log2(chip.signal.zt16.2[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



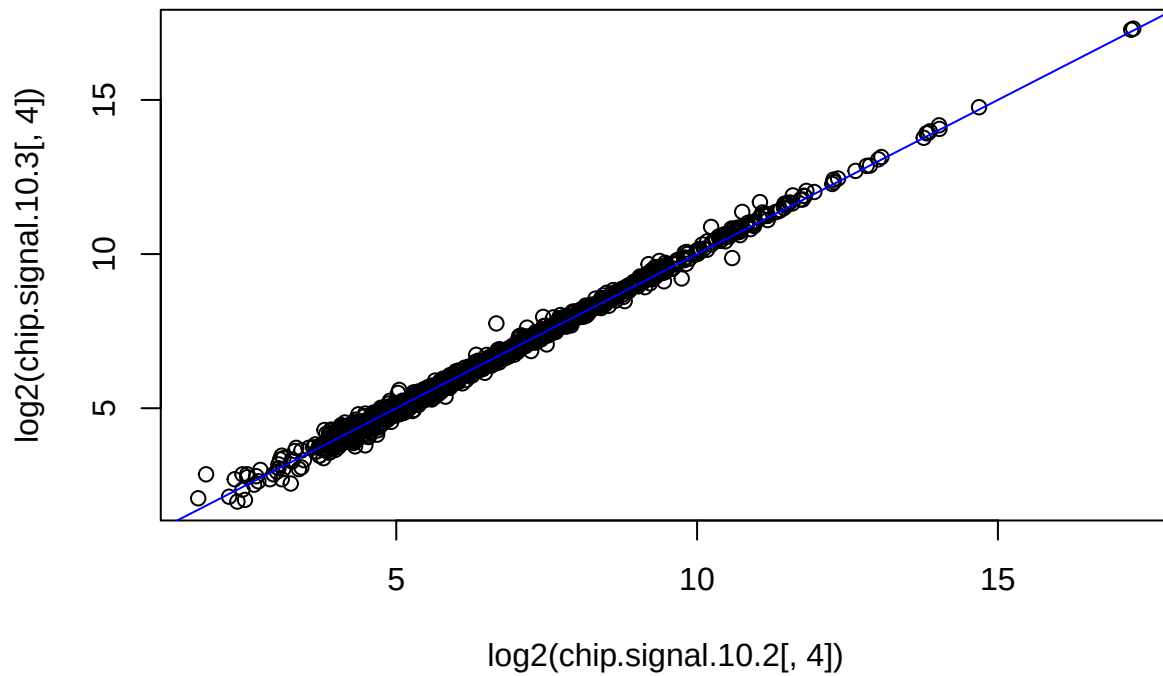
```
plot(log2(chip.signal.10.1[,4]),log2(chip.signal.10.2[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



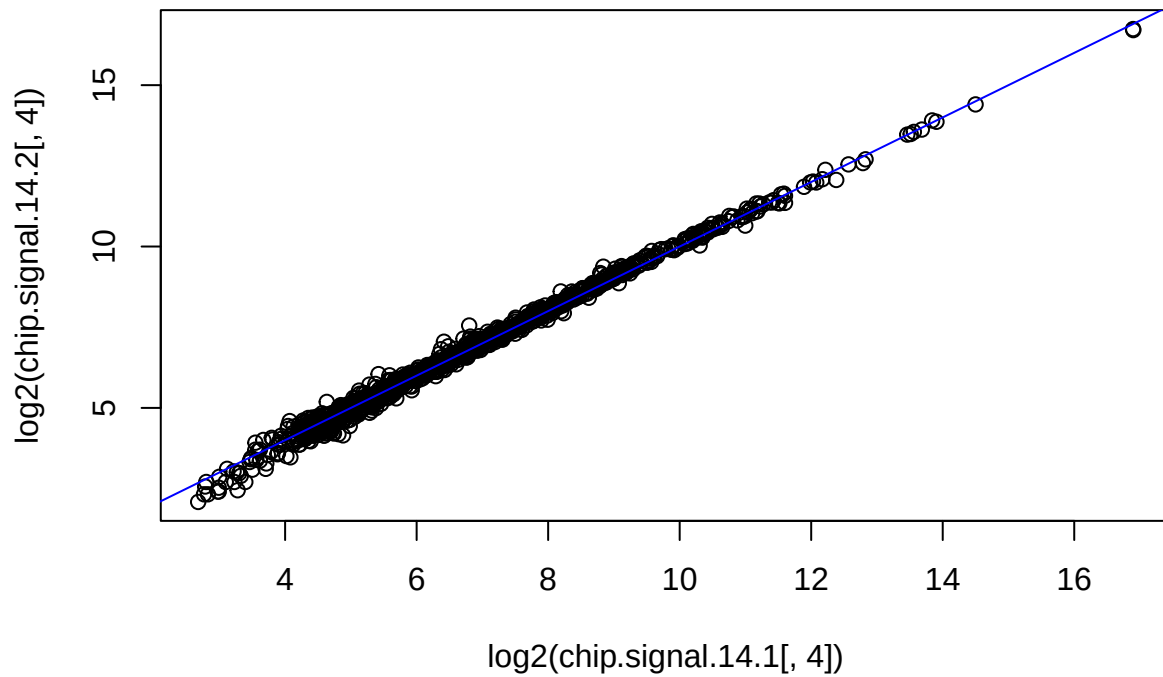
```
plot(log2(chip.signal.10.1[,4]),log2(chip.signal.10.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



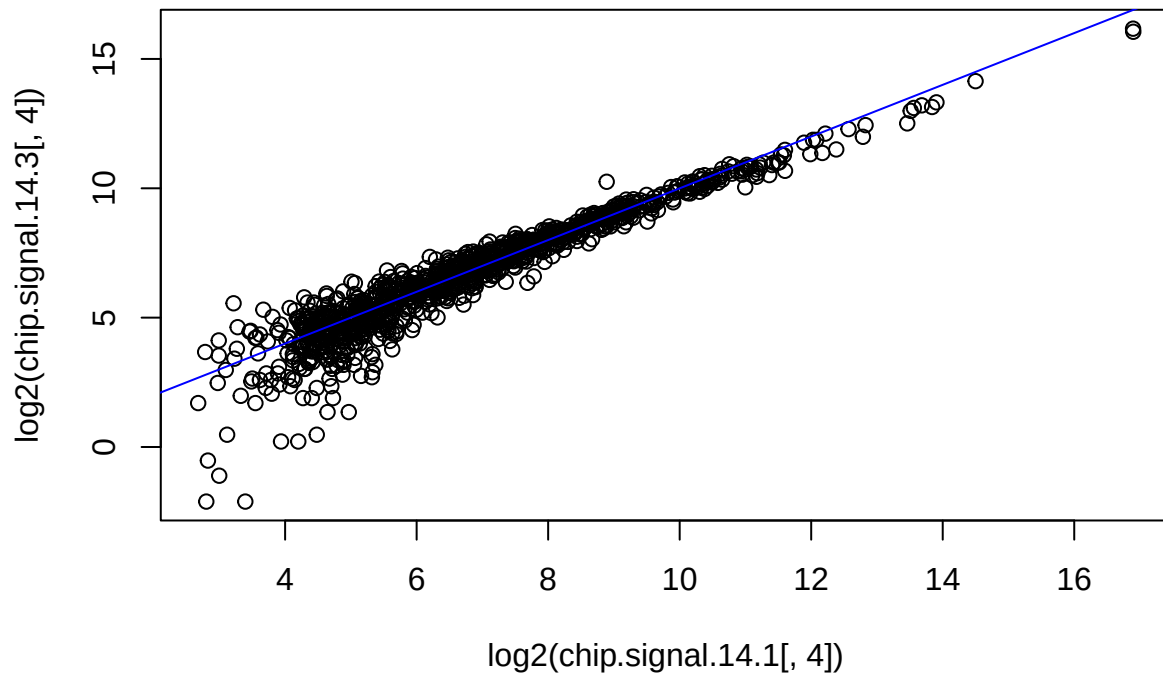
```
plot(log2(chip.signal.10.2[,4]),log2(chip.signal.10.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



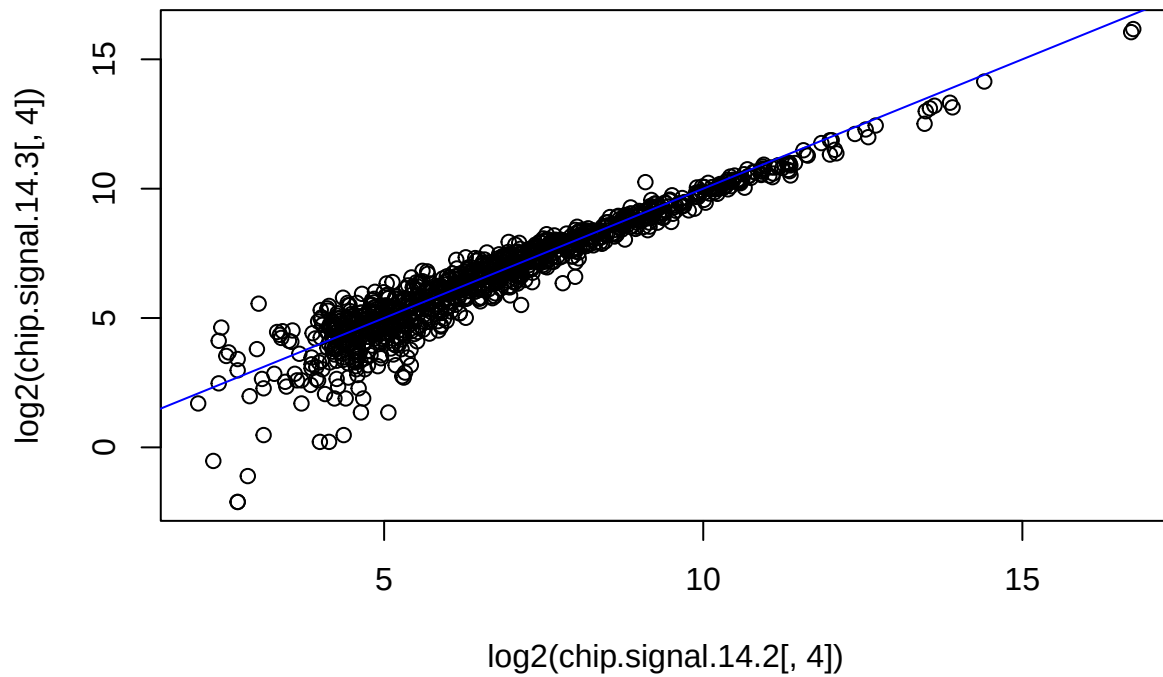
```
plot(log2(chip.signal.14.1[,4]),log2(chip.signal.14.2[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



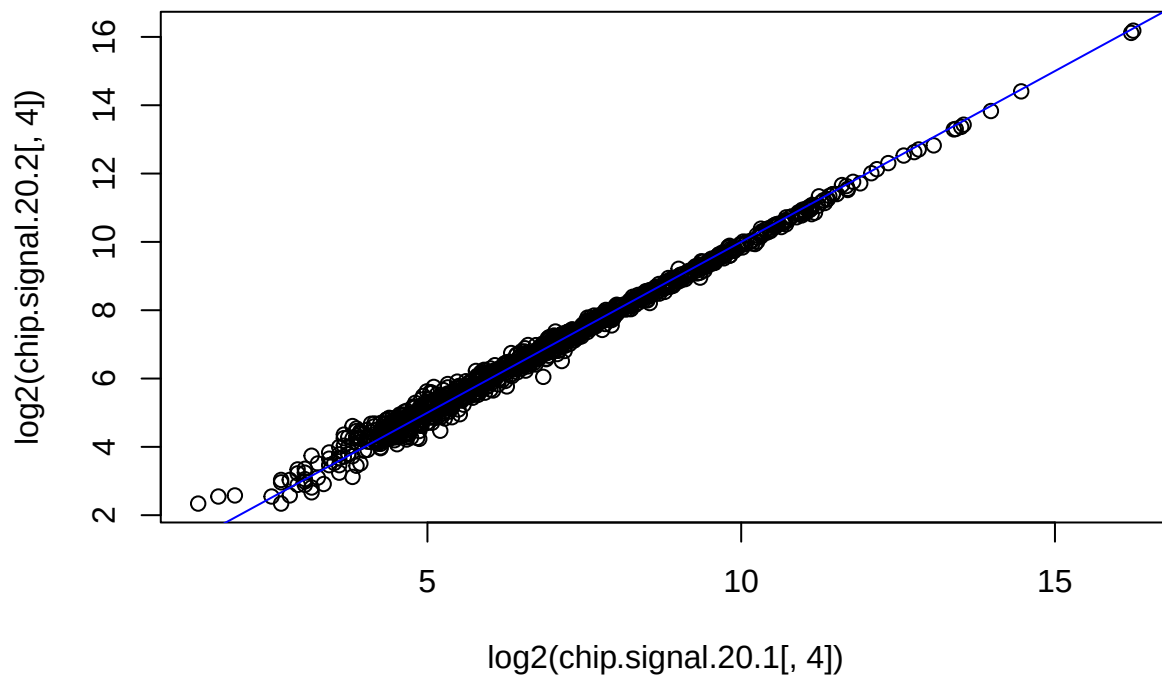
```
plot(log2(chip.signal.14.1[,4]),log2(chip.signal.14.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



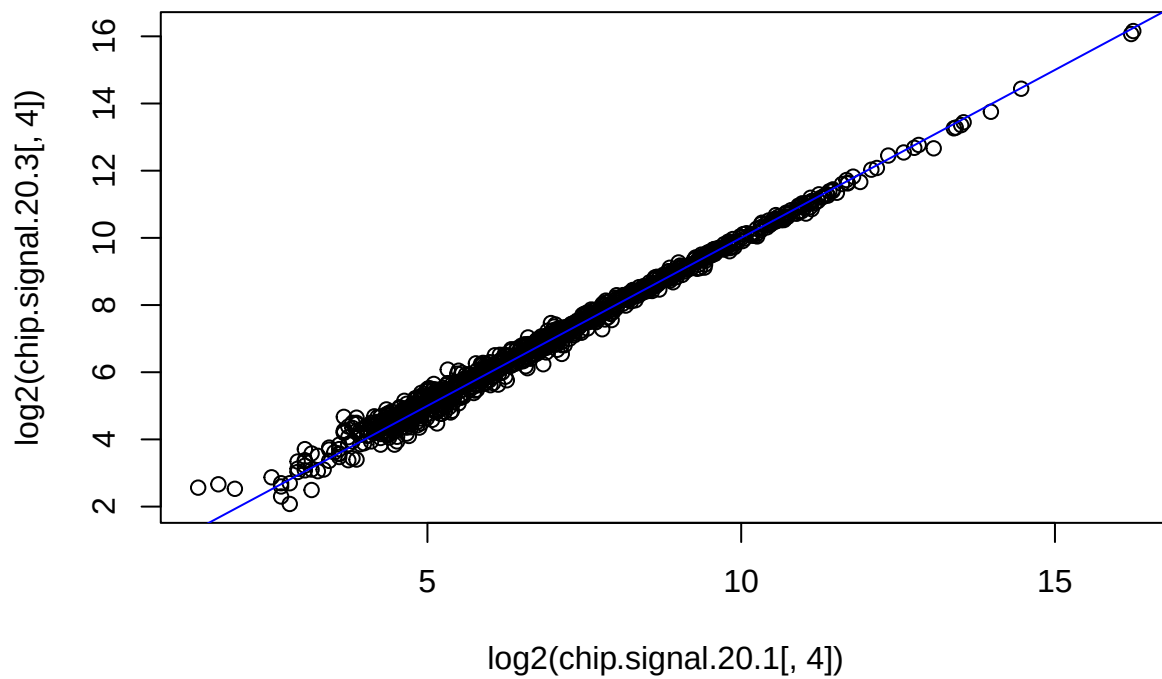
```
plot(log2(chip.signal.14.2[,4]),log2(chip.signal.14.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



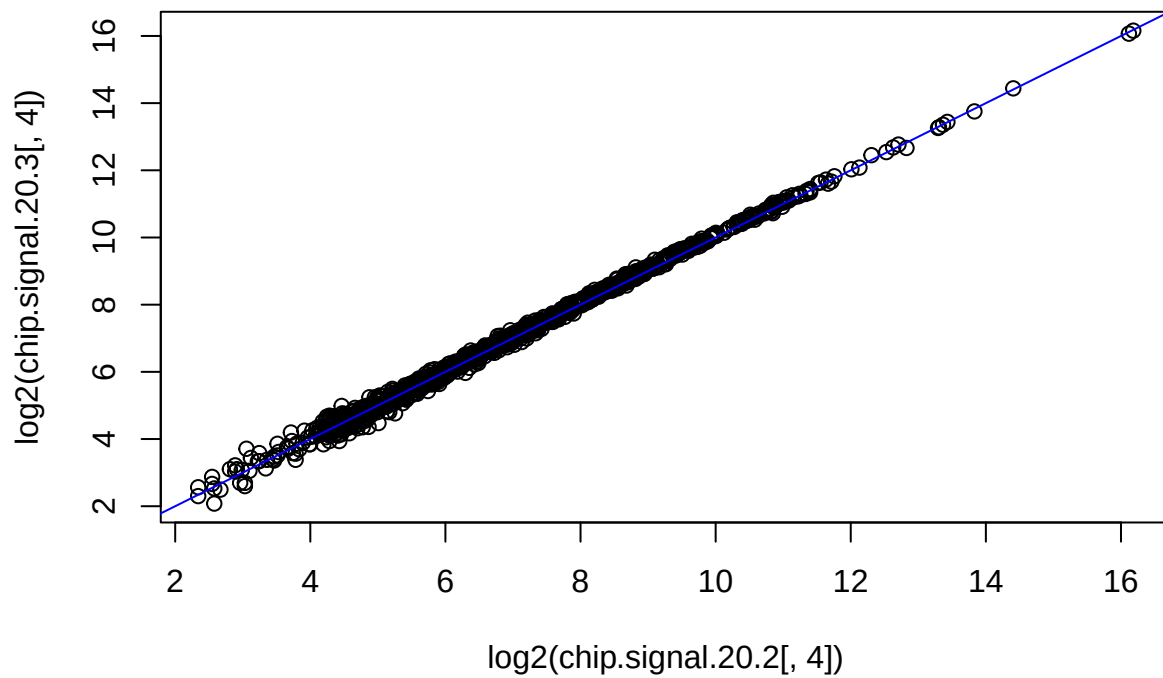
```
plot(log2(chip.signal.20.1[,4]),log2(chip.signal.20.2[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



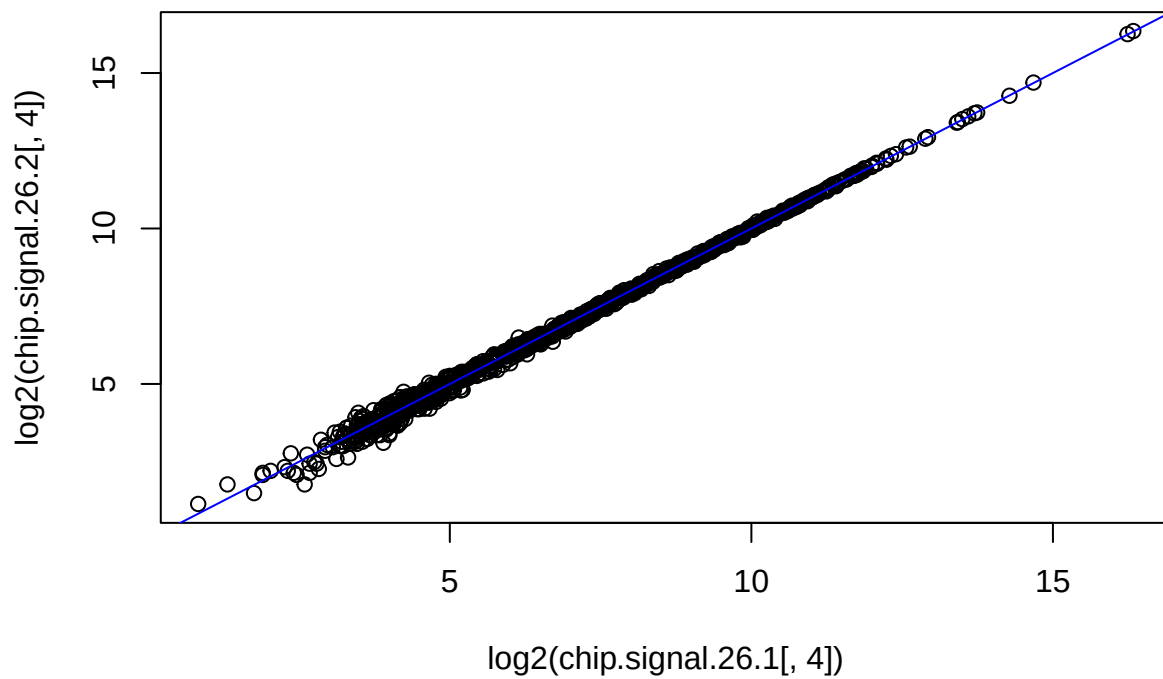
```
plot(log2(chip.signal.20.1[,4]),log2(chip.signal.20.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



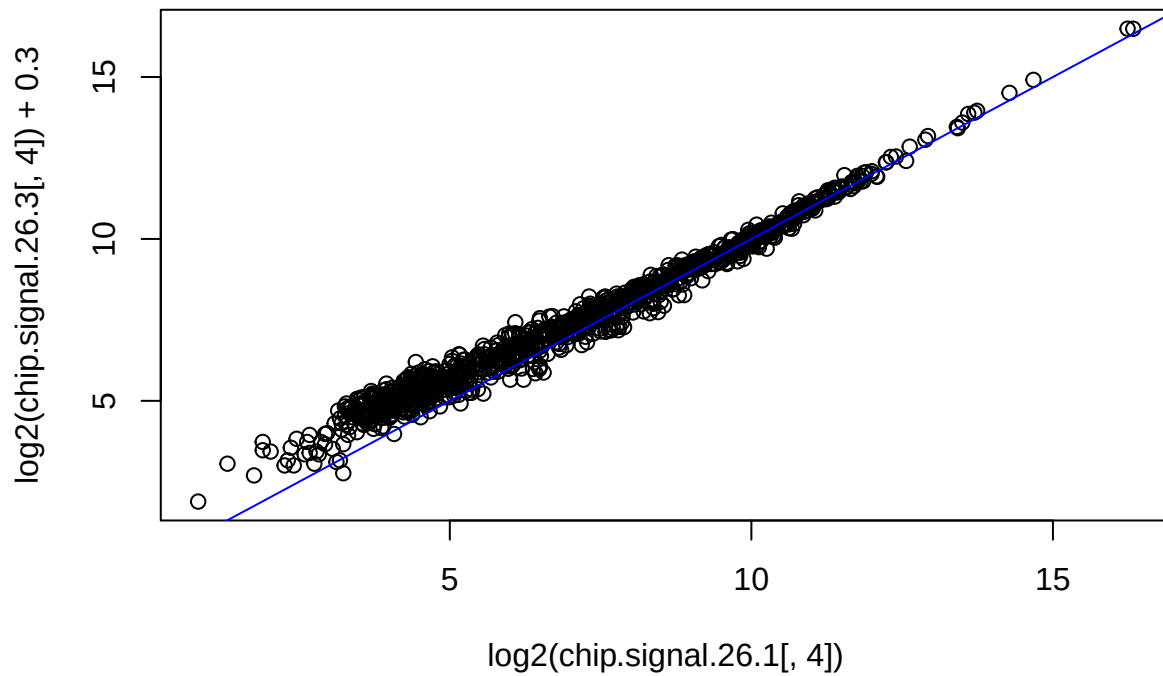
```
plot(log2(chip.signal.20.2[,4]),log2(chip.signal.20.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



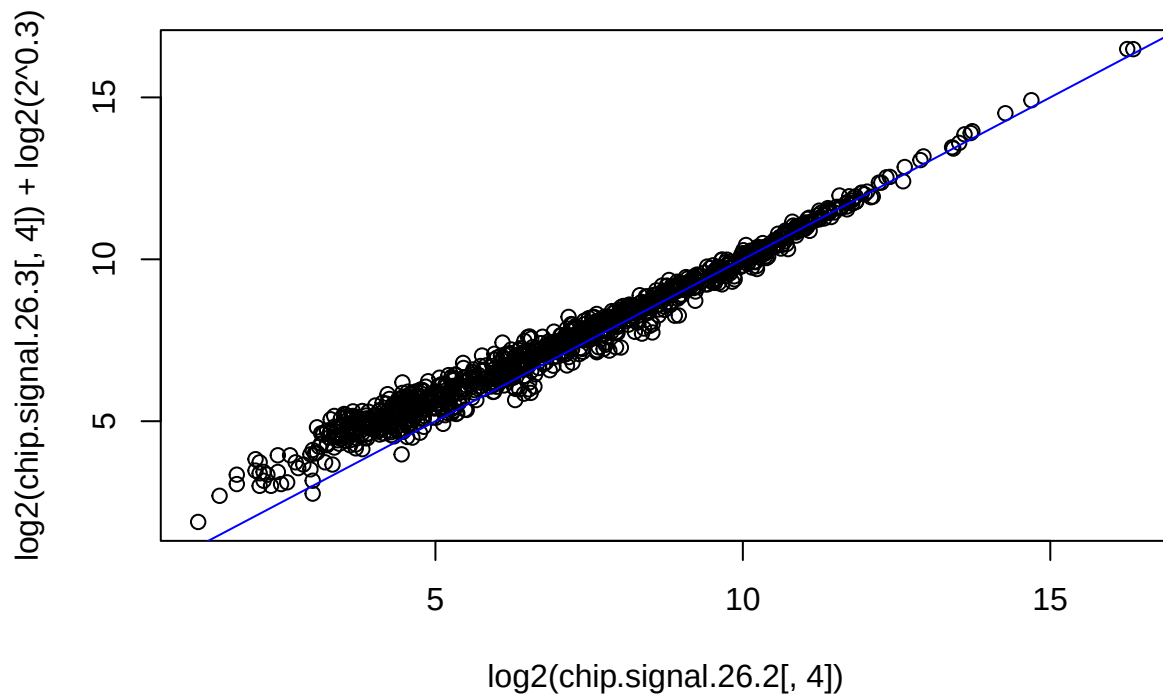
```
plot(log2(chip.signal.26.1[,4]),log2(chip.signal.26.2[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



```
plot(log2(chip.signal.26.1[,4]),log2(chip.signal.26.3[,4])+0.3)
lines(x=c(0,20),y=c(0,20),col="blue")
```



```
plot(log2(chip.signal.26.2[,4]),log2(chip.signal.26.3[,4])+log2(2^0.3))
lines(x=c(0,20),y=c(0,20),col="blue")
```



We generate the ChIP signal matrix.

```
chip.signal <- matrix(data = c(chip.signal.zt8.1[,4],
                                chip.signal.zt8.2[,4],
                                chip.signal.zt16.1[,4],
                                chip.signal.zt16.2[,4],
                                chip.signal.10.1[,4],
                                chip.signal.10.2[,4],
```

```

chip.signal.10.3[,4],
chip.signal.14.1[,4],
chip.signal.14.2[,4],
chip.signal.14.3[,4],
chip.signal.20.1[,4],
chip.signal.20.2[,4],
chip.signal.20.3[,4],
chip.signal.26.1[,4],
chip.signal.26.2[,4],
chip.signal.26.3[,4]),ncol=16)
colnames(chip.signal) <- c("ZT8_1","ZT8_2",
  "ZT16_1","ZT16_2",
  "t10C_1","t10C_2","t10C_3",
  "t14C_1","t14C_2","t14C_3",
  "t20C_1","t20C_2","t20C_3",
  "t26C_1","t26C_2","t26C_3")
rownames(chip.signal) <- apply(X = chip.signal.10.1[,1:3],
  MARGIN = 1,
  FUN =
    function(x)
    {
      paste(x[1],paste(x[2],x[3],sep="-"),sep=":")
    }
  )
head(chip.signal)

```

```

##           ZT8_1      ZT8_2      ZT16_1      ZT16_2      t10C_1      t10C_2
## 1:0-1529      180.95726 294.42781 301.72915 314.754776 170.01606 158.47952
## 1:2910-6526   175.19147 179.99126 162.51221 150.917868 262.26825 254.28264
## 1:17915-18081  10.64454  10.61001  11.64763   7.927592  20.25409  18.66295
## 1:25485-25705  14.19273  14.02037  12.47961  14.093497  19.81057  17.26323
## 1:38421-41511 547.75054 570.66703 630.63610 631.858468 652.71382 680.73096
## 1:92611-93775  55.88386  70.48079  67.94452  68.705800  89.88675  88.64899
##           t10C_3      t14C_1      t14C_2      t14C_3      t20C_1      t20C_2
## 1:0-1529      140.51485 157.10415 158.81322 118.354324 197.03419 179.30140
## 1:2910-6526   244.80676 259.06902 272.57628 291.369353 303.81401 298.09994
## 1:17915-18081  14.94197  20.49893  15.77514   9.727753  19.06782  17.00832
## 1:25485-25705  18.41333  21.39551  23.51103   9.727753  18.43223  18.04699
## 1:38421-41511 725.81733 596.54747 628.12374 641.568447 762.71300 737.58961
## 1:92611-93775  83.46371  87.86421  92.83065 133.409179 109.32220 107.11344
##           t20C_3      t26C_1      t26C_2      t26C_3
## 1:0-1529      184.43825 299.90605 298.200549 242.18751
## 1:2910-6526   302.56321 430.16111 419.325722 421.94780
## 1:17915-18081  19.57017  15.97133  10.227457  18.55267
## 1:25485-25705  18.44383  10.82501   8.823688  20.05694
## 1:38421-41511 813.49942 1307.87433 1298.285377 953.45662
## 1:92611-93775 105.17204 113.57389 119.520866 114.07383

```

Principal component analysis to analyze the underlying data structure.

```

library(FactoMineR)
library(factoextra)

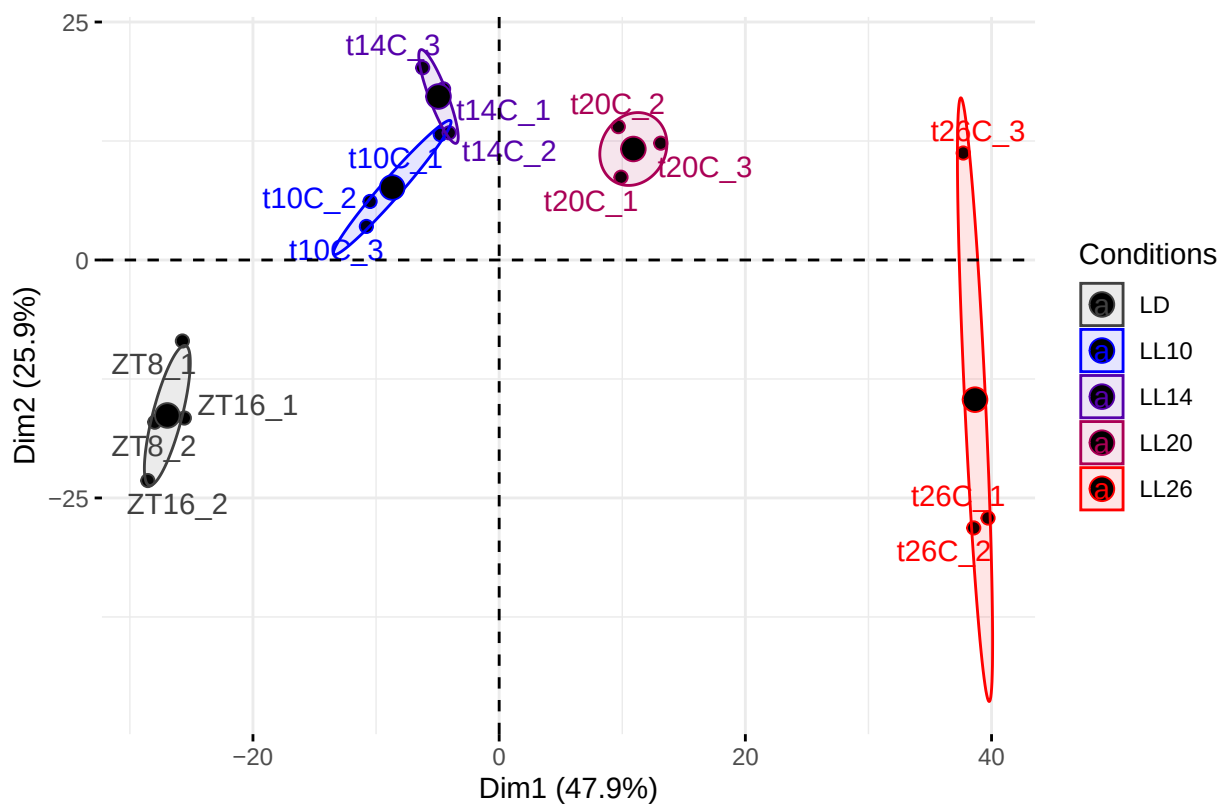
```

```
## Loading required package: ggplot2
```

Welcome! Want to learn more? See two factoextra-related books at <https://goo.gl/ve3WBa>

```
pca.temp.chip.signal <- data.frame(colnames(chip.signal),
                                   t(chip.signal))

colnames(pca.temp.chip.signal)[1] <- "Sample"
res.pca <- PCA(pca.temp.chip.signal,
              graph = FALSE,
              scale.unit = TRUE,
              quali.sup = 1 )
#jpeg(filename = "images/h3k27me3_chip_signal_pca.jpg",res = 160,width = 750,height = 750)
fviz_pca_ind(res.pca, col.ind = c("LD","LD","LD","LD",
                                   rep("LL10",3),rep("LL14",3),
                                   rep("LL20",3),rep("LL26",3)),
             palette = c("gray25",colorRampPalette(c("blue", "red"))(4)),
             pointsize=2, pointshape=21,fill="black",
             repel = TRUE,
             addEllipses = TRUE,ellipse.type = "confidence",
             legend.title="Conditions",
             title="",
             show_legend=TRUE,show_guide=TRUE)
```



```
#dev.off()

res.hcpc <- HCPC(res.pca, graph=FALSE,nb.clust = 4)

fviz_dend(res.hcpc,k=4,
          cex = 0.75,
          palette = c("gray25",rev(colorRampPalette(c("blue", "red"))(3))),# "jco",
          # Color
```

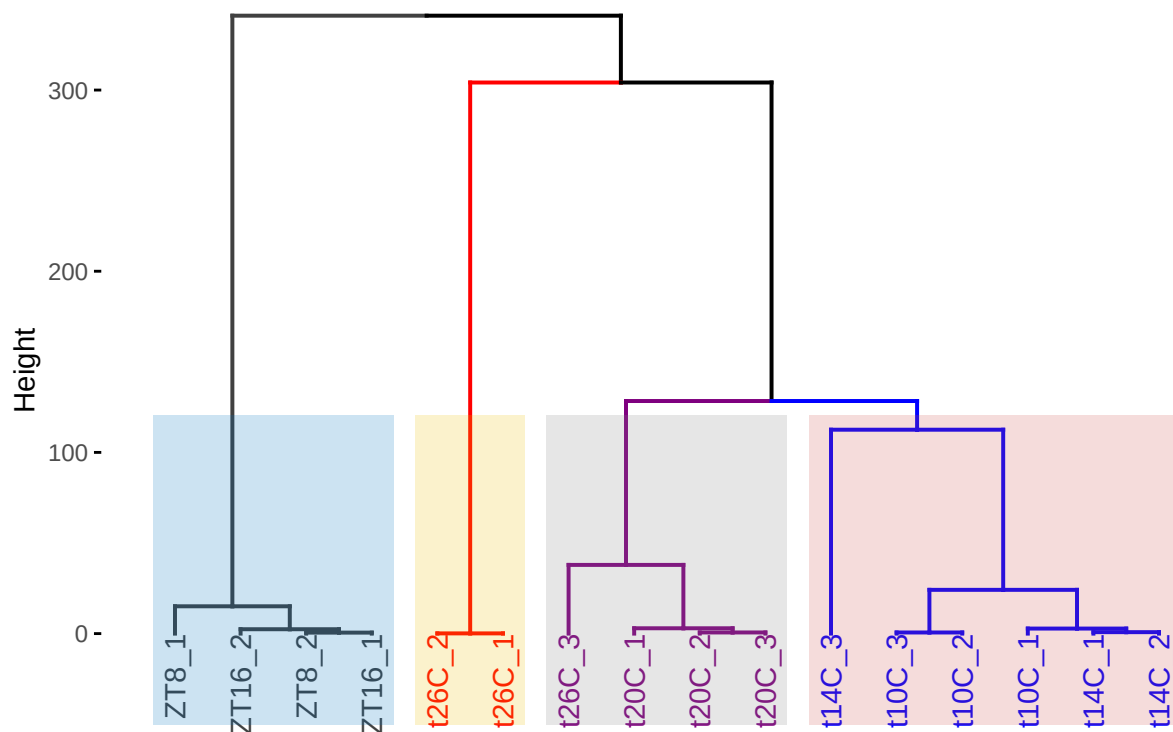
```

    rect = TRUE, rect_fill = TRUE, # Add rectangle around groups
    rect_border = "jco",          # Rectangle color
    type="rectangle",
    labels_track_height = 50      # Augment the room for labels
  )

## Warning: The `<scale>` argument of `guides()` cannot be `FALSE`. Use "none" instead as
## of ggplot2 3.3.4.
## i The deprecated feature was likely used in the factoextra package.
## Please report the issue at <https://github.com/kassambara/factoextra/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

Cluster Dendrogram



Principal Components Analysis is repeated without the samples for ZT8 and ZT16.

```

pca.temp.chip.signal <- data.frame(colnames(chip.signal[,5:16]),
                                   t(chip.signal[,5:16]))

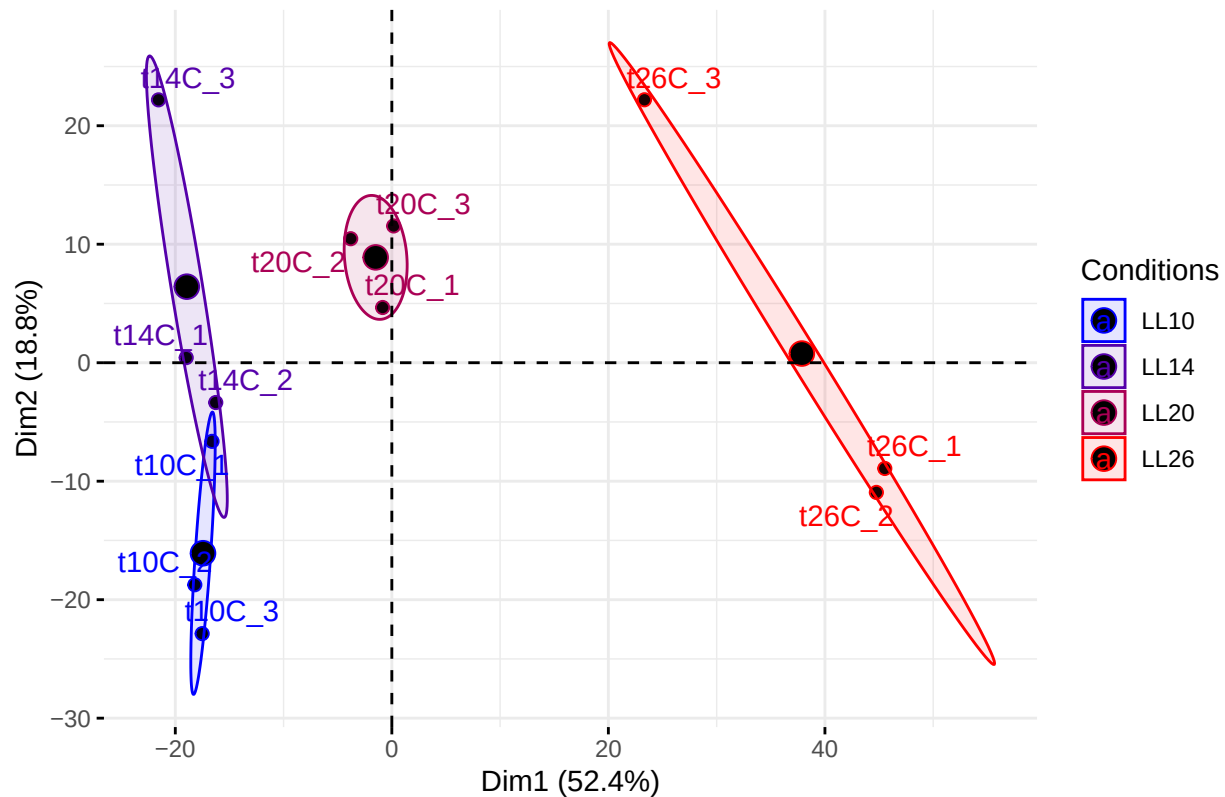
colnames(pca.temp.chip.signal)[1] <- "Sample"
res.pca <- PCA(pca.temp.chip.signal,
              graph = FALSE,
              scale.unit = TRUE,
              quali.sup = 1 )
#jpeg(filename = "images/h3k27me3_chip_signal_pca_2.jpg",res = 160,width = 750,height = 750)
fviz_pca_ind(res.pca, col.ind = c(rep("LL10",3),rep("LL14",3),
                                   rep("LL20",3),rep("LL26",3)),
             palette = colorRampPalette(c("blue", "red"))(4),
             pointsize=2, pointshape=21,fill="black",

```

```

repel = TRUE,
addEllipses = TRUE,ellipse.type = "confidence",
legend.title="Conditions",
title="",
show_legend=TRUE,show_guide=TRUE)

```



```

#dev.off()

```

```

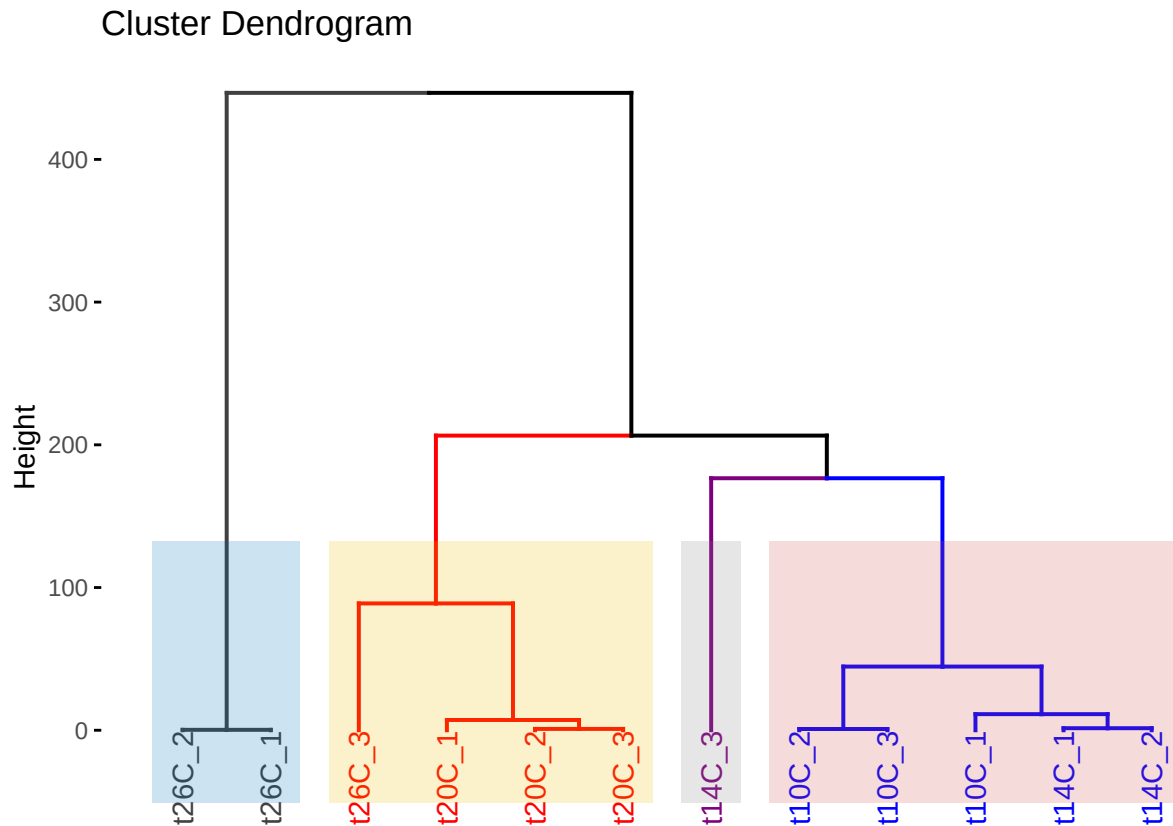
res.hcpc <- HCPC(res.pca, graph=FALSE,nb.clust = 4)

```

```

fviz_dend(res.hcpc,k=4,
  cex = 0.75,                                # Label size
  palette = c("gray25",rev(colorRampPalette(c("blue", "red"))(3))),#"jco",          # Color
  rect = TRUE, rect_fill = TRUE, # Add rectangle around groups
  rect_border = "jco",                        # Rectangle color
  type="rectangle",
  labels_track_height = 50                    # Augment the room for labels
)

```



We determine the H3K27me3 peaks that significantly change their ChIP signal as a response to temperature and the corresponding differentially H3k27me3 marked genes.

```
log2.chip.signal <- log2(chip.signal)
```

```
library(limma)
```

```
##
```

```
## Attaching package: 'limma'
```

```
## The following object is masked from 'package:BiocGenerics':
```

```
##
```

```
## plotMA
```

```
## The following object is masked from 'package:seqinr':
```

```
##
```

```
## zscore
```

```
limma.experimental.design <- model.matrix(~ -1+factor(c(1,1,2,2,3,3,3,4,4,4,5,5,5,6,6,6)))
colnames(limma.experimental.design) <- c("LDZT8", "LDZT16",
                                           "LL10", "LL14", "LL20", "LL26")
```

```
linear.fit <- lmFit(log2.chip.signal, limma.experimental.design)
```

```
contrast.matrix <- makeContrasts(LDZT16-LDZT8,
                                LL26-LL10,
                                LL20-LL10,
                                LL14-LL10,
                                levels = c("LDZT8", "LDZT16", "LL10", "LL14", "LL20", "LL26"))
```

```

contrast.linear.fit <- contrasts.fit(linear.fit, contrast.matrix)
contrast.results <- eBayes(contrast.linear.fit)

zt16.zt8 <- topTable(contrast.results,
                     number=nrow(log2.chip.signal),
                     coef = 1, sort.by = "logFC")

write.table(x = zt16.zt8,file = "tables/diff_peaks_zt16_zt8.tsv",
            quote = F,sep = "\t")

zt16.zt8.logfc <- zt16.zt8$logFC
zt16.zt8.adj.p.val <- zt16.zt8$adj.P.Val
zt16.zt8.peak.ids <- rownames(zt16.zt8)

zt16.zt8.higher.peaks <- zt16.zt8.peak.ids[zt16.zt8.logfc > log2(1) &
                                           zt16.zt8.adj.p.val < 0.05]

zt16.zt8.lower.peaks <- zt16.zt8.peak.ids[zt16.zt8.logfc < -log2(1) &
                                           zt16.zt8.adj.p.val < 0.05]

t26.t10 <- topTable(contrast.results,
                    number=nrow(log2.chip.signal),
                    coef = 2, sort.by = "logFC")

write.table(x = t26.t10,file = "tables/diff_peaks_26c_10c.tsv",
            quote = F,sep = "\t")

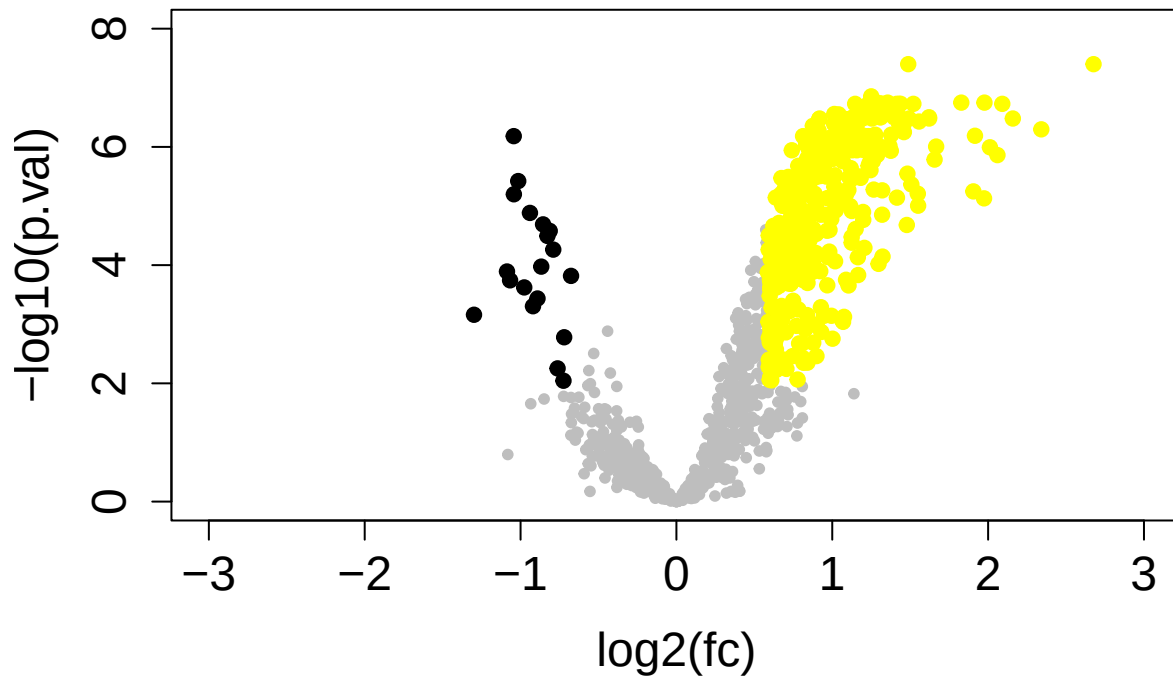
t26.t10.logfc <- t26.t10$logFC
t26.t10.adj.p.val <- t26.t10$adj.P.Val
t26.t10.peak.ids <- rownames(t26.t10)
names(t26.t10.logfc) <- t26.t10.peak.ids
names(t26.t10.adj.p.val) <- t26.t10.peak.ids

t26.t10.higher.peaks <- t26.t10.peak.ids[t26.t10.logfc > log2(1.5) &
                                           t26.t10.adj.p.val < 0.01]

t26.t10.lower.peaks <- t26.t10.peak.ids[t26.t10.logfc < -log2(1.5) &
                                           t26.t10.adj.p.val < 0.01]

#jpeg(filename = "images/h3k27me3_chip_signal_volcano_26_10.jpg",res = 160,width = 750,height = 750)
plot(x = t26.t10.logfc,y = -log10(t26.t10.adj.p.val),
     pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,col="grey",
     xlab="log2(fc)",ylab="-log10(p.val)",cex.lab=1.5)
points(x = t26.t10.logfc[t26.t10.higher.peaks],y = -log10(t26.t10.adj.p.val)[t26.t10.higher.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="yellow")
points(x = t26.t10.logfc[t26.t10.lower.peaks],y = -log10(t26.t10.adj.p.val)[t26.t10.lower.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="black")

```



```
#dev.off()

t26.t10.higher.peaks.genes <- unique(unlist(
  strsplit(x = subset(k27.consensus.peak.annotation,
    rownames(k27.consensus.peak.annotation) %in%
    t26.t10.higher.peaks)$target.genes,split=",")))

t26.t10.lower.peaks.genes <- unique(unlist(
  strsplit(x = subset(k27.consensus.peak.annotation,
    rownames(k27.consensus.peak.annotation) %in%
    t26.t10.lower.peaks)$target.genes,split=",")))

t20.t10 <- topTable(contrast.results,
  number=nrow(log2.chip.signal),
  coef = 3, sort.by = "logFC")

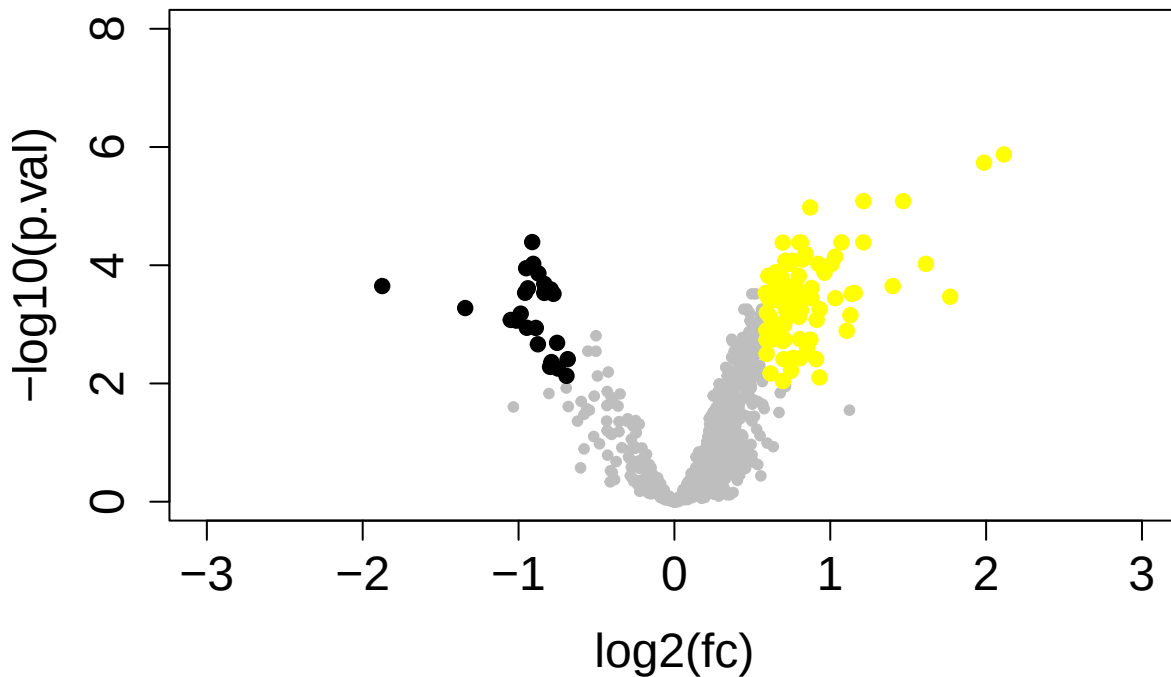
write.table(x = t20.t10,file = "tables/diff_peaks_20c_10c.tsv",
  quote = F,sep = "\t")

t20.t10.logfc <- t20.t10$logFC
t20.t10.adj.p.val <- t20.t10$adj.P.Val
t20.t10.peak.ids <- rownames(t20.t10)
names(t20.t10.logfc) <- t20.t10.peak.ids
names(t20.t10.adj.p.val) <- t20.t10.peak.ids

t20.t10.higher.peaks <- t20.t10.peak.ids[t20.t10.logfc > log2(1.5) &
  t20.t10.adj.p.val < 0.01]

t20.t10.lower.peaks <- t20.t10.peak.ids[t20.t10.logfc < -log2(1.5) &
  t20.t10.adj.p.val < 0.01]
```

```
#jpeg(filename = "images/h3k27me3_chip_signal_volcano_20_10.jpg",res = 160,width = 750,height = 750)
plot(x = t20.t10.logfc,y = -log10(t20.t10.adj.p.val),
     pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,col="grey",
     xlab="log2(fc)",ylab="-log10(p.val)",cex.lab=1.5)
points(x = t20.t10.logfc[t20.t10.higher.peaks],y = -log10(t20.t10.adj.p.val)[t20.t10.higher.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="yellow")
points(x = t20.t10.logfc[t20.t10.lower.peaks],y = -log10(t20.t10.adj.p.val)[t20.t10.lower.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="black")
```



```
#dev.off()

t20.t10.higher.peaks.genes <- unique(unlist(
  strsplit(x = subset(k27.consensus.peak.annotation,
                     rownames(k27.consensus.peak.annotation) %in%
                     t20.t10.higher.peaks)$target.genes,split=",")))

t20.t10.lower.peaks.genes <- unique(unlist(
  strsplit(x = subset(k27.consensus.peak.annotation,
                     rownames(k27.consensus.peak.annotation) %in%
                     t20.t10.lower.peaks)$target.genes,split=",")))

t14.t10 <- topTable(contrast.results,
                    number=nrow(log2.chip.signal),
                    coef = 4, sort.by = "logFC")

write.table(x = t14.t10,file = "tables/diff_peaks_14c_10c.tsv",
           quote = F,sep = "\t")

t14.t10.logfc <- t14.t10$logFC
t14.t10.adj.p.val <- t14.t10$adj.P.Val
```

```

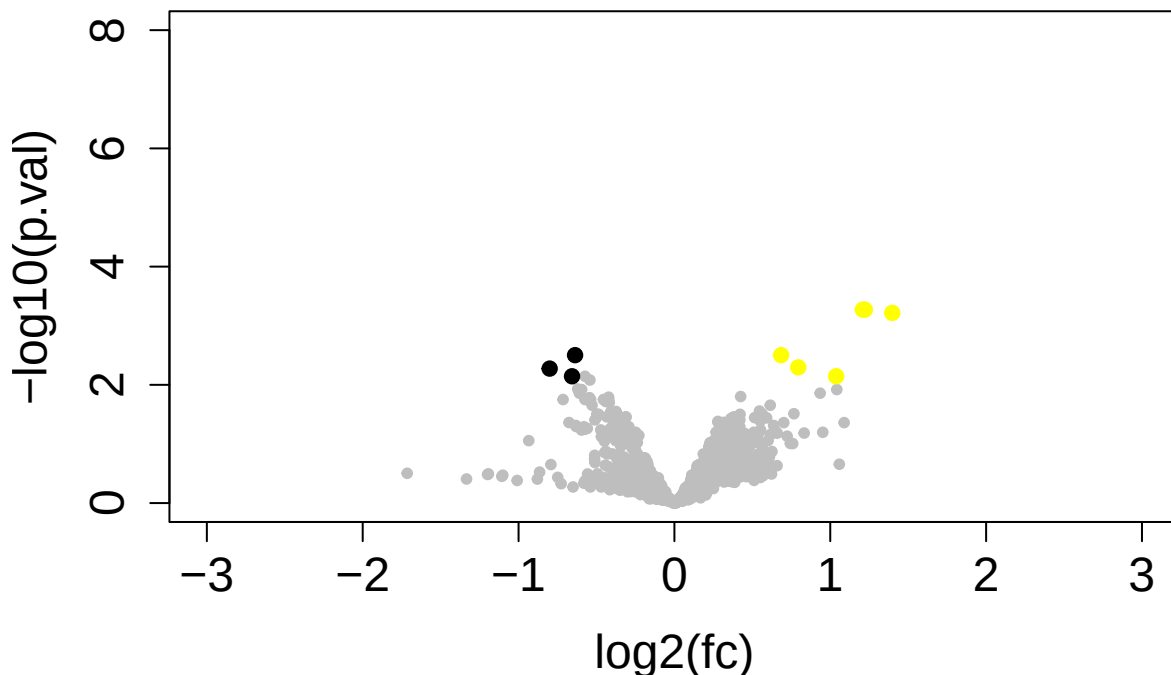
t14.t10.peak.ids <- rownames(t14.t10)
names(t14.t10.logfc) <- t14.t10.peak.ids
names(t14.t10.adj.p.val) <- t14.t10.peak.ids

t14.t10.higher.peaks <- t14.t10.peak.ids[t14.t10.logfc > log2(1.5) &
                                         t14.t10.adj.p.val < 0.01]

t14.t10.lower.peaks <- t14.t10.peak.ids[t14.t10.logfc < -log2(1.5) &
                                         t14.t10.adj.p.val < 0.01]

#jpeg(filename = "images/h3k27me3_chip_signal_volcano_14_10.jpg",res = 160,width = 750,height = 750)
plot(x = t14.t10.logfc,y = -log10(t14.t10.adj.p.val),
     pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,col="grey",
     xlab="log2(fc)",ylab="-log10(p.val)",cex.lab=1.5)
points(x = t14.t10.logfc[t14.t10.higher.peaks],y = -log10(t14.t10.adj.p.val)[t14.t10.higher.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="yellow")
points(x = t14.t10.logfc[t14.t10.lower.peaks],y = -log10(t14.t10.adj.p.val)[t14.t10.lower.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="black")

```



```

#dev.off()

t14.t10.higher.peaks.genes <- unique(unlist(
  strsplit(x = subset(k27.consensus.peak.annotation,
                     rownames(k27.consensus.peak.annotation) %in%
                     t14.t10.higher.peaks)$target.genes,split=",")
))

t14.t10.lower.peaks.genes <- unique(unlist(
  strsplit(x = subset(k27.consensus.peak.annotation,
                     rownames(k27.consensus.peak.annotation) %in%
                     t14.t10.lower.peaks)$target.genes,split=",")
))

```

```

k27.temp.higher.peaks <- unique(c(t26.t10.higher.peaks,
                                t20.t10.higher.peaks,
                                t14.t10.higher.peaks))

length(k27.temp.higher.peaks)

## [1] 402

length(k27.temp.higher.peaks)/nrow(k27.consensus.peak.annotation)

## [1] 0.3806818

k27.temp.lower.peaks <- unique(c(t26.t10.lower.peaks,
                                t20.t10.lower.peaks,
                                t14.t10.lower.peaks))

length(k27.temp.lower.peaks)/nrow(k27.consensus.peak.annotation)

## [1] 0.02746212

k27.temp.higher.genes <- unique(c(t26.t10.higher.peaks.genes,
                                t20.t10.higher.peaks.genes,
                                t14.t10.higher.peaks.genes))

k27.temp.lower.genes <- unique(c(t26.t10.lower.peaks.genes,
                                t20.t10.lower.peaks.genes,
                                t14.t10.lower.peaks.genes))

length(k27.temp.higher.genes)/length(all.k27.genes)

## [1] 0.4084507

length(intersect(k27.temp.higher.genes,
                 c(k27.gene.body,k27.tss)))/length(c(k27.gene.body,k27.tss))

## [1] 0.4876033

length(k27.temp.lower.genes)/length(all.k27.genes)

## [1] 0.0583501

length(intersect(k27.temp.lower.genes,c(
  k27.gene.body,k27.tss)))/length(c(k27.gene.body,k27.tss))

## [1] 0.1349862

```

Over 40.8% of the H3K27me3 marked genes significantly increase more than 50% their H3K27me3 ChIP signal as a response to temperature and 5.8% significantly decrease their H3K27me3 ChIP signal.

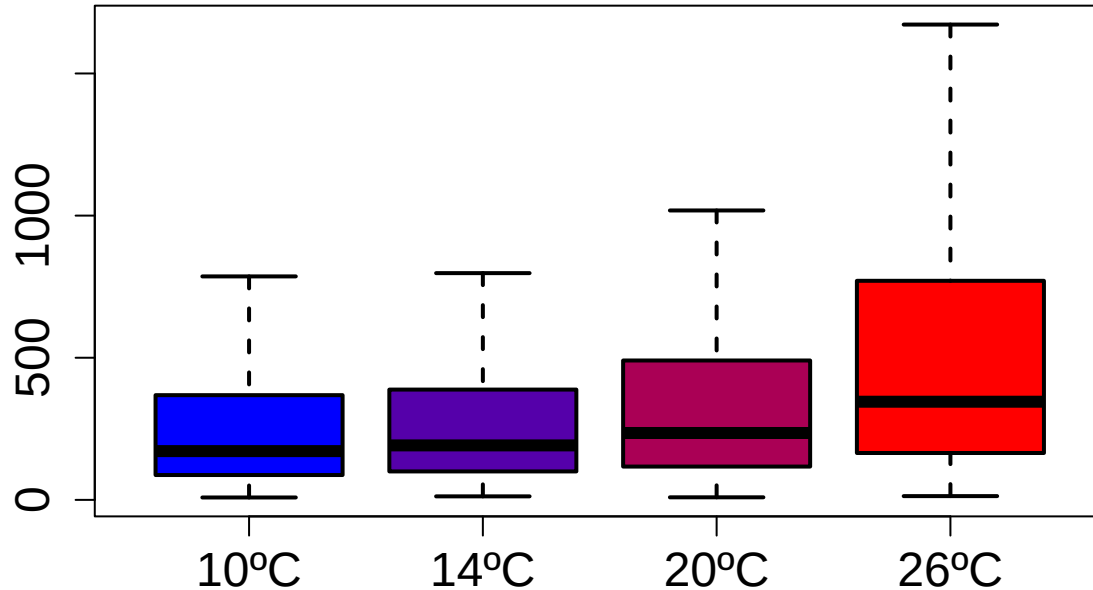
```

chip.signal.10C <- rowMeans(chip.signal[,paste("t10C",1:3,sep="_")])
chip.signal.14C <- rowMeans(chip.signal[,paste("t14C",1:3,sep="_")])
chip.signal.20C <- rowMeans(chip.signal[,paste("t20C",1:3,sep="_")])
chip.signal.26C <- rowMeans(chip.signal[,paste("t26C",1:3,sep="_")])

#jpeg(filename = "images/h3k27me3_higher.jpg",res = 160,width = 750,height = 900)
boxplot(chip.signal.10C[k27.temp.higher.peaks],
        chip.signal.14C[k27.temp.higher.peaks],

```

```
chip.signal.20C[k27.temp.higher.peaks],
chip.signal.26C[k27.temp.higher.peaks],
outline=F,
names=c("10°C", "14°C", "20°C", "26°C"),
col = colorRampPalette(c("blue", "red"))(4),lwd=2,cex.axis=1.5)
```



```
#dev.off()
```

```
wilcox.test(x = chip.signal.10C[k27.temp.higher.peaks],
            y = chip.signal.14C[k27.temp.higher.peaks],
            alternative = "less",paired = T)
```

```
##
## Wilcoxon signed rank test with continuity correction
##
## data: chip.signal.10C[k27.temp.higher.peaks] and chip.signal.14C[k27.temp.higher.peaks]
## V = 30222, p-value = 5.18e-06
## alternative hypothesis: true location shift is less than 0
```

```
wilcox.test(x = chip.signal.14C[k27.temp.higher.peaks],
            y = chip.signal.20C[k27.temp.higher.peaks],
            alternative = "less",paired = T)$p.value
```

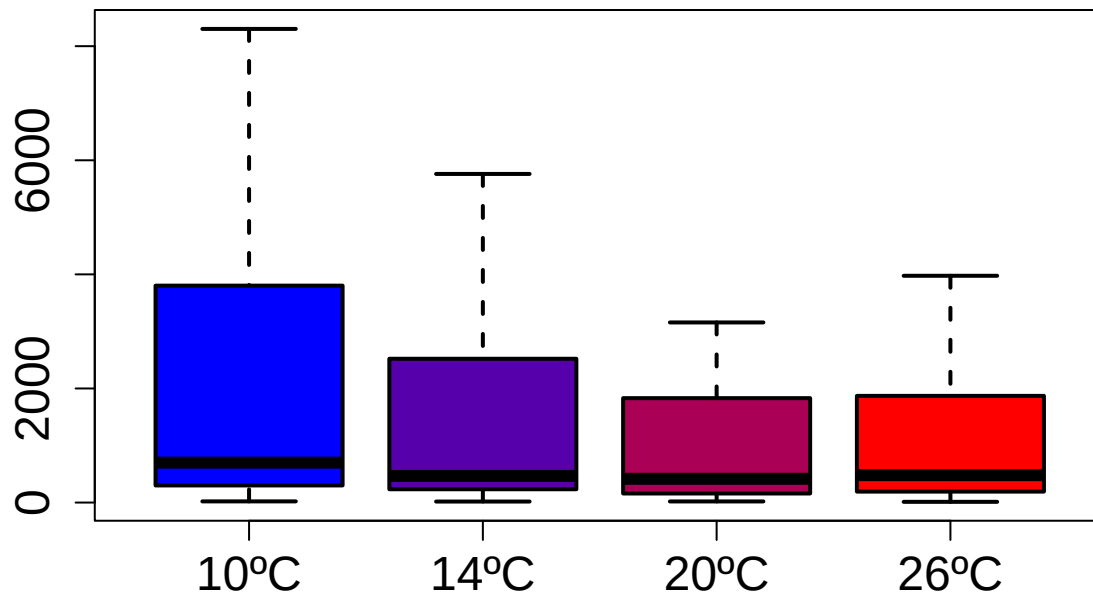
```
## [1] 1.597737e-60
```

```
wilcox.test(x = chip.signal.20C[k27.temp.higher.peaks],
            y = chip.signal.26C[k27.temp.higher.peaks],
            alternative = "less",paired = T)$p.value
```

```
## [1] 4.070955e-67
```

```
#jpeg(filename = "images/h3k27me3_lower.jpg",res = 160,width = 750,height = 900)
boxplot(chip.signal.10C[k27.temp.lower.peaks],
        chip.signal.14C[k27.temp.lower.peaks],
        chip.signal.20C[k27.temp.lower.peaks],
        chip.signal.26C[k27.temp.lower.peaks],
        outline=F,
```

```
names=c("10°C","14°C","20°C","26°C"),
col = colorRampPalette(c("blue", "red"))(4),lwd=2,cex.axis=1.5)
```



```
#dev.off()
```

```
wilcox.test(x = chip.signal.10C[k27.temp.lower.peaks],
            y = chip.signal.14C[k27.temp.lower.peaks],
            alternative = "greater",paired = T)
```

```
##
## Wilcoxon signed rank exact test
##
## data: chip.signal.10C[k27.temp.lower.peaks] and chip.signal.14C[k27.temp.lower.peaks]
## V = 435, p-value = 1.863e-09
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = chip.signal.14C[k27.temp.lower.peaks],
            y = chip.signal.20C[k27.temp.lower.peaks],
            alternative = "greater",paired = T)
```

```
##
## Wilcoxon signed rank exact test
##
## data: chip.signal.14C[k27.temp.lower.peaks] and chip.signal.20C[k27.temp.lower.peaks]
## V = 398, p-value = 1.057e-05
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = chip.signal.20C[k27.temp.lower.peaks],
            y = chip.signal.26C[k27.temp.lower.peaks],
            alternative = "greater",paired = T)
```

```
##
## Wilcoxon signed rank exact test
##
## data: chip.signal.20C[k27.temp.lower.peaks] and chip.signal.26C[k27.temp.lower.peaks]
## V = 81, p-value = 0.9989
```

alternative hypothesis: true location shift is greater than 0

We observe a temperature dependent gradual increase in the higher peaks. This increase is significant according to wilcox test with p values of 5.2×10^{-6} between 10°C and 14°C, 1.6×10^{-60} between 14°C and 20°C and 4.1×10^{-67} between 20°C and 24°C.

Similarly in the lower peaks, we observe a temperature dependent gradual decrease. This decrease is significant according to wilcox test with p values of 1.9×10^{-9} between 10°C and 14°C, 1.1×10^{-5} between 14°C and 20 and no significant difference between 20°C and 26°C.

Next, we performed a differential gene expression analysis to integrate our ChIP-seq data with gene expression data.

```
## Read gene expression files
ostta.10C.1 <- read.table(file="gene_expression_data/ostta_10C_100uE_1.tsv",
                        header=T, sep="\t")
gene.expression.ostta.10C.1 <- ostta.10C.1$TPM
names(gene.expression.ostta.10C.1) <- ostta.10C.1$Gene.ID

ostta.10C.2 <- read.table(file="gene_expression_data/ostta_10C_100uE_2.tsv",
                        header=T, sep="\t")
gene.expression.ostta.10C.2 <- ostta.10C.2$TPM
names(gene.expression.ostta.10C.2) <- ostta.10C.2$Gene.ID

ostta.10C.3 <- read.table(file="gene_expression_data/ostta_10C_100uE_3.tsv",
                        header=T, sep="\t")
gene.expression.ostta.10C.3 <- ostta.10C.3$TPM
names(gene.expression.ostta.10C.3) <- ostta.10C.3$Gene.ID

ostta.14C.1 <- read.table(file="gene_expression_data/ostta_14C_100uE_1.tsv",
                        header=T, sep="\t")
gene.expression.ostta.14C.1 <- ostta.14C.1$TPM
names(gene.expression.ostta.14C.1) <- ostta.14C.1$Gene.ID

ostta.14C.2 <- read.table(file="gene_expression_data/ostta_14C_100uE_2.tsv",
                        header=T, sep="\t")
gene.expression.ostta.14C.2 <- ostta.14C.2$TPM
names(gene.expression.ostta.14C.2) <- ostta.14C.2$Gene.ID

ostta.14C.3 <- read.table(file="gene_expression_data/ostta_14C_100uE_3.tsv",
                        header=T, sep="\t")
gene.expression.ostta.14C.3 <- ostta.14C.3$TPM
names(gene.expression.ostta.14C.3) <- ostta.14C.3$Gene.ID

ostta.20C.1 <- read.table(file="gene_expression_data/ostta_20C_100uE_1.tsv",
                        header=T, sep="\t")
gene.expression.ostta.20C.1 <- ostta.20C.1$TPM
names(gene.expression.ostta.20C.1) <- ostta.20C.1$Gene.ID

ostta.20C.2 <- read.table(file="gene_expression_data/ostta_20C_100uE_2.tsv",
                        header=T, sep="\t")
gene.expression.ostta.20C.2 <- ostta.20C.2$TPM
names(gene.expression.ostta.20C.2) <- ostta.20C.2$Gene.ID

ostta.20C.3 <- read.table(file="gene_expression_data/ostta_20C_100uE_3.tsv",
                        header=T, sep="\t")
```

```

gene.expression.ostta.20C.3 <- ostta.20C.3$TPM
names(gene.expression.ostta.20C.3) <- ostta.20C.3$Gene.ID

ostta.26C.1 <- read.table(file="gene_expression_data/ostta_26C_100uE_1.tsv",
                        header=T,sep="\t")
gene.expression.ostta.26C.1 <- ostta.26C.1$TPM
names(gene.expression.ostta.26C.1) <- ostta.26C.1$Gene.ID

ostta.26C.2 <- read.table(file="gene_expression_data/ostta_26C_100uE_2.tsv",
                        header=T,sep="\t")
gene.expression.ostta.26C.2 <- ostta.26C.2$TPM
names(gene.expression.ostta.26C.2) <- ostta.26C.2$Gene.ID

ostta.26C.3 <- read.table(file="gene_expression_data/ostta_26C_100uE_3.tsv",
                        header=T,sep="\t")
gene.expression.ostta.26C.3 <- ostta.26C.3$TPM
names(gene.expression.ostta.26C.3) <- ostta.26C.3$Gene.ID

gene.ids <- ostta.26C.3$Gene.ID

gene.expression <- matrix(data = c(gene.expression.ostta.10C.1[gene.ids],
                                gene.expression.ostta.10C.2[gene.ids],
                                gene.expression.ostta.10C.3[gene.ids],
                                gene.expression.ostta.14C.1[gene.ids],
                                gene.expression.ostta.14C.2[gene.ids],
                                gene.expression.ostta.14C.3[gene.ids],
                                gene.expression.ostta.20C.1[gene.ids],
                                gene.expression.ostta.20C.2[gene.ids],
                                gene.expression.ostta.20C.3[gene.ids],
                                gene.expression.ostta.26C.1[gene.ids],
                                gene.expression.ostta.26C.2[gene.ids],
                                gene.expression.ostta.26C.3[gene.ids]),
                        nrow = length(gene.ids),
                        ncol = 12)

rownames(gene.expression) <- gene.ids
colnames(gene.expression) <- c("t10C_1","t10C_2","t10C_3",
                              "t14C_1","t14C_2","t14C_3",
                              "t20C_1","t20C_2","t20C_3",
                              "t26C_1","t26C_2","t26C_3")

head(gene.expression)

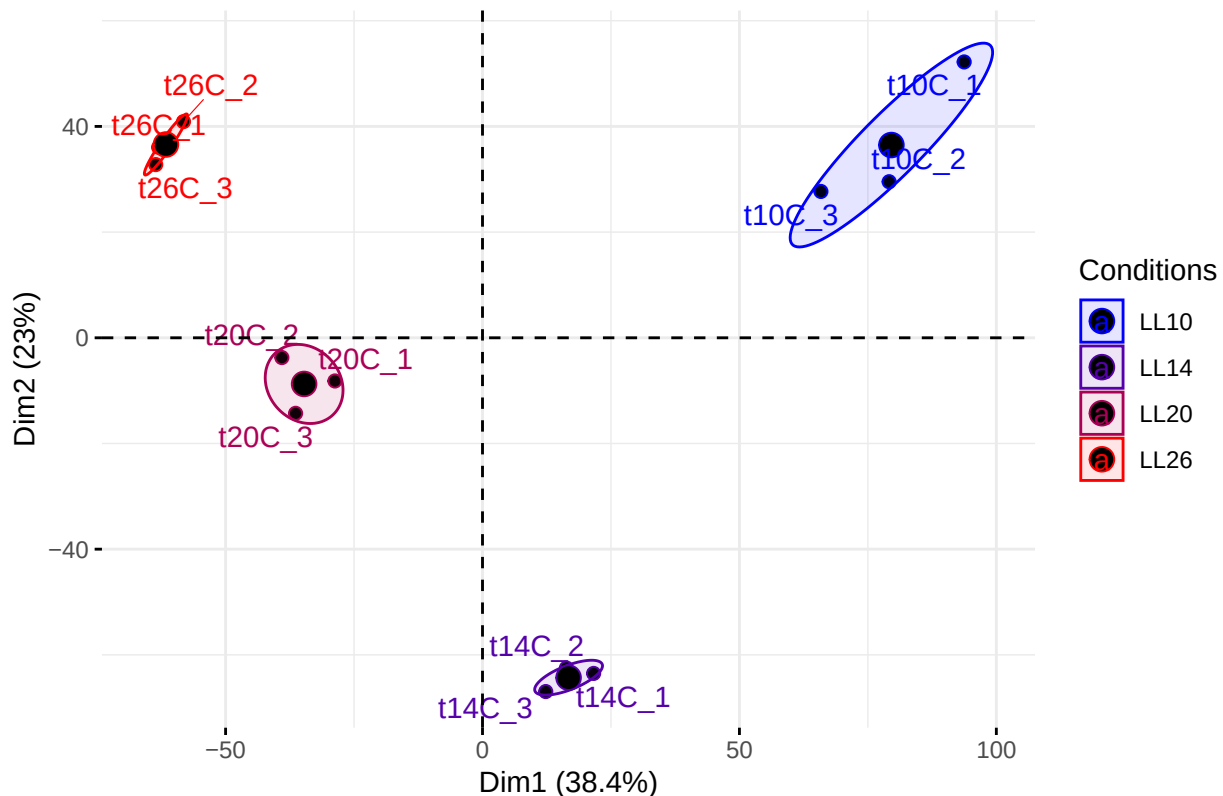
##           t10C_1  t10C_2  t10C_3  t14C_1  t14C_2  t14C_3
## ostta01g00010  43.99131  39.58039  38.63458  38.72335  35.989841  38.272041
## ostta01g00020  26.47679  23.34998  22.08406  19.80787  18.337345  16.679943
## ostta01g00030  15.07906  15.93047  14.52224  10.63125   7.941051   8.693375
## ostta01g00040 117.22766 115.20618 112.28110  81.08285  79.161140  81.916862
## ostta01g00050  29.81894  30.76012  32.92440  27.33033  26.426275  28.375336
## ostta01g00060 277.69348 250.38902 235.78523 170.02148 177.248367 193.834045
##           t20C_1  t20C_2  t20C_3  t26C_1  t26C_2  t26C_3
## ostta01g00010  34.791672  30.202894  29.067356  45.34333  46.015034  44.321796
## ostta01g00020  25.442566  19.781862  17.439911  27.53025  29.908211  27.591972
## ostta01g00030   7.299343   6.777994   6.423476   8.77062   9.945183   8.240674

```

```
## ostta01g00040 93.010406 102.530823 99.107620 88.69782 88.700920 80.888710
## ostta01g00050 36.986897 41.585278 34.467091 29.38928 27.855453 28.767195
## ostta01g00060 177.767761 163.940140 183.768082 195.94548 191.679077 189.283966

pca.gene.expression <- data.frame(colnames(gene.expression),
                                   t(gene.expression))

colnames(pca.gene.expression)[1] <- "Sample"
res.pca <- PCA(pca.gene.expression,
               graph = FALSE,
               scale.unit = TRUE,
               quali.sup = 1 )
#jpeg(filename = "images/temp_gene_expression_pca.jpg",res = 160,width = 750,height = 750)
fviz_pca_ind(res.pca, col.ind = c(rep("LL10",3),rep("LL14",3),
                                   rep("LL20",3),rep("LL26",3)),
              palette = colorRampPalette(c("blue", "red"))(4),
              pointsize=2, pointshape=21,fill="black",
              repel = TRUE,
              addEllipses = TRUE,ellipse.type = "confidence",
              legend.title="Conditions",
              title="",
              show_legend=TRUE,show_guide=TRUE)
```



```
#dev.off()

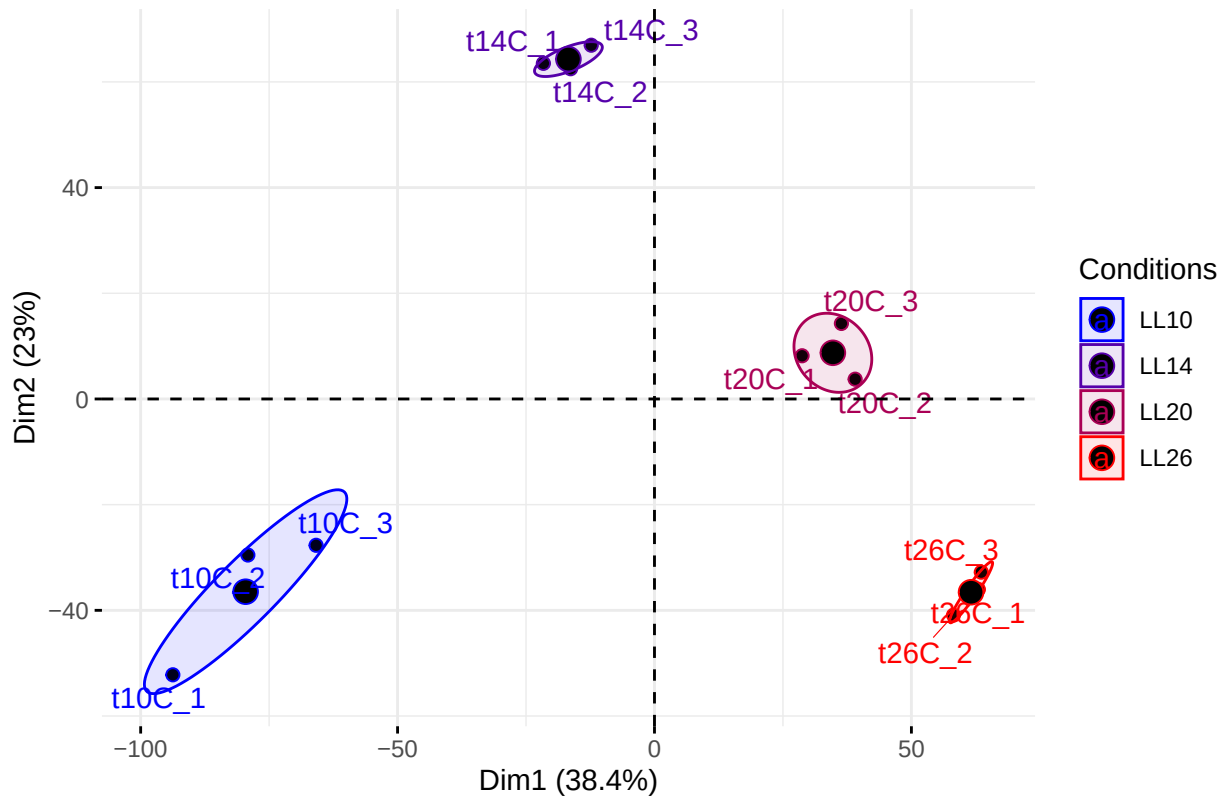
res.pca$ind$coord <- - res.pca$ind$coord

#jpeg(filename = "images/temp_gene_expression_pca.jpg",res = 160,width = 750,height = 750)
fviz_pca_ind(res.pca, col.ind = c(rep("LL10",3),rep("LL14",3),
```

```

rep("LL20",3),rep("LL26",3)),
palette = colorRampPalette(c("blue", "red"))(4),
pointsize=2, pointshape=21,fill="black",
repel = TRUE,
addEllipses = TRUE,ellipse.type = "confidence",
legend.title="Conditions",
title="",
show_legend=TRUE,show_guide=TRUE)

```



```

#dev.off()

gene.expression.10 <- rowMeans(gene.expression[,paste("t10C",1:3,sep="_")])
gene.expression.14 <- rowMeans(gene.expression[,paste("t14C",1:3,sep="_")])
gene.expression.20 <- rowMeans(gene.expression[,paste("t20C",1:3,sep="_")])
gene.expression.26 <- rowMeans(gene.expression[,paste("t26C",1:3,sep="_")])

library(DESeq2)

```

```

## Loading required package: SummarizedExperiment
## Loading required package: MatrixGenerics
## Loading required package: matrixStats
##
## Attaching package: 'matrixStats'
## The following objects are masked from 'package:Biobase':
##
##      anyMissing, rowMedians

```

```
## The following object is masked from 'package:seqinr':
##
##     count
##
## Attaching package: 'MatrixGenerics'
## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars
## The following object is masked from 'package:Biobase':
##
##     rowMedians
## Load the gene expression estimation based on read count.
gene.count.matrix <- read.table(file = "gene_expression_data/gene_count_matrix.csv",header = T,sep = ",
head(gene.count.matrix)
```

	gene_id	sample01	sample02	sample03	sample04	sample05	sample06	sample07
## 1	ostta06g00070	193	243	241	746	523	699	694
## 2	ostta15g01110	960	1325	1078	5808	5363	5851	1382
## 3	ostta10g01190	743	1020	825	2990	2818	3214	984
## 4	ostta15g01440	254	395	646	925	870	1028	936
## 5	ostta02g04020	594	879	925	2087	1974	2099	1917
## 6	ostta06g00075	26	30	41	41	47	45	52

```
## sample08 sample09 sample10 sample11 sample12
## 1      812      715      883      728      992
## 2     1530     3259     5928     5323     4691
## 3      722      898     1072      999      938
## 4      868      816      699      573      830
## 5     1926     2088     1791     1747     1654
## 6       59       52       34       47       49
## Extract gene ids in the order present in the gene count matrix
deseq.gene.ids <- gene.count.matrix$gene_id
## Generate a numerical matrix by removing the first columns with gene names.
gene.count.matrix <- gene.count.matrix[,-1]
rownames(gene.count.matrix) <- deseq.gene.ids
colnames(gene.count.matrix) <- colnames(gene.expression)
head(gene.count.matrix)
```

```
##          t10C_1 t10C_2 t10C_3 t14C_1 t14C_2 t14C_3 t20C_1 t20C_2 t20C_3
## ostta06g00070    193    243    241    746    523    699    694    812    715
## ostta15g01110    960   1325   1078   5808   5363   5851   1382   1530   3259
## ostta10g01190    743   1020    825   2990   2818   3214    984    722    898
## ostta15g01440    254    395    646    925    870   1028    936    868    816
## ostta02g04020    594    879    925   2087   1974   2099   1917   1926   2088
## ostta06g00075     26     30     41     41     47     45     52     59     52
##          t26C_1 t26C_2 t26C_3
## ostta06g00070    883    728    992
## ostta15g01110   5928   5323   4691
## ostta10g01190   1072    999    938
## ostta15g01440    699    573    830
## ostta02g04020   1791   1747   1654
## ostta06g00075     34     47     49
```

```
experimental.design <- data.frame(sample=colnames(gene.expression),
                                   temperature=c(rep("t10C",3),
                                                  rep("t14C",3),
                                                  rep("t20C",3),
                                                  rep("t26C",3)))
```

```
## Generate object needed by DESeq2 to perform DEGs analysis
dds <- DESeqDataSetFromMatrix(countData=gene.count.matrix,
                              colData=experimental.design,
                              design = ~ temperature)
```

```
## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

```
## Perform DEGs analysis
dds <- DESeq(dds)
```

```
## estimating size factors
## estimating dispersions
## gene-wise dispersion estimates
## mean-dispersion relationship
## final dispersion estimates
## fitting model and testing
```

```
## Extract results for comparison treatment=t14C and control=t10C
res <- results(dds, contrast = c("temperature","t14C","t10C"))
res
```

```
## log2 fold change (MLE): temperature t14C vs t10C
## Wald test p-value: temperature t14C vs t10C
## DataFrame with 7668 rows and 6 columns
##          baseMean log2FoldChange    lfcSE      stat      pvalue
##          <numeric>      <numeric> <numeric> <numeric> <numeric>
## ostta06g00070    551.901    -0.0129251 0.1328453 -0.097294 0.92249294
## ostta15g01110   3069.217     0.7891119 0.2934634  2.688962 0.00716746
## ostta10g01190   1404.718     0.2508187 0.1292536  1.940517 0.05231692
## ostta15g01440    708.273    -0.3760826 0.1789573 -2.101522 0.03559515
## ostta02g04020   1530.488    -0.1687498 0.0757952 -2.226392 0.02598793
```

```
## ...
## ostta07g00240 145.961 -0.4187228 0.1771658 -2.363452 0.01810557
## ostta09g03890 2930.494 -0.1584050 0.0784882 -2.018200 0.04357039
## ostta10g03460 668.908 0.0802104 0.1168297 0.686558 0.49236139
## ostta06g01690 9947.346 -0.2395645 0.0682067 -3.512333 0.00044419
## ostta02g05370 535.817 0.4234591 0.1296938 3.265069 0.00109438
## padj
## <numeric>
## ostta06g00070 0.9485295
## ostta15g01110 0.0178349
## ostta10g01190 0.0971781
## ostta15g01440 0.0702010
## ostta02g04020 0.0536128
## ...
## ostta07g00240 0.03956689
## ostta09g03890 0.08292115
## ostta10g03460 0.60106048
## ostta06g01690 0.00155022
## ostta02g05370 0.00342898
```

```
## Extract gene ids, log2 fc and q-value for this comparison
```

```
gene.ids.14.vs.10 <- rownames(res)
log.fold.change <- res$log2FoldChange
adjusted.pval <- res$padj
```

```
## Name the vectors with the gene ids
```

```
names(log.fold.change) <- gene.ids.14.vs.10
names(adjusted.pval) <- gene.ids.14.vs.10
```

```
## Determine activated and repressed genes removing NA
```

```
activated.genes.14.vs.10 <- gene.ids.14.vs.10[log.fold.change > 1 &
                                              adjusted.pval < 0.05]
activated.genes.14.vs.10 <- activated.genes.14.vs.10[!is.na(activated.genes.14.vs.10)]
length(activated.genes.14.vs.10)
```

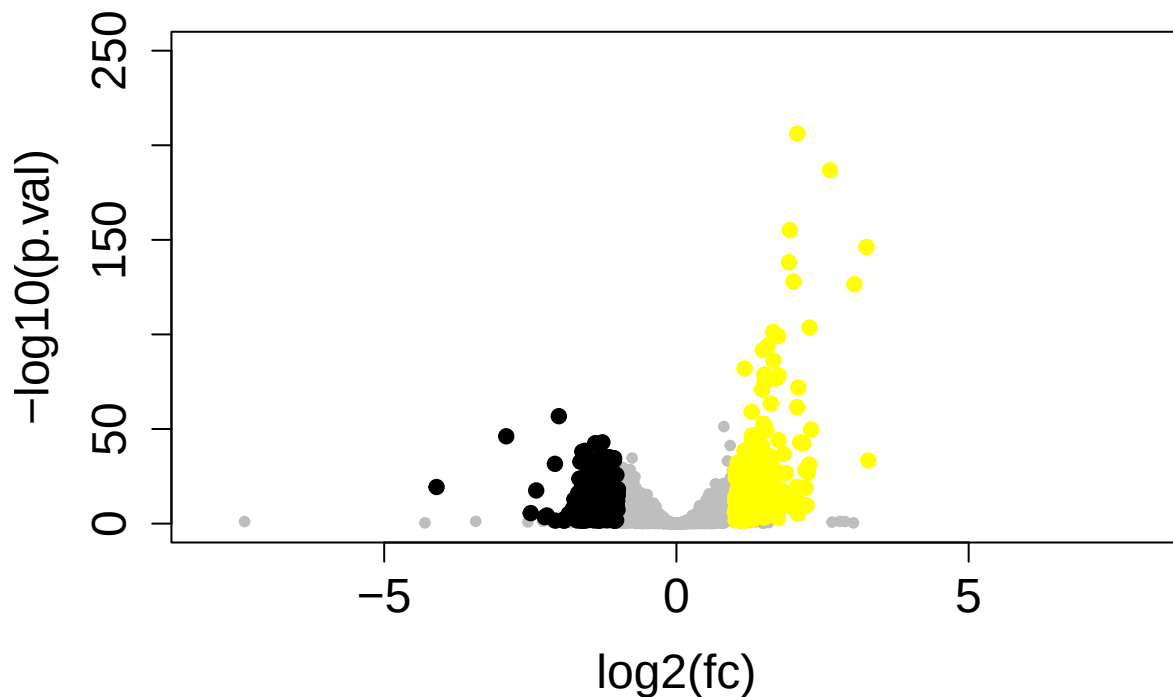
```
## [1] 232
```

```
repressed.genes.14.vs.10 <- gene.ids.14.vs.10[log.fold.change < -1 &
                                              adjusted.pval < 0.05]
repressed.genes.14.vs.10 <- repressed.genes.14.vs.10[!is.na(repressed.genes.14.vs.10)]
length(repressed.genes.14.vs.10)
```

```
## [1] 181
```

```
#jpeg(filename = "images/gene_expression_volcano_14_10.jpg",res = 160,width = 750,height = 750)
```

```
plot(x = log.fold.change,y = -log10(adjusted.pval),
     pch=20,cex.axis=1.5,col="grey",xlim=c(-8,8),ylim=c(0,250),
     xlab="log2(fc)",ylab="-log10(p.val)",cex.lab=1.5)
points(x = log.fold.change[activated.genes.14.vs.10],y = -log10(adjusted.pval)[activated.genes.14.vs.10],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="yellow")
points(x = log.fold.change[repressed.genes.14.vs.10],y = -log10(adjusted.pval)[repressed.genes.14.vs.10],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="black")
```



```
#dev.off()
```

```
## Extract results for comparison treatment=20C and control=t10C
```

```
res <- results(dds, contrast = c("temperature", "t20C", "t10C"))
res
```

```
## log2 fold change (MLE): temperature t20C vs t10C
```

```
## Wald test p-value: temperature t20C vs t10C
```

```
## DataFrame with 7668 rows and 6 columns
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
## ostta06g00070	551.901	0.208896	0.1323604	1.57824	1.14511e-01
## ostta15g01110	3069.217	-0.654044	0.2938364	-2.22588	2.60224e-02
## ostta10g01190	1404.718	-1.500714	0.1314294	-11.41841	3.38381e-30
## ostta15g01440	708.273	-0.440426	0.1791060	-2.45902	1.39315e-02
## ostta02g04020	1530.488	-0.186487	0.0758803	-2.45765	1.39851e-02
## ...	...	...	...	...	...
## ostta07g00240	145.961	-0.2192803	0.1757658	-1.247571	2.12188e-01
## ostta09g03890	2930.494	-0.1539482	0.0785127	-1.960807	4.99016e-02
## ostta10g03460	668.908	0.0662574	0.1169494	0.566548	5.71022e-01
## ostta06g01690	9947.346	-0.5658014	0.0683089	-8.282987	1.20123e-16
## ostta02g05370	535.817	0.8664284	0.1285861	6.738121	1.60448e-11
##	padj				
##	<numeric>				
## ostta06g00070	1.65368e-01				
## ostta15g01110	4.41522e-02				
## ostta10g01190	1.21185e-28				
## ostta15g01440	2.53192e-02				
## ostta02g04020	2.54106e-02				
## ...	...				
## ostta07g00240	2.82328e-01				

```

## ostta09g03890 7.90994e-02
## ostta10g03460 6.43291e-01
## ostta06g01690 1.42954e-15
## ostta02g05370 1.10881e-10

## Extract gene ids, log2 fc and q-value for this comparison
gene.ids.20.vs.10 <- rownames(res)
log.fold.change <- res$log2FoldChange
adjusted.pval <- res$padj

## Name the vectors with the gene ids
names(log.fold.change) <- gene.ids.20.vs.10
names(adjusted.pval) <- gene.ids.20.vs.10

## Determine activated and repressed genes removing NA
activated.genes.20.vs.10 <- gene.ids.20.vs.10[log.fold.change > 1 &
                                             adjusted.pval < 0.05]
activated.genes.20.vs.10 <- activated.genes.20.vs.10[!is.na(activated.genes.20.vs.10)]
length(activated.genes.20.vs.10)

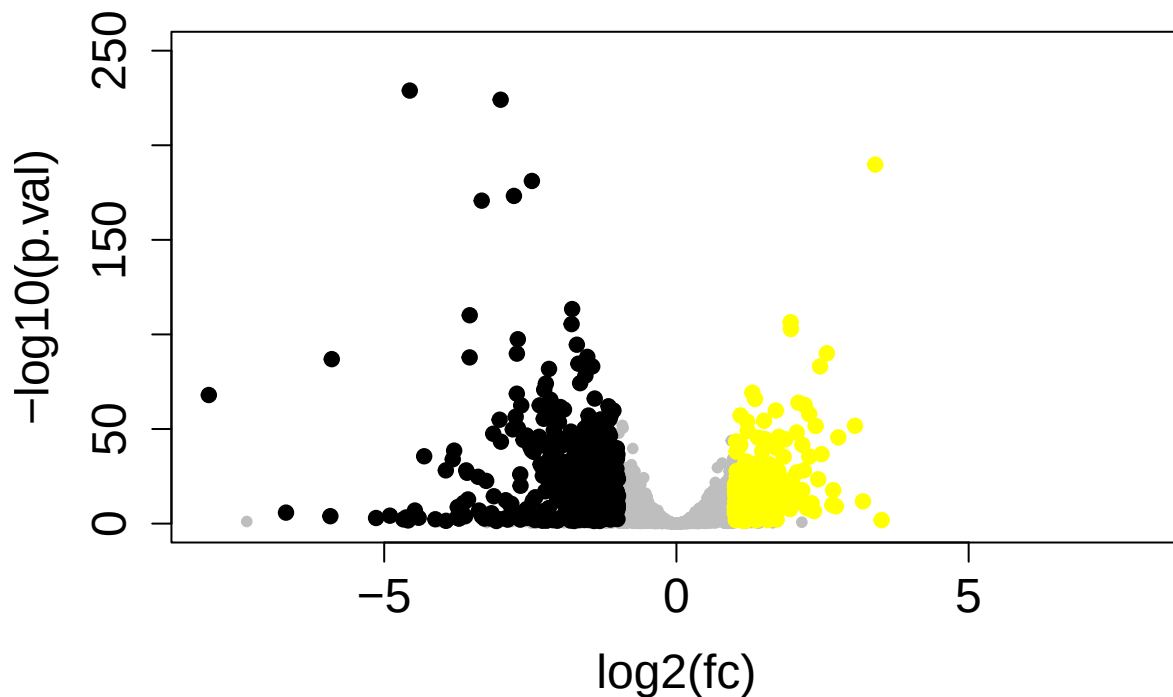
## [1] 388

repressed.genes.20.vs.10 <- gene.ids.20.vs.10[log.fold.change < - 1 &
                                             adjusted.pval < 0.05]
repressed.genes.20.vs.10 <- repressed.genes.20.vs.10[!is.na(repressed.genes.20.vs.10)]
length(repressed.genes.20.vs.10)

## [1] 460

#jpeg(filename = "images/gene_expression_volcano_20_10.jpg",res = 160,width = 750,height = 750)
plot(x = log.fold.change,y = -log10(adjusted.pval),
     pch=20,cex.axis=1.5,col="grey",xlim=c(-8,8),ylim=c(0,250),
     xlab="log2(fc)",ylab="-log10(p.val)",cex.lab=1.5)
points(x = log.fold.change[activated.genes.20.vs.10],y = -log10(adjusted.pval)[activated.genes.20.vs.10],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="yellow")
points(x = log.fold.change[repressed.genes.20.vs.10],y = -log10(adjusted.pval)[repressed.genes.20.vs.10],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="black")

```



```
#dev.off()
```

```
## Extract results for comparison treatment=26C and control=t10C
```

```
res <- results(dds, contrast = c("temperature", "t26C", "t10C"))
res
```

```
## log2 fold change (MLE): temperature t26C vs t10C
```

```
## Wald test p-value: temperature t26C vs t10C
```

```
## DataFrame with 7668 rows and 6 columns
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
## ostta06g00070	551.901	0.338988	0.1318480	2.57106	1.01389e-02
## ostta15g01110	3069.217	0.637036	0.2934773	2.17065	2.99578e-02
## ostta10g01190	1404.718	-1.389629	0.1310244	-10.60588	2.79827e-26
## ostta15g01440	708.273	-0.860367	0.1796643	-4.78875	1.67823e-06
## ostta02g04020	1530.488	-0.472268	0.0762068	-6.19719	5.74785e-10
## ...	...	...	...	...	...
## ostta07g00240	145.961	0.319810	0.1715130	1.86464	6.22315e-02
## ostta09g03890	2930.494	-0.602862	0.0788266	-7.64795	2.04214e-14
## ostta10g03460	668.908	0.173796	0.1164908	1.49193	1.35719e-01
## ostta06g01690	9947.346	-0.747170	0.0683373	-10.93357	7.96520e-28
## ostta02g05370	535.817	1.126543	0.1278380	8.81227	1.22632e-18
##	padj				
##	<numeric>				
## ostta06g00070	1.74617e-02				
## ostta15g01110	4.70871e-02				
## ostta10g01190	5.23071e-25				
## ostta15g01440	5.20937e-06				
## ostta02g04020	2.72427e-09				
## ...	...				
## ostta07g00240	9.06389e-02				

```

## ostta09g03890 1.51656e-13
## ostta10g03460 1.82578e-01
## ostta06g01690 1.61069e-26
## ostta02g05370 1.29994e-17

## Extract gene ids, log2 fc and q-value for this comparison
gene.ids.26.vs.10 <- rownames(res)
log.fold.change <- res$log2FoldChange
adjusted.pval <- res$padj

## Name the vectors with the gene ids
names(log.fold.change) <- gene.ids.26.vs.10
names(adjusted.pval) <- gene.ids.26.vs.10

## Determine activated and repressed genes removing NA
activated.genes.26.vs.10 <- gene.ids.26.vs.10[log.fold.change > 1 &
                                             adjusted.pval < 0.05]
activated.genes.26.vs.10 <- activated.genes.26.vs.10[!is.na(activated.genes.26.vs.10)]
length(activated.genes.26.vs.10)

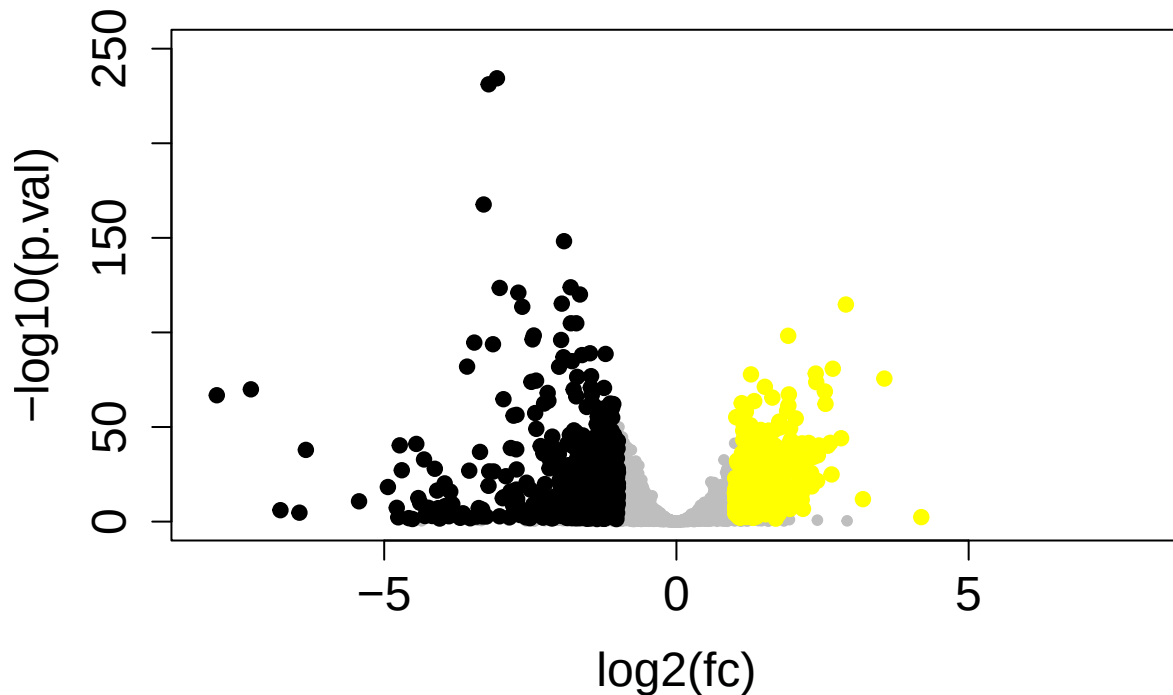
## [1] 545

repressed.genes.26.vs.10 <- gene.ids.26.vs.10[log.fold.change < - 1 &
                                             adjusted.pval < 0.05]
repressed.genes.26.vs.10 <- repressed.genes.26.vs.10[!is.na(repressed.genes.26.vs.10)]
length(repressed.genes.26.vs.10)

## [1] 438

#jpeg(filename = "images/gene_expression_volcano_26_10.jpg",res = 160,width = 750,height = 750)
plot(x = log.fold.change,y = -log10(adjusted.pval),
     pch=20,cex.axis=1.5,col="grey",xlim=c(-8,8),ylim=c(0,250),
     xlab="log2(fc)",ylab="-log10(p.val)",cex.lab=1.5)
points(x = log.fold.change[activated.genes.26.vs.10],y = -log10(adjusted.pval)[activated.genes.26.vs.10],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="yellow")
points(x = log.fold.change[repressed.genes.26.vs.10],y = -log10(adjusted.pval)[repressed.genes.26.vs.10],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="black")

```



```
#dev.off()
```

```
temp.repressed <- unique(c(repressed.genes.14.vs.10, repressed.genes.20.vs.10, repressed.genes.26.vs.10))
length(temp.repressed)
```

```
## [1] 751
```

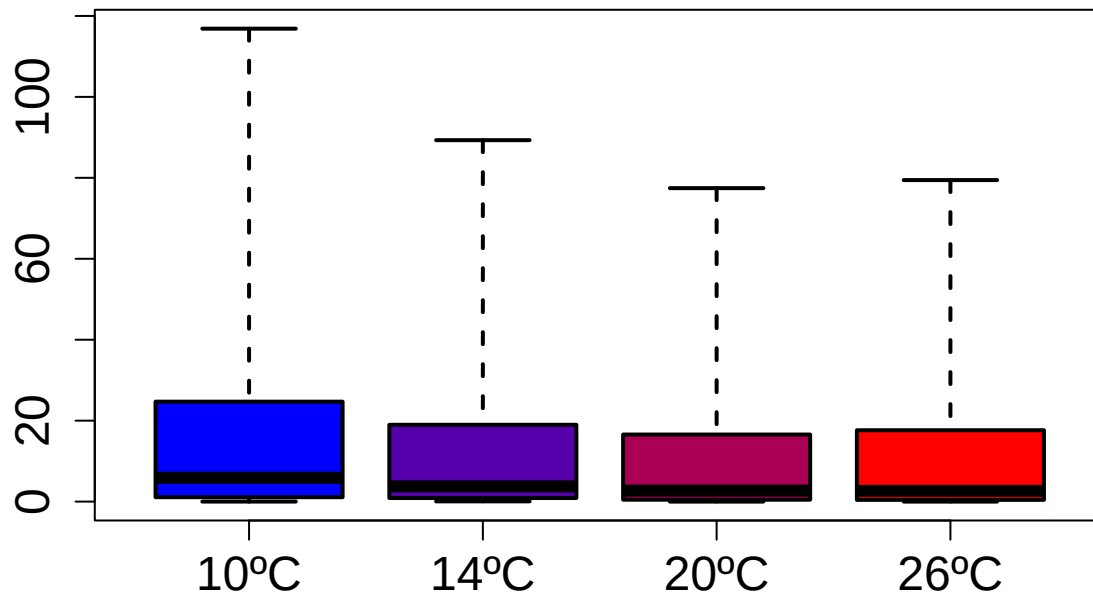
```
temp.activated <- unique(c(activated.genes.14.vs.10, activated.genes.20.vs.10, activated.genes.26.vs.10))
length(temp.activated)
```

```
## [1] 738
```

We found 751 genes that significantly increase more than twice their expression as a response to temperature and 138 genes that significantly decrease more than half their expression as a response to temperature.

Next we study the expression level of H3K27me3 genes that significantly increase their ChIP signal as a response to temperature more than 50%.

```
#jpeg(filename = "images/gene_expression_h3k27me3_higher.jpg",res = 160,width = 750,height = 900)
boxplot(gene.expression.10[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
        gene.expression.14[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
        gene.expression.20[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
        gene.expression.26[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
        outline=F,
        names=c("10°C","14°C","20°C","26°C"),
        col = colorRampPalette(c("blue", "red"))(4),lwd=2,cex.axis=1.5,range=4)
```



```
#dev.off()
```

```
wilcox.test(x = gene.expression.10[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
            y = gene.expression.14[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
            alternative = "greater",paired = T)
```

```
##
```

```
## Wilcoxon signed rank test with continuity correction
```

```
##
```

```
## data: gene.expression.10[intersect(k27.temp.higher.genes, c(k27.tss, k27.gene.body))] and gene.expr
```

```
## V = 13087, p-value = 1.157e-14
```

```
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = gene.expression.14[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
            y = gene.expression.20[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
            alternative = "greater",paired = T)
```

```
##
```

```
## Wilcoxon signed rank test with continuity correction
```

```
##
```

```
## data: gene.expression.14[intersect(k27.temp.higher.genes, c(k27.tss, k27.gene.body))] and gene.expr
```

```
## V = 10432, p-value = 9.102e-05
```

```
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = gene.expression.20[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
            y = gene.expression.26[intersect(k27.temp.higher.genes,c(k27.tss,k27.gene.body))],
            alternative = "greater",paired = T)
```

```
##
```

```
## Wilcoxon signed rank test with continuity correction
```

```
##
```

```
## data: gene.expression.20[intersect(k27.temp.higher.genes, c(k27.tss, k27.gene.body))] and gene.expr
```

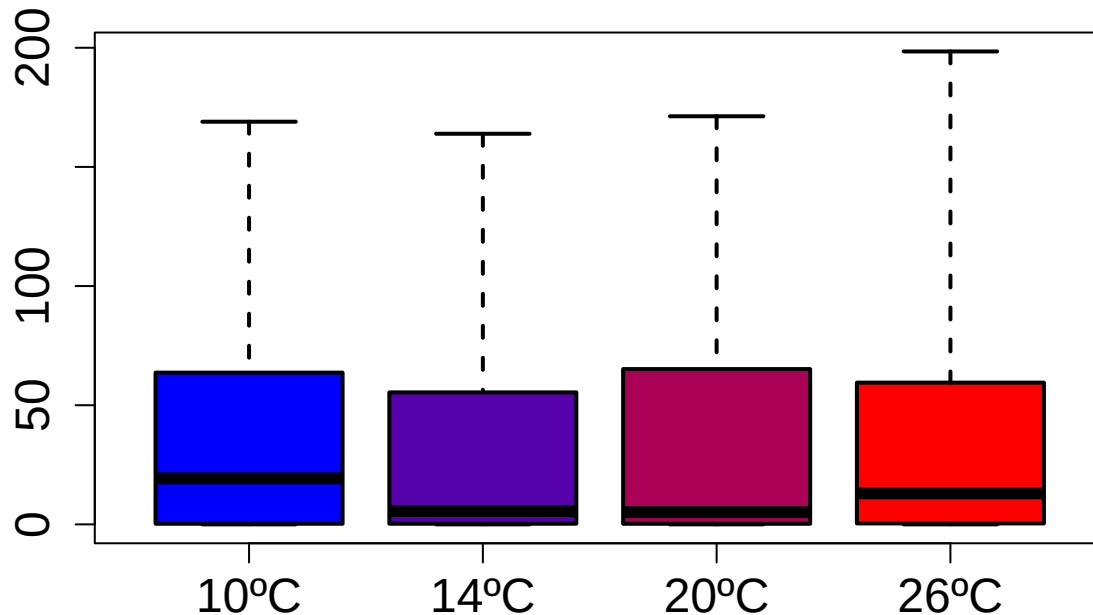
```
## V = 9716, p-value = 0.002203
```

```
## alternative hypothesis: true location shift is greater than 0
```

We observe that globally genes that significantly increase their H3K27me3 ChIP signal more than 50% exhibit

significant decrease in their expression levels that significant with p-values of 1.157×10^{-14} for 10°C vs 14°C, 9.102×10^{-5} for 14°C vs 20°C and 2.203×10^{-3} .

```
#jpeg(filename = "images/gene_expression_h3k27me3_lower.jpg",res = 160,width = 750,height = 900)
boxplot(gene.expression.10[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
        gene.expression.14[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
        gene.expression.20[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
        gene.expression.26[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
        outline=F,
        names=c("10°C","14°C","20°C","26°C"),
        col = colorRampPalette(c("blue", "red"))(4),lwd=2,cex.axis=1.5,range=2.5)
```



```
#dev.off()

wilcox.test(x = gene.expression.10[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            y = gene.expression.14[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            alternative = "less",paired = T)

## Warning in wilcox.test.default(x =
## gene.expression.10[intersect(k27.temp.lower.genes, : cannot compute exact
## p-value with zeroes
##
## Wilcoxon signed rank test with continuity correction
##
## data: gene.expression.10[intersect(k27.temp.lower.genes, c(k27.tss, k27.gene.body))] and gene.expression.14[intersect(k27.temp.lower.genes, c(k27.tss, k27.gene.body))]
## V = 714, p-value = 0.9444
## alternative hypothesis: true location shift is less than 0

wilcox.test(x = gene.expression.14[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            y = gene.expression.20[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            alternative = "less",paired = T)

## Warning in wilcox.test.default(x =
## gene.expression.14[intersect(k27.temp.lower.genes, : cannot compute exact
## p-value with zeroes
```

```
##
## Wilcoxon signed rank test with continuity correction
##
## data: gene.expression.14[intersect(k27.temp.lower.genes, c(k27.tss, k27.gene.body))] and gene.expre
## V = 329, p-value = 0.004009
## alternative hypothesis: true location shift is less than 0
wilcox.test(x = gene.expression.20[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            y = gene.expression.26[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            alternative = "less",paired = T)

##
## Wilcoxon signed rank exact test
##
## data: gene.expression.20[intersect(k27.temp.lower.genes, c(k27.tss, k27.gene.body))] and gene.expre
## V = 495, p-value = 0.1235
## alternative hypothesis: true location shift is less than 0

Similarly we observe that globally genes that significantly decrease their H3K27me3 ChIP signal more than
50% exhibit increase in their expression levels. However, this increase is only significant for 20°C vs 14°C
with a p value of  $4 \times 10^{-3}$ .

temp.repressed.k27.higher.genes <- intersect(temp.repressed,k27.temp.higher.genes)
length(temp.repressed.k27.higher.genes)

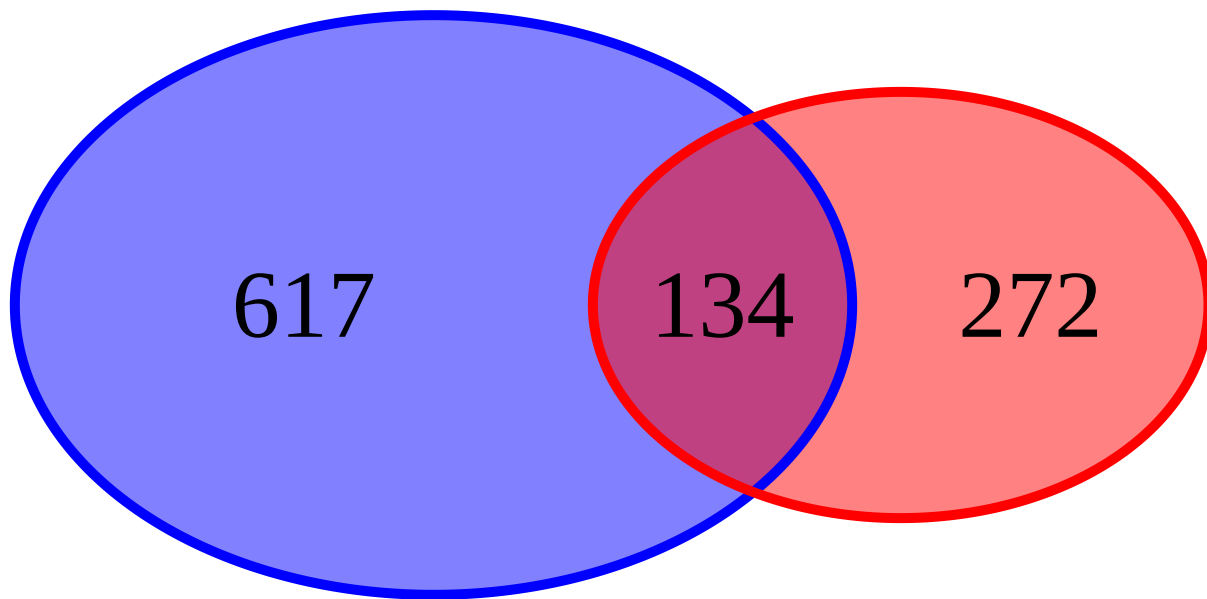
## [1] 134
length(temp.repressed.k27.higher.genes)/length(temp.repressed)

## [1] 0.1784288
length(temp.repressed.k27.higher.genes)/length(k27.temp.higher.genes)

## [1] 0.3300493
sum(as.data.frame(res)[setdiff(k27.temp.higher.genes,temp.repressed),"log2FoldChange"] < 0)/length(setd

## [1] 0.5808824
library(VennDiagram)

#jpeg(filename = "images/venn_diagram_k27_increase_repressed.jpg",res = 160,width = 1000,height = 1000)
grid.newpage()
draw.pairwise.venn(area1 = length(temp.repressed),
                  area2 = length(k27.temp.higher.genes),
                  cross.area = length(temp.repressed.k27.higher.genes),
                  lwd = 5,category = c("", ""),
                  euler.d = T,
                  col = c("blue","red"),
                  fill = c("blue","red"),
                  alpha = 0.5,cex = 3,cat.cex = 2)
```



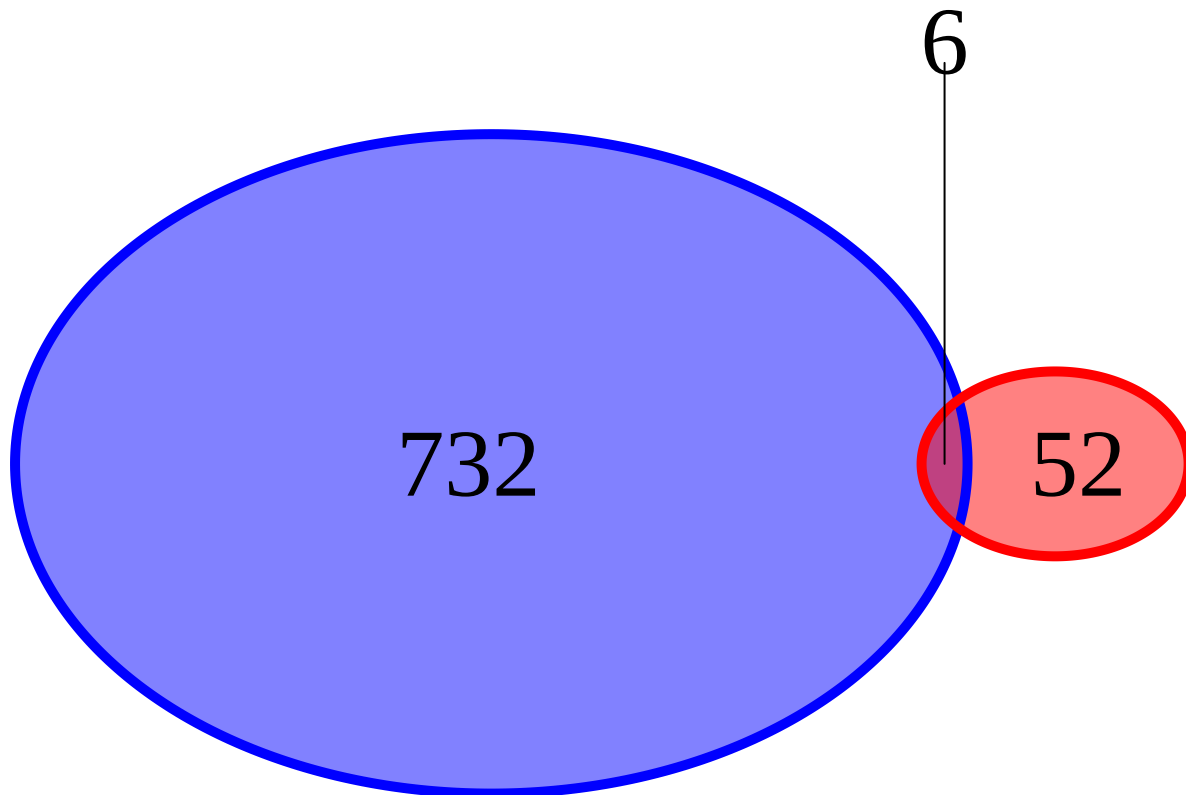
```
## (polygon[GRID.polygon.955], polygon[GRID.polygon.956], polygon[GRID.polygon.957], polygon[GRID.polygon.958])
#dev.off()
```

```
overlap <- length(temp.repressed.k27.higher.genes)
size.1 <- length(temp.repressed)
size.2 <- length(k27.temp.higher.genes)
total.genes <- length(gene.ids)
phyper(overlap - 1, size.1, total.genes - size.1, size.2, lower.tail = FALSE)
```

```
## [1] 1.147667e-40
```

```
temp.activated.k27.lower.genes <- intersect(temp.activated,k27.temp.lower.genes)
```

```
#jpeg(filename = "images/venn_diagram_k27_decrease_activated.jpg",res = 160,width = 1000,height = 1000)
grid.newpage()
draw.pairwise.venn(area1 = length(temp.activated),
                    area2 = length(k27.temp.lower.genes),
                    cross.area = length(temp.activated.k27.lower.genes),
                    lwd = 5,category = c("", ""),
                    euler.d = T,
                    col = c("blue","red"),
                    fill = c("blue","red"),
                    alpha = 0.5,cex = 3,cat.cex = 2)
```



```
## (polygon[GRID.polygon.964], polygon[GRID.polygon.965], polygon[GRID.polygon.966], polygon[GRID.polygon.967],
#dev.off()
```

```
sum(as.data.frame(res)[setdiff(k27.temp.lower.genes,temp.activated),"log2FoldChange"] > 0)/length(setdiff(k27.temp.lower.genes,temp.activated))
```

```
## [1] 0.4807692
```

```
overlap <- length(temp.activated.k27.lower.genes)
size.1 <- length(temp.activated)
size.2 <- length(k27.temp.lower.genes)
total.genes <- length(gene.ids)
phyper(overlap - 1, size.1, total.genes - size.1, size.2, lower.tail = FALSE)
```

```
## [1] 0.4904345
```

We identified 134 genes that, in response to high temperature, exhibited a more than 50% significant increase in H3K27me3 ChIP signal and a greater than twofold reduction in expression levels. These genes represent 17.8% of the total set of genes that showed significant downregulation in response to temperature increase. The overlap between genes with a significant increase in H3K27me3 ChIP signal and those significantly downregulated is highly significant with a p-value of 1.15×10^{-40} . This suggests that H3K27me3 plays a key role in temperature response being involved in the regulation of nearly 18% of the differentially repressed genes.

Next we explore the biological processes affected by changes in H3K27me3 as a response to temperature.

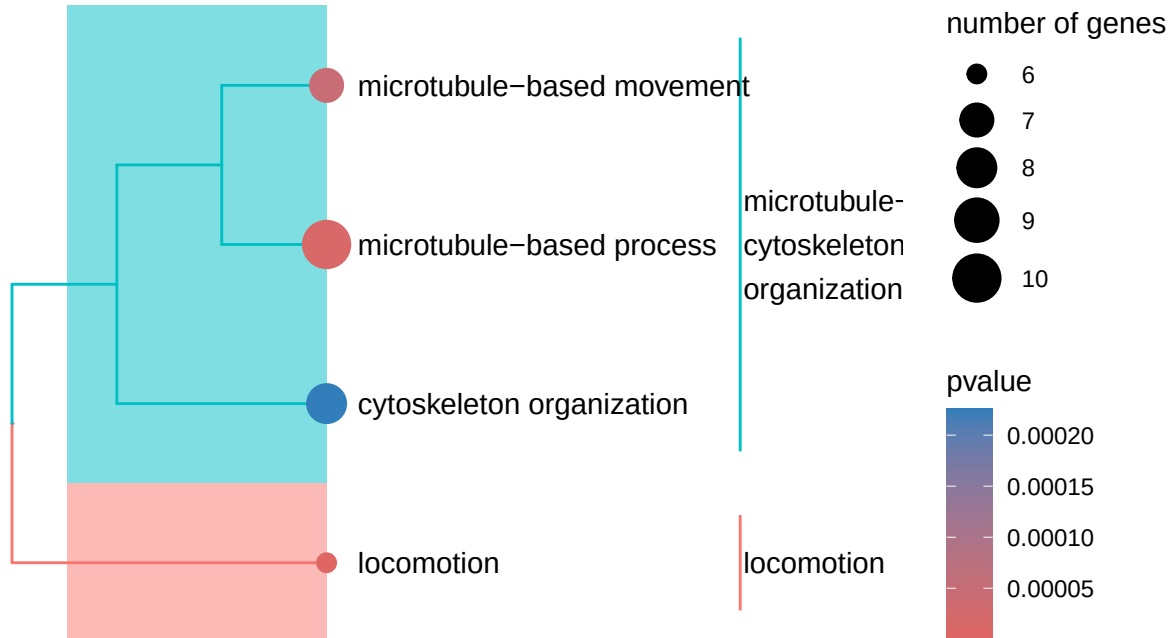
```
temp.repressed.k27.higher.genes.enrichment <- enrichGO(gene = temp.repressed.k27.higher.genes,
                                                         pvalueCutoff = 1,
                                                         qvalueCutoff = 1,
                                                         OrgDb = org.Otauriv5.eg.db,
                                                         ont = "BP",
```

```

keyType = "GID")
k27.go.terms.filtered <- k27.go.terms
k27.go.terms.filtered@result <- subset(k27.go.terms@result, pvalue < 0.001)

treeplot(x = pairwise_termsim(k27.go.terms.filtered), cluster.params = list(n = 2), color = "pvalue")

```



Our analysis identified cytoskeleton organization and locomotion as significantly enriched processes.

The following two functions are used to represent the ChIP signal and expression level for specific genes.

```

gene.profile.complete <- function(gene.name,
                                   promoter_length=1000,
                                   selected.bigwig.files,
                                   cols,
                                   selected.bed.files,
                                   txdb)
{
  genes.data.df <- as.data.frame(genes(txdb))
  cds.data <- as.data.frame(cds(txdb))
  exons.data <- as.data.frame(exons(txdb))
  target.gene.body <- genes.data.df[gene.name,]
  target.gene.chr <- as.character(target.gene.body$seqnames)
  target.gene.start <- target.gene.body$start
  target.gene.end <- target.gene.body$end

  target.gene.strand <- as.character(target.gene.body$strand)

  ## Extract cds annotation
  cds.data.target.gene <- subset(cds.data,
                                seqnames == target.gene.chr &
                                (start >= target.gene.start &
                                 end <= target.gene.end))
}

```

```

## Extract exons annotation
exons.data.target.gene <- subset(exons.data,
                                seqnames == target.gene.chr &
                                (start >= target.gene.start &
                                 end <= target.gene.end))

## Determine the genome range to plot including promoter, gene body and 5' UTR
## This depends on whether the gene is on the forward or reverse strand
range.to.plot <- target.gene.body

if(target.gene.strand == "+")
{
  range.to.plot$start <- range.to.plot$start - promoter_length
  range.to.plot$end <- range.to.plot$end + promoter_length
} else if (target.gene.strand == "-")
{
  range.to.plot$end <- range.to.plot$end + promoter_length
  range.to.plot$start <- range.to.plot$start - promoter_length
}

## Compute the length of the genome range to represent
current.length <- range.to.plot$end - range.to.plot$start

## Compute profile in gene
## Since ChIPpeakAnno needs more than one region to plot our region
## is duplicated
regions.plot <- GRanges(rbind(range.to.plot,range.to.plot))

## Import signal from the bigwig files
cvglists <- sapply(selected.bigwig.files, import,
                   format="BigWig",
                   which=regions.plot,
                   as="RleList")

## Compute signal in the region to plot
chip.signal <- featureAlignedSignal(cvglists, regions.plot,
                                   upstream=ceiling(current.length/2),
                                   downstream=ceiling(current.length/2),
                                   n.tile=current.length)

chip.signal[[1]][1:2,1:5]

chip.signal.mean <- matrix(nrow=length(selected.bigwig.files),ncol=dim(chip.signal[[1]])[2])

## Compute mean signal
for(i in 1:length(selected.bigwig.files))
{
  if(target.gene.strand == "+")
  {
    chip.signal.mean[i,] <- colMeans(chip.signal[[i]],na.rm = TRUE)
  }
}

```

```

    } else if (target.gene.strand == "-")
    {
      chip.signal.mean[i,] <- rev(colMeans(chip.signal[[i]],na.rm = TRUE))
    }
  }

  ## Normalization
  #chip.signal.mean <- 20 * chip.signal.mean / max(chip.signal.mean)
  upper.lim <- 1.1*max(chip.signal.mean)
  ## Colors to draw signal
  line.colors <- cols[i]#"blue"
  area.colors <- cols[i]#"lightblue"

  gene.height <- -25
  cord.x <- 1:current.length

  ## Exon width to plot
  exon.width <- 2#1#2

  ## Cds width to plot
  cds.width <- 3#1.5#3

  ## Width of the rectangle representing the peak reagon
  peak.width <- 1#0.5#1

  ## Extract exons for target gene
  exons.data.target.gene <- subset(exons.data, seqnames == target.gene.chr & (start >= target.gene.start & end <= target.gene.end))

  ## Transform exon coordinates to current range
  min.pos <- min(exons.data.target.gene$start)

  exons.data.target.gene$start <- exons.data.target.gene$start - min.pos + promoter_length
  exons.data.target.gene$end <- exons.data.target.gene$end - min.pos + promoter_length

  ## Extract cds for target gene
  cds.data.target.gene <- subset(cds.data, seqnames == target.gene.chr & (start >= target.gene.start & end <= target.gene.end))

  cds.data.target.gene$start <- cds.data.target.gene$start - min.pos + promoter_length
  cds.data.target.gene$end <- cds.data.target.gene$end - min.pos + promoter_length

  plot(cord.x, rep(gene.height,length(cord.x)),type="l",col="black",lwd=3,ylab="",
       cex.lab=2,axes=FALSE,xlab="",main="",cex.main=2,
       ylim=c(-35,upper.lim),xlim=c(0,max(cord.x)))

  ## Represent exons
  for(i in 1:nrow(exons.data.target.gene))
  {
    # Determine start/end for each exon
    current.exon.start <- exons.data.target.gene$start[i]
    current.exon.end <- exons.data.target.gene$end[i]

    ## Determine coordinates for each exon polygon and represent it
    exon.x <- c(current.exon.start,current.exon.end,current.exon.end,current.exon.start)
  }

```

```

    exon.y <- c(gene.height + exon.width, gene.height + exon.width, gene.height - exon.width, gene.height - exon.width)

    polygon(x = exon.x, y = exon.y, col = "blue",border = "blue")
  }

  if(nrow(cds.data.target.gene) > 0)
  {
    for(i in 1:nrow(cds.data.target.gene))
    {
      # Determine current cds start/end
      current.cds.start <- cds.data.target.gene$start[i]
      current.cds.end <- cds.data.target.gene$end[i]

      # Determine current cds coordinates for the polygon and represent it
      cds.x <- c(current.cds.start,current.cds.end,current.cds.end,current.cds.start)
      cds.y <- c(gene.height + cds.width, gene.height + cds.width, gene.height - cds.width, gene.height - cds.width)
      polygon(x = cds.x, y = cds.y, col = "blue",border = "blue")
    }
  }

  ## Draw arrow to represent transcription direction
  if(target.gene.strand == "+")
  {
    lines(c(promoter_length,promoter_length,promoter_length+100),
          y=c(gene.height,gene.height+5,gene.height+5),lwd=3)
    lines(c(promoter_length+50,promoter_length+100),y=c(gene.height+6,gene.height+5),lwd=3)
    lines(c(promoter_length+50,promoter_length+100),y=c(gene.height+4,gene.height+5),lwd=3)
  } else if (target.gene.strand == "-")
  {
    lines(c(current.length - promoter_length, current.length - promoter_length, current.length - promoter_length - 100),
          y=c(gene.height,gene.height+5,gene.height+5),lwd=3)
    lines(c(current.length - promoter_length-50, current.length - promoter_length - 100),y=c(gene.height+6,gene.height+5),lwd=3)
    lines(c(current.length - promoter_length-50, current.length - promoter_length - 100),y=c(gene.height+4,gene.height+5),lwd=3)
  }

  ## Draw promoter range
  if(target.gene.strand == "+")
  {
    axis(side = 1,labels = c(- promoter_length, - promoter_length / 2,"TSS"),at = c(1,promoter_length/2))
  } else if(target.gene.strand == "-")
  {
    axis(side = 1,labels = c("TSS",- promoter_length / 2,- promoter_length),at = c(current.length-promoter_length,
    current.length - promoter_length / 2, - promoter_length))
  }

  ## Draw gene name
  text(x = current.length / 2, y = -33 ,
       labels = bquote(italic(. (gene.name))),cex = 1.7,font = 3)

  ## Extract bed file name 1 and read it
  current.peaks <- read.table(file=selected.bed.files,header = F, as.is = T)
  current.peaks <- current.peaks[,1:3]
  colnames(current.peaks) <- c("seqnames","start","end")
  peak.coordinates <- subset(current.peaks, seqnames == as.character(range.to.plot$seqnames) & start >=
  current.peaks.to.plot <- peak.coordinates[,2:3]

```

```

## Transform coordinates
current.peaks.to.plot <- current.peaks.to.plot - range.to.plot$start

## Check if there are peaks for the target gene
if(nrow(current.peaks.to.plot) > 0)
{
  #motifs.in.peaks <- vector(mode="list", length=nrow(current.peaks.to.plot))
  for(j in 1:nrow(current.peaks.to.plot))
  {
    ## Extract start and end point of each peak region
    current.peak.start <- current.peaks.to.plot[j,1]
    current.peak.end <- current.peaks.to.plot[j,2]

    ## Computer coordinates for polygon and draw it
    peak.x <- c(current.peak.start,current.peak.end,
               current.peak.end,current.peak.start)
    peak.y <- c(peak.width - 12, peak.width - 12,
               - peak.width - 12, - peak.width - 12)

    polygon(x = peak.x, y = peak.y, col = "blue", border = "blue",lwd=2)
  }
}

## Draw profiles if bw file is provided
## Compute base line for current TF
current.base.line <- - 10

i <- 4

for(i in 1:length(selected.bigwig.files))
{
  ## Represent signal from the current TF
  lines(chip.signal.mean[i,]+current.base.line,type="l",col=cols[i],lwd=3)
  ## Determine polygon coordinates and represent it
  cord.y <- c(current.base.line,chip.signal.mean[i,]+current.base.line,current.base.line)
  cord.x <- 1:length(cord.y)
  polygon(cord.x,cord.y,col=cols[i])
}

axis(side = 2,lwd = 2,
     at = seq(from=0,to=ceiling(upper.lim/10)*10,by=10)+current.base.line,
     labels = seq(from=0,to=ceiling(upper.lim/10)*10,by=10))
}

gene.expression.barplot <- function(gene.id, gene.expression)
{
  gene.expression.10 <- unlist(gene.expression[gene.id,paste("t10C",1:3,sep="_")])
  gene.expression.14 <- unlist(gene.expression[gene.id,paste("t14C",1:3,sep="_")])
  gene.expression.20 <- unlist(gene.expression[gene.id,paste("t20C",1:3,sep="_")])
  gene.expression.26 <- unlist(gene.expression[gene.id,paste("t26C",1:3,sep="_")])

  means <- c(mean(gene.expression.10),
             mean(gene.expression.14),

```

```

        mean(gene.expression.20),
        mean(gene.expression.26))

sds <- c(sd(gene.expression.10),
        sd(gene.expression.14),
        sd(gene.expression.20),
        sd(gene.expression.26))

y.max <- 1.2*max(means + sds)

par(lwd=2)
xpos <- barplot(means,ylim=c(0,y.max),
               col = colorRampPalette(c("blue", "red"))(4),cex.axis = 1.5,
               names.arg = c("10°C", "14°C", "20°C", "26°C"),cex.names=2,lwd=2)
par(lwd=1)
arrows(x0 = xpos,y0 = means-sds/sqrt(3), x1 = xpos,y1=means+sds/sqrt(3),code=3,angle=90,length=0.07,lwd=1)

points(x = rep(xpos+0.2,each=3),y=c(gene.expression.10,
                                     gene.expression.14,
                                     gene.expression.20,
                                     gene.expression.26),lwd=2)

}

```

Specifically we found increased levels of H3K27me3 and gene expression repression in genes codifying for proteins involved in microtubule development tubulin alpha and beta:

```

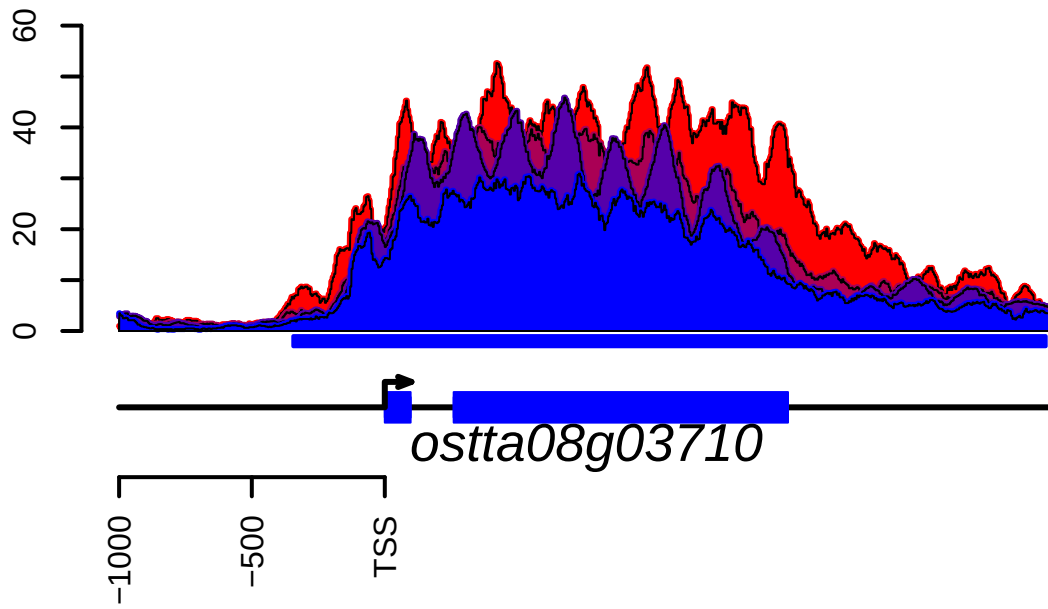
#jpeg(filename = "images/tuba_ostta08g03710.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta08g03710", #TUBA
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip.2.bw",
                                                "h3k27me3/h3k27me3_11_20_chip.2.bw",
                                                "h3k27me3/h3k27me3_11_14_chip.1.bw",
                                                "h3k27me3/h3k27me3_11_10_chip.2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb=TxDb.Otauri.JGI)

```

```

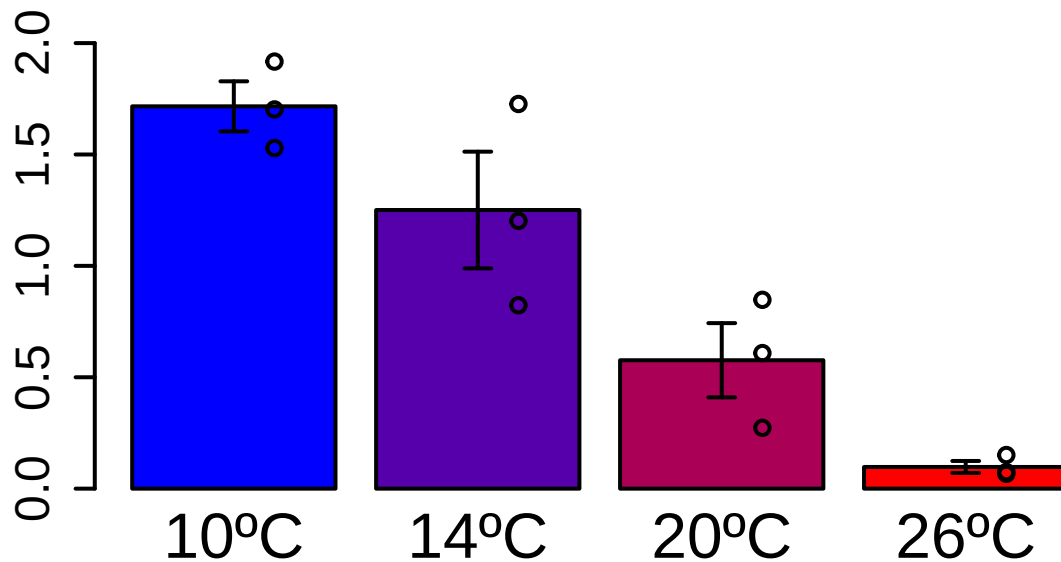
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

```



```
#dev.off()

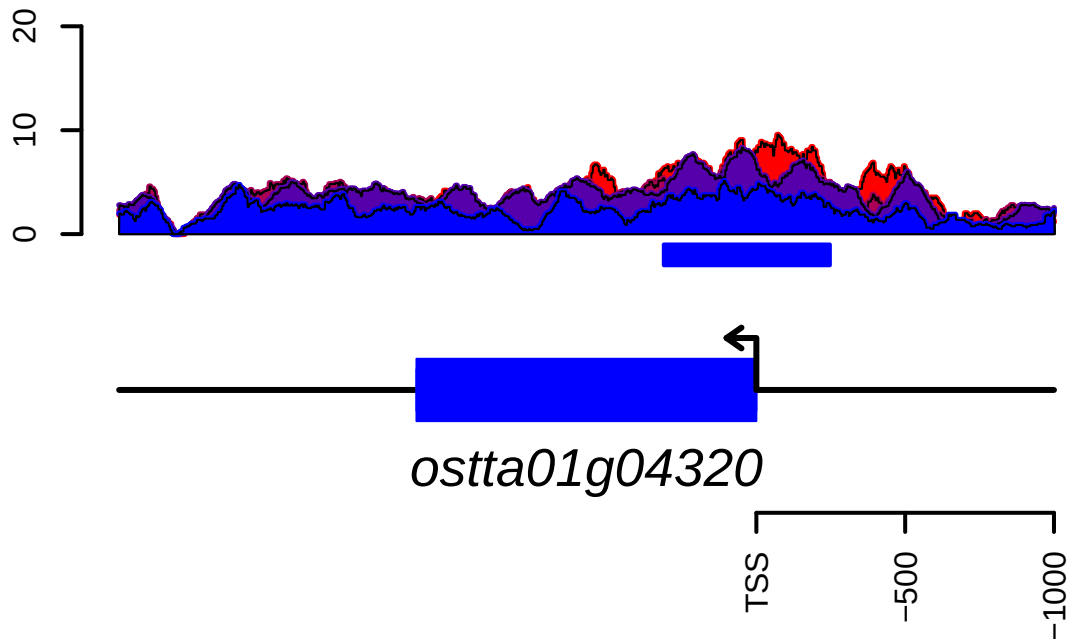
#jpeg(filename = "images/tuba_ostta08g03710_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta08g03710", gene.expression=gene.expression)
```



```
#dev.off()

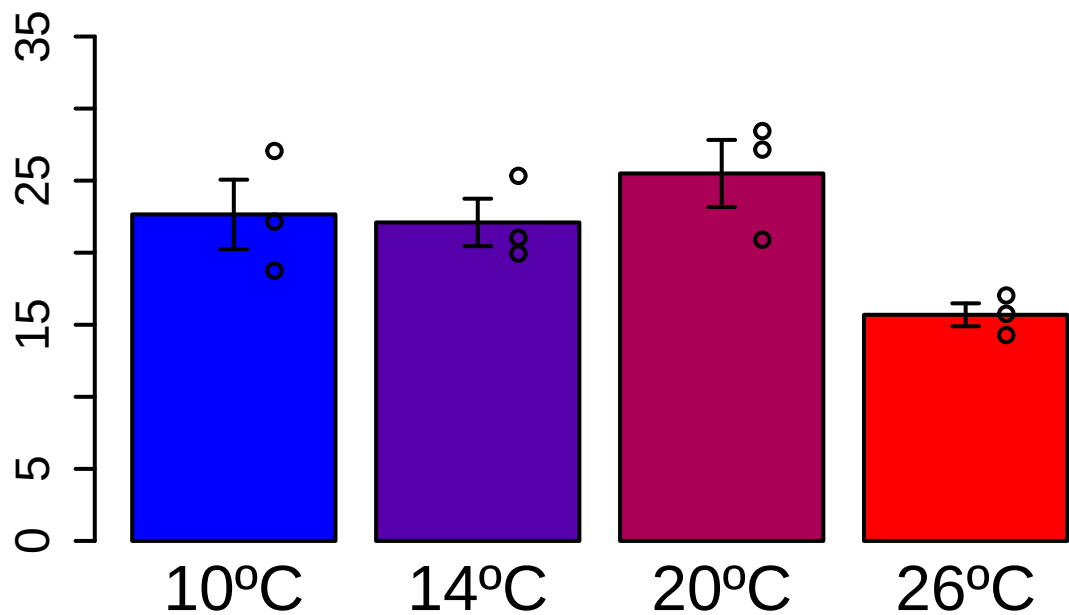
#jpeg(filename = "images/tubb_ostta01g04320.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta01g04320", #TUBB
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_20_chip_2.bw",
    "h3k27me3/h3k27me3_ll_14_chip_1.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/tubb_ostta01g04320_expression.jpg", res = 100, width = 600, height = 580)
gene.expression.barplot(gene.id = "ostta01g04320", gene.expression=expression)
```



```
#dev.off()

gene.profile.complete(gene.name = "ostta08g03710", #TUBA
promoter_length=1000,
selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
"h3k27me3/h3k27me3_ll_10_chip_2.bw",
```

```

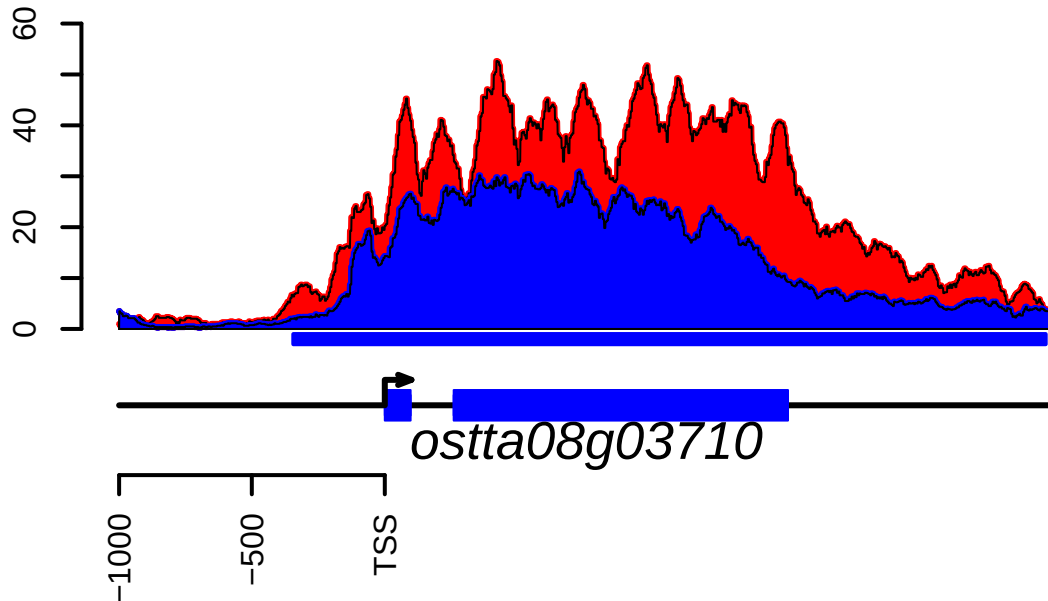
        "h3k27me3/h3k27me3_11_10_chip_2.bw",
        "h3k27me3/h3k27me3_11_10_chip_2.bw"),
cols = rev(colorRampPalette(c("blue", "red"))(4)),
selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
txdb=TxDb.Otauri.JGI)

```

```

## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

```



```

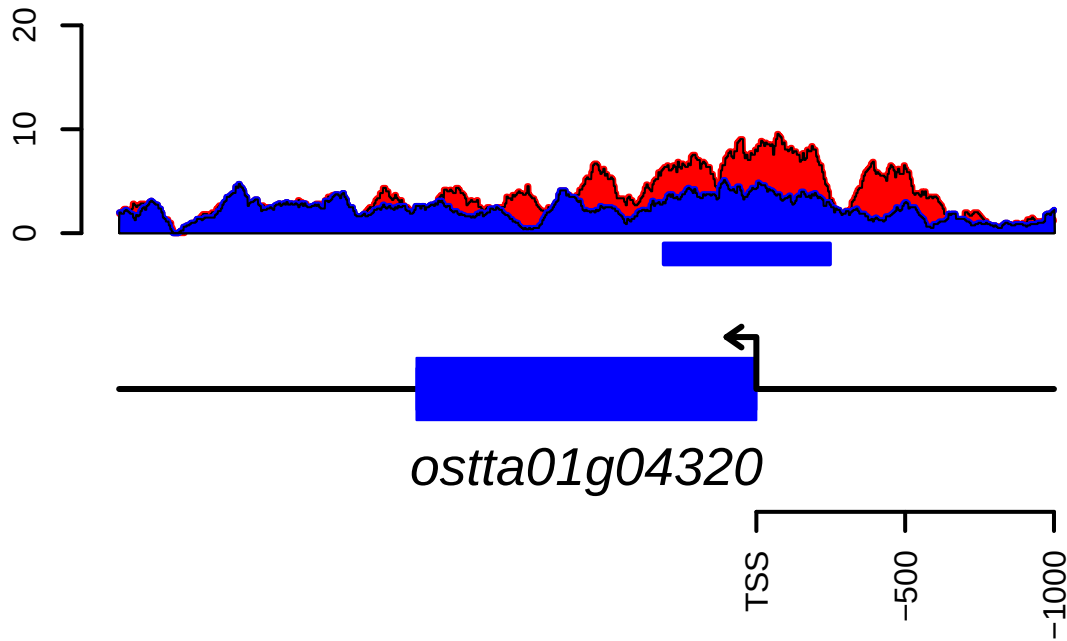
gene.profile.complete(gene.name = "ostta01g04320", #TUBB
    promoter_length=1000,
    selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
        "h3k27me3/h3k27me3_11_10_chip_2.bw",
        "h3k27me3/h3k27me3_11_10_chip_2.bw",
        "h3k27me3/h3k27me3_11_10_chip_2.bw"),
    cols = rev(colorRampPalette(c("blue", "red"))(4)),
    selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
    txdb=TxDb.Otauri.JGI)

```

```

## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

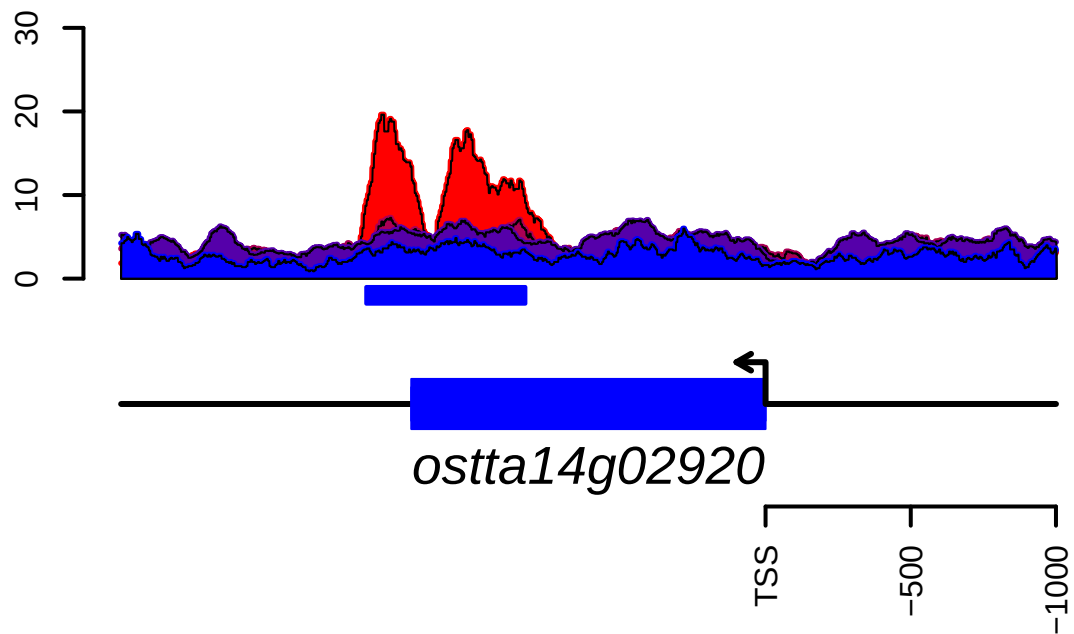
```



Genes involved in cortical actin network remodelling. ARP2 (ostta14g02920) is involved in the branching of actin filaments. FIL-MORN-Invasing (ostta09g01910) proteins is involved in the crosslinking of actin filaments (filamin domain), interactions between actin filaments and membranes (MORN Membrane Occupation and Recognition Nexus) and membrane penetration or remodeling to assist in vesicle budding (Invasin D3 domain).

```
#jpeg(filename = "images/actin_ostta14g02920.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta14g02920", #ARP2
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_20_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_14_chip_1.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb=TxDb.Otauri.JGI)
```

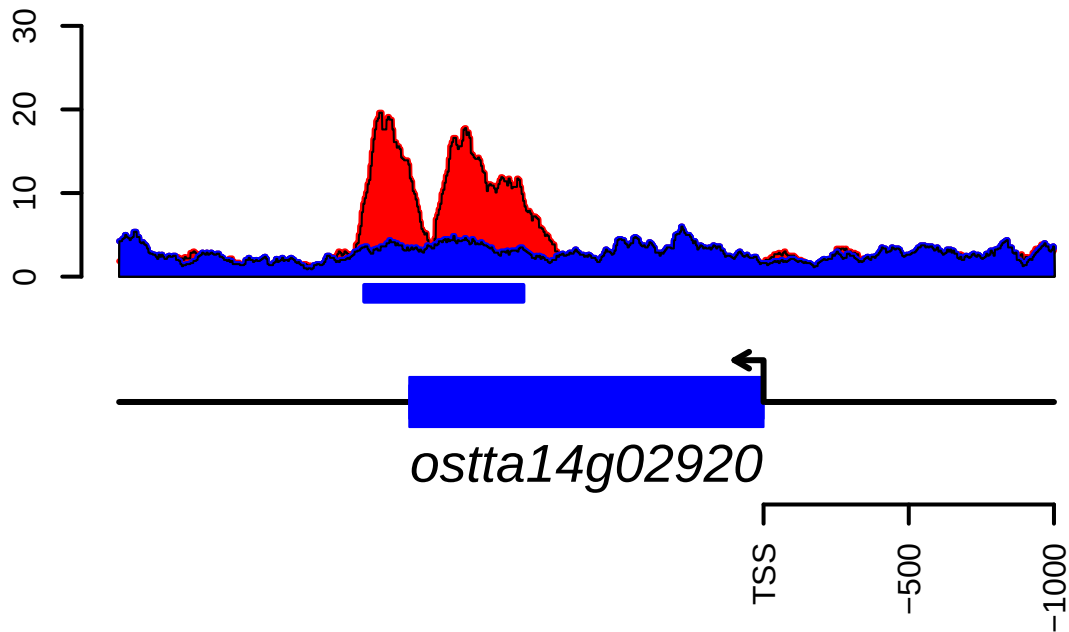
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



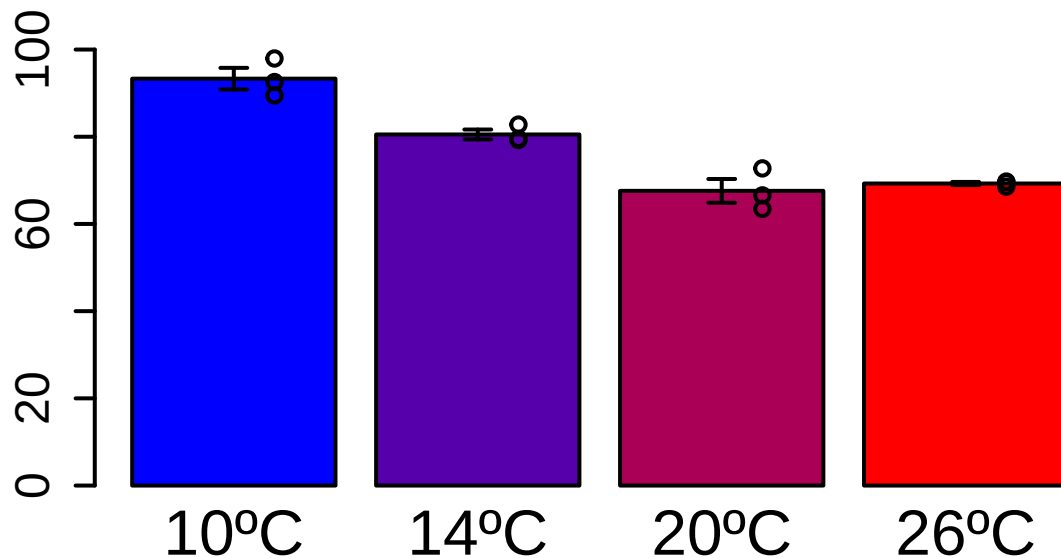
```
#dev.off()

gene.profile.complete(gene.name = "ostta14g02920", #ARP2
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



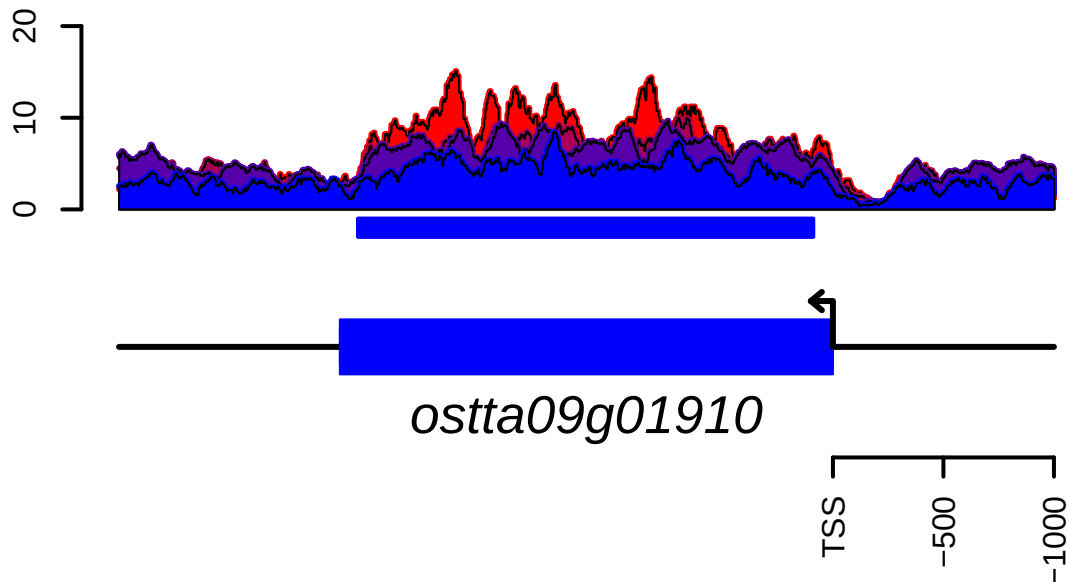
```
#jpeg(filename = "images/actin_ostta14g02920_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta14g02920", gene.expression=gene.expression)
```



```
#dev.off()

#jpeg(filename = "images/filamin_ostta09g01910.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta09g01910", #Filamin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_20_chip_2.bw",
    "h3k27me3/h3k27me3_ll_14_chip_1.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

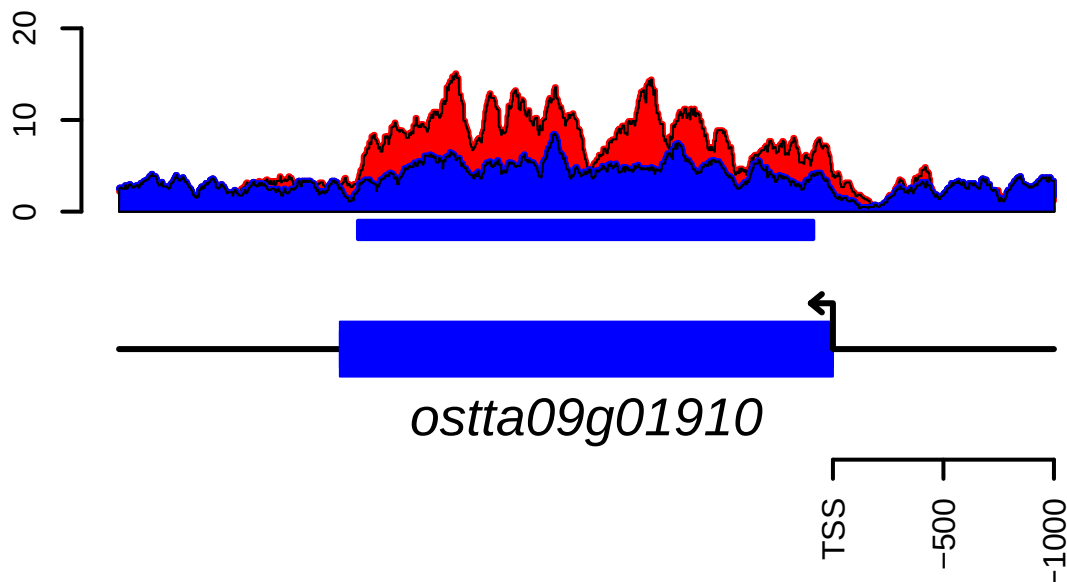
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



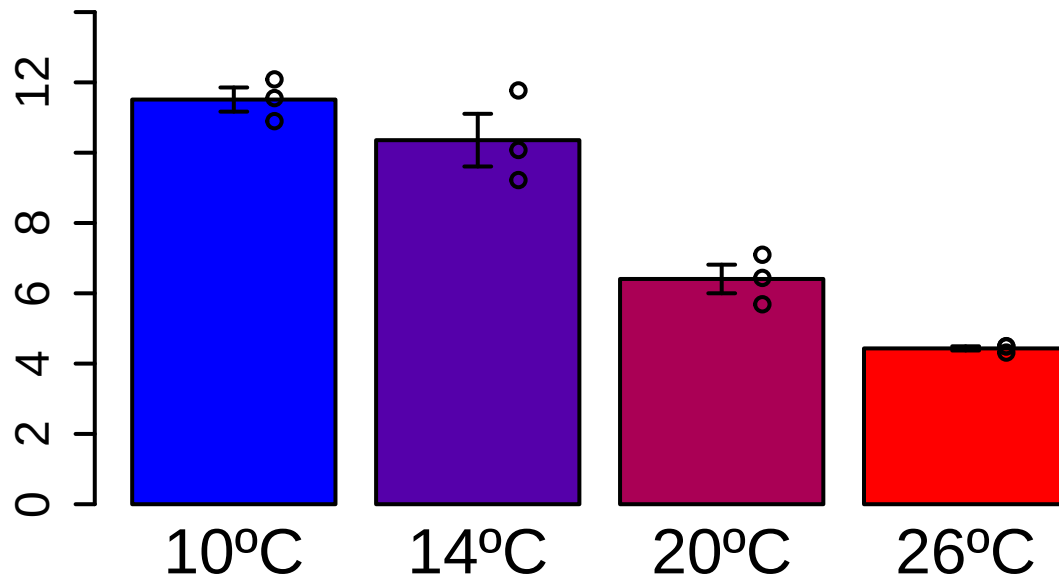
```
#dev.off()
```

```
gene.profile.complete(gene.name = "ostta09g01910", #Filamin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#jpeg(filename = "images/filamin_ostta09g01910_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta09g01910", gene.expression=gene.expression)
```

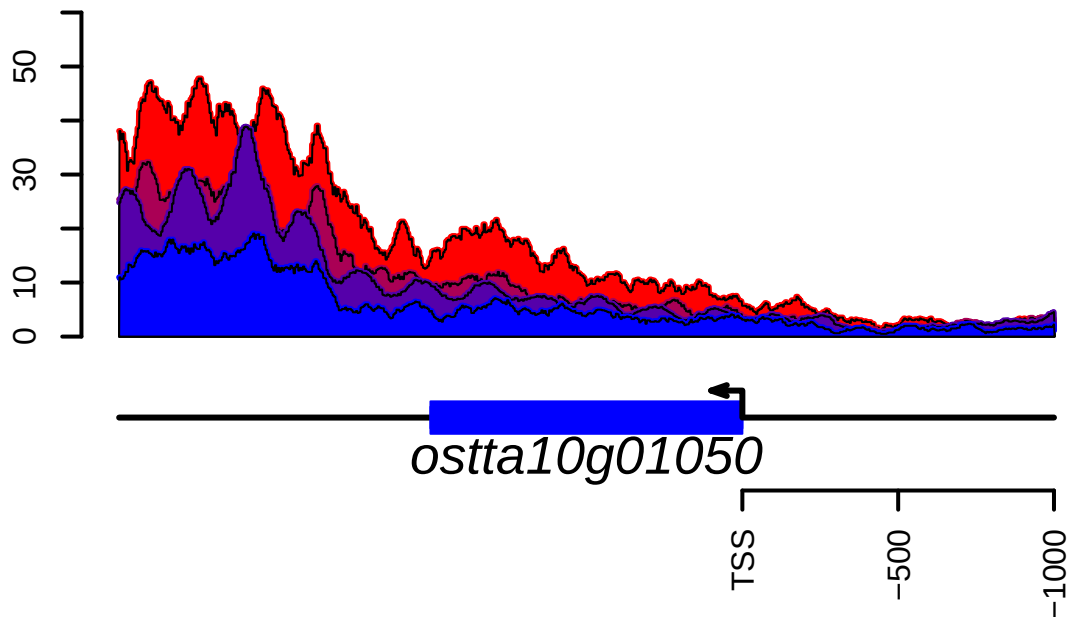


```
#dev.off()
```

We also found increase H3K27me3 signal increase and expression repression in the gene codifying for AURORA kinase (ostta10g01050) that is involved in microtubule organization by phosphorylating tubulins and actins.

```
#jpeg(filename = "images/aurora_ostta10g01050.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta10g01050", #Aurora
    promoter_length=1000,
    selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
                             "h3k27me3/h3k27me3_ll_20_chip_2.bw",
                             "h3k27me3/h3k27me3_ll_14_chip_1.bw",
                             "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
    cols = rev(colorRampPalette(c("blue", "red"))(4)),
    selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
    txdb=TxDb.Otauri.JGI)
```

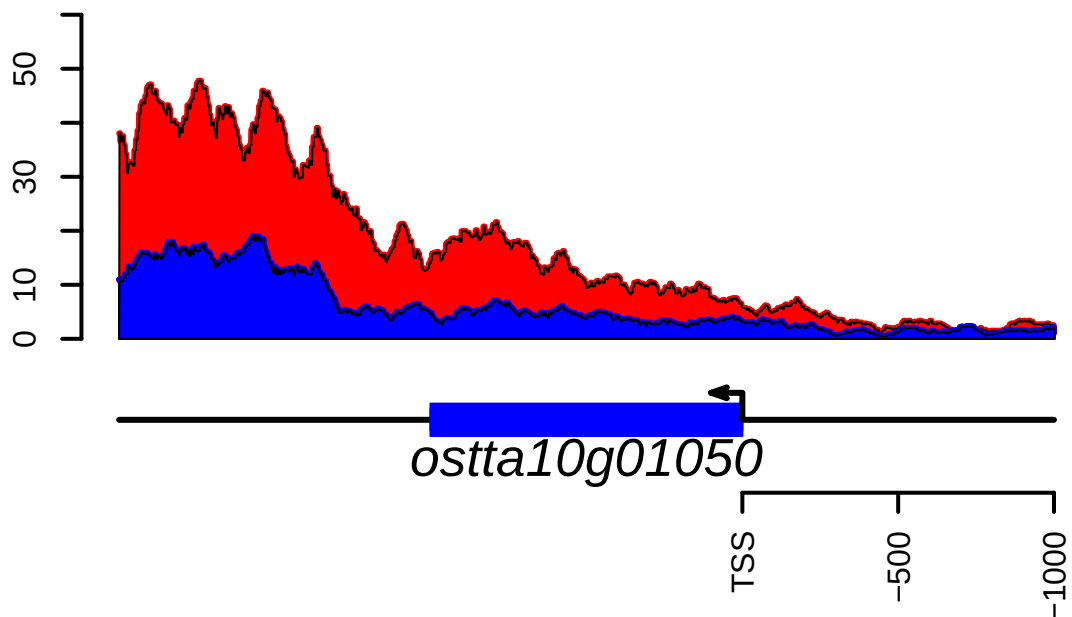
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

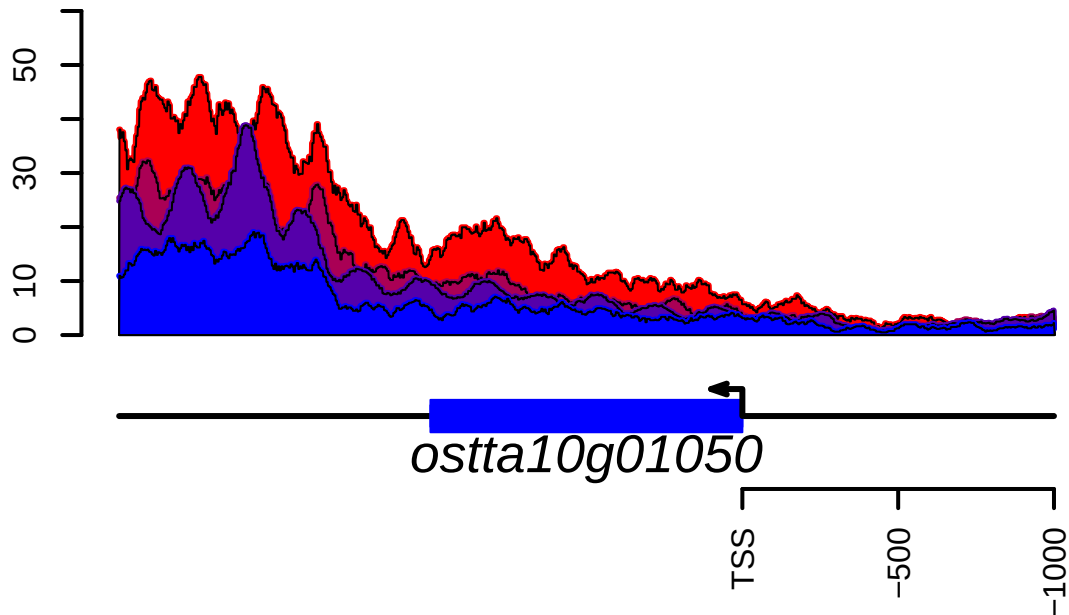
gene.profile.complete(gene.name = "ostta10g01050", #Aurora
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

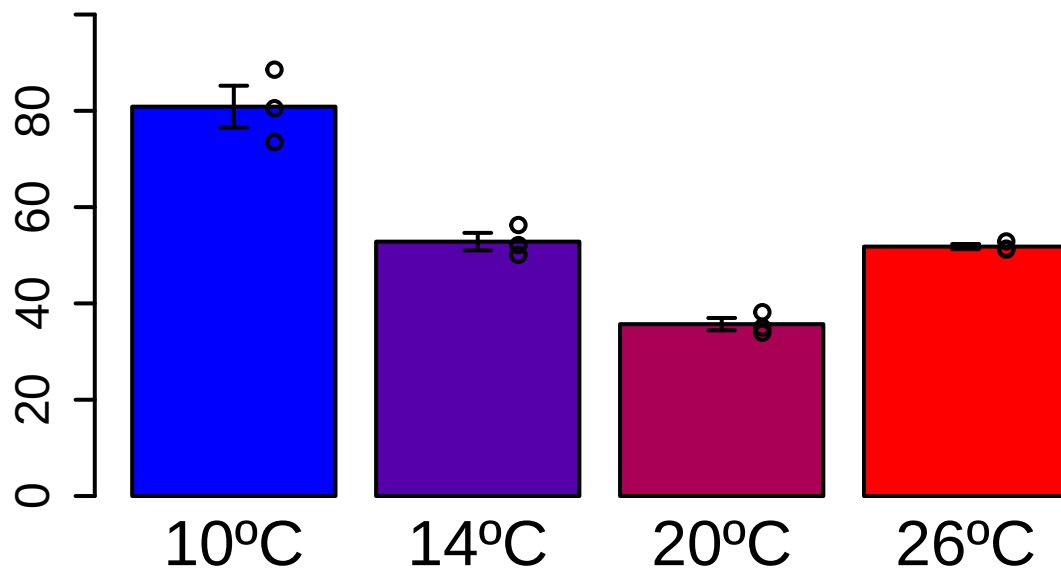


```
gene.profile.complete(gene.name = "ostta10g01050", #Aurora
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_20_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_14_chip_1.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#jpeg(filename = "images/aurora_ostta10g01050_expression.jpg", res = 100, width = 600, height = 580)
gene.expression.barplot(gene.id = "ostta10g01050", gene.expression=gene.expression)
```

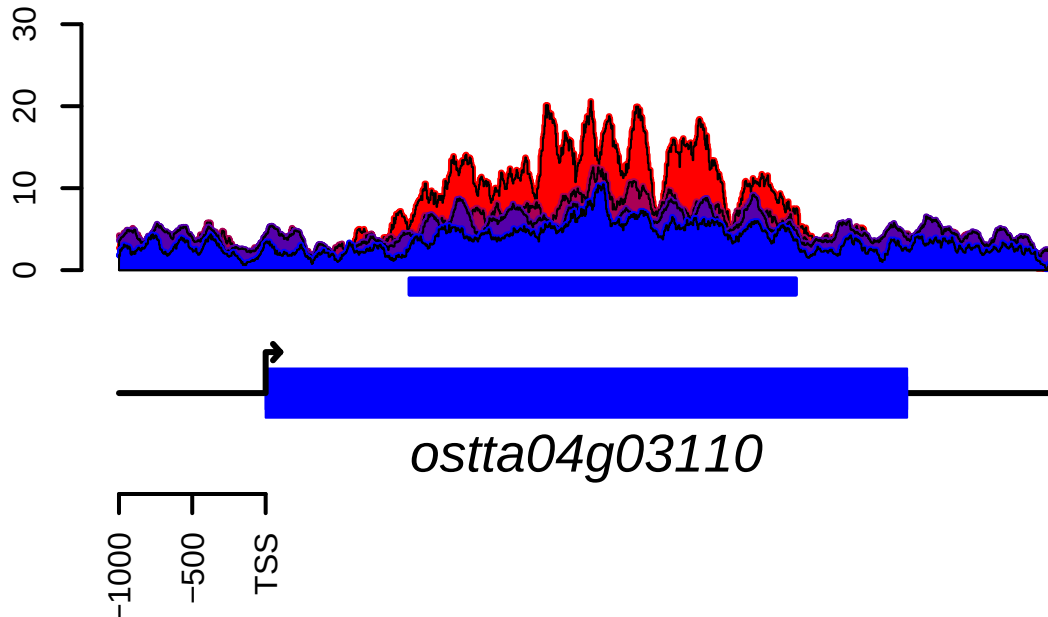


```
#dev.off()
```

This significant regulation over cytoskeleton remodelling is coupled with the activation of motor proteins. In genes codifying for kinesin (ostta04g03110) and myosin (ostta06g03320), we detected a significantly increased H3K27me3 ChIP signal and decreased expression as a response to elevated temperatures.

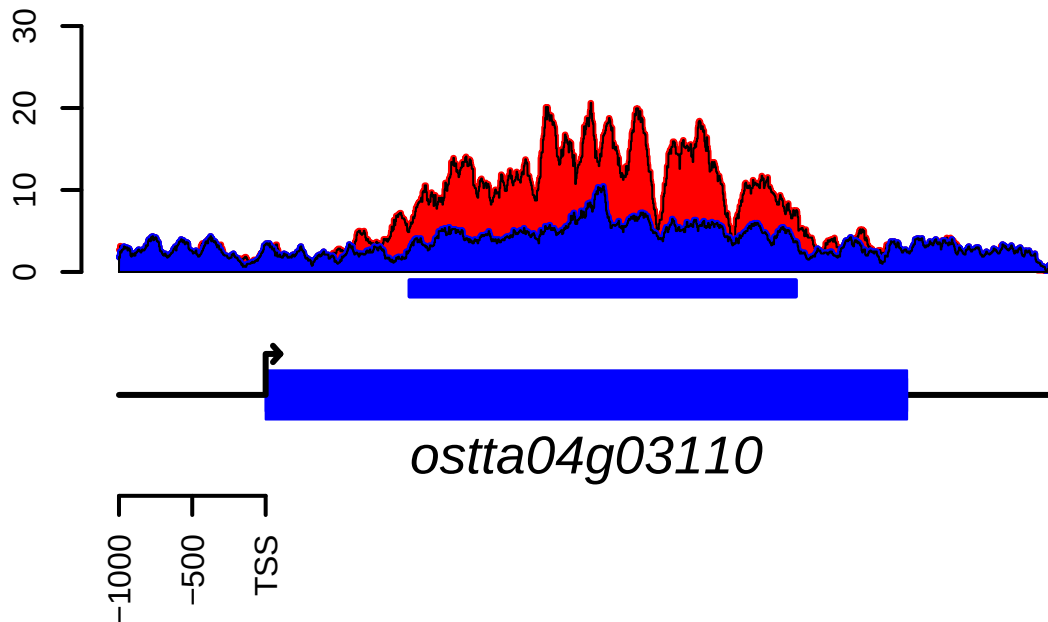
```
#jpeg(filename = "images/kinesin_ostta04g03110.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta04g03110", #Kinesin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_20_chip_2.bw",
    "h3k27me3/h3k27me3_ll_14_chip_1.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



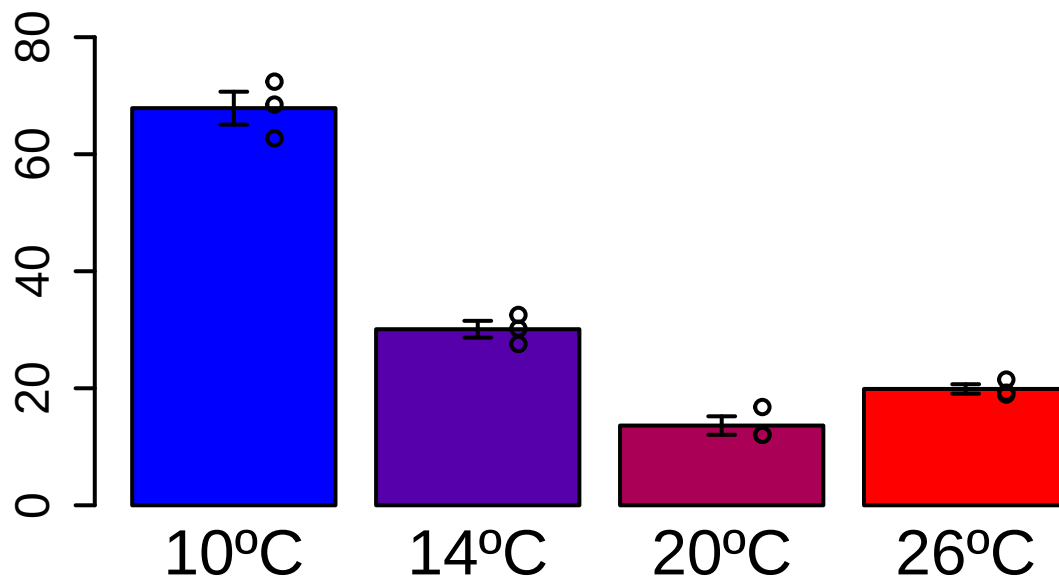
```
gene.profile.complete(gene.name = "ostta04g03110", #Kinesin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/kinesin_ostta04g03110_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta04g03110", gene.expression=gene.expression)
```

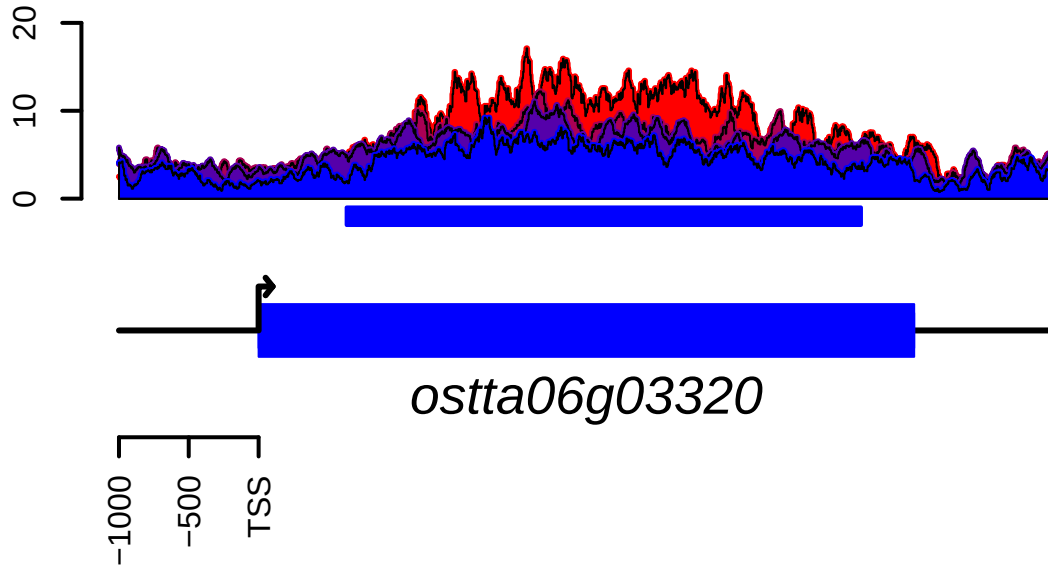


```
#dev.off()

#jpeg(filename = "images/myosin_ostta06g03320.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta06g03320", #Myosin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_1l_26_chip_2.bw",
    "h3k27me3/h3k27me3_1l_20_chip_2.bw",
    "h3k27me3/h3k27me3_1l_14_chip_1.bw",
    "h3k27me3/h3k27me3_1l_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
```

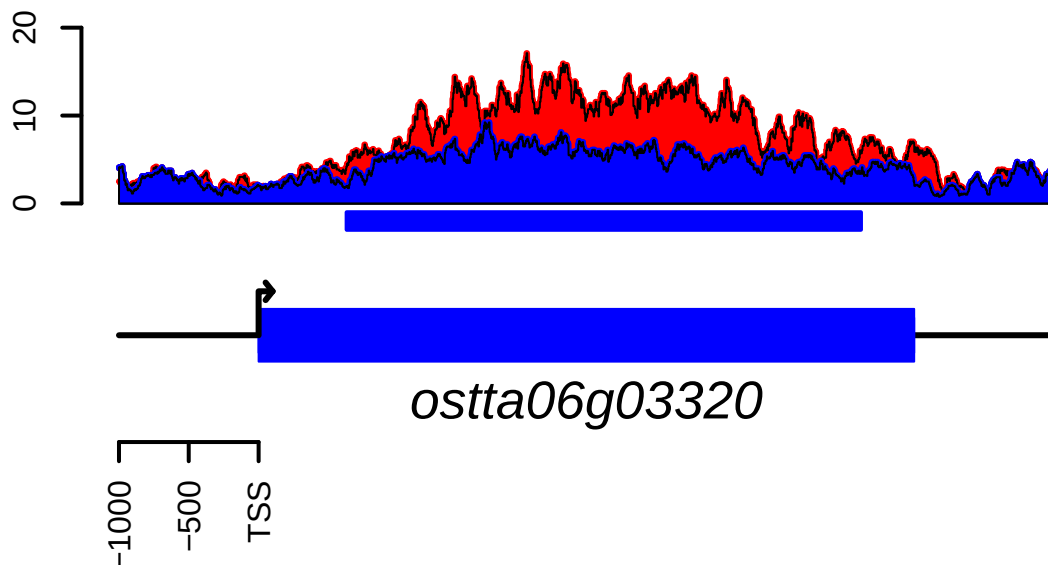
```
txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =  
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



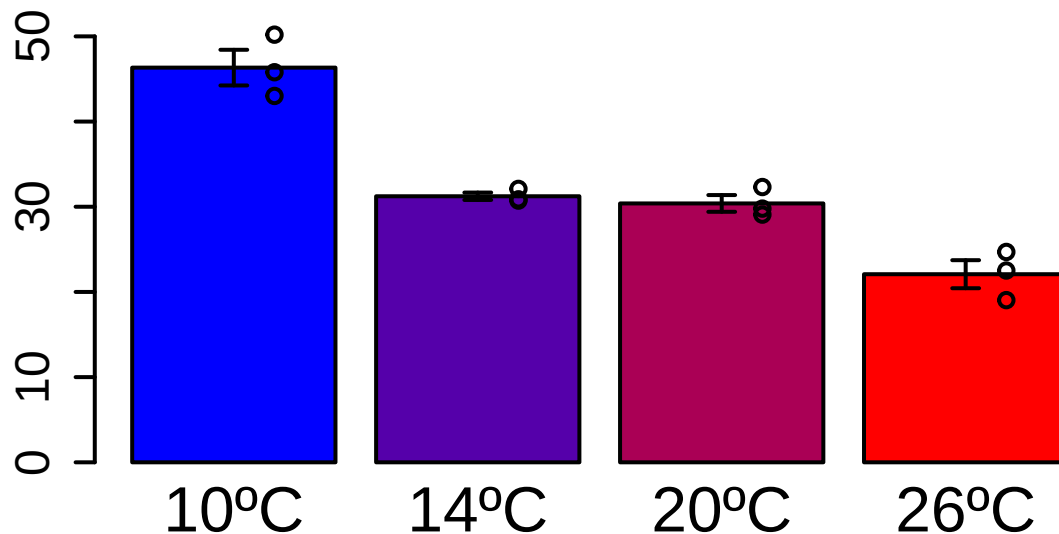
```
gene.profile.complete(gene.name = "ostta06g03320", #Myosin  
                      promoter_length=1000,  
                      selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",  
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",  
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",  
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw"),  
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),  
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",  
                      txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =  
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

```
#jpeg(filename = "images/myosin_ostta06g03320_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta06g03320", gene.expression=gene.expression)
```



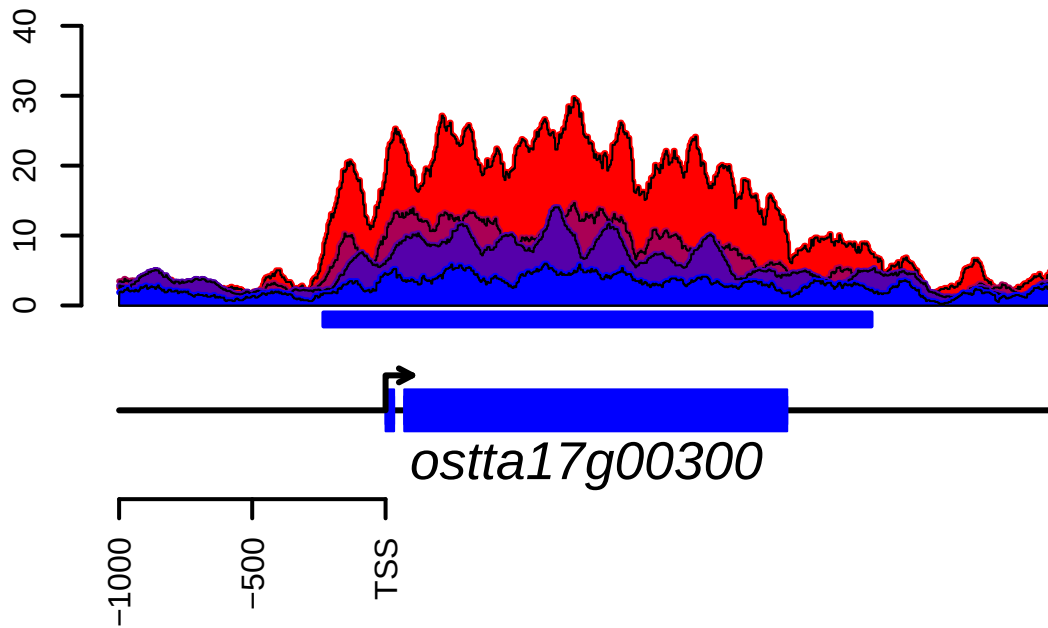
```
#dev.off()
```

Kinesins are involved in long range vesicle transport through microtubules and myosins are involved in short range transport through actin filaments. Kinesins (microtubules) and myosins (actin filaments) combination suggests a two-phase vesicle transport system activated at low temperatures. Long-range transport along microtubules by kinesins followed by short-range delivery to the membrane along the cortical actin network by myosins.

Although no significant enrichment was found for lipid metabolism in the genes that significantly increase their H3K27me3 ChIP-signal and decrease their expression level, key genes in lipid metabolism and transport were identified in this set of genes: enoyl-ACP reductase (ENR, ostta17g00300), diacylglycerol acyltransferase (DGAT, ostta16g00910) and ABC type 2 transporter (ABC, ostta09g01615).

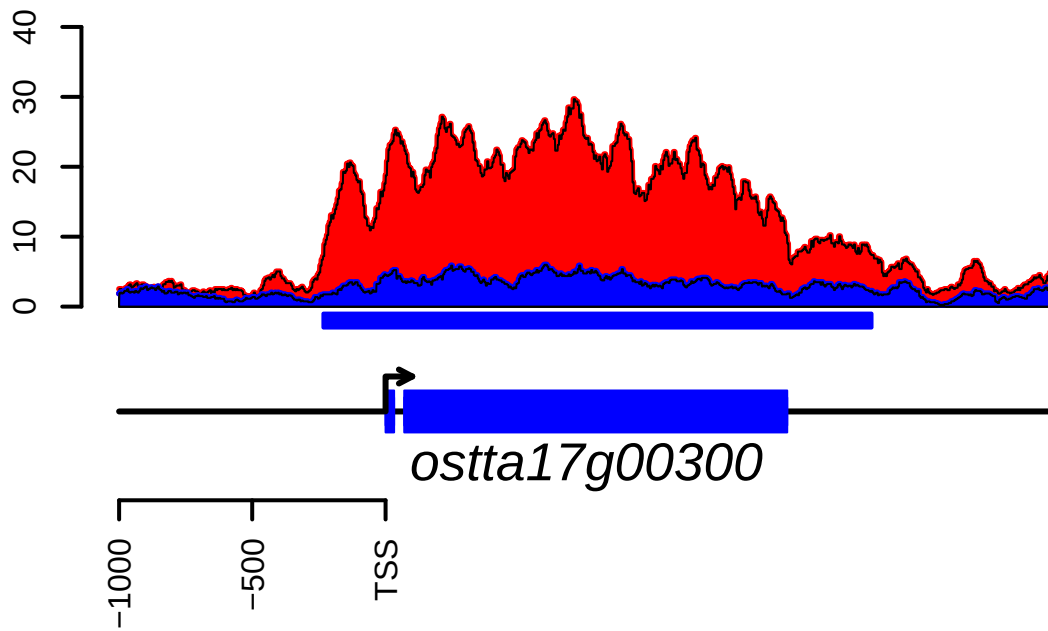
```
#jpeg(filename = "images/ENR_ostta17g00300.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta17g00300", #ENR
    promoter_length=1000,
    selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
                             "h3k27me3/h3k27me3_11_20_chip_2.bw",
                             "h3k27me3/h3k27me3_11_14_chip_1.bw",
                             "h3k27me3/h3k27me3_11_10_chip_2.bw"),
    cols = rev(colorRampPalette(c("blue", "red"))(4)),
    selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
    txdb=TxDb.Otari.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



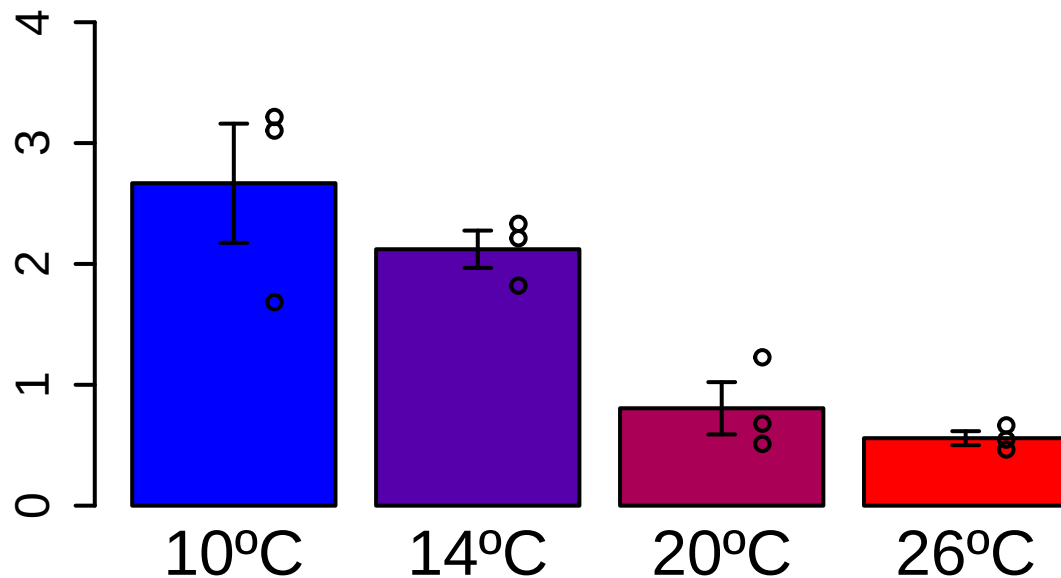
```
gene.profile.complete(gene.name = "ostta17g00300", #ENR
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

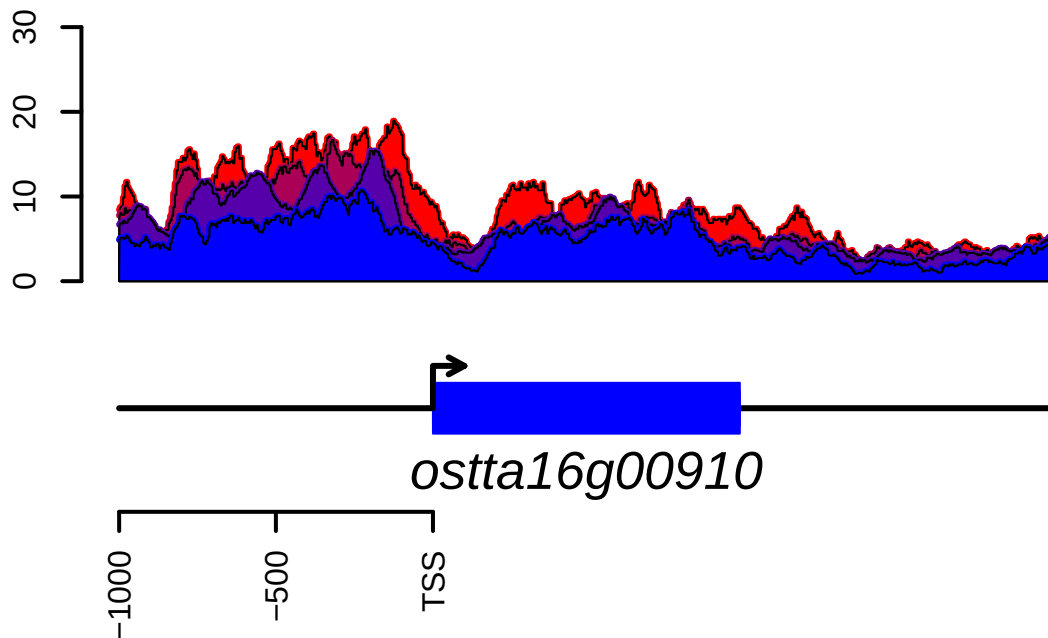
```
#jpeg(filename = "images/ENR_ostta17g00300_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta17g00300", gene.expression=gene.expression)
```



```
#dev.off()
```

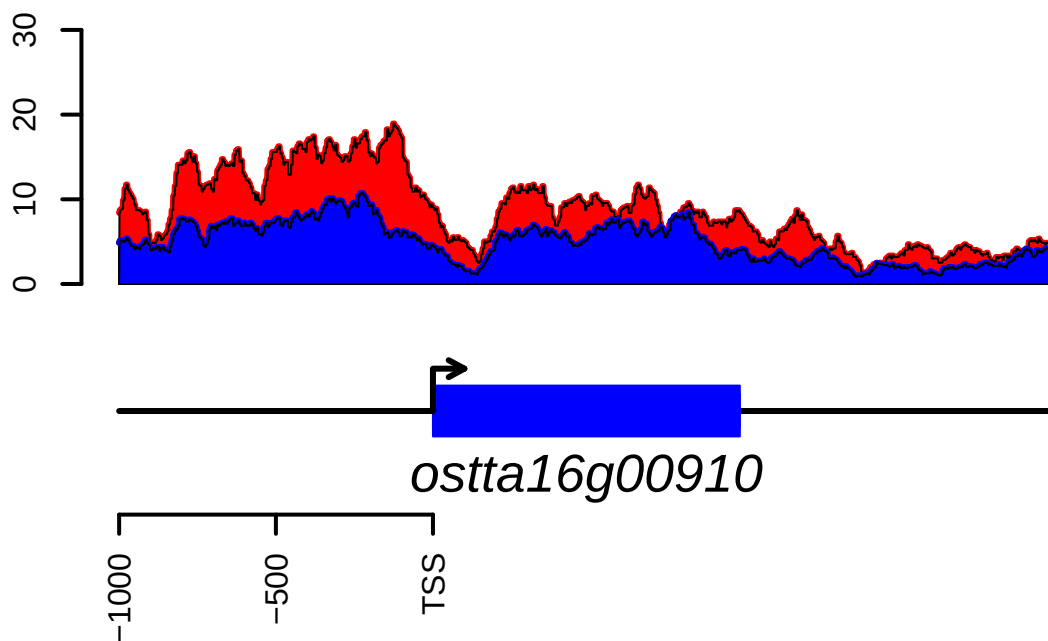
```
#jpeg(filename = "images/DGAT_ostta16g00910.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta16g00910", #DGAT
    promoter_length=1000,
    selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
                             "h3k27me3/h3k27me3_ll_20_chip_2.bw",
                             "h3k27me3/h3k27me3_ll_14_chip_1.bw",
                             "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
    cols = rev(colorRampPalette(c("blue", "red"))(4)),
    selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
    txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



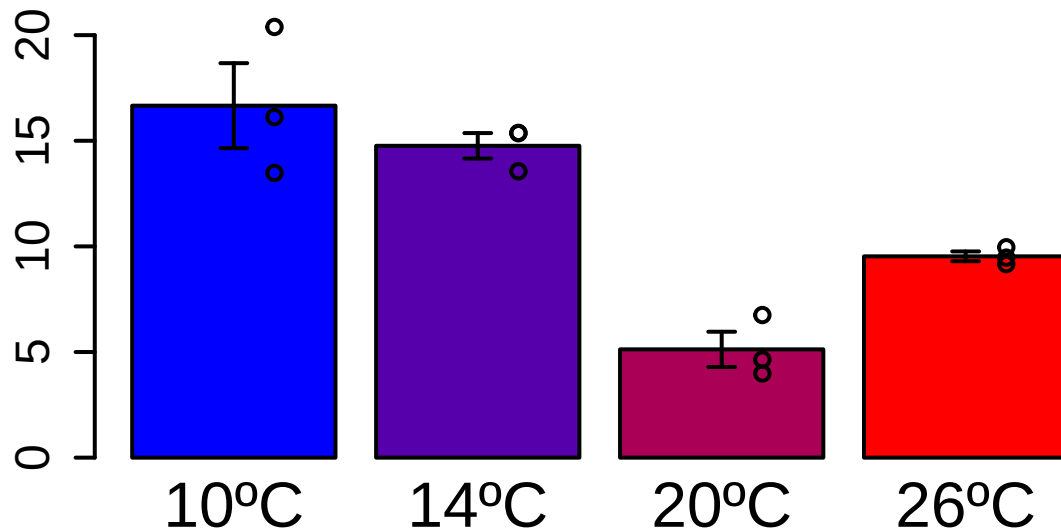
```
gene.profile.complete(gene.name = "ostta16g00910", #DGAT
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

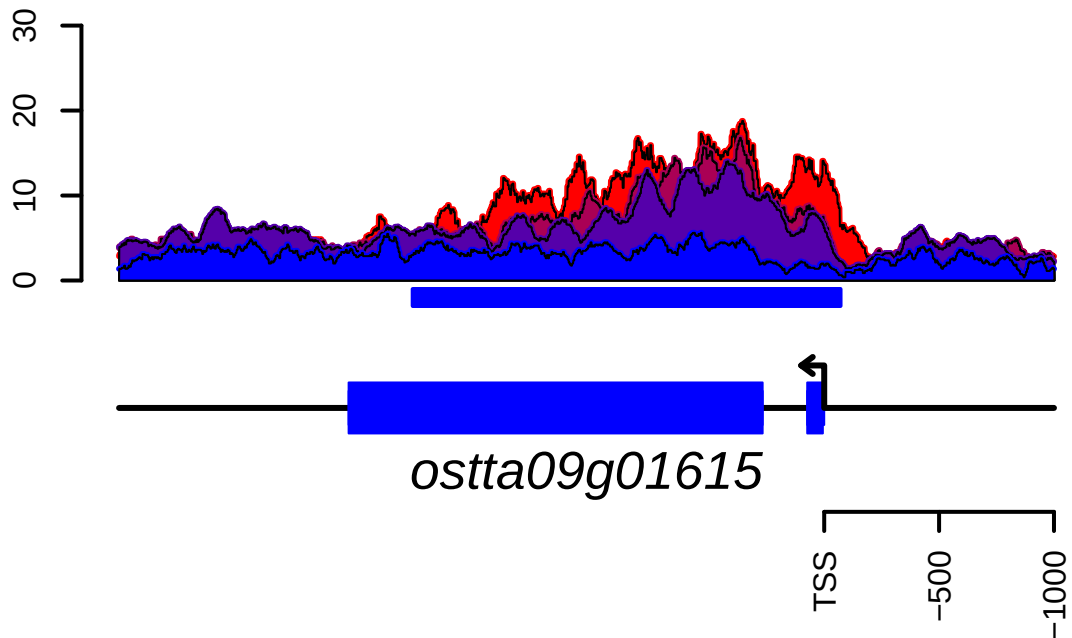
```
#jpeg(filename = "images/DGAT_ostta16g00910_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta16g00910", gene.expression=gene.expression)
```



```
#dev.off()
```

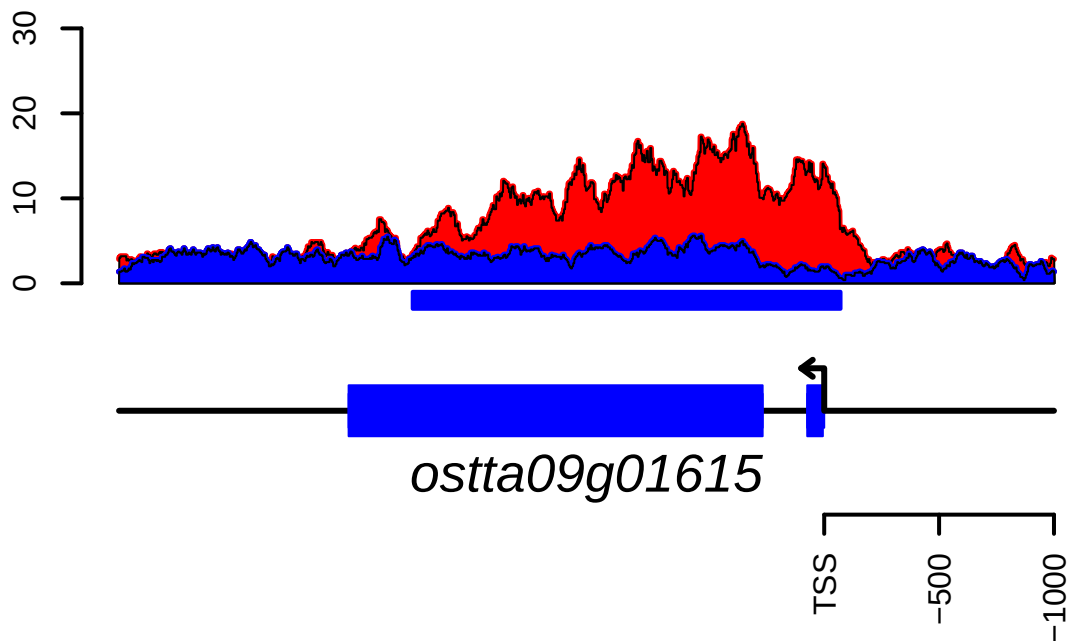
```
#jpeg(filename = "images/ABC_ostta09g01615.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta09g01615", #ABC
    promoter_length=1000,
    selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
                             "h3k27me3/h3k27me3_ll_20_chip_2.bw",
                             "h3k27me3/h3k27me3_ll_14_chip_1.bw",
                             "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
    cols = rev(colorRampPalette(c("blue", "red"))(4)),
    selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
    txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



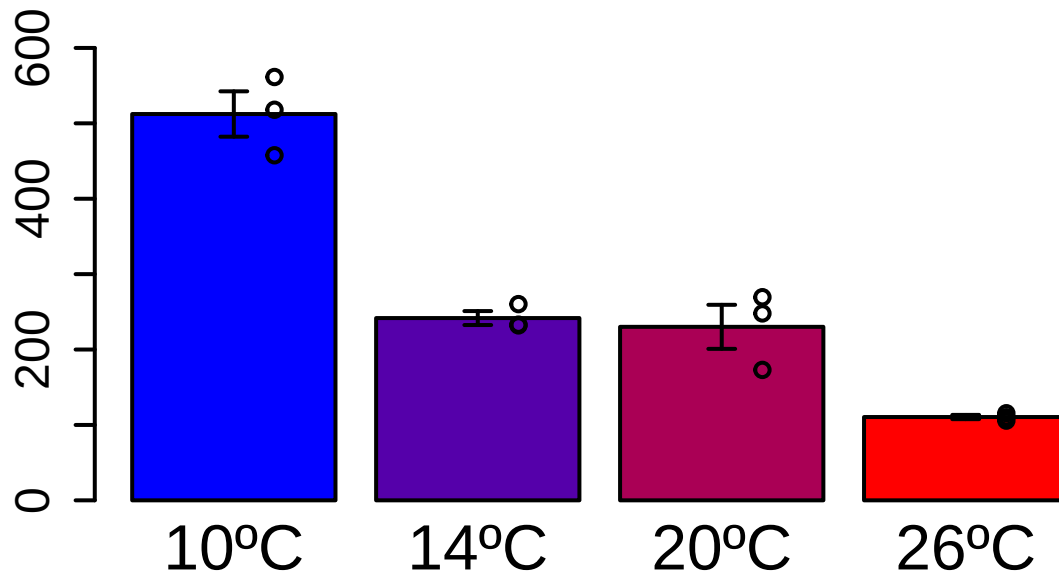
```
gene.profile.complete(gene.name = "ostta09g01615", #ABC
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
    "h3k27me3/h3k27me3_11_10_chip_2.bw",
    "h3k27me3/h3k27me3_11_10_chip_2.bw",
    "h3k27me3/h3k27me3_11_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

```
#jpeg(filename = "images/ABC_ostta09g01615_expression.jpg",res = 100,width = 600,height = 580)  
gene.expression.barplot(gene.id = "ostta09g01615", gene.expression=gene.expression)
```



```
#dev.off()
```

H3K4me3 response to different temperatures

We study the changes in H3K4me3 as a response to temperature. We estimate ChIP signals using CPM (Counts Per Million of mapped reads).

```
## Million of mapped reads for each sample
```

```
mapped.reads.k4.10c.1 <- 4660652/1E6
```

```
mapped.reads.k4.10c.2 <- 4572532/1E6
```

```
mapped.reads.k4.10c.3 <- 4490999/1E6
```

```
mapped.reads.k4.14c.1 <- 4609469/1E6
```

```
mapped.reads.k4.14c.2 <- 5034594/1E6
```

```
mapped.reads.k4.14c.3 <- 5174666/1E6
```

```
mapped.reads.k4.20c.1 <- 6832850/1E6
```

```
mapped.reads.k4.20c.2 <- 5993486/1E6
```

```
mapped.reads.k4.20c.3 <- 6701422/1E6
```

```
mapped.reads.k4.26c.1 <- 6059862/1E6
```

```
mapped.reads.k4.26c.2 <- 5263311/1E6
```

```
mapped.reads.k4.26c.3 <- 4251439/1E6
```

```
## Load read counts and normalization by dividing by the millions  
## of mapped reads
```

```
chip.signal.10.1 <- read.table(  
  file = "h3k4me3/h3k4me3_1l_10_chip_1_count.bed",  
  header = F, sep = "\t")
```

```

chip.signal.10.1[,4] <- chip.signal.10.1[,4]/mapped.reads.k4.10c.1

chip.signal.10.2 <- read.table(
  file = "h3k4me3/h3k4me3_ll_10_chip_2_count.bed",
  header = F, sep = "\t")
chip.signal.10.2[,4] <- chip.signal.10.2[,4]/mapped.reads.k4.10c.2

chip.signal.10.3 <- read.table(
  file = "h3k4me3/h3k4me3_ll_10_chip_3_count.bed",
  header = F, sep = "\t")
chip.signal.10.3[,4] <- chip.signal.10.3[,4]/mapped.reads.k4.10c.3


chip.signal.14.1 <- read.table(
  file = "h3k4me3/h3k4me3_ll_14_chip_1_count.bed",
  header = F, sep = "\t")
chip.signal.14.1[,4] <- chip.signal.14.1[,4]/mapped.reads.k4.14c.1

chip.signal.14.2 <- read.table(
  file = "h3k4me3/h3k4me3_ll_14_chip_2_count.bed",
  header = F, sep = "\t")
chip.signal.14.2[,4] <- chip.signal.14.2[,4]/mapped.reads.k4.14c.2

chip.signal.14.3 <- read.table(
  file = "h3k4me3/h3k4me3_ll_14_chip_3_count.bed",
  header = F, sep = "\t")
chip.signal.14.3[,4] <- chip.signal.14.3[,4]/mapped.reads.k4.14c.3


chip.signal.20.1 <- read.table(
  file = "h3k4me3/h3k4me3_ll_20_chip_1_count.bed",
  header = F, sep = "\t")
chip.signal.20.1[,4] <- chip.signal.20.1[,4]/mapped.reads.k4.20c.1

chip.signal.20.2 <- read.table(
  file = "h3k4me3/h3k4me3_ll_20_chip_2_count.bed",
  header = F, sep = "\t")
chip.signal.20.2[,4] <- chip.signal.20.2[,4]/mapped.reads.k4.20c.2

chip.signal.20.3 <- read.table(
  file = "h3k4me3/h3k4me3_ll_20_chip_3_count.bed",
  header = F, sep = "\t")
chip.signal.20.3[,4] <- chip.signal.20.3[,4]/mapped.reads.k4.20c.3


chip.signal.26.1 <- read.table(
  file = "h3k4me3/h3k4me3_ll_26_chip_1_count.bed",
  header = F, sep = "\t")
chip.signal.26.1[,4] <- chip.signal.26.1[,4]/mapped.reads.k4.26c.1

```

```

chip.signal.26.2 <- read.table(
  file = "h3k4me3/h3k4me3_11_26_chip_2_count.bed",
  header = F, sep = "\t")
chip.signal.26.2[,4] <- chip.signal.26.2[,4]/mapped.reads.k4.26c.2

chip.signal.26.3 <- read.table(
  file = "h3k4me3/h3k4me3_11_26_chip_3_count.bed",
  header = F, sep = "\t")
chip.signal.26.3[,4] <- chip.signal.26.3[,4]/mapped.reads.k4.26c.3

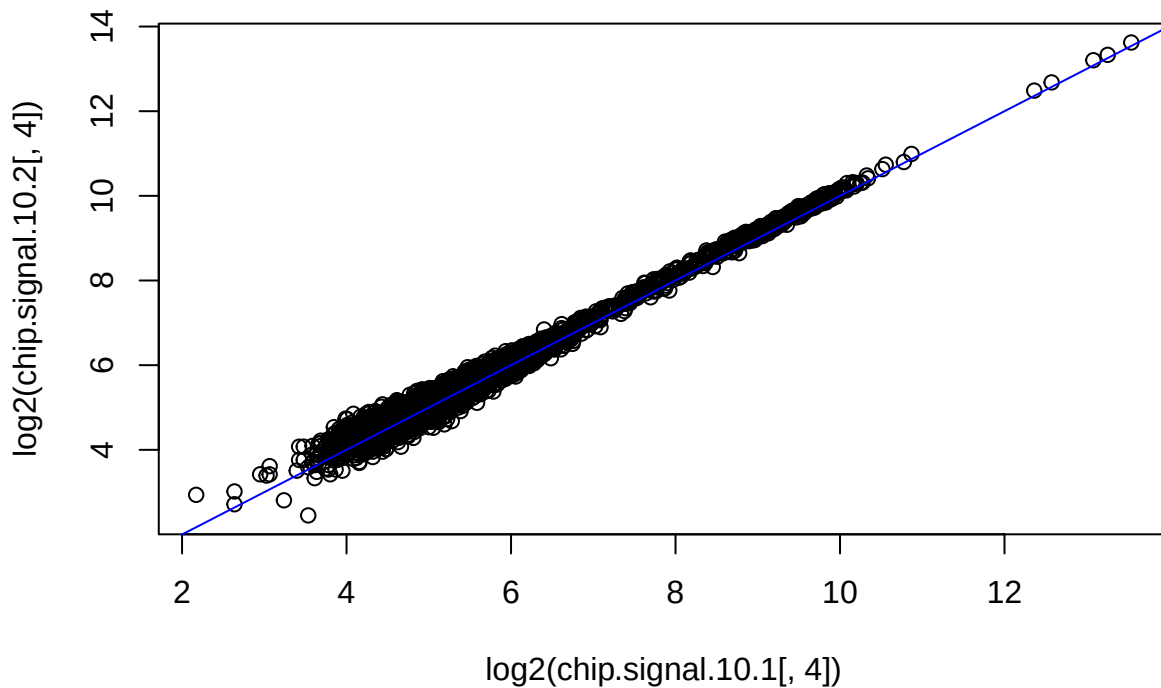
```

Scatter plots between replicates to check similarity.

```

plot(log2(chip.signal.10.1[,4]), log2(chip.signal.10.2[,4]))
lines(x=c(0,20), y=c(0,20), col="blue")

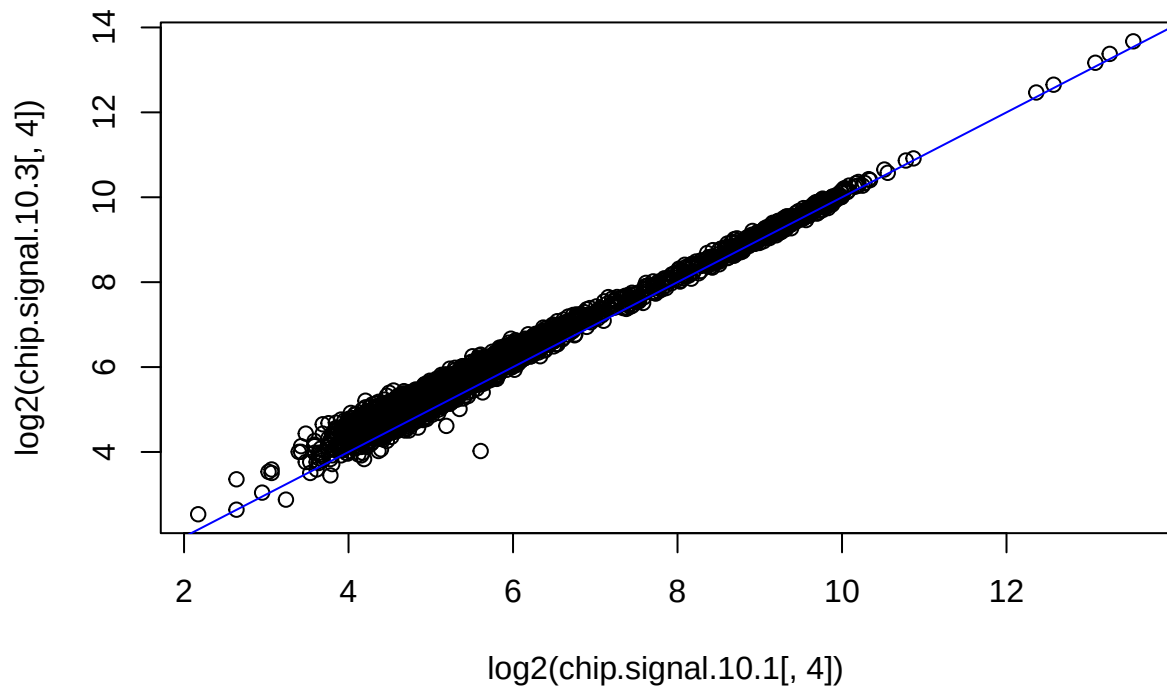
```



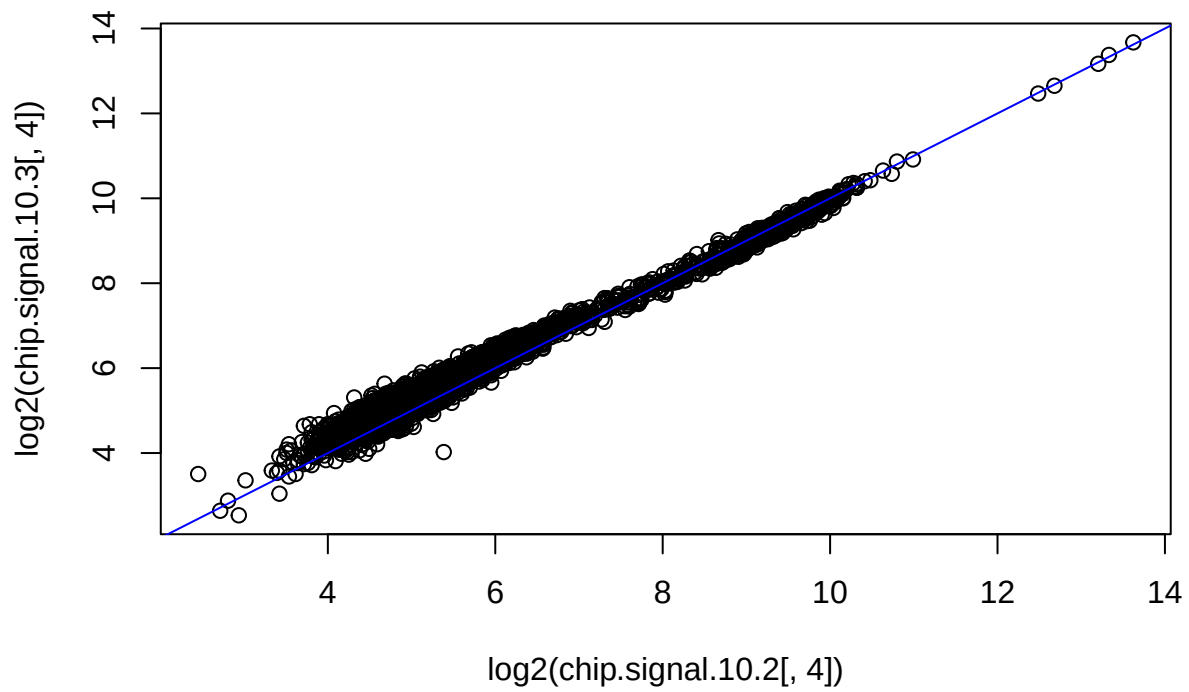
```

plot(log2(chip.signal.10.1[,4]), log2(chip.signal.10.3[,4]))
lines(x=c(0,20), y=c(0,20), col="blue")

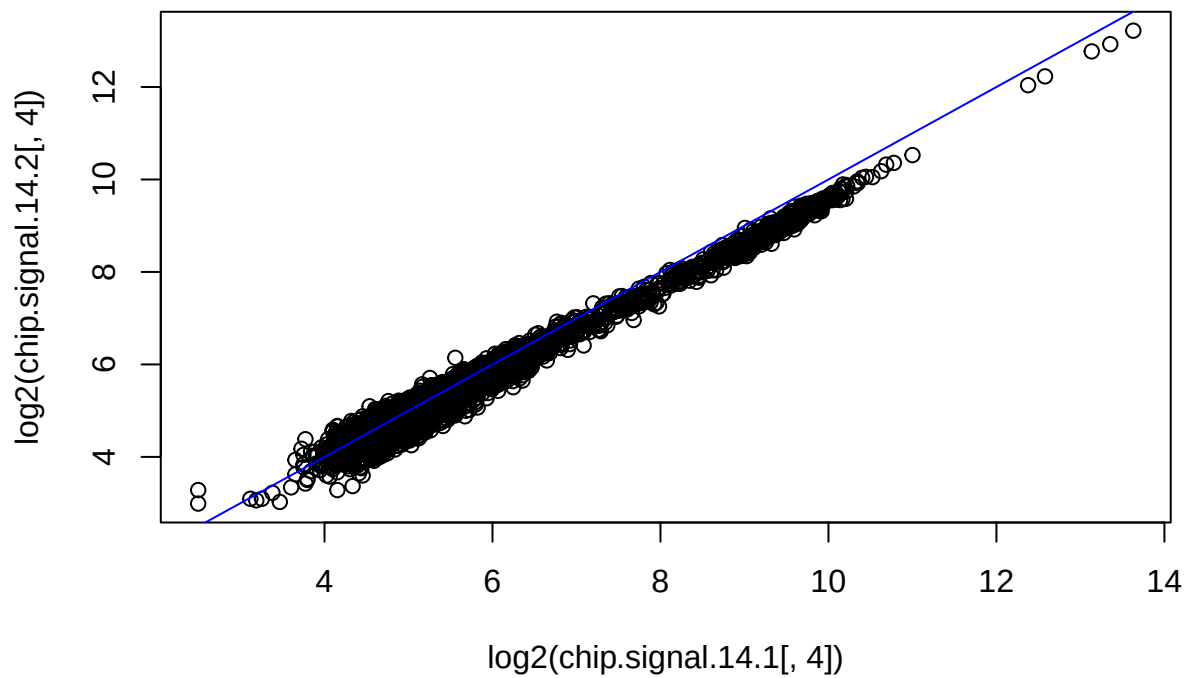
```



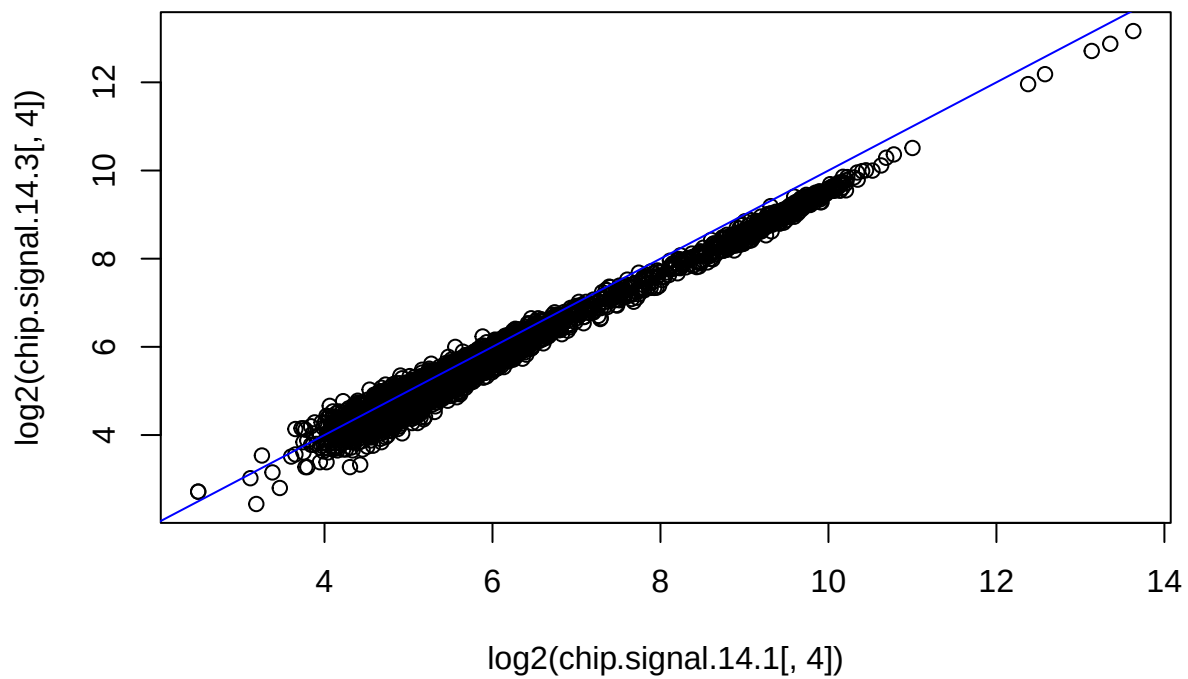
```
plot(log2(chip.signal.10.2[,4]),log2(chip.signal.10.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



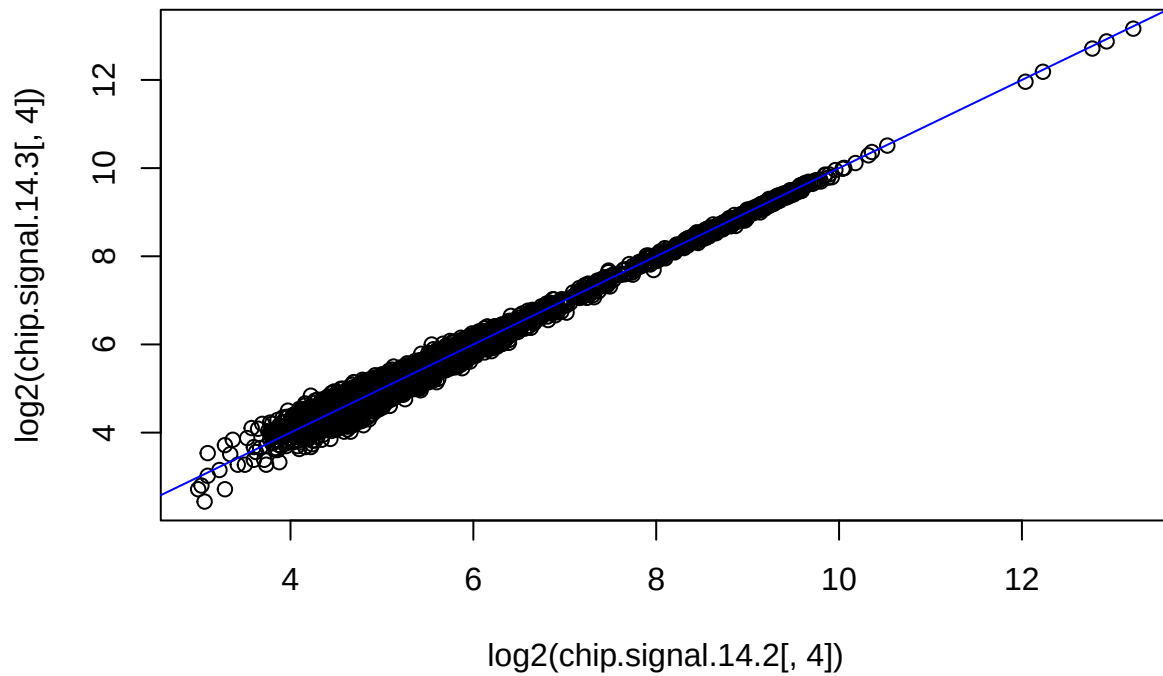
```
plot(log2(chip.signal.14.1[,4]),log2(chip.signal.14.2[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



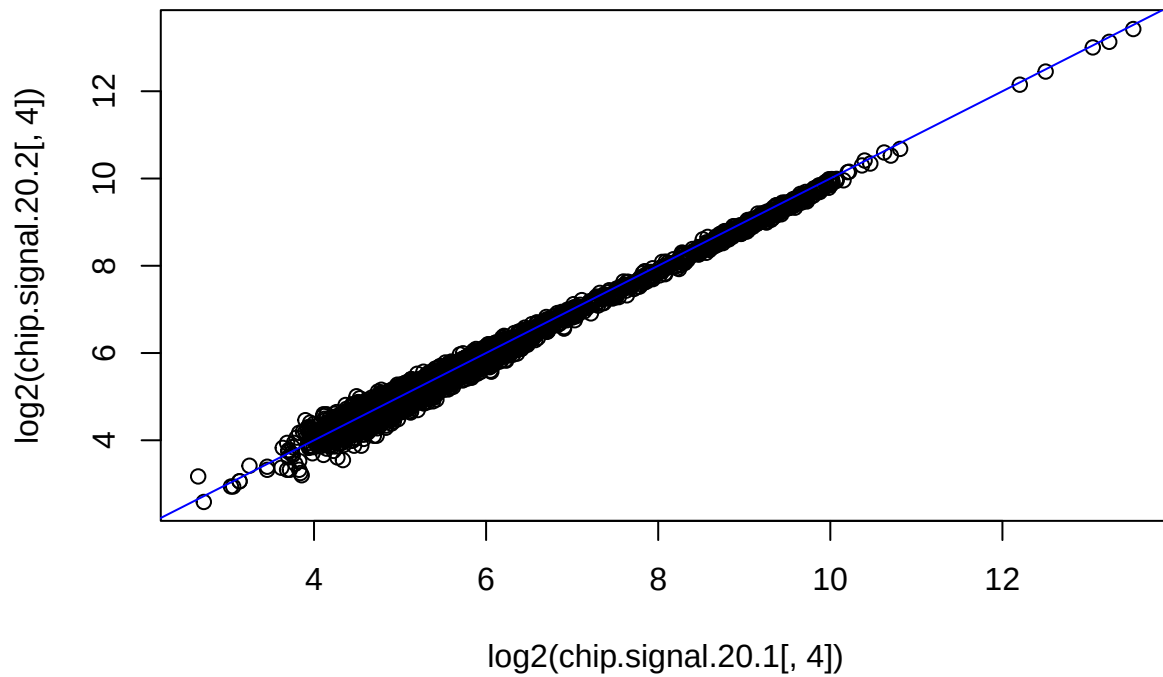
```
plot(log2(chip.signal.14.1[,4]),log2(chip.signal.14.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



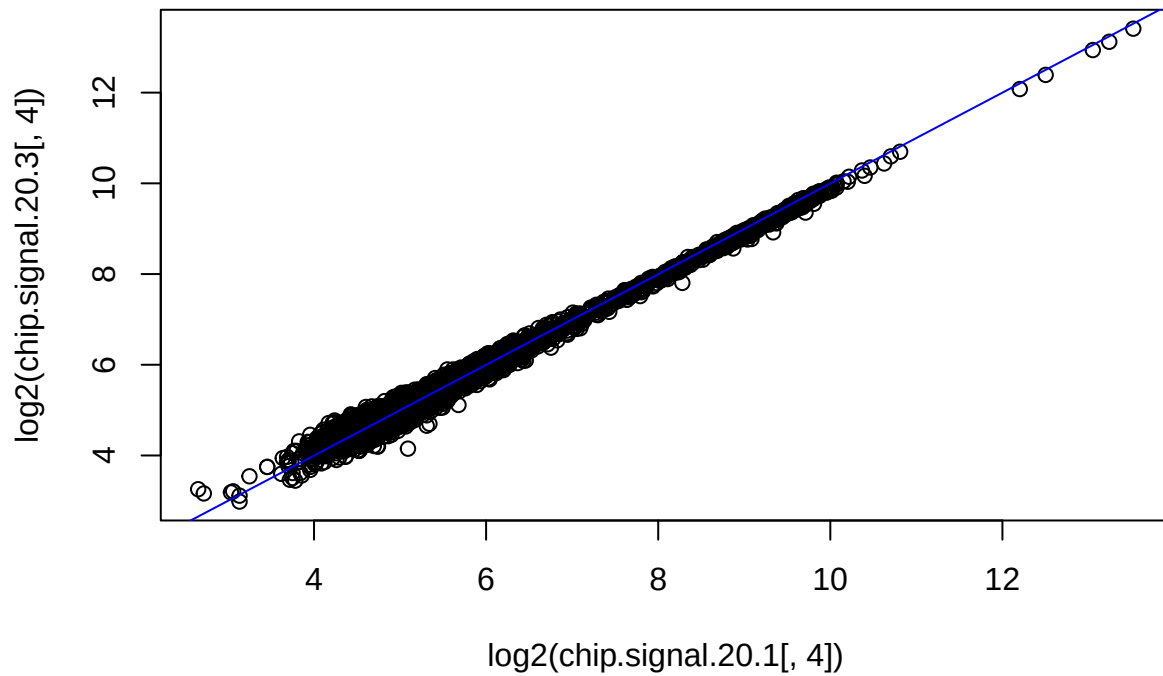
```
plot(log2(chip.signal.14.2[,4]),log2(chip.signal.14.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



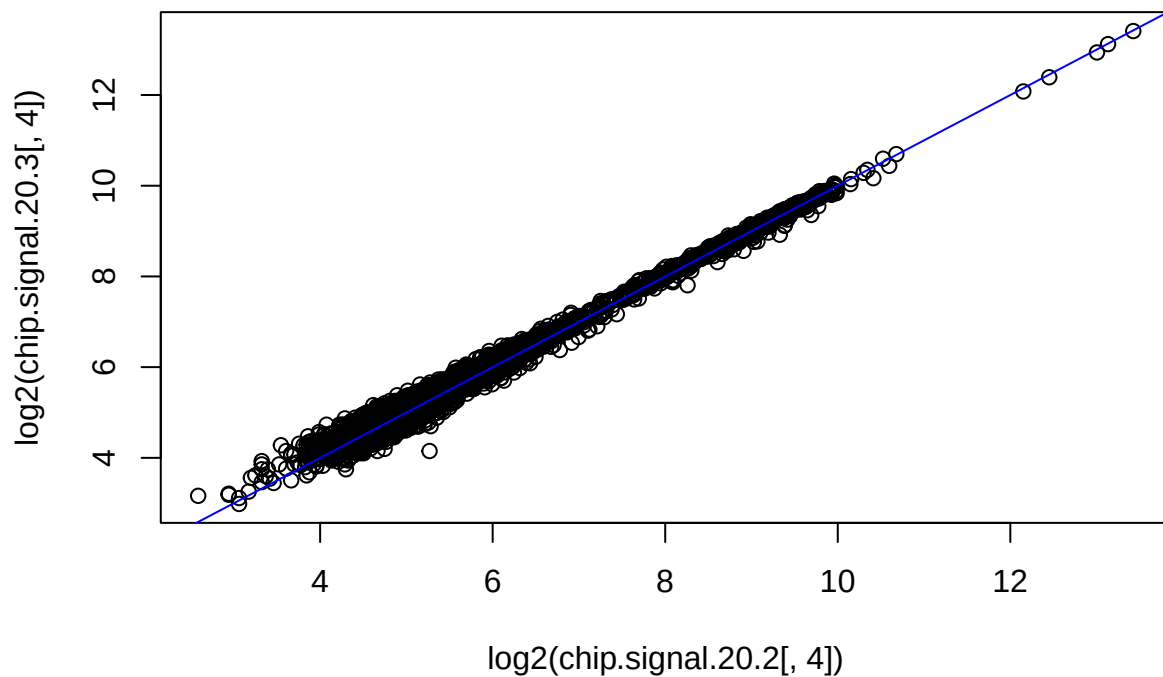
```
plot(log2(chip.signal.20.1[,4]),log2(chip.signal.20.2[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



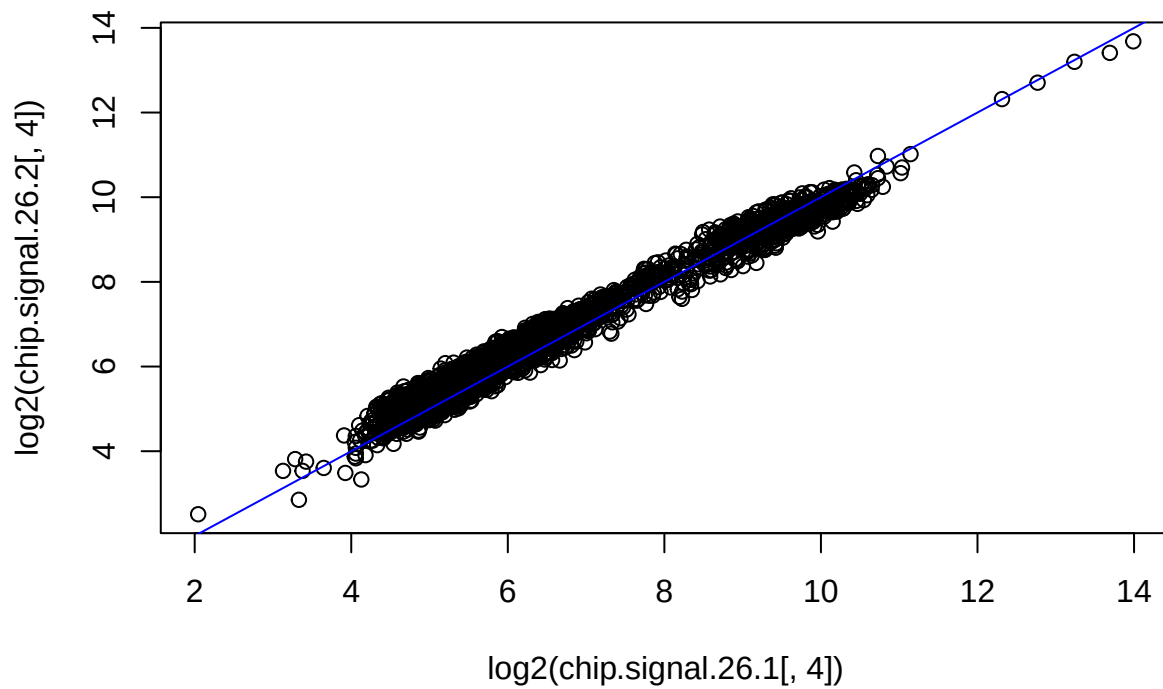
```
plot(log2(chip.signal.20.1[,4]),log2(chip.signal.20.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



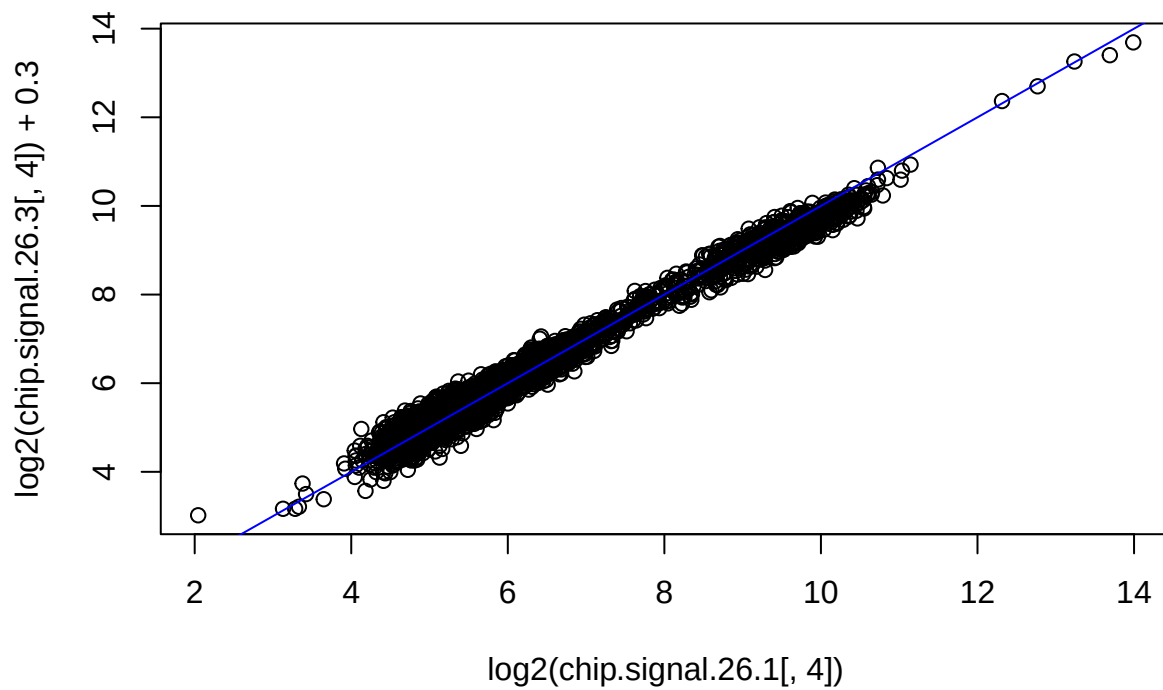
```
plot(log2(chip.signal.20.2[,4]),log2(chip.signal.20.3[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



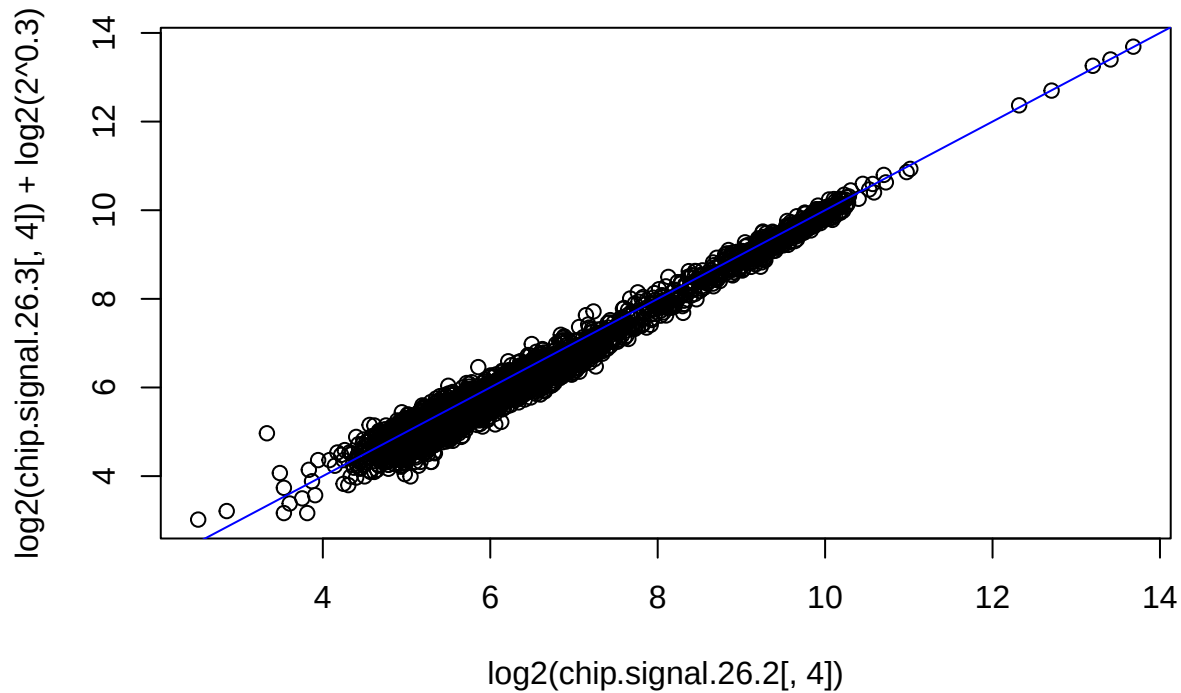
```
plot(log2(chip.signal.26.1[,4]),log2(chip.signal.26.2[,4]))
lines(x=c(0,20),y=c(0,20),col="blue")
```



```
plot(log2(chip.signal.26.1[,4]),log2(chip.signal.26.3[,4])+0.3)
lines(x=c(0,20),y=c(0,20),col="blue")
```



```
plot(log2(chip.signal.26.2[,4]),log2(chip.signal.26.3[,4])+log2(2^0.3))
lines(x=c(0,20),y=c(0,20),col="blue")
```



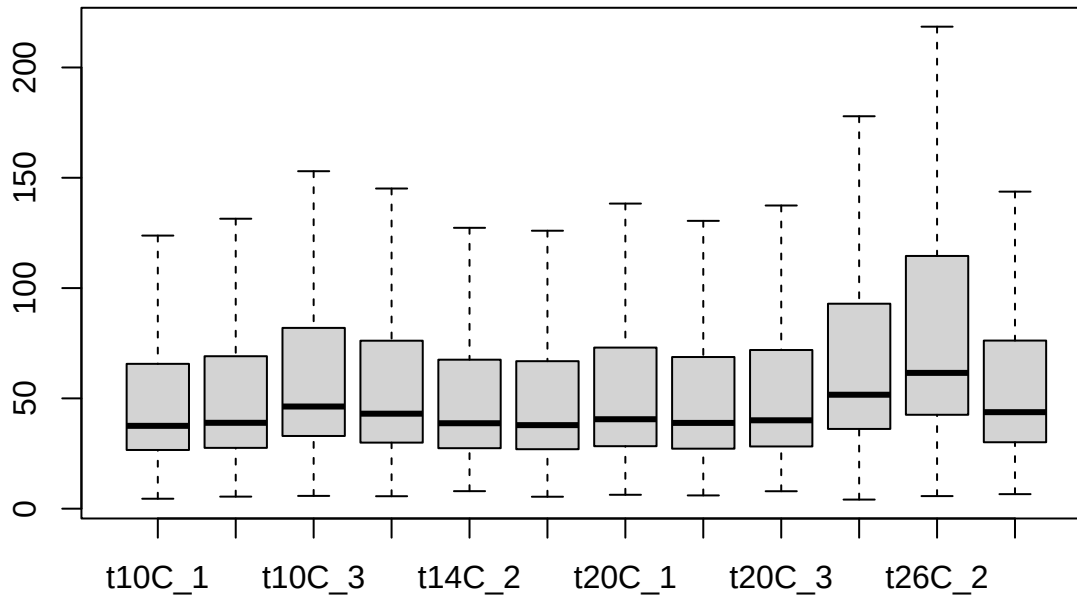
We generate the ChIP signal matrix.

```
chip.signal <- matrix(data = c(chip.signal.10.1[,4],
                                chip.signal.10.2[,4],
                                chip.signal.10.3[,4],
                                chip.signal.14.1[,4],
                                chip.signal.14.2[,4],
                                chip.signal.14.3[,4],
                                chip.signal.20.1[,4],
                                chip.signal.20.2[,4],
                                chip.signal.20.3[,4],
                                chip.signal.26.1[,4],
                                chip.signal.26.2[,4],
                                chip.signal.26.3[,4]), ncol=12)

colnames(chip.signal) <- c("t10C_1", "t10C_2", "t10C_3",
                           "t14C_1", "t14C_2", "t14C_3",
                           "t20C_1", "t20C_2", "t20C_3",
                           "t26C_1", "t26C_2", "t26C_3")

rownames(chip.signal) <- apply(X = chip.signal.10.1[,1:3],
                              MARGIN = 1,
                              FUN =
                                function(x)
                                {
                                  paste(x[1], paste(x[2], x[3], sep="-"), sep=":")
                                }
                              )

boxplot(chip.signal, outline=F)
```



```
head(chip.signal)
```

```
##           t10C_1    t10C_2    t10C_3    t14C_1    t14C_2    t14C_3
## 1:1456-2448   87.32684  83.76103 103.76310  86.99484  91.36784  92.17986
## 1:9308-10501  97.41126  82.88624 102.42710 105.86903  85.21045  89.47437
## 1:11126-11452 31.96978  30.83631  34.95881  37.31449  26.21860  32.46586
## 1:12564-13452 499.28637 511.97017 551.77033 530.64681 388.90921 387.46462
## 1:16087-16450 27.89309  30.39891  37.40816  32.32476  29.79386  27.05489
## 1:20482-20649 17.80867  15.09011  18.70408  19.95892  14.69831  13.14094
##           t20C_1    t20C_2    t20C_3    t26C_1    t26C_2    t26C_3
## 1:1456-2448   84.73770  82.42282  82.51980  84.98543 117.60658 104.20001
## 1:9308-10501 102.59262  93.60162  97.44201 116.83434 145.53577 101.14223
## 1:11126-11452 32.19740  26.02826  30.14286  35.47936  40.08883  29.87224
## 1:12564-13452 467.44770 435.80647 447.96463 683.67894 537.87435 481.48403
## 1:16087-16450 31.02658  30.86684  32.23197  36.96454  39.70884  29.87224
## 1:20482-20649 13.61072  15.35000  17.16054  19.30737  24.88928  18.11151
```

We perform median normalization:

```
apply(X = chip.signal,MARGIN = 2,FUN = median)
```

```
##    t10C_1    t10C_2    t10C_3    t14C_1    t14C_2    t14C_3    t20C_1    t20C_2
## 37.54839 38.92810 46.31486 43.06353 38.73202 37.87684 40.53945 38.87554
##    t20C_3    t26C_1    t26C_2    t26C_3
## 40.06612 51.65134 61.55821 43.74989
```

```
apply(X = chip.signal,MARGIN = 2,FUN = median)[1:3]
```

```
##    t10C_1    t10C_2    t10C_3
## 37.54839 38.92810 46.31486
```

```
apply(X = chip.signal,MARGIN = 2,FUN = median)[1:3][3]/apply(X = chip.signal,MARGIN = 2,FUN = median)[1
```

```
##    t10C_3
## 1.233471
```

```
apply(X = chip.signal,MARGIN = 2,FUN = median)[4:6]
```

```

##      t14C_1      t14C_2      t14C_3
## 43.06353 38.73202 37.87684
apply(X = chip.signal,MARGIN = 2,FUN = median)[4:6][1]/apply(X = chip.signal,MARGIN = 2,FUN = median)[4

##      t14C_1
## 1.111833
apply(X = chip.signal,MARGIN = 2,FUN = median)[4:6][3]/apply(X = chip.signal,MARGIN = 2,FUN = median)[4

##      t14C_3
## 0.9779206
apply(X = chip.signal,MARGIN = 2,FUN = median)[7:9]

##      t20C_1      t20C_2      t20C_3
## 40.53945 38.87554 40.06612
apply(X = chip.signal,MARGIN = 2,FUN = median)[7:9][1]/apply(X = chip.signal,MARGIN = 2,FUN = median)[7

##      t20C_1
## 1.042801
apply(X = chip.signal,MARGIN = 2,FUN = median)[7:9][1]/apply(X = chip.signal,MARGIN = 2,FUN = median)[7

##      t20C_1
## 1.011814
apply(X = chip.signal,MARGIN = 2,FUN = median)[10:12]

##      t26C_1      t26C_2      t26C_3
## 51.65134 61.55821 43.74989
apply(X = chip.signal,MARGIN = 2,FUN = median)[10:12][1]/apply(X = chip.signal,MARGIN = 2,FUN = median)

##      t26C_1
## 1.180605
apply(X = chip.signal,MARGIN = 2,FUN = median)[10:12][2]/apply(X = chip.signal,MARGIN = 2,FUN = median)

##      t26C_2
## 1.407048
chip.signal <- matrix(data = c(chip.signal.10.1[,4],
                                chip.signal.10.2[,4],
                                chip.signal.10.3[,4]/1.2,
                                chip.signal.14.1[,4]/1.1,
                                chip.signal.14.2[,4],
                                chip.signal.14.3[,4]/0.97,
                                chip.signal.20.1[,4],
                                chip.signal.20.2[,4]*1.04,
                                chip.signal.20.3[,4]*1.01,
                                chip.signal.26.1[,4]/1.2,
                                chip.signal.26.2[,4]/1.4,
                                chip.signal.26.3[,4]),ncol=12)
colnames(chip.signal) <- c("t10C_1","t10C_2","t10C_3",
                           "t14C_1","t14C_2","t14C_3",
                           "t20C_1","t20C_2","t20C_3",
                           "t26C_1","t26C_2","t26C_3")
rownames(chip.signal) <- apply(X = chip.signal.10.1[,1:3],

```

```

MARGIN = 1,
FUN =
  function(x)
  {
    paste(x[1],paste(x[2],x[3],sep="-"),sep=":")
  }
)
head(chip.signal)

```

```

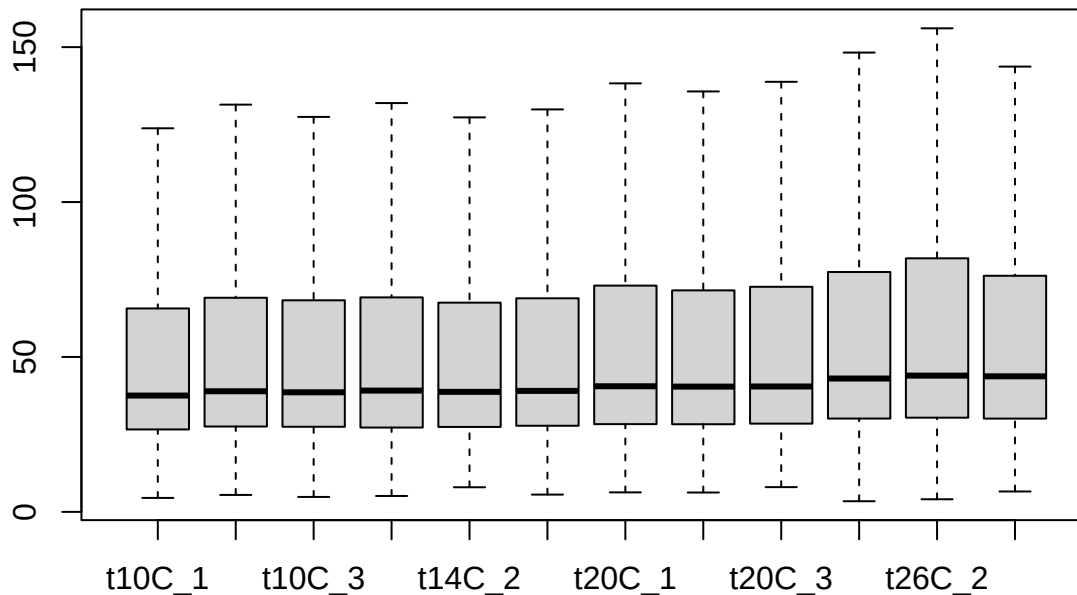
##           t10C_1    t10C_2    t10C_3    t14C_1    t14C_2    t14C_3
## 1:1456-2448   87.32684  83.76103  86.46925  79.08621  91.36784  95.03079
## 1:9308-10501  97.41126  82.88624  85.35592  96.24457  85.21045  92.24162
## 1:11126-11452 31.96978  30.83631  29.13235  33.92227  26.21860  33.46996
## 1:12564-13452 499.28637 511.97017 459.80861 482.40619 388.90921 399.44806
## 1:16087-16450 27.89309  30.39891  31.17346  29.38615  29.79386  27.89164
## 1:20482-20649 17.80867  15.09011  15.58673  18.14447  14.69831  13.54737
##           t20C_1    t20C_2    t20C_3    t26C_1    t26C_2    t26C_3
## 1:1456-2448   84.73770  85.71973  83.34500  70.82119  84.00470 104.20001
## 1:9308-10501 102.59262  97.34568  98.41643  97.36195 103.95412 101.14223
## 1:11126-11452 32.19740  27.06939  30.44428  29.56613  28.63488  29.87224
## 1:12564-13452 467.44770 453.23873 452.44427 569.73245 384.19597 481.48403
## 1:16087-16450 31.02658  32.10152  32.55428  30.80378  28.36346  29.87224
## 1:20482-20649 13.61072  15.96400  17.33214  16.08948  17.77805  18.11151

```

```

boxplot(chip.signal,outline=F)

```



Principal component analysis to analyze the underlying data structure.

```

library(FactoMineR)
library(factoextra)

pca.temp.chip.signal <- data.frame(colnames(chip.signal),
                                   t(chip.signal))

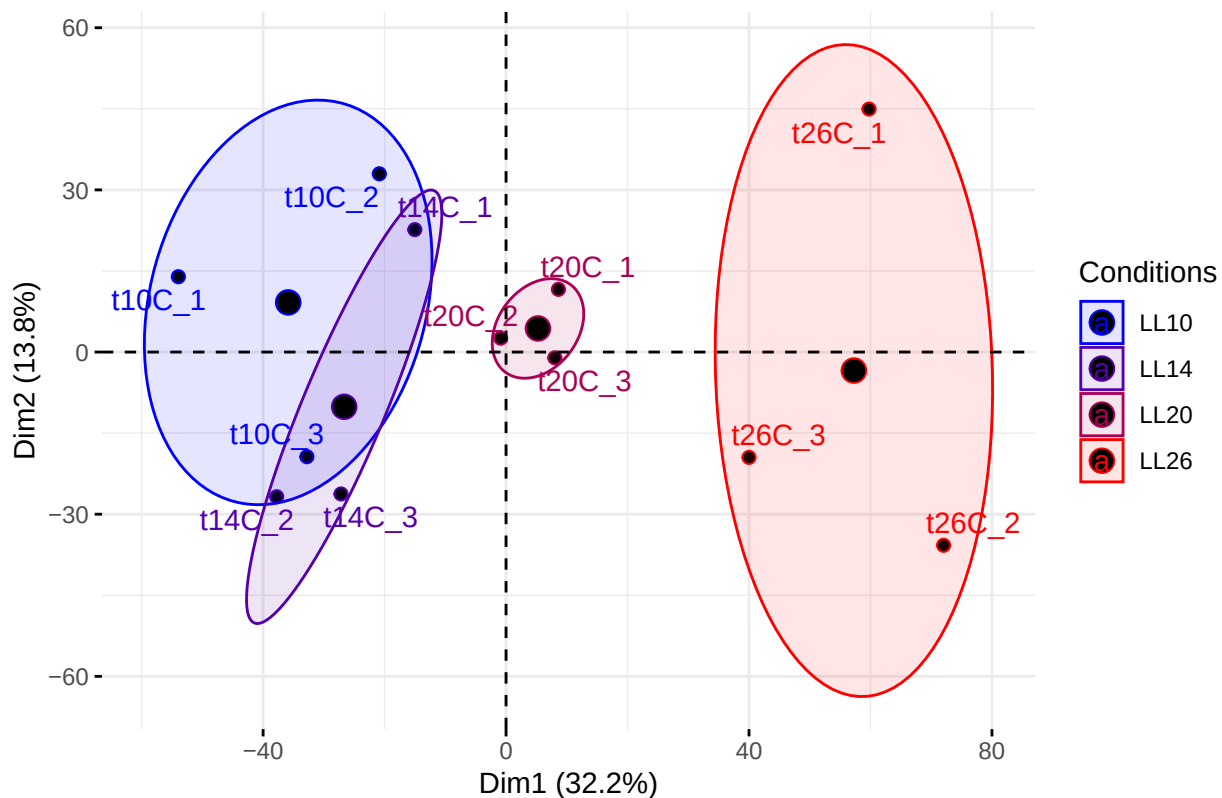
colnames(pca.temp.chip.signal)[1] <- "Sample"
res.pca <- PCA(pca.temp.chip.signal,

```

```

graph = FALSE,
scale.unit = TRUE,
quali.sup = 1 )
#jpeg(filename = "images/h3k4me3_chip_signal_pca.jpg",res = 160,width = 750,height = 750)
fviz_pca_ind(res.pca, col.ind = c(rep("LL10",3),rep("LL14",3),
                                rep("LL20",3),rep("LL26",3)),
palette = c(colorRampPalette(c("blue", "red"))(4)),
pointsize=2, pointshape=21,fill="black",
repel = TRUE,
addEllipses = TRUE,ellipse.type = "confidence",
legend.title="Conditions",
title="",
show_legend=TRUE,show_guide=TRUE)

```



```

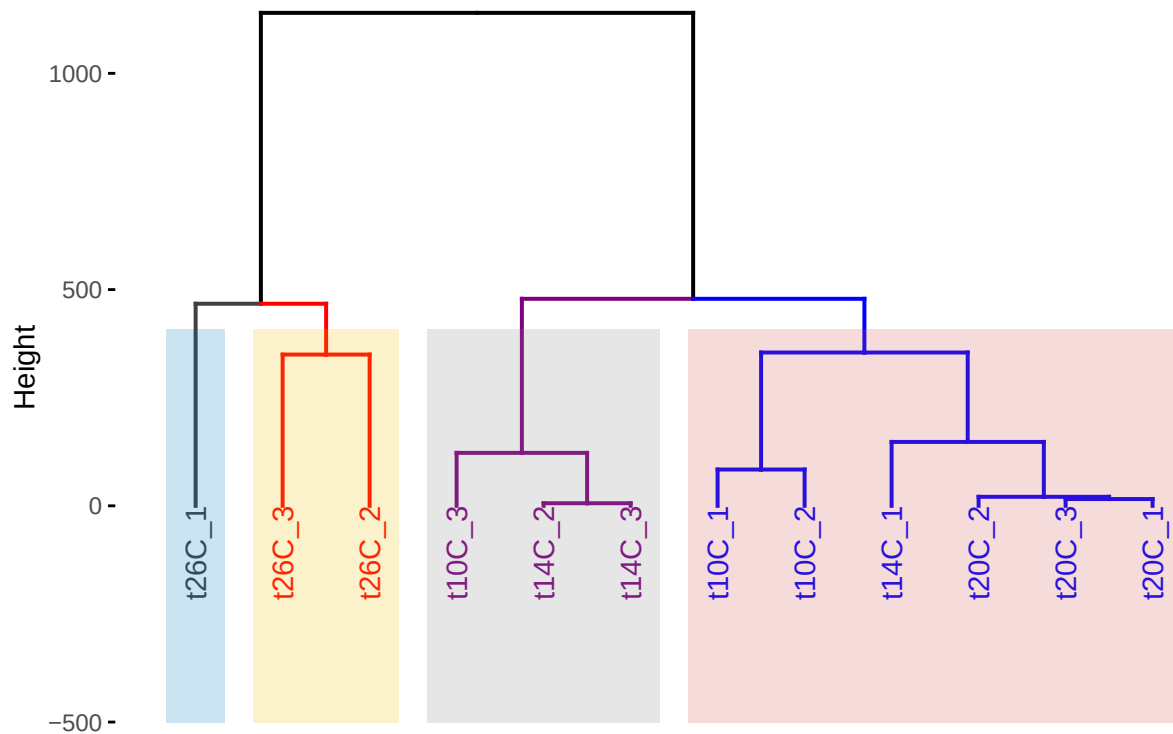
#dev.off()

res.hcpc <- HCPC(res.pca, graph=FALSE,nb.clust = 4)

fviz_dend(res.hcpc,k=4,
cex = 0.75,
palette = c("gray25",rev(colorRampPalette(c("blue", "red"))(3))),# "jco",
rect = TRUE, rect_fill = TRUE, # Add rectangle around groups
rect_border = "jco",
type="rectangle",
labels_track_height = 500
)

```

Cluster Dendrogram



We determine the H3K4me3 peaks that significantly change their ChIP signal as a response to temperature and the corresponding differentially H3k4me3 marked genes.

```
log2.chip.signal <- log2(chip.signal)

library(limma)
limma.experimental.design <- model.matrix(~ -1+factor(c(1,1,1,2,2,2,3,3,3,4,4,4)))
colnames(limma.experimental.design) <- c("LL10", "LL14", "LL20", "LL26")

linear.fit <- lmFit(log2.chip.signal, limma.experimental.design)

contrast.matrix <- makeContrasts(LL26-LL10,
                                LL20-LL10,
                                LL14-LL10,
                                levels = c("LL10", "LL14", "LL20", "LL26"))

contrast.linear.fit <- contrasts.fit(linear.fit, contrast.matrix)
contrast.results <- eBayes(contrast.linear.fit)

t26.t10 <- topTable(contrast.results,
                    number=nrow(log2.chip.signal),
                    coef = 1, sort.by = "logFC")

write.table(x = t26.t10, file = "tables/k4_diff_peaks_26c_10c.tsv",
            quote = F, sep = "\t")

t26.t10.logfc <- t26.t10$logFC
t26.t10.adj.p.val <- t26.t10$adj.P.Val
```

```

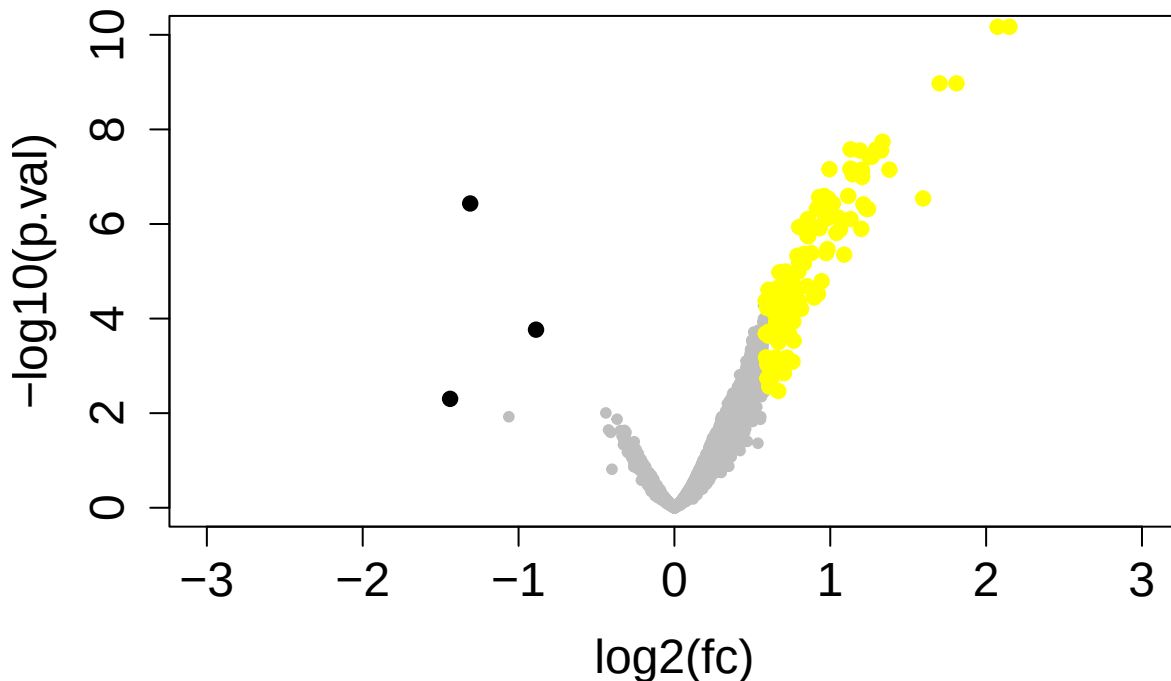
t26.t10.peak.ids <- rownames(t26.t10)
names(t26.t10.logfc) <- t26.t10.peak.ids
names(t26.t10.adj.p.val) <- t26.t10.peak.ids

t26.t10.higher.peaks <- t26.t10.peak.ids[t26.t10.logfc > log2(1.5) &
                                         t26.t10.adj.p.val < 0.01]

t26.t10.lower.peaks <- t26.t10.peak.ids[t26.t10.logfc < -log2(1.5) &
                                         t26.t10.adj.p.val < 0.01]

#jpeg(filename = "images/h3k4me3_chip_signal_volcano_26_10.jpg",res = 160,width = 750,height = 750)
plot(x = t26.t10.logfc,y = -log10(t26.t10.adj.p.val),
     pch=20,xlim=c(-3,3),ylim=c(0,10),cex.axis=1.5,col="grey",
     xlab="log2(fc)",ylab="-log10(p.val)",cex.lab=1.5)
points(x = t26.t10.logfc[t26.t10.higher.peaks],y = -log10(t26.t10.adj.p.val)[t26.t10.higher.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="yellow")
points(x = t26.t10.logfc[t26.t10.lower.peaks],y = -log10(t26.t10.adj.p.val)[t26.t10.lower.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="black")

```



```

#dev.off()

t26.t10.higher.peaks.genes <- unique(unlist(
  strsplit(x = subset(k4.consensus.peak.annotation,
                     rownames(k4.consensus.peak.annotation) %in%
                     t26.t10.higher.peaks)$target.genes,split=",")
))

t26.t10.lower.peaks.genes <- unique(unlist(
  strsplit(x = subset(k4.consensus.peak.annotation,
                     rownames(k4.consensus.peak.annotation) %in%
                     t26.t10.lower.peaks)$target.genes,split=",")
))

```

```

t20.t10 <- topTable(contrast.results,
                    number=nrow(log2.chip.signal),
                    coef = 2, sort.by = "logFC")

write.table(x = t20.t10,file = "tables/k4_diff_peaks_20c_10c.tsv",
            quote = F,sep = "\t")

t20.t10.logfc <- t20.t10$logFC
t20.t10.adj.p.val <- t20.t10$adj.P.Val
t20.t10.peak.ids <- rownames(t20.t10)
names(t20.t10.logfc) <- t20.t10.peak.ids
names(t20.t10.adj.p.val) <- t20.t10.peak.ids

t20.t10.higher.peaks <- t20.t10.peak.ids[t20.t10.logfc > log2(1.5) &
                                         t20.t10.adj.p.val < 0.01]

t20.t10.lower.peaks <- t20.t10.peak.ids[t20.t10.logfc < -log2(1.5) &
                                         t20.t10.adj.p.val < 0.01]

jpeg(filename = "images/h3k4me3_chip_signal_volcano_20_10.jpg",res = 160,width = 750,height = 750)
plot(x = t20.t10.logfc,y = -log10(t20.t10.adj.p.val),
     pch=20,xlim=c(-3,3),ylim=c(0,10),cex.axis=1.5,col="grey",
     xlab="log2(fc)",ylab="-log10(p.val)",cex.lab=1.5)
points(x = t20.t10.logfc[t20.t10.higher.peaks],y = -log10(t20.t10.adj.p.val)[t20.t10.higher.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="yellow")
points(x = t20.t10.logfc[t20.t10.lower.peaks],y = -log10(t20.t10.adj.p.val)[t20.t10.lower.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="black")
dev.off()

## pdf
## 2

t20.t10.higher.peaks.genes <- unique(unlist(
  strsplit(x = subset(k4.consensus.peak.annotation,
                     rownames(k4.consensus.peak.annotation) %in%
                     t20.t10.higher.peaks)$target.genes,split=",")
))

t20.t10.lower.peaks.genes <- unique(unlist(
  strsplit(x = subset(k4.consensus.peak.annotation,
                     rownames(k4.consensus.peak.annotation) %in%
                     t20.t10.lower.peaks)$target.genes,split=",")
))

t14.t10 <- topTable(contrast.results,
                    number=nrow(log2.chip.signal),
                    coef = 3, sort.by = "logFC")

write.table(x = t14.t10,file = "tables/k4_diff_peaks_14c_10c.tsv",
            quote = F,sep = "\t")

t14.t10.logfc <- t14.t10$logFC
t14.t10.adj.p.val <- t14.t10$adj.P.Val
t14.t10.peak.ids <- rownames(t14.t10)

```

```

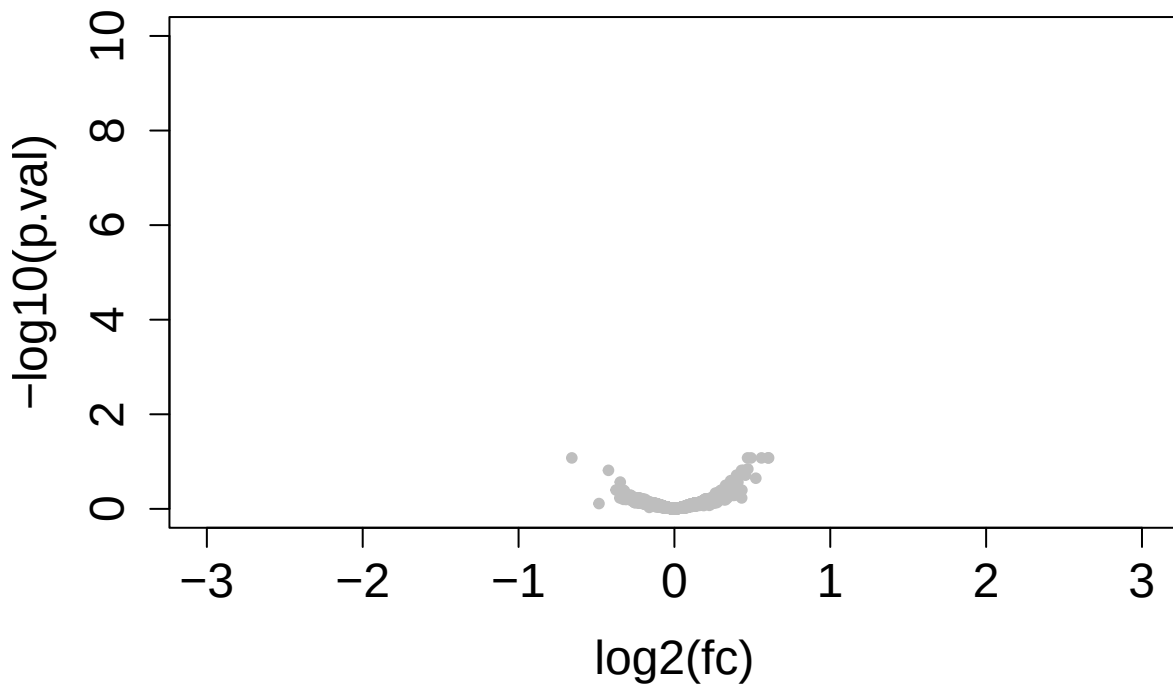
names(t14.t10.logfc) <- t14.t10.peak.ids
names(t14.t10.adj.p.val) <- t14.t10.peak.ids

t14.t10.higher.peaks <- t14.t10.peak.ids[t14.t10.logfc > log2(1.5) &
                                         t14.t10.adj.p.val < 0.01]

t14.t10.lower.peaks <- t14.t10.peak.ids[t14.t10.logfc < -log2(1.5) &
                                         t14.t10.adj.p.val < 0.01]

#jpeg(filename = "images/h3k4me3_chip_signal_volcano_14_10.jpg",res = 160,width = 750,height = 750)
plot(x = t14.t10.logfc,y = -log10(t14.t10.adj.p.val),
     pch=20,xlim=c(-3,3),ylim=c(0,10),cex.axis=1.5,col="grey",
     xlab="log2(fc)",ylab="-log10(p.val)",cex.lab=1.5)
points(x = t14.t10.logfc[t14.t10.higher.peaks],y = -log10(t14.t10.adj.p.val)[t14.t10.higher.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="yellow")
points(x = t14.t10.logfc[t14.t10.lower.peaks],y = -log10(t14.t10.adj.p.val)[t14.t10.lower.peaks],
       pch=20,xlim=c(-3,3),ylim=c(0,8),cex.axis=1.5,cex=1.5,col="black")

```



```

#dev.off()

t14.t10.higher.peaks.genes <- unique(unlist(
  strsplit(x = subset(k4.consensus.peak.annotation,
                      rownames(k4.consensus.peak.annotation) %in%
                        t14.t10.higher.peaks)$target.genes,split=",")
))

t14.t10.lower.peaks.genes <- unique(unlist(
  strsplit(x = subset(k4.consensus.peak.annotation,
                      rownames(k4.consensus.peak.annotation) %in%
                        t14.t10.lower.peaks)$target.genes,split=",")
))

k4.temp.higher.peaks <- unique(c(t26.t10.higher.peaks,

```

```

        t20.t10.higher.peaks,
        t14.t10.higher.peaks))

length(k4.temp.higher.peaks)

## [1] 128

length(k4.temp.higher.peaks)/nrow(k4.consensus.peak.annotation)

## [1] 0.02859696

k4.temp.lower.peaks <- unique(c(t26.t10.lower.peaks,
                               t20.t10.lower.peaks,
                               t14.t10.lower.peaks))

length(k4.temp.lower.peaks)/nrow(k4.consensus.peak.annotation)

## [1] 0.0006702413

k4.temp.higher.genes <- unique(c(t26.t10.higher.peaks.genes,
                                t20.t10.higher.peaks.genes,
                                t14.t10.higher.peaks.genes))

k4.temp.lower.genes <- unique(c(t26.t10.lower.peaks.genes,
                                t20.t10.lower.peaks.genes,
                                t14.t10.lower.peaks.genes))

length(k4.temp.higher.genes)/length(all.k4.genes)

## [1] 0.0269222

length(intersect(k4.temp.higher.genes,
                 c(k4.gene.body,k4.tss)))/length(c(k4.gene.body,k4.tss))

## [1] 0.03292308

length(k4.temp.lower.genes)/length(all.k4.genes)

## [1] 0.0004563085

length(intersect(k4.temp.lower.genes,c(
  k4.gene.body,k4.tss)))/length(c(k4.gene.body,k4.tss))

## [1] 0.0003076923

```

Only 2.8% of the H3K4me3 marked genes significantly increase more than 50% their H3K27me3 ChIP signal as a response to temperature and only two genes significantly decrease their H3K4me3 ChIP signal.

```

chip.signal.10C <- rowMeans(chip.signal[,paste("t10C",1:3,sep="_")])
chip.signal.14C <- rowMeans(chip.signal[,paste("t14C",1:3,sep="_")])
chip.signal.20C <- rowMeans(chip.signal[,paste("t20C",1:3,sep="_")])
chip.signal.26C <- rowMeans(chip.signal[,paste("t26C",1:3,sep="_")])

jpeg(filename = "images/h3k4me3_higher.jpg",res = 160,width = 750,height = 900)
boxplot(chip.signal.10C[k4.temp.higher.peaks],
        chip.signal.14C[k4.temp.higher.peaks],
        chip.signal.20C[k4.temp.higher.peaks],

```

```

chip.signal.26C[k4.temp.higher.peaks],
outline=F,
names=c("10°C","14°C","20°C","26°C"),
col = colorRampPalette(c("blue", "red"))(4),lwd=2,cex.axis=1.5)
dev.off()

## pdf
## 2

wilcox.test(x = chip.signal.10C[k4.temp.higher.peaks],
            y = chip.signal.14C[k4.temp.higher.peaks],
            alternative = "less",paired = T)$p.value

## [1] 1.033851e-19

wilcox.test(x = chip.signal.14C[k4.temp.higher.peaks],
            y = chip.signal.20C[k4.temp.higher.peaks],
            alternative = "less",paired = T)$p.value

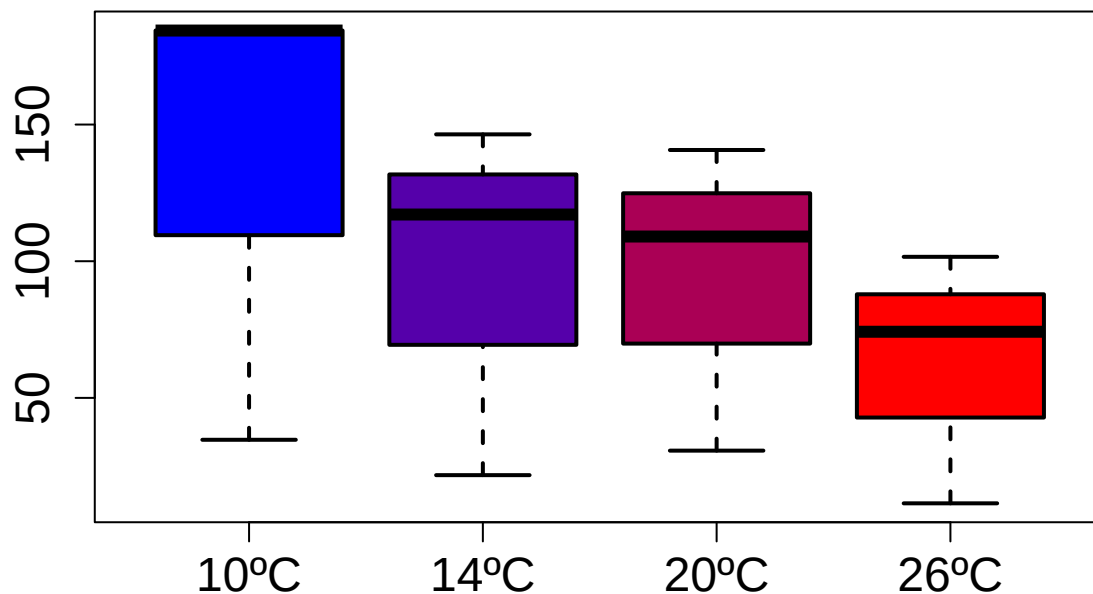
## [1] 5.543846e-23

wilcox.test(x = chip.signal.20C[k4.temp.higher.peaks],
            y = chip.signal.26C[k4.temp.higher.peaks],
            alternative = "less",paired = T)$p.value

## [1] 6.386458e-10

#jpeg(filename = "images/h3k27me3_lower.jpg",res = 160,width = 750,height = 900)
boxplot(chip.signal.10C[k4.temp.lower.peaks],
        chip.signal.14C[k4.temp.lower.peaks],
        chip.signal.20C[k4.temp.lower.peaks],
        chip.signal.26C[k4.temp.lower.peaks],
        outline=F,
        names=c("10°C","14°C","20°C","26°C"),
        col = colorRampPalette(c("blue", "red"))(4),lwd=2,cex.axis=1.5)

```



```

#dev.off()

```

We observe a temperature dependent gradual increase in the higher peaks. This increase is significant according to wilcox test with p values of 5.2×10^{-6} between 10°C and 14°C, 1.6×10^{-60} between 14°C and 20°C and 4.1×10^{-67} between 20°C and 24°C.

Similarly in the lower peaks, we observe a temperature dependent gradual decrease. This decrease is significant according to wilcox text with p values of 1.9×10^{-9} between 10°C and 14°C, 1.1×10^{-5} between 14°C and 20 and no significant difference between 20°C and 26°C.

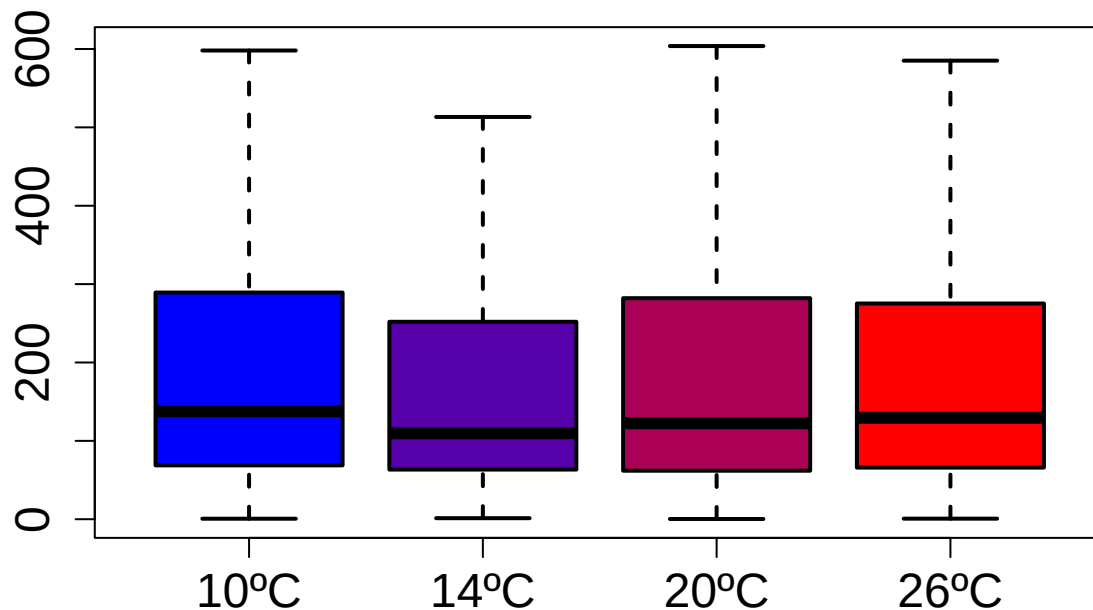
Next, we performed a differential gene expression analysis to integrate our ChIP-seq data with gene expression data.

We found 751 genes that significantly increase more than twice their expression as a response to temperature and 138 genes that significantly decrease more than half their expression as a response to temperature.

Next we study the expression level of H3K27me3 genes that significantly increase their ChIP signal as a response to temperature more than 50%.

```
#jpeg(filename = "images/gene_expression_h3k4me3_higher.jpg",res = 160,width = 750,height = 900)
#boxplot(gene.expression.10[intersect(k4.temp.higher.genes,c(k4.tss,k4.gene.body))],
#         gene.expression.14[intersect(k4.temp.higher.genes,c(k4.tss,k4.gene.body))],
#         gene.expression.20[intersect(k4.temp.higher.genes,c(k4.tss,k4.gene.body))],
#         gene.expression.26[intersect(k4.temp.higher.genes,c(k4.tss,k4.gene.body))],
#         outline=F,
#         names=c("10°C","14°C","20°C","26°C"),
#         col = colorRampPalette(c("blue", "red"))(4),lwd=2,cex.axis=1.5,range=4)

boxplot(gene.expression.10[k4.temp.higher.genes],
gene.expression.14[k4.temp.higher.genes],
gene.expression.20[k4.temp.higher.genes],
gene.expression.26[k4.temp.higher.genes],
outline=F,
names=c("10°C","14°C","20°C","26°C"),
col = colorRampPalette(c("blue", "red"))(4),lwd=2,cex.axis=1.5)#,range=4)
```



```
#dev.off()
```

```

wilcox.test(x = gene.expression.10[k4.temp.higher.genes],
            y = gene.expression.14[k4.temp.higher.genes],
            alternative = "two.sided")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.10[k4.temp.higher.genes] and gene.expression.14[k4.temp.higher.genes]
## W = 7393, p-value = 0.4117
## alternative hypothesis: true location shift is not equal to 0

wilcox.test(x = gene.expression.14[intersect(k4.temp.higher.genes,c(k4.tss,k4.gene.body))],
            y = gene.expression.20[intersect(k4.temp.higher.genes,c(k4.tss,k4.gene.body))],
            alternative = "two.sided")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.14[intersect(k4.temp.higher.genes, c(k4.tss, k4.gene.body))] and gene.expression.20[intersect(k4.temp.higher.genes, c(k4.tss, k4.gene.body))]
## W = 5537, p-value = 0.6797
## alternative hypothesis: true location shift is not equal to 0

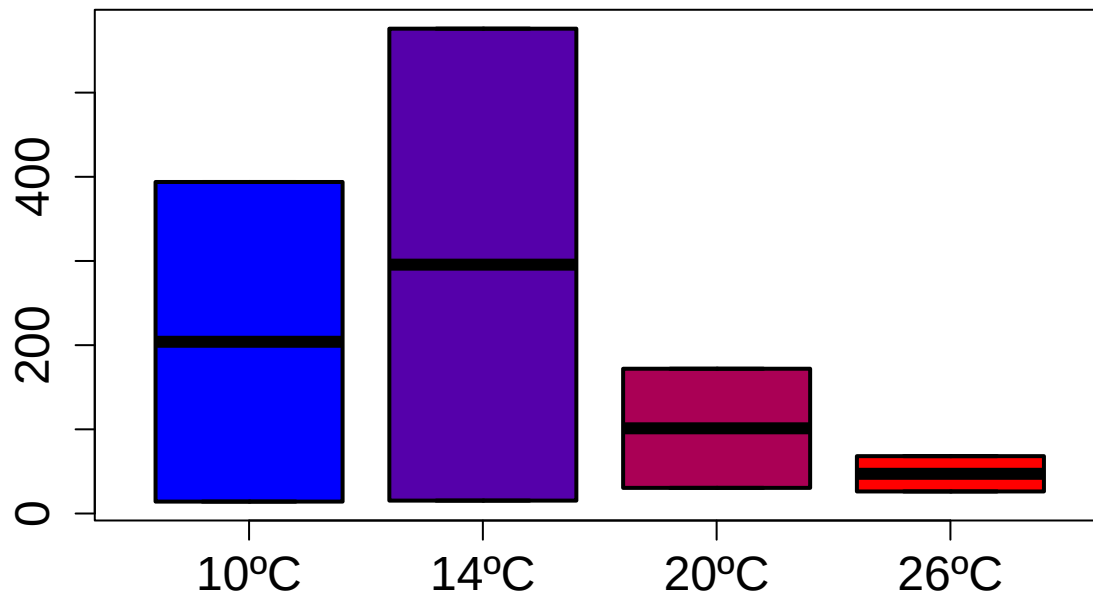
wilcox.test(x = gene.expression.20[intersect(k4.temp.higher.genes,c(k4.tss,k4.gene.body))],
            y = gene.expression.26[intersect(k4.temp.higher.genes,c(k27.tss,k4.gene.body))],
            alternative = "two.sided")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[intersect(k4.temp.higher.genes, c(k4.tss, k4.gene.body))] and gene.expression.26[intersect(k4.temp.higher.genes, c(k27.tss, k4.gene.body))]
## W = 941, p-value = 0.8798
## alternative hypothesis: true location shift is not equal to 0

We observe that globally genes that significantly increase their H3K27me3 ChIP signal more than 50% exhibit significant decrease in their expression levels that significant with p-values of  $1.157 \times 10^{-14}$  for 10°C vs 14°C,  $9.102 \times 10^{-5}$  for 14°C vs 20°C and  $2.203 \times 10^{-3}$ .

#jpeg(filename = "images/gene_expression_h3k27me3_lower.jpg",res = 160,width = 750,height = 900)
boxplot(gene.expression.10[k4.temp.lower.genes],
        gene.expression.14[k4.temp.lower.genes],
        gene.expression.20[k4.temp.lower.genes],
        gene.expression.26[k4.temp.lower.genes],
        outline=F,
        names=c("10°C","14°C","20°C","26°C"),
        col = colorRampPalette(c("blue", "red"))(4),lwd=2,cex.axis=1.5)

```



```
#dev.off()
```

```
wilcox.test(x = gene.expression.10[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            y = gene.expression.14[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            alternative = "less",paired = T)
```

```
## Warning in wilcox.test.default(x =
## gene.expression.10[intersect(k27.temp.lower.genes, : cannot compute exact
## p-value with zeroes
```

```
##
```

```
## Wilcoxon signed rank test with continuity correction
```

```
##
```

```
## data: gene.expression.10[intersect(k27.temp.lower.genes, c(k27.tss, k27.gene.body))] and gene.expre
```

```
## V = 714, p-value = 0.9444
```

```
## alternative hypothesis: true location shift is less than 0
```

```
wilcox.test(x = gene.expression.14[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            y = gene.expression.20[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            alternative = "less",paired = T)
```

```
## Warning in wilcox.test.default(x =
```

```
## gene.expression.14[intersect(k27.temp.lower.genes, : cannot compute exact
```

```
## p-value with zeroes
```

```
##
```

```
## Wilcoxon signed rank test with continuity correction
```

```
##
```

```
## data: gene.expression.14[intersect(k27.temp.lower.genes, c(k27.tss, k27.gene.body))] and gene.expre
```

```
## V = 329, p-value = 0.004009
```

```
## alternative hypothesis: true location shift is less than 0
```

```
wilcox.test(x = gene.expression.20[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            y = gene.expression.26[intersect(k27.temp.lower.genes,c(k27.tss,k27.gene.body))],
            alternative = "less",paired = T)
```

```
##
```

```
## Wilcoxon signed rank exact test
```

```
##
## data: gene.expression.20[intersect(k27.temp.lower.genes, c(k27.tss, k27.gene.body))] and gene.expre
## V = 495, p-value = 0.1235
## alternative hypothesis: true location shift is less than 0
```

Similarly we observe that globally genes that significantly decrease their H3K27me3 ChIP signal more than 50% exhibit increase in their expression levels. However, this increase is only significant for 20°C vs 14°C with a p value of 4×10^{-3} .

```
library(VennDiagram)
temp.activated.k4.higher.genes <- intersect(temp.activated,k4.temp.higher.genes)

intersect(all.k27.genes,k4.temp.higher.genes)
```

```
## [1] "ostta01g02810" "ostta13g01250" "ostta02g00700" "ostta02g00770"
## [5] "ostta02g00800" "ostta02g01247" "ostta02g01720" "ostta02g02580"
## [9] "ostta03g01920" "ostta03g03260" "ostta04g02090" "ostta05g03090"
## [13] "ostta06g02290" "ostta08g00680" "ostta09g01040"
```

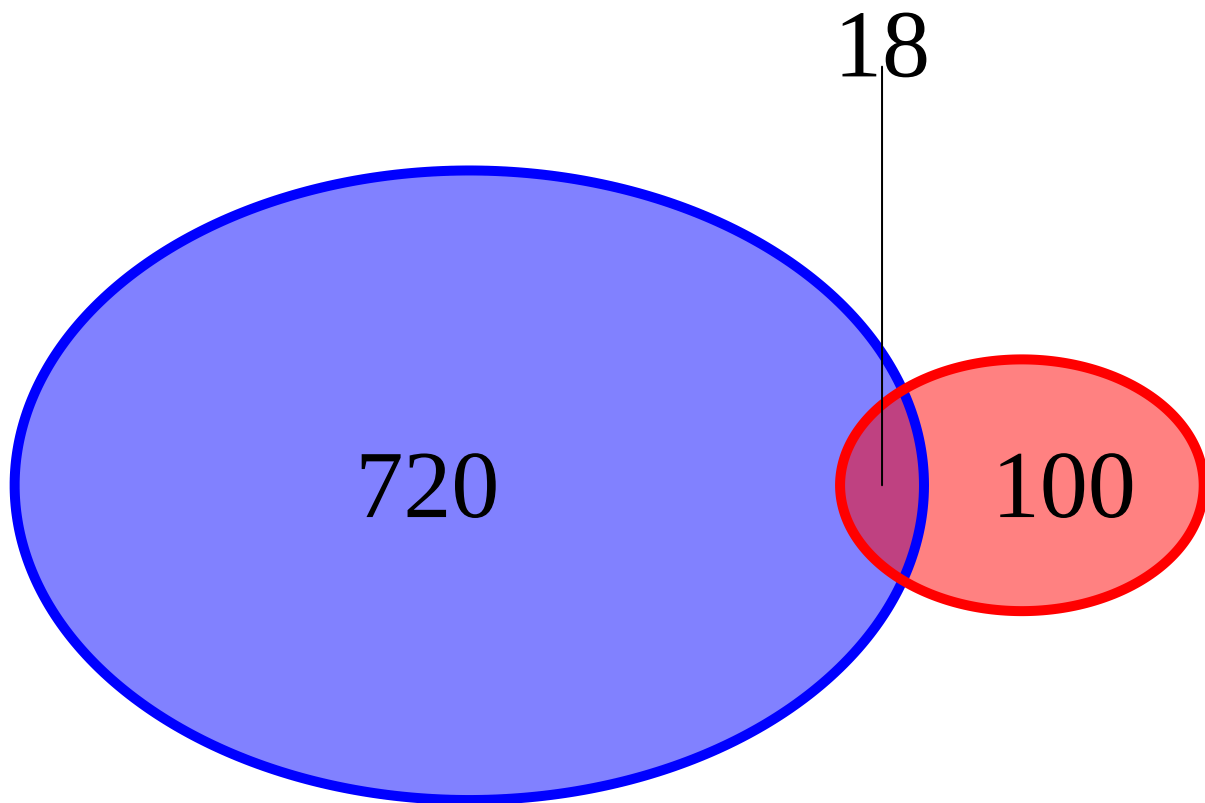
```
intersect(temp.activated,all.k4.genes)
```

```
## [1] "ostta14g00630" "ostta06g04170" "ostta01g03450" "ostta04g00520"
## [5] "ostta16g01795" "ostta05g00320" "ostta11g02030" "ostta03g01020"
## [9] "ostta04g02550" "ostta13g01250" "ostta17g02180" "ostta01g00140"
## [13] "ostta10g02015" "ostta11g01360" "ostta05g03520" "ostta04g02510"
## [17] "ostta03g02080" "ostta16g02200" "ostta03g02500" "ostta15g00710"
## [21] "ostta14g02420" "ostta05g04900" "ostta06g03390" "ostta08g02030"
## [25] "ostta03g02710" "ostta04g00040" "ostta03g05970" "ostta15g02900"
## [29] "ostta11g00580" "ostta05g01950" "ostta18g00350" "ostta03g04550"
## [33] "ostta03g04080" "ostta14g01890" "ostta01g04940" "ostta05g01890"
## [37] "ostta11g02500" "ostta17g01590" "ostta04g04190" "ostta02g04800"
## [41] "ostta06g01860" "ostta01g00290" "ostta02g02460" "ostta12g01400"
## [45] "ostta06g00250" "ostta03g03790" "ostta01g05650" "ostta20g00220"
## [49] "ostta12g01240" "ostta03g05860" "ostta16g00770" "ostta01g04730"
## [53] "ostta16g01220" "ostta07g02150" "ostta05g03870" "ostta11g02820"
## [57] "ostta04g04700" "ostta11g00500" "ostta03g02050" "ostta09g00080"
## [61] "ostta01g03030" "ostta12g00420" "ostta02g04750" "ostta17g02110"
## [65] "ostta16g00420" "ostta14g00065" "ostta10g02780" "ostta07g00230"
## [69] "ostta04g00300" "ostta06g02930" "ostta04g04170" "ostta17g02150"
## [73] "ostta04g01440" "ostta07g04180" "ostta11g03180" "ostta12g02350"
## [77] "ostta15g00780" "ostta18g01090" "ostta03g01635" "ostta05g02810"
## [81] "ostta18g00620" "ostta03g03300" "ostta01g03400" "ostta05g04560"
## [85] "ostta06g04120" "ostta09g01500" "ostta01g03820" "ostta05g01630"
## [89] "ostta10g01620" "ostta01g01200" "ostta06g02400" "ostta17g01810"
## [93] "ostta04g05060" "ostta05g00800" "ostta10g02680" "ostta16g02610"
## [97] "ostta18g00800" "ostta15g00045" "ostta18g00770" "ostta17g01870"
## [101] "ostta06g01150" "ostta17g01460" "ostta12g00390" "ostta02g00920"
## [105] "ostta01g02070" "ostta01g03670" "ostta12g01450" "ostta15g01210"
## [109] "ostta10g01990" "ostta07g02530" "ostta20g00170" "ostta02g04360"
## [113] "ostta13g00230" "ostta08g04240" "ostta05g02310" "ostta07g01205"
## [117] "ostta05g01460" "ostta08g02960" "ostta03g02160" "ostta05g01260"
## [121] "ostta06g01650" "ostta06g03220" "ostta03g04335" "ostta03g04520"
## [125] "ostta17g00570" "ostta08g01490" "ostta05g01100" "ostta10g00620"
## [129] "ostta14g02270" "ostta05g02740" "ostta17g00290" "ostta02g03260"
## [133] "ostta04g00130" "ostta08g01750" "ostta08g02710" "ostta14g00010"
## [137] "ostta07g03990" "ostta01g03010" "ostta03g01340" "ostta05g04130"
```

[141] "ostta15g02170" "ostta12g00970" "ostta16g02580" "ostta06g00030"
 ## [145] "ostta15g01735" "ostta05g04850" "ostta14g02170" "ostta16g00450"
 ## [149] "ostta10g00330" "ostta09g01010" "ostta12g02940" "ostta04g01640"
 ## [153] "ostta10g03200" "ostta11g02080" "ostta16g01530" "ostta03g03720"
 ## [157] "ostta13g01375" "ostta01g05910" "ostta14g01940" "ostta17g00090"
 ## [161] "ostta06g03830" "ostta04g04400" "ostta17g01700" "ostta02g01070"
 ## [165] "ostta08g00570" "ostta06g01260" "ostta02g02860" "ostta15g01320"
 ## [169] "ostta04g00540" "ostta05g03730" "ostta10g00350" "ostta02g03870"
 ## [173] "ostta15g00200" "ostta01g02160" "ostta02g05020" "ostta01g06210"
 ## [177] "ostta10g02030" "ostta06g03850" "ostta15g00540" "ostta01g02400"
 ## [181] "ostta03g01210" "ostta11g01940" "ostta16g00940" "ostta11g01170"
 ## [185] "ostta06g01570" "ostta15g00195" "ostta10g03540" "ostta06g02290"
 ## [189] "ostta06g03400" "ostta13g00885" "ostta04g05030" "ostta14g01105"
 ## [193] "ostta08g00840" "ostta08g02340" "ostta07g01270" "ostta04g01340"
 ## [197] "ostta06g02310" "ostta01g01740" "ostta06g00430" "ostta03g04510"
 ## [201] "ostta07g02560" "ostta03g05480" "ostta08g00970" "ostta16g01400"
 ## [205] "ostta01g01760" "ostta02g02700" "ostta20g00940" "ostta01g01020"
 ## [209] "ostta09g00730" "ostta08g00680" "ostta05g04160" "ostta09g02830"
 ## [213] "ostta02g00190" "ostta05g00060" "ostta17g02090" "ostta09g00670"
 ## [217] "ostta02g00410" "ostta12g01200" "ostta03g00370" "ostta03g01350"
 ## [221] "ostta06g00870" "ostta20g00240" "ostta09g00650" "ostta14g02160"
 ## [225] "ostta08g01395" "ostta12g01010" "ostta15g02290" "ostta05g04690"
 ## [229] "ostta09g01760" "ostta06g02580" "ostta08g00050" "ostta14g01930"
 ## [233] "ostta03g02460" "ostta11g02240" "ostta16g00070" "ostta05g00960"
 ## [237] "ostta15g02360" "ostta03g04880" "ostta10g01160" "ostta06g04180"
 ## [241] "ostta03g04960" "ostta01g02615" "ostta16g01730" "ostta04g02970"
 ## [245] "ostta04g02020" "ostta17g00795" "ostta11g00620" "ostta11g03270"
 ## [249] "ostta06g02070" "ostta02g05220" "ostta15g01620" "ostta14g02475"
 ## [253] "ostta09g02770" "ostta16g01770" "ostta12g02310" "ostta17g00820"
 ## [257] "ostta03g05070" "ostta02g05050" "ostta11g03230" "ostta12g02870"
 ## [261] "ostta15g02390" "ostta15g01370" "ostta13g02310" "ostta08g02270"
 ## [265] "ostta02g03040" "ostta13g01950" "ostta04g01480" "ostta09g03150"
 ## [269] "ostta09g03235" "ostta03g05410" "ostta09g01820" "ostta07g00470"
 ## [273] "ostta07g00910" "ostta16g01030" "ostta03g05920" "ostta15g00120"
 ## [277] "ostta15g01660" "ostta14g02840" "ostta10g01390" "ostta05g00075"
 ## [281] "ostta07g00430" "ostta08g04150" "ostta20g00930" "ostta07g00950"
 ## [285] "ostta11g02580" "ostta12g00640" "ostta11g02870" "ostta02g01960"
 ## [289] "ostta14g02250" "ostta07g03830" "ostta05g00010" "ostta06g01420"
 ## [293] "ostta03g04290" "ostta09g02780" "ostta02g00890" "ostta20g00270"
 ## [297] "ostta13g02550" "ostta02g04930" "ostta07g04510" "ostta03g03530"
 ## [301] "ostta10g03220" "ostta06g04500" "ostta05g02210" "ostta02g02000"
 ## [305] "ostta06g03500" "ostta07g02080" "ostta16g01700" "ostta02g04560"
 ## [309] "ostta17g02140" "ostta01g00180" "ostta18g00630" "ostta05g00950"
 ## [313] "ostta10g00590" "ostta10g00900" "ostta09g00950" "ostta05g03690"
 ## [317] "ostta04g00770" "ostta14g01355" "ostta11g01480" "ostta17g02075"
 ## [321] "ostta17g00230" "ostta11g01090" "ostta07g01530" "ostta02g01245"
 ## [325] "ostta02g01247" "ostta07g03020" "ostta08g03540" "ostta15g02530"
 ## [329] "ostta20g00570" "ostta02g01010" "ostta12g00890" "ostta02g04290"
 ## [333] "ostta13g00070" "ostta08g01000" "ostta14g00330" "ostta11g02550"
 ## [337] "ostta17g01350" "ostta04g04930" "ostta17g02220" "ostta04g00970"
 ## [341] "ostta07g03520" "ostta16g00880" "ostta07g00170" "ostta18g00760"
 ## [345] "ostta16g00500" "ostta12g02450" "ostta15g02590" "ostta07g02900"
 ## [349] "ostta01g05760" "ostta07g04680" "ostta05g00220" "ostta08g03240"
 ## [353] "ostta17g01310" "ostta03g05700" "ostta05g03140" "ostta09g01150"

```
## [357] "ostta11g02640" "ostta03g05760" "ostta06g01470" "ostta03g03960"
## [361] "ostta10g01730" "ostta03g02770" "ostta08g03410" "ostta05g04180"
## [365] "ostta06g00230" "ostta05g00450" "ostta03g01170" "ostta14g01060"
## [369] "ostta15g01180" "ostta09g04010" "ostta02g05115" "ostta03g05690"
## [373] "ostta11g02320" "ostta04g04870" "ostta04g00640" "ostta03g05440"
## [377] "ostta10g00545" "ostta14g02795" "ostta15g01160" "ostta01g02570"
## [381] "ostta17g00240" "ostta07g03260" "ostta08g03615" "ostta08g00640"
## [385] "ostta02g05090" "ostta03g03240" "ostta16g01465" "ostta17g02305"
## [389] "ostta01g03150" "ostta07g04450" "ostta12g01480" "ostta05g02880"
## [393] "ostta07g03650" "ostta03g04380" "ostta15g00290" "ostta09g03180"
## [397] "ostta04g04150" "ostta02g04770" "ostta18g00590" "ostta06g00530"
## [401] "ostta14g02775" "ostta03g01070" "ostta15g00930" "ostta03g02690"
## [405] "ostta15g02870" "ostta09g00230" "ostta05g00500" "ostta04g00550"
## [409] "ostta01g00560" "ostta02g05480" "ostta02g05225" "ostta06g00120"
## [413] "ostta18g00640" "ostta14g02310" "ostta01g03190" "ostta16g01090"
## [417] "ostta17g00640" "ostta04g01170" "ostta07g04710" "ostta08g01430"
## [421] "ostta11g00780" "ostta14g00490" "ostta17g01270" "ostta02g02790"
## [425] "ostta16g00530" "ostta06g01920" "ostta05g00110" "ostta05g04910"
## [429] "ostta06g02170" "ostta07g02870" "ostta14g02370" "ostta04g01270"
## [433] "ostta07g01990" "ostta18g01340" "ostta11g00210" "ostta01g01910"
## [437] "ostta01g06000" "ostta16g00870" "ostta07g04360" "ostta07g04020"
## [441] "ostta06g02130" "ostta11g02380" "ostta15g02010" "ostta01g03060"
## [445] "ostta04g01940" "ostta07g03170" "ostta14g02820" "ostta08g03760"
## [449] "ostta15g02200" "ostta20g00130" "ostta12g01410" "ostta10g01230"
## [453] "ostta15g02070" "ostta04g01700" "ostta09g02180" "ostta18g01450"
## [457] "ostta01g02895" "ostta17g02290" "ostta14g01070" "ostta09g01220"
## [461] "ostta07g01590" "ostta03g05530" "ostta06g04390" "ostta17g01250"
## [465] "ostta02g04650" "ostta05g03820" "ostta11g01890" "ostta04g00810"
## [469] "ostta02g04680" "ostta12g02920" "ostta02g03140" "ostta11g03140"
## [473] "ostta10g02210" "ostta12g01760" "ostta03g02470" "ostta01g05330"
## [477] "ostta03g05715" "ostta14g02760" "ostta04g01860" "ostta17g00400"
## [481] "ostta06g01030" "ostta04g03360" "ostta02g04390" "ostta02g05370"
```

```
#jpeg(filename = "images/venn_diagram_k4_increase_activated.jpg",res = 160,width = 1000,height = 1000)
grid.newpage()
draw.pairwise.venn(area1 = length(temp.activated),
                    area2 = length(k4.temp.higher.genes),
                    cross.area = length(temp.activated.k4.higher.genes),
                    lwd = 5,category = c("",""),
                    euler.d = T,
                    col = c("blue","red"),
                    fill = c(" blue","red"),
                    alpha = 0.5,cex = 3,cat.cex = 2)
```



```
## (polygon[GRID.polygon.1178], polygon[GRID.polygon.1179], polygon[GRID.polygon.1180], polygon[GRID.polygon.1181])
#dev.off()

overlap <- length(temp.activated.k4.higher.genes)
size.1 <- length(temp.activated)
size.2 <- length(k4.temp.higher.genes)
total.genes <- length(gene.ids)
phyper(overlap - 1, size.1, total.genes - size.1, size.2)

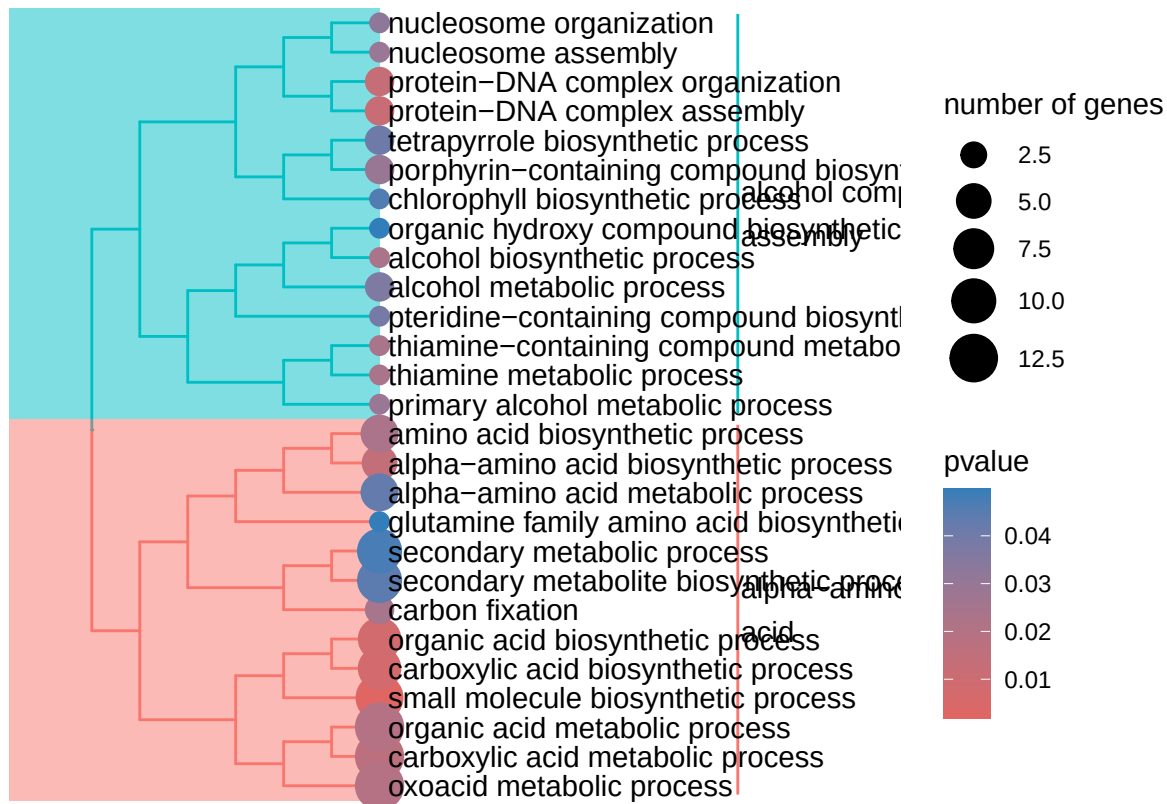
## [1] 0.9674412
```

We identified 134 genes that, in response to high temperature, exhibited a more than 50% significant increase in H3K27me3 ChIP signal and a greater than twofold reduction in expression levels. These genes represent 17.8% of the total set of genes that showed significant downregulation in response to temperature increase. The overlap between genes with a significant increase in H3K27me3 ChIP signal and those significantly downregulated is highly significant with a p-value of 1.15×10^{-40} . This suggests that H3K27me3 plays a key role in temperature response being involved in the regulation of nearly 18% of the differentially repressed genes.

Next we explore the biological processes affected by changes in H3K27me3 as a response to temperature.

```
temp.repressed.k27.higher.genes.enrichment <- enrichGO(gene = k4.temp.higher.genes,
                                                         pvalueCutoff = 1,
                                                         qvalueCutoff = 1,
                                                         OrgDb = org.Otauriv5.eg.db,
                                                         ont = "BP",
                                                         keyType = "GID")
k27.go.terms.filtered <- temp.repressed.k27.higher.genes.enrichment
k27.go.terms.filtered@result <- subset(k27.go.terms.filtered@result, pvalue < 0.05)
```

```
treeplot(x = pairwise_termsim(k27.go.terms.filtered), cluster.params = list(n = 2), color = "pvalue")
```



Our analysis identified cytoskeleton organization and locomotion as significantly enriched processes.

The following two functions are used to represent the ChIP signal and expression level for specific genes.

```
gene.profile.complete <- function(gene.name,
                                  promoter_length=1000,
                                  selected.bigwig.files,
                                  cols,
                                  selected.bed.files,
                                  txdb)
{
  genes.data.df <- as.data.frame(genes(txdb))
  cds.data <- as.data.frame(cds(txdb))
  exons.data <- as.data.frame(exons(txdb))
  target.gene.body <- genes.data.df[gene.name,]
  target.gene.chr <- as.character(target.gene.body$seqnames)
  target.gene.start <- target.gene.body$start
  target.gene.end <- target.gene.body$end

  target.gene.strand <- as.character(target.gene.body$strand)

  ## Extract cds annotation
  cds.data.target.gene <- subset(cds.data,
                                seqnames == target.gene.chr &
                                (start >= target.gene.start &
```

```

        end <= target.gene.end))

## Extract exons annotation
exons.data.target.gene <- subset(exons.data,
                                seqnames == target.gene.chr &
                                (start >= target.gene.start &
                                 end <= target.gene.end))

## Determine the genome range to plot including promoter, gene body and 5' UTR
## This depends on whether the gene is on the forward or reverse strand
range.to.plot <- target.gene.body

if(target.gene.strand == "+")
{
  range.to.plot$start <- range.to.plot$start - promoter_length
  range.to.plot$end <- range.to.plot$end + promoter_length
} else if (target.gene.strand == "-")
{
  range.to.plot$end <- range.to.plot$end + promoter_length
  range.to.plot$start <- range.to.plot$start - promoter_length
}

## Compute the length of the genome range to represent
current.length <- range.to.plot$end - range.to.plot$start

## Compute profile in gene
## Since ChIPpeakAnno needs more than one region to plot our region
## is duplicated
regions.plot <- GRanges(rbind(range.to.plot,range.to.plot))

## Import signal from the bigwig files
cvglists <- sapply(selected.bigwig.files, import,
                   format="BigWig",
                   which=regions.plot,
                   as="RleList")

## Compute signal in the region to plot
chip.signal <- featureAlignedSignal(cvglists, regions.plot,
                                   upstream=ceiling(current.length/2),
                                   downstream=ceiling(current.length/2),
                                   n.tile=current.length)

chip.signal[[1]][1:2,1:5]

chip.signal.mean <- matrix(nrow=length(selected.bigwig.files),ncol=dim(chip.signal[[1]])[2])

## Compute mean signal
for(i in 1:length(selected.bigwig.files))
{
  if(target.gene.strand == "+")

```

```

{
  chip.signal.mean[i,] <- colMeans(chip.signal[[i]],na.rm = TRUE)
} else if (target.gene.strand == "-")
{
  chip.signal.mean[i,] <- rev(colMeans(chip.signal[[i]],na.rm = TRUE))
}
}

## Normalization
#chip.signal.mean <- 20 * chip.signal.mean / max(chip.signal.mean)
upper.lim <- 1.1*max(chip.signal.mean)
## Colors to draw signal
line.colors <- cols[i]#"blue"
area.colors <- cols[i]#"lightblue"

gene.height <- -25
cord.x <- 1:current.length

## Exon width to plot
exon.width <- 2#1#2

## Cds width to plot
cds.width <- 3#1.5#3

## Width of the rectangle representing the peak reagon
peak.width <- 1#0.5#1

## Extract exons for target gene
exons.data.target.gene <- subset(exons.data, seqnames == target.gene.chr & (start >= target.gene.start & end <= target.gene.end))

## Transform exon coordinates to current range
min.pos <- min(exons.data.target.gene$start)

exons.data.target.gene$start <- exons.data.target.gene$start - min.pos + promoter_length
exons.data.target.gene$end <- exons.data.target.gene$end - min.pos + promoter_length

## Extract cds for target gene
cds.data.target.gene <- subset(cds.data, seqnames == target.gene.chr & (start >= target.gene.start & end <= target.gene.end))

cds.data.target.gene$start <- cds.data.target.gene$start - min.pos + promoter_length
cds.data.target.gene$end <- cds.data.target.gene$end - min.pos + promoter_length

plot(cord.x, rep(gene.height,length(cord.x)),type="l",col="black",lwd=3,ylab="",
      cex.lab=2,axes=FALSE,xlab="",main="",cex.main=2,
      ylim=c(-35,upper.lim),xlim=c(0,max(cord.x)))

## Represent exons
for(i in 1:nrow(exons.data.target.gene))
{
  # Determine start/end for each exon
  current.exon.start <- exons.data.target.gene$start[i]
  current.exon.end <- exons.data.target.gene$end[i]

```

```

## Determine coordinates for each exon polygon and represent it
exon.x <- c(current.exon.start,current.exon.end,current.exon.end,current.exon.start)
exon.y <- c(gene.height + exon.width, gene.height + exon.width, gene.height - exon.width, gene.height - exon.width)

polygon(x = exon.x, y = exon.y, col = "blue",border = "blue")
}

if(nrow(cds.data.target.gene) > 0)
{
  for(i in 1:nrow(cds.data.target.gene))
  {
    # Determine current cds start/end
    current.cds.start <- cds.data.target.gene$start[i]
    current.cds.end <- cds.data.target.gene$end[i]

    # Determine current cds coordinates for the polygon and represent it
    cds.x <- c(current.cds.start,current.cds.end,current.cds.end,current.cds.start)
    cds.y <- c(gene.height + cds.width, gene.height + cds.width, gene.height - cds.width, gene.height - cds.width)
    polygon(x = cds.x, y = cds.y, col = "blue",border = "blue")
  }
}

## Draw arrow to represent transcription direction
if(target.gene.strand == "+")
{
  lines(c(promoter_length,promoter_length,promoter_length+100),
        y=c(gene.height,gene.height+5,gene.height+5),lwd=3)
  lines(c(promoter_length+50,promoter_length+100),y=c(gene.height+6,gene.height+5),lwd=3)
  lines(c(promoter_length+50,promoter_length+100),y=c(gene.height+4,gene.height+5),lwd=3)
} else if (target.gene.strand == "-")
{
  lines(c(current.length - promoter_length, current.length - promoter_length, current.length - promoter_length - 100),
        y=c(gene.height,gene.height+5,gene.height+5),lwd=3)
  lines(c(current.length - promoter_length-50, current.length - promoter_length - 100),y=c(gene.height+6,gene.height+5),lwd=3)
  lines(c(current.length - promoter_length-50, current.length - promoter_length - 100),y=c(gene.height+4,gene.height+5),lwd=3)
}

## Draw promoter range
if(target.gene.strand == "+")
{
  axis(side = 1,labels = c(- promoter_length, - promoter_length / 2,"TSS"),at = c(1,promoter_length/2))
} else if(target.gene.strand == "-")
{
  axis(side = 1,labels = c("TSS",- promoter_length / 2,- promoter_length),at = c(current.length-promoter_length,
current.length - promoter_length / 2, - promoter_length))
}

## Draw gene name
text(x = current.length / 2, y = -33 ,
     labels = bquote(italic(.(gene.name))),cex = 1.7,font = 3)

## Extract bed file name 1 and read it
current.peaks <- read.table(file=selected.bed.files,header = F, as.is = T)
current.peaks <- current.peaks[,1:3]
colnames(current.peaks) <- c("seqnames","start","end")

```

```

peak.coordinates <- subset(current.peaks, seqnames == as.character(range.to.plot$seqnames) & start >=
current.peaks.to.plot <- peak.coordinates[,2:3]

## Transform coordinates
current.peaks.to.plot <- current.peaks.to.plot - range.to.plot$start

## Check if there are peaks for the target gene
if(nrow(current.peaks.to.plot) > 0)
{
  #motifs.in.peaks <- vector(mode="list", length=nrow(current.peaks.to.plot))
  for(j in 1:nrow(current.peaks.to.plot))
  {
    ## Extract start and end point of each peak region
    current.peak.start <- current.peaks.to.plot[j,1]
    current.peak.end <- current.peaks.to.plot[j,2]

    ## Computer coordinates for polygon and draw it
    peak.x <- c(current.peak.start,current.peak.end,
               current.peak.end,current.peak.start)
    peak.y <- c(peak.width - 12, peak.width - 12,
               - peak.width - 12, - peak.width - 12)

    polygon(x = peak.x, y = peak.y, col = "blue", border = "blue",lwd=2)
  }
}

## Draw profiles if bw file is provided
## Compute base line for current TF
current.base.line <- - 10

i <- 4

for(i in 1:length(selected.bigwig.files))
{
  ## Represent signal from the current TF
  lines(chip.signal.mean[i,]+current.base.line,type="l",col=cols[i],lwd=3)
  ## Determine polygon coordinates and represent it
  cord.y <- c(current.base.line,chip.signal.mean[i,]+current.base.line,current.base.line)
  cord.x <- 1:length(cord.y)
  polygon(cord.x,cord.y,col=cols[i])
}

axis(side = 2,lwd = 2,
     at = seq(from=0,to=ceiling(upper.lim/10)*10,by=10)+current.base.line,
     labels = seq(from=0,to=ceiling(upper.lim/10)*10,by=10))
}

gene.expression.barplot <- function(gene.id, gene.expression)
{
  gene.expression.10 <- unlist(gene.expression[gene.id,paste("t10C",1:3,sep="_")])
  gene.expression.14 <- unlist(gene.expression[gene.id,paste("t14C",1:3,sep="_")])
  gene.expression.20 <- unlist(gene.expression[gene.id,paste("t20C",1:3,sep="_")])
  gene.expression.26 <- unlist(gene.expression[gene.id,paste("t26C",1:3,sep="_")])
}

```

```

means <- c(mean(gene.expression.10),
           mean(gene.expression.14),
           mean(gene.expression.20),
           mean(gene.expression.26))

sds <- c(sd(gene.expression.10),
         sd(gene.expression.14),
         sd(gene.expression.20),
         sd(gene.expression.26))

y.max <- 1.2*max(means + sds)

par(lwd=2)
xpos <- barplot(means,ylim=c(0,y.max),
               col = colorRampPalette(c("blue", "red"))(4),cex.axis = 1.5,
               names.arg = c("10°C","14°C","20°C","26°C"),cex.names=2,lwd=2)

par(lwd=1)
arrows(x0 = xpos,y0 = means-sds/sqrt(3), x1 = xpos,y1=means+sds/sqrt(3),code=3,angle=90,length=0.07,lwd=1)

points(x = rep(xpos+0.2,each=3),y=c(gene.expression.10,
                                     gene.expression.14,
                                     gene.expression.20,
                                     gene.expression.26),lwd=2)

}

```

Specifically we found increased levels of H3K27me3 and gene expression repression in genes condensing for proteins involved in microtubule development tubulin alpha and beta:

```

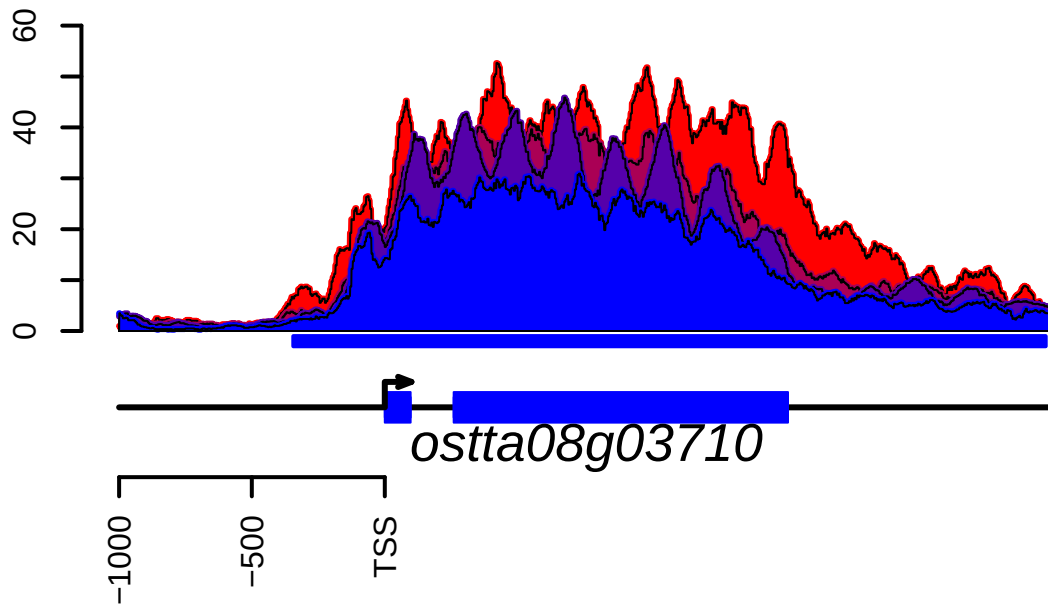
#jpeg(filename = "images/tuba_ostta08g03710.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta08g03710", #TUBA
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_20_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_14_chip_1.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb=TxDb.Otauri.JGI)

```

```

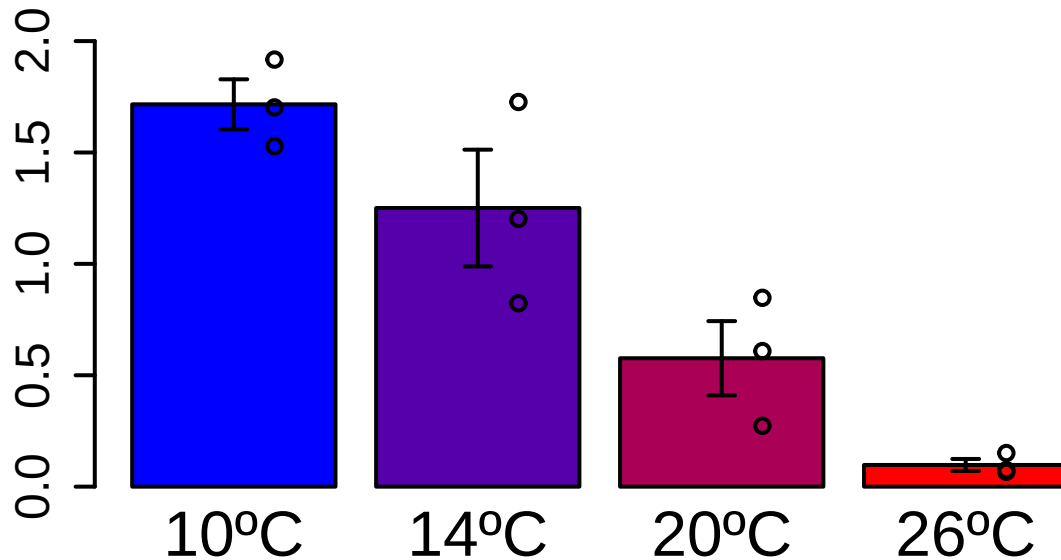
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

```



```
#dev.off()

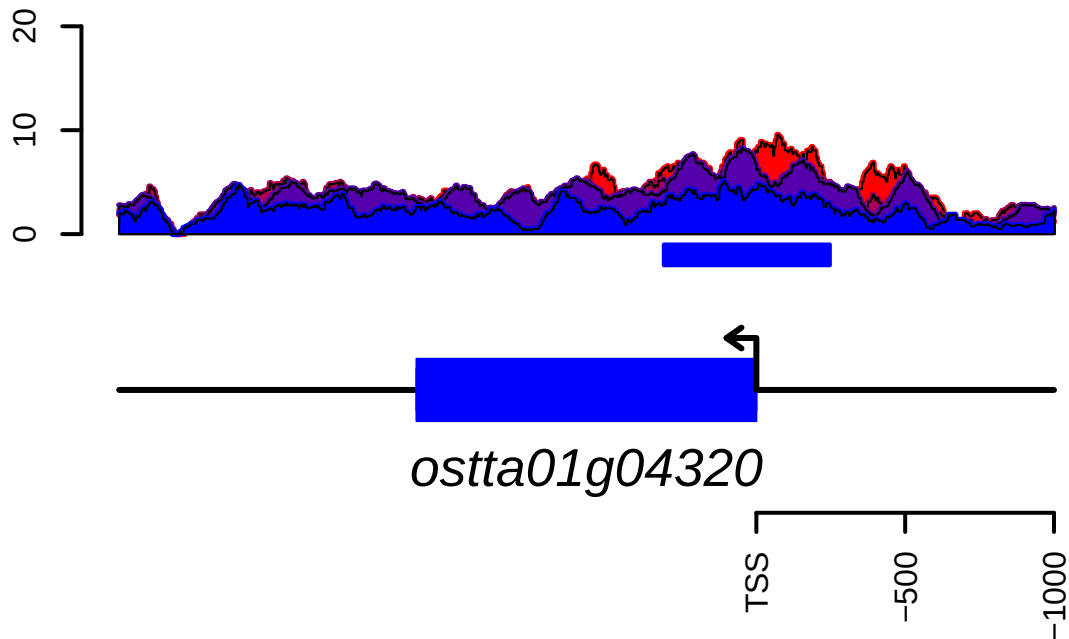
#jpeg(filename = "images/tuba_ostta08g03710_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta08g03710", gene.expression=gene.expression)
```



```
#dev.off()

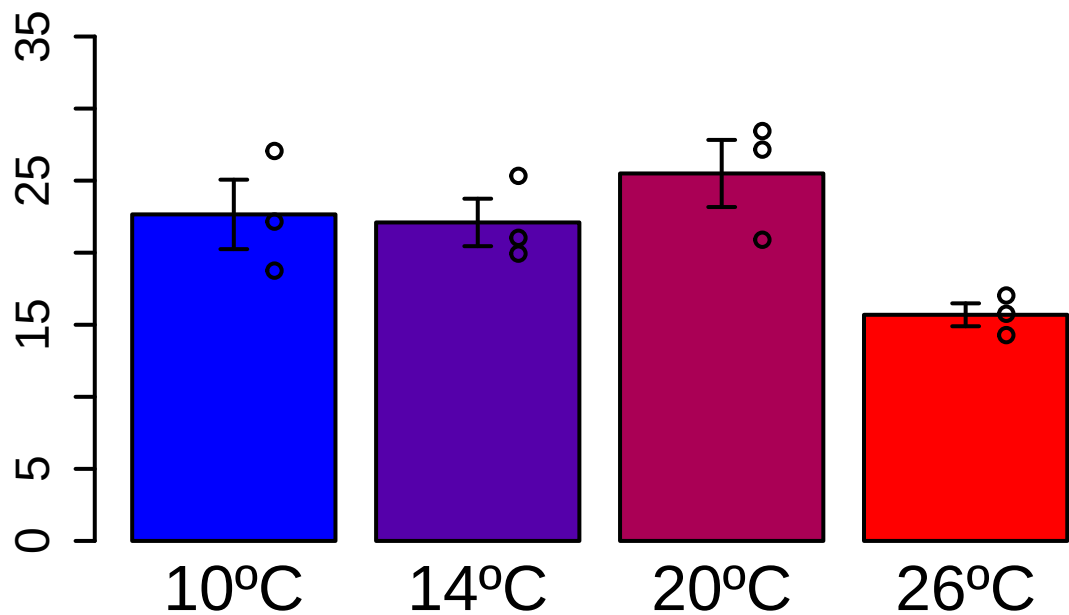
#jpeg(filename = "images/tubb_ostta01g04320.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta01g04320", #TUBB
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_20_chip_2.bw",
    "h3k27me3/h3k27me3_ll_14_chip_1.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/tubb_ostta01g04320_expression.jpg", res = 100, width = 600, height = 580)
gene.expression.barplot(gene.id = "ostta01g04320", gene.expression=gene.expression)
```



```
#dev.off()

gene.profile.complete(gene.name = "ostta08g03710", #TUBA
promoter_length=1000,
selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
"h3k27me3/h3k27me3_ll_10_chip_2.bw",
```

```

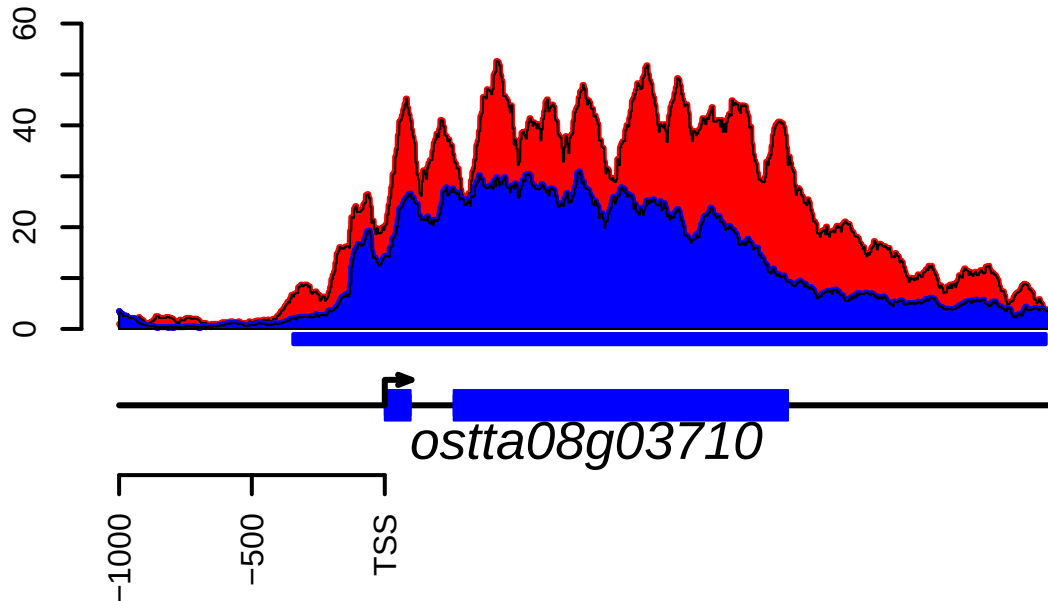
        "h3k27me3/h3k27me3_11_10_chip_2.bw",
        "h3k27me3/h3k27me3_11_10_chip_2.bw"),
cols = rev(colorRampPalette(c("blue", "red"))(4)),
selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
txdb=TxDb.Otauri.JGI)

```

```

## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

```



```

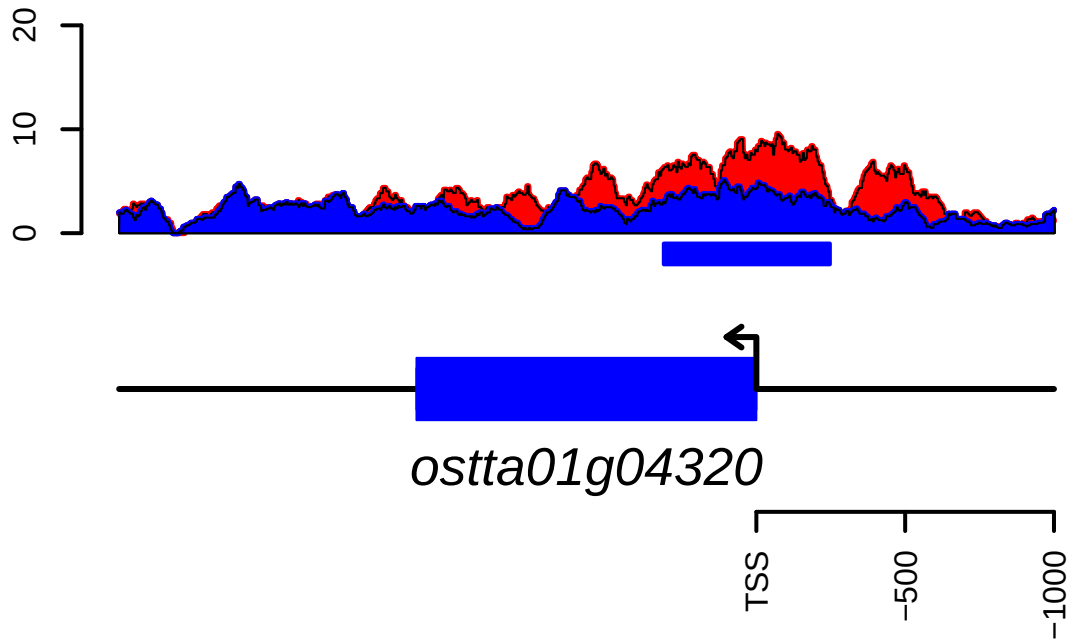
gene.profile.complete(gene.name = "ostta01g04320", #TUBB
    promoter_length=1000,
    selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
        "h3k27me3/h3k27me3_11_10_chip_2.bw",
        "h3k27me3/h3k27me3_11_10_chip_2.bw",
        "h3k27me3/h3k27me3_11_10_chip_2.bw"),
    cols = rev(colorRampPalette(c("blue", "red"))(4)),
    selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
    txdb=TxDb.Otauri.JGI)

```

```

## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

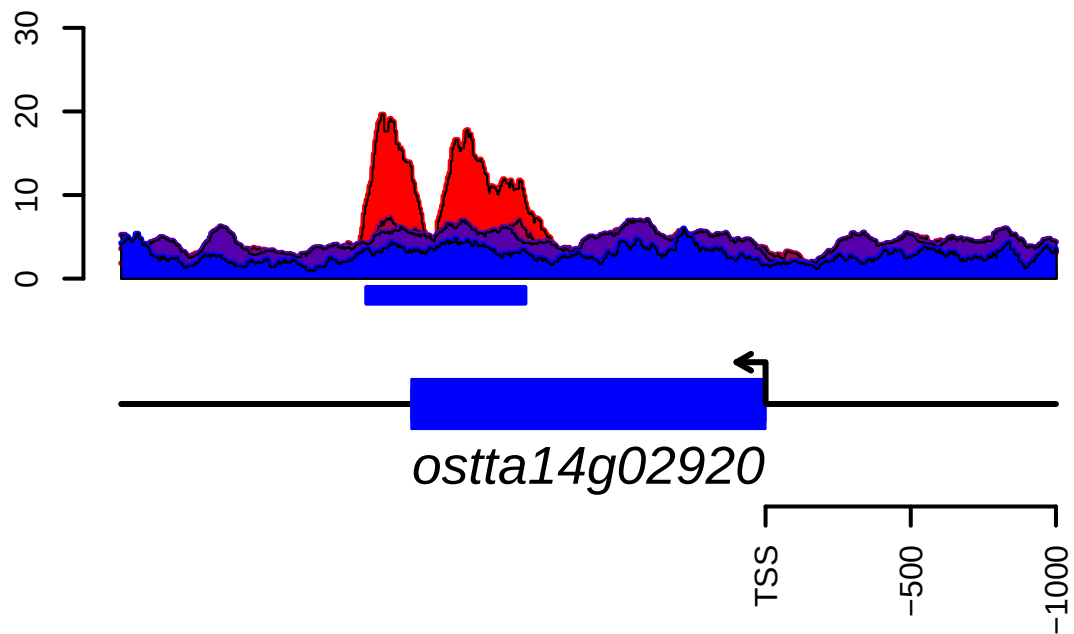
```



Genes involved in cortical actin network remodelling. ARP2 (ostta14g02920) is involved in the branching of actin filaments. FIL-MORN-Invasing (ostta09g01910) proteins is involved in the crosslinking of actin filaments (filamin domain), interactions between actin filaments and membranes (MORN Membrane Occupation and Recognition Nexus) and membrane penetration or remodeling to assist in vesicle budding (Invasin D3 domain).

```
#jpeg(filename = "images/actin_ostta14g02920.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta14g02920", #ARP2
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_20_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_14_chip_1.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb=TxDb.Otauri.JGI)
```

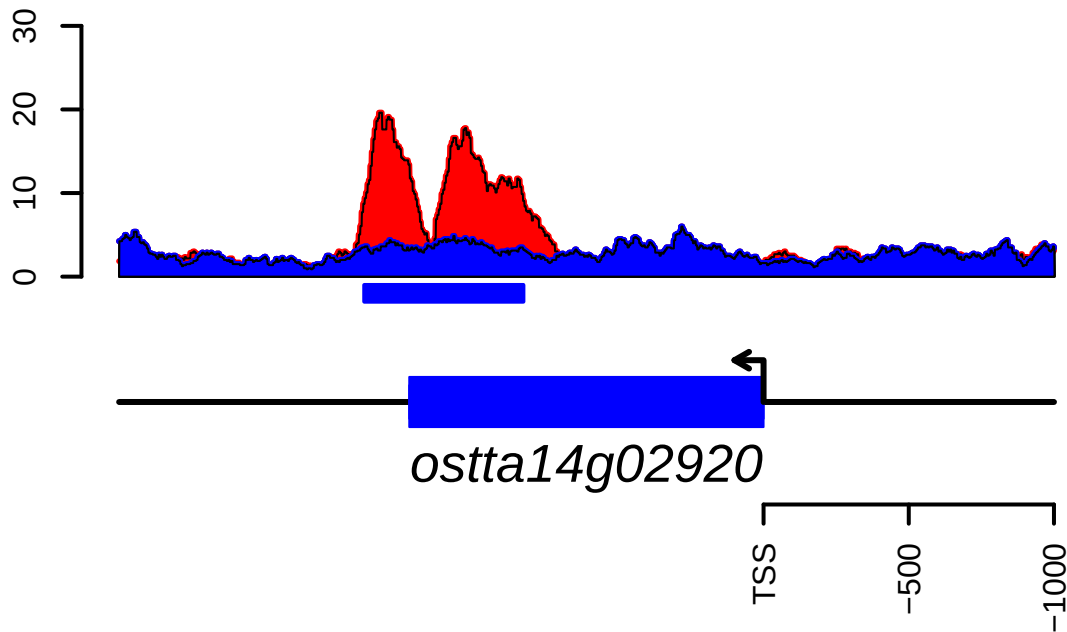
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



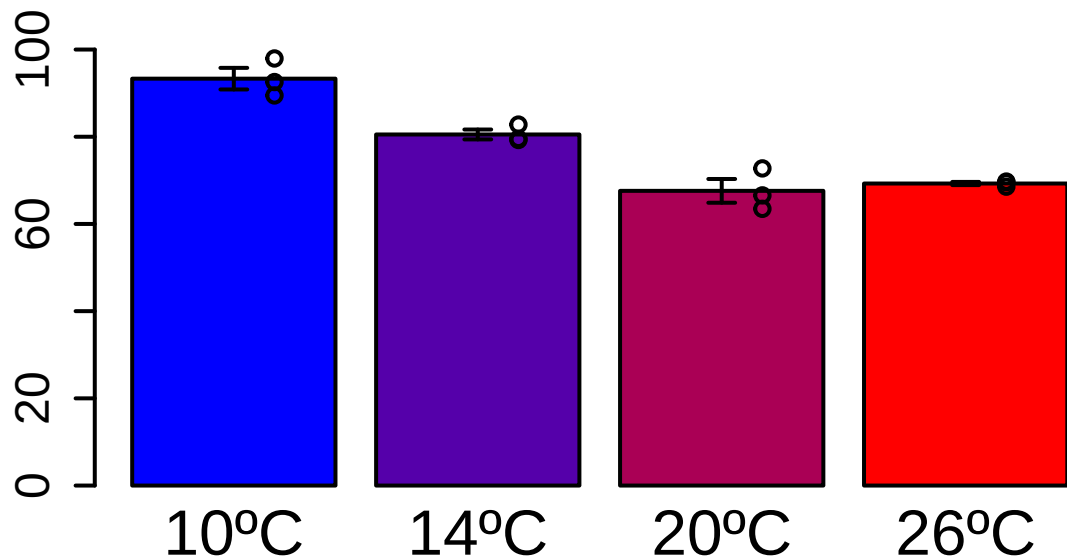
```
#dev.off()

gene.profile.complete(gene.name = "ostta14g02920", #ARP2
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



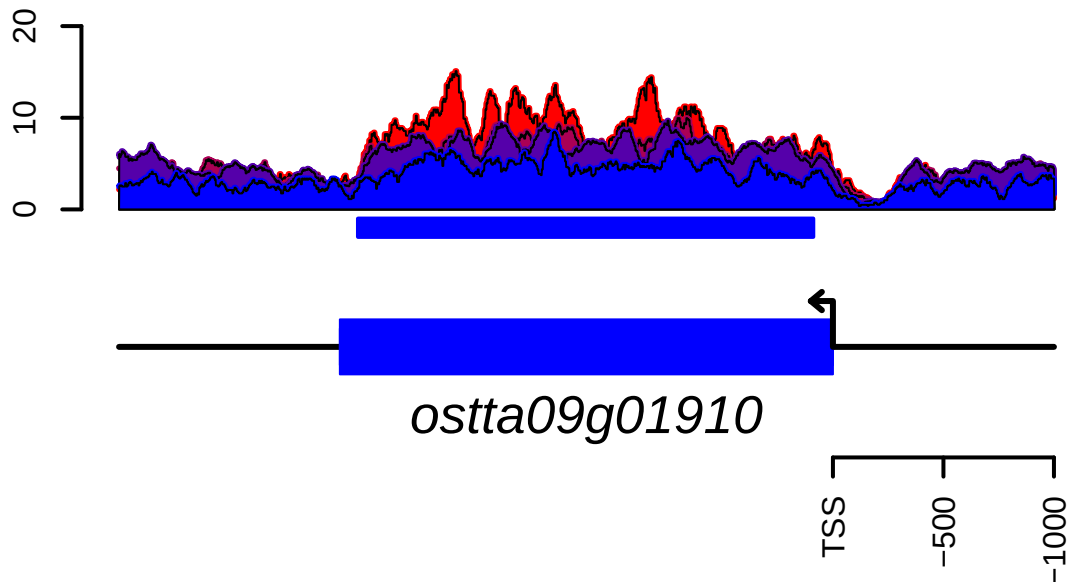
```
#jpeg(filename = "images/actin_ostta14g02920_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta14g02920", gene.expression=gene.expression)
```



```
#dev.off()

#jpeg(filename = "images/filamin_ostta09g01910.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta09g01910", #Filamin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_20_chip_2.bw",
    "h3k27me3/h3k27me3_ll_14_chip_1.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

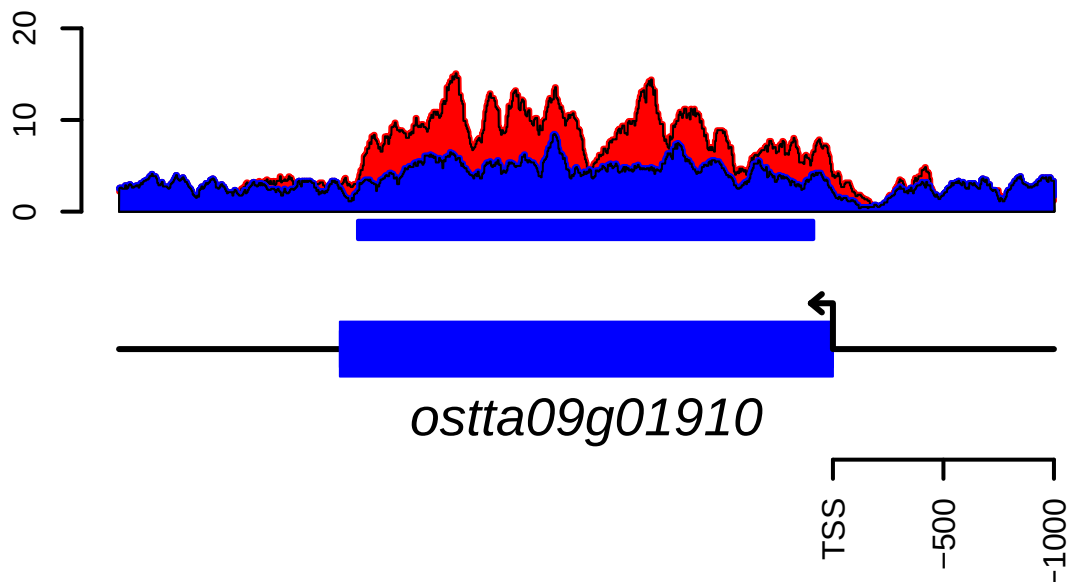
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



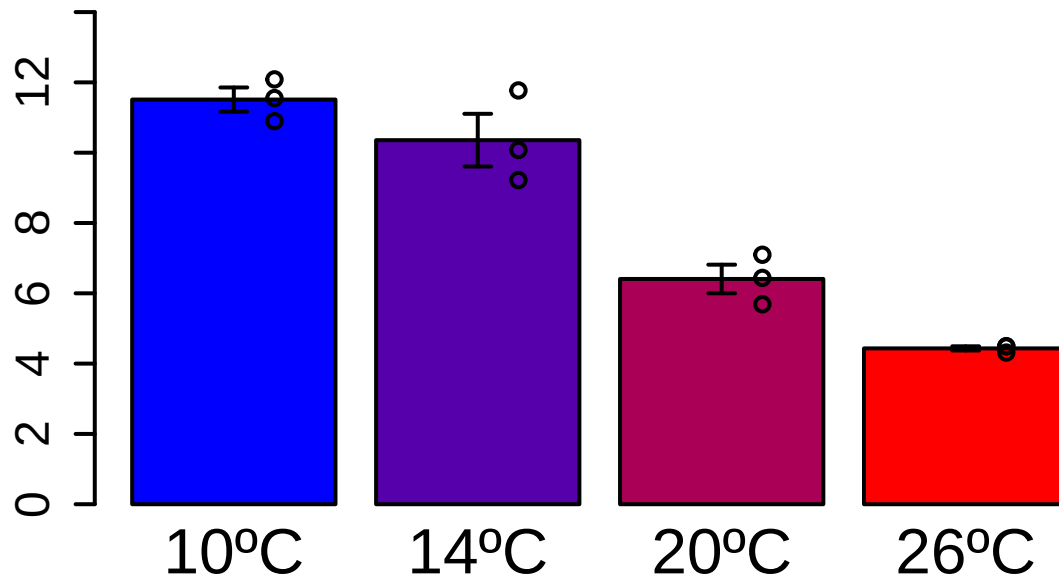
```
#dev.off()
```

```
gene.profile.complete(gene.name = "ostta09g01910", #Filamin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#jpeg(filename = "images/filamin_ostta09g01910_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta09g01910", gene.expression=gene.expression)
```

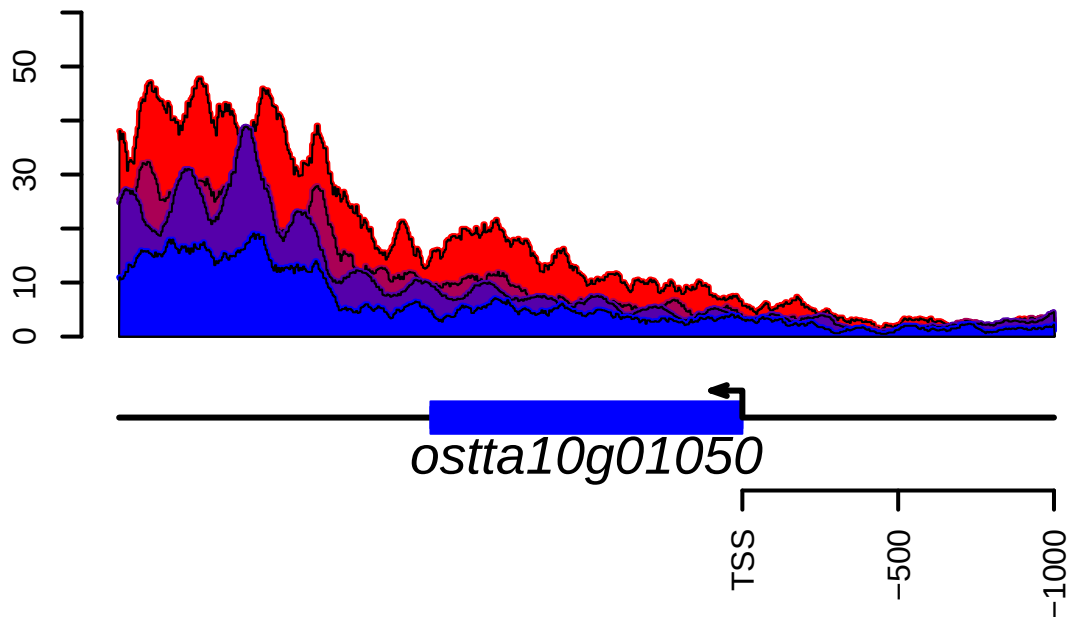


```
#dev.off()
```

We also found increase H3K27me3 signal increase and expression repression in the gene codifying for AURORA kinase (ostta10g01050) that is involved in microtubule organization by phosphorylating tubulins and actins.

```
#jpeg(filename = "images/aurora_ostta10g01050.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta10g01050", #Aurora
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_20_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_14_chip_1.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb=TxDb.Otauri.JGI)
```

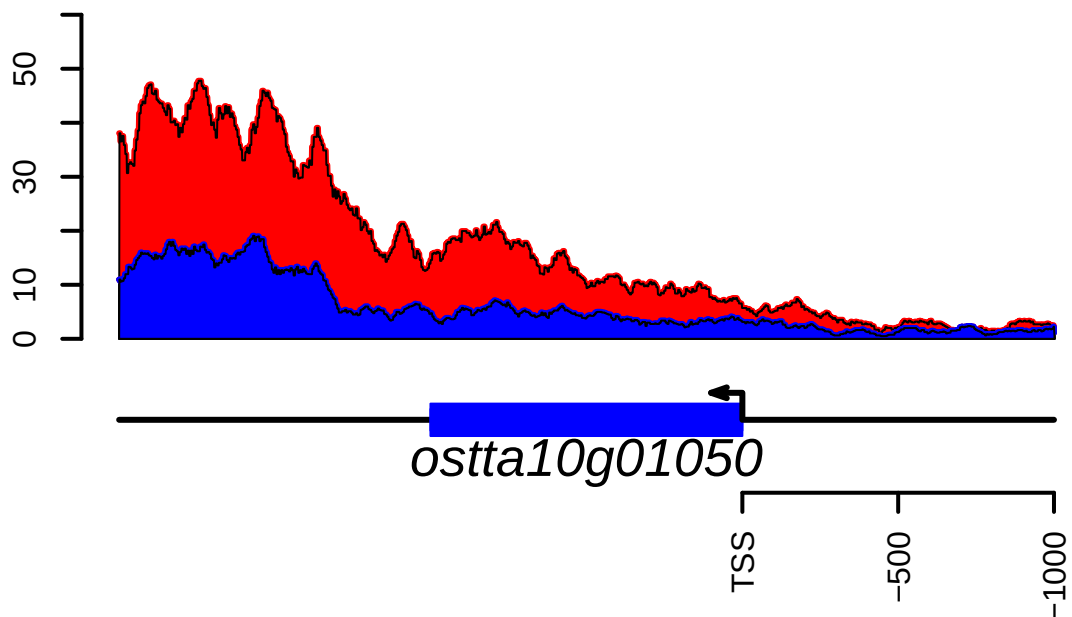
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

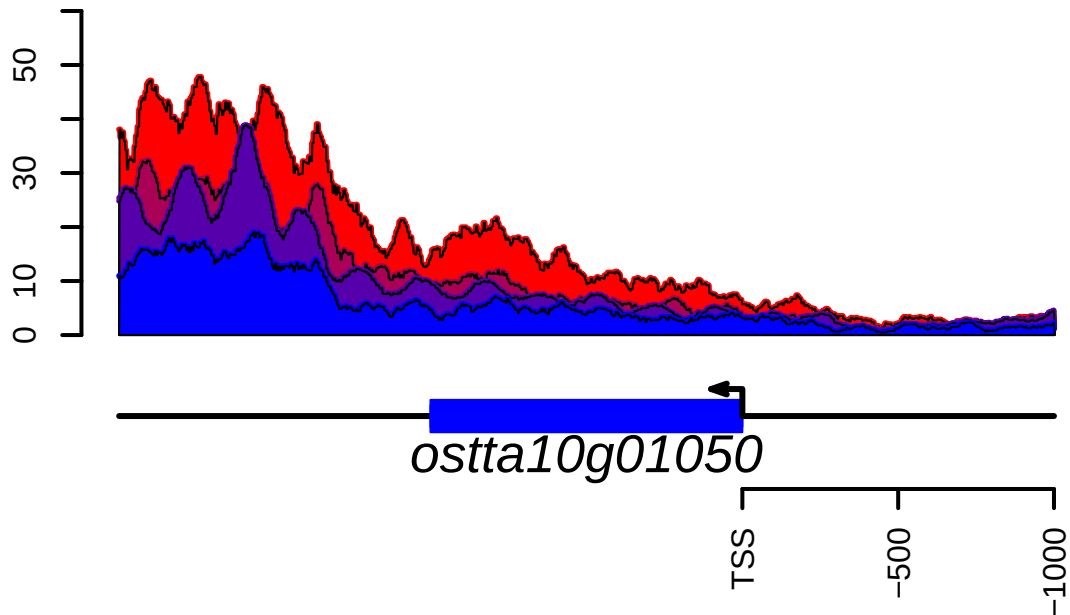
gene.profile.complete(gene.name = "ostta10g01050", #Aurora
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

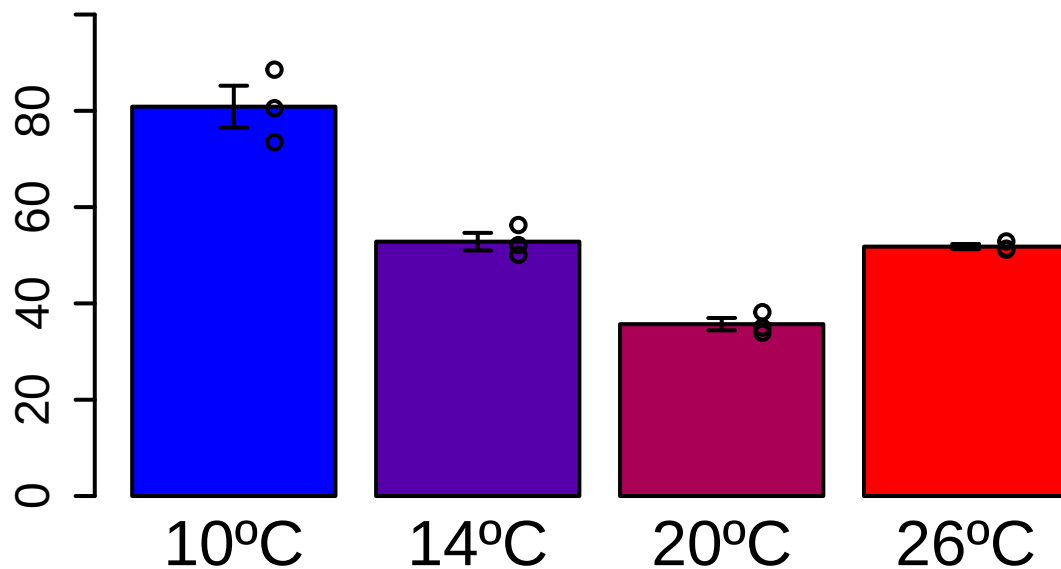


```
gene.profile.complete(gene.name = "ostta10g01050", #Aurora
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
    "h3k27me3/h3k27me3_11_20_chip_2.bw",
    "h3k27me3/h3k27me3_11_14_chip_1.bw",
    "h3k27me3/h3k27me3_11_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#jpeg(filename = "images/aurora_ostta10g01050_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta10g01050", gene.expression=gene.expression)
```

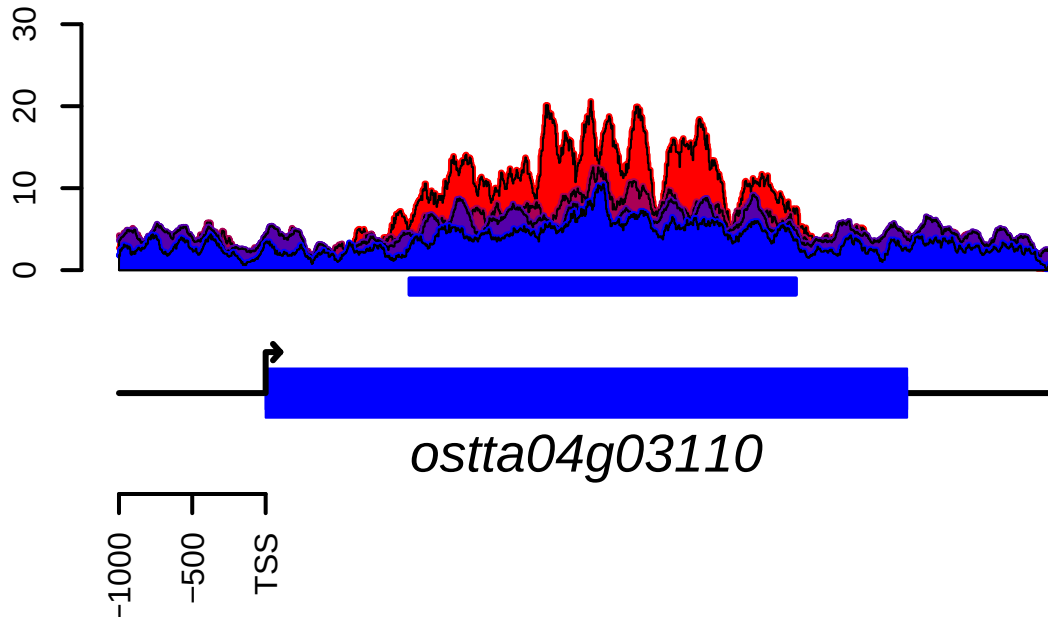


```
#dev.off()
```

This significant regulation over cytoskeleton remodelling is coupled with the activation of motor proteins. In genes codifying for kinesin (ostta04g03110) and myosin (ostta06g03320), we detected a significantly increased H3K27me3 ChIP signal and decreased expression as a response to elevated temperatures.

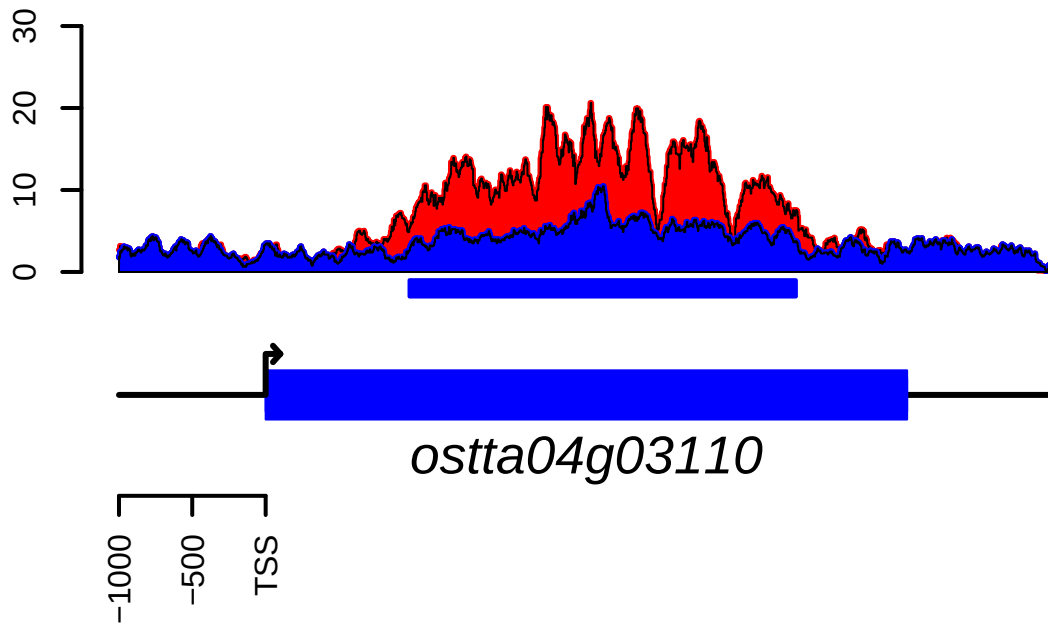
```
#jpeg(filename = "images/kinesin_ostta04g03110.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta04g03110", #Kinesin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_20_chip_2.bw",
    "h3k27me3/h3k27me3_ll_14_chip_1.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



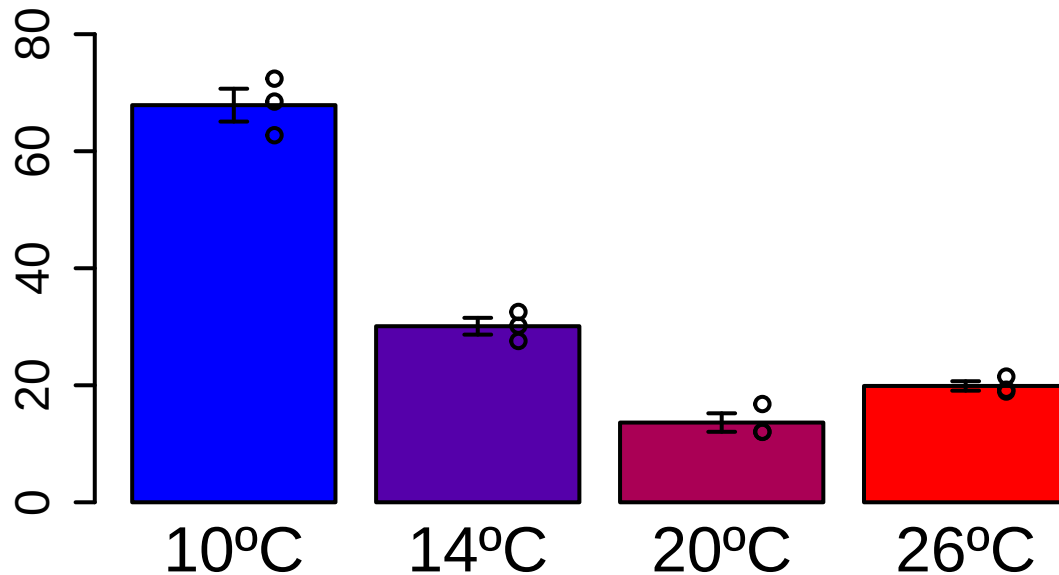
```
gene.profile.complete(gene.name = "ostta04g03110", #Kinesin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

#jpeg(filename = "images/kinesin_ostta04g03110_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta04g03110", gene.expression=expression)
```

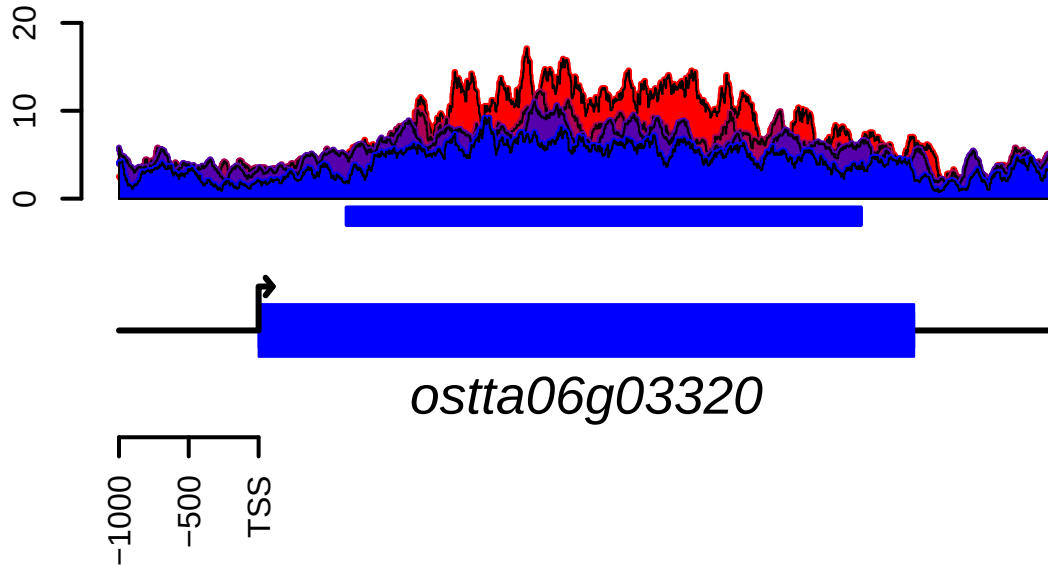


```
#dev.off()

#jpeg(filename = "images/myosin_ostta06g03320.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta06g03320", #Myosin
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
    "h3k27me3/h3k27me3_11_20_chip_2.bw",
    "h3k27me3/h3k27me3_11_14_chip_1.bw",
    "h3k27me3/h3k27me3_11_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
```

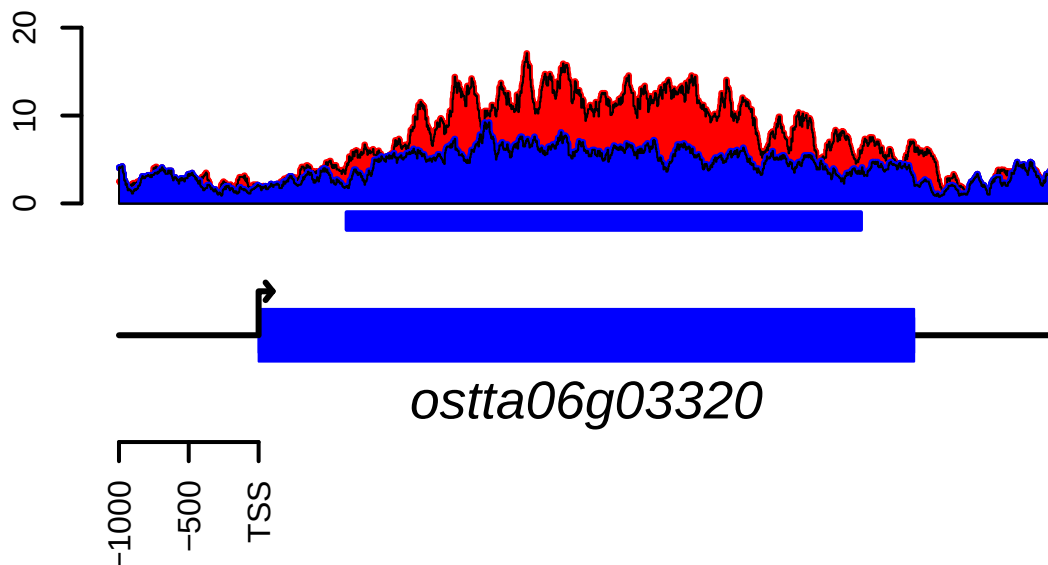
```
txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =  
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



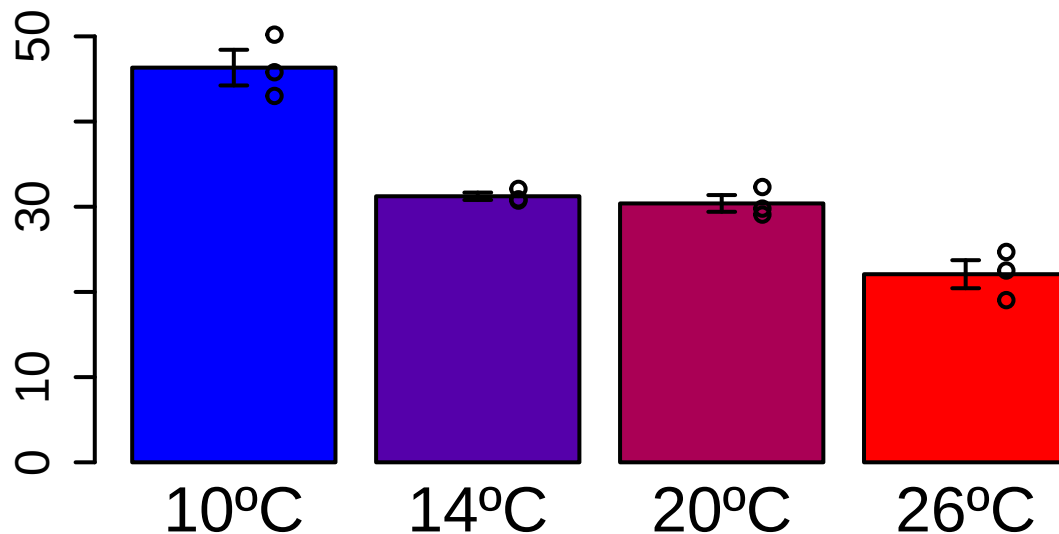
```
gene.profile.complete(gene.name = "ostta06g03320", #Myosin  
                      promoter_length=1000,  
                      selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",  
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",  
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",  
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw"),  
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),  
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",  
                      txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =  
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

```
#jpeg(filename = "images/myosin_ostta06g03320_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta06g03320", gene.expression=gene.expression)
```



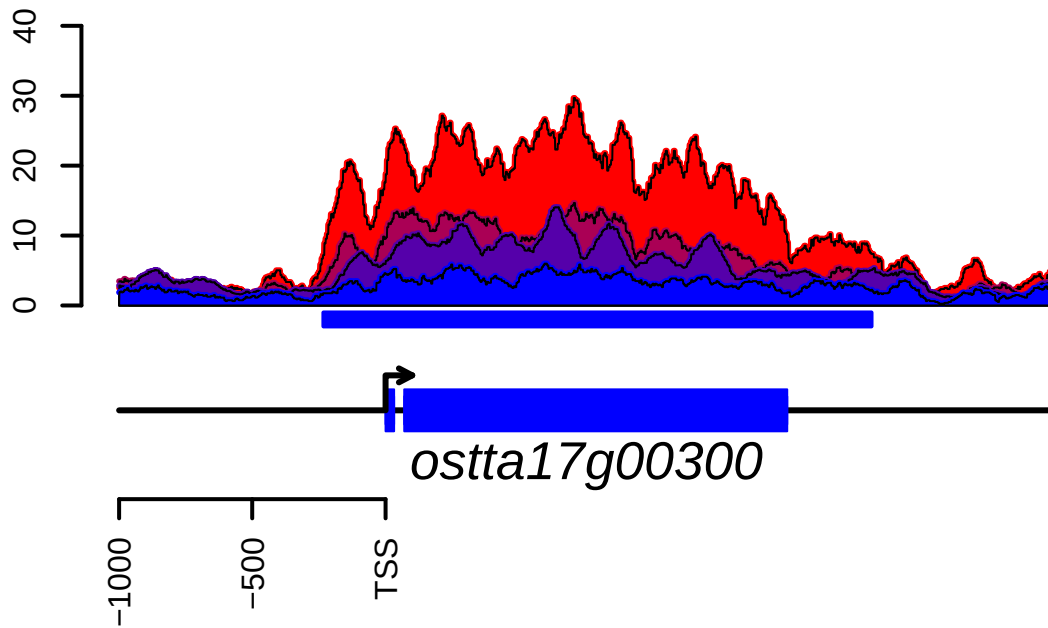
```
#dev.off()
```

Kinesins are involved in long range vesicle transport through microtubules and myosins are involved in short range transport through actin filaments. Kinesins (microtubules) and myosins (actin filaments) combination suggests a two-phase vesicle transport system activated at low temperatures. Long-range transport along microtubules by kinesins followed by short-range delivery to the membrane along the cortical actin network by myosins.

Although no significant enrichment was found for lipid metabolism in the genes that significantly increase their H3K27me3 ChIP-signal and decrease their expression level, key genes in lipid metabolism and transport were identified in this set of genes: enoyl-ACP reductase (ENR, ostta17g00300), diacylglycerol acyltransferase (DGAT, ostta16g00910) and ABC type 2 transporter (ABC, ostta09g01615).

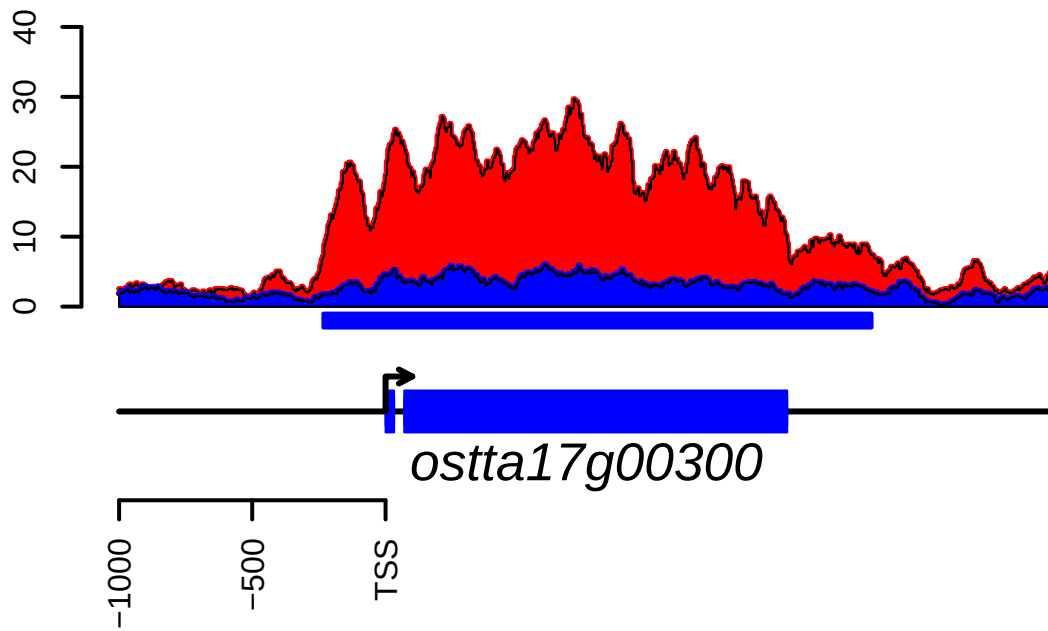
```
#jpeg(filename = "images/ENR_ostta17g00300.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta17g00300", #ENR
    promoter_length=1000,
    selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
                             "h3k27me3/h3k27me3_11_20_chip_2.bw",
                             "h3k27me3/h3k27me3_11_14_chip_1.bw",
                             "h3k27me3/h3k27me3_11_10_chip_2.bw"),
    cols = rev(colorRampPalette(c("blue", "red"))(4)),
    selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
    txdb=TxDb.Otari.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



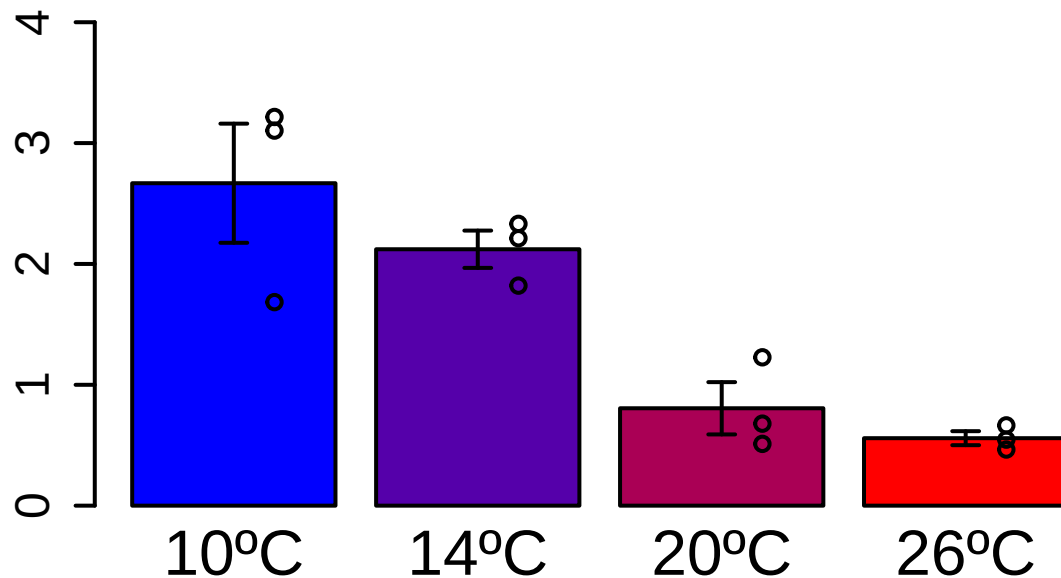
```
gene.profile.complete(gene.name = "ostta17g00300", #ENR
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

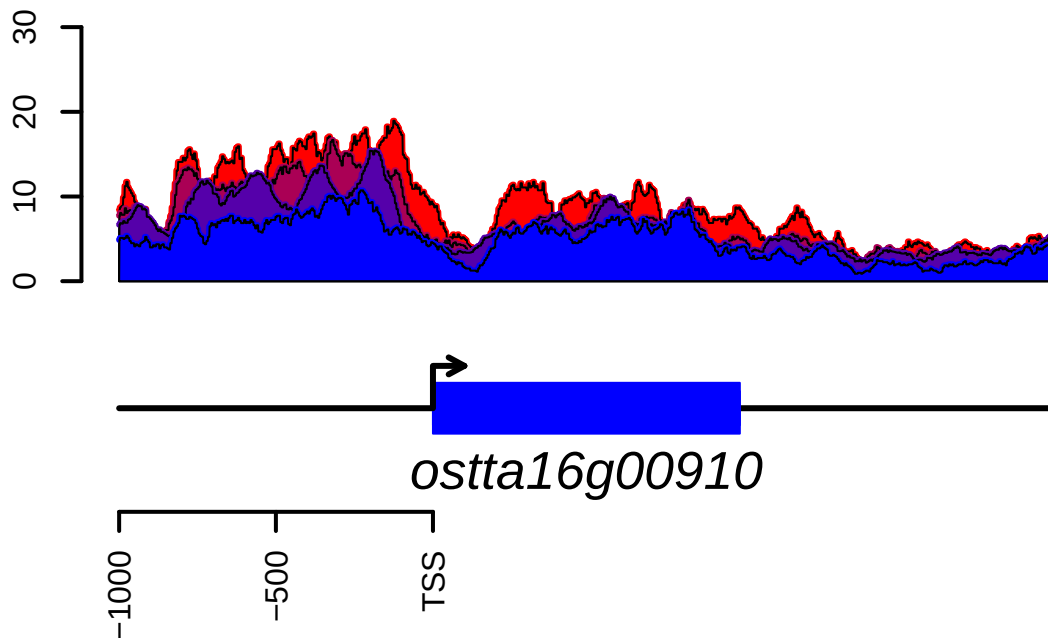
```
#jpeg(filename = "images/ENR_ostta17g00300_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta17g00300", gene.expression=gene.expression)
```



```
#dev.off()
```

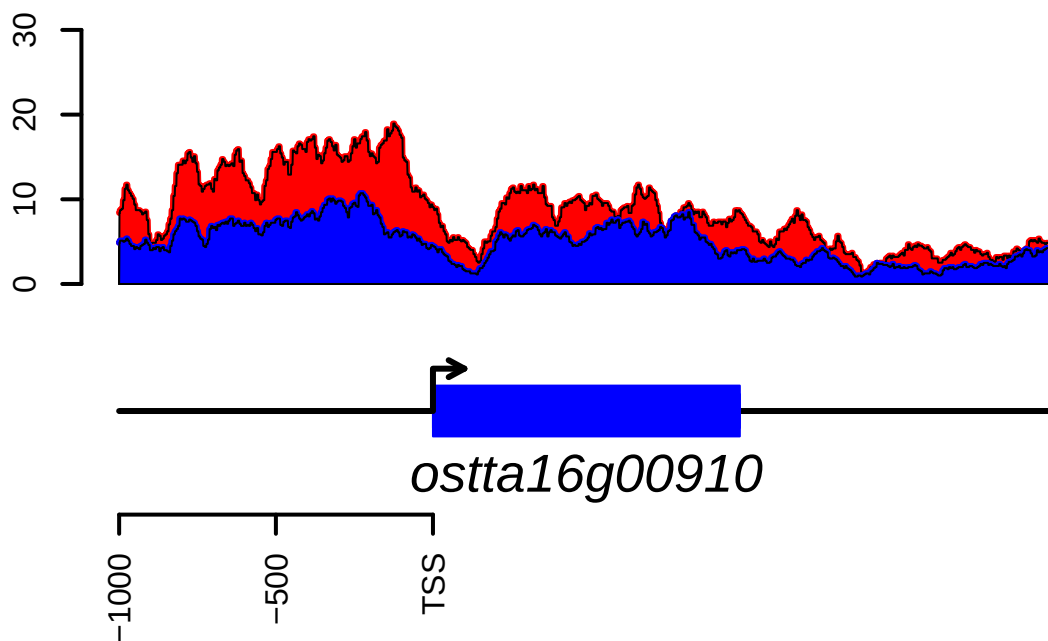
```
#jpeg(filename = "images/DGAT_ostta16g00910.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta16g00910", #DGAT
    promoter_length=1000,
    selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
                             "h3k27me3/h3k27me3_ll_20_chip_2.bw",
                             "h3k27me3/h3k27me3_ll_14_chip_1.bw",
                             "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
    cols = rev(colorRampPalette(c("blue", "red"))(4)),
    selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
    txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



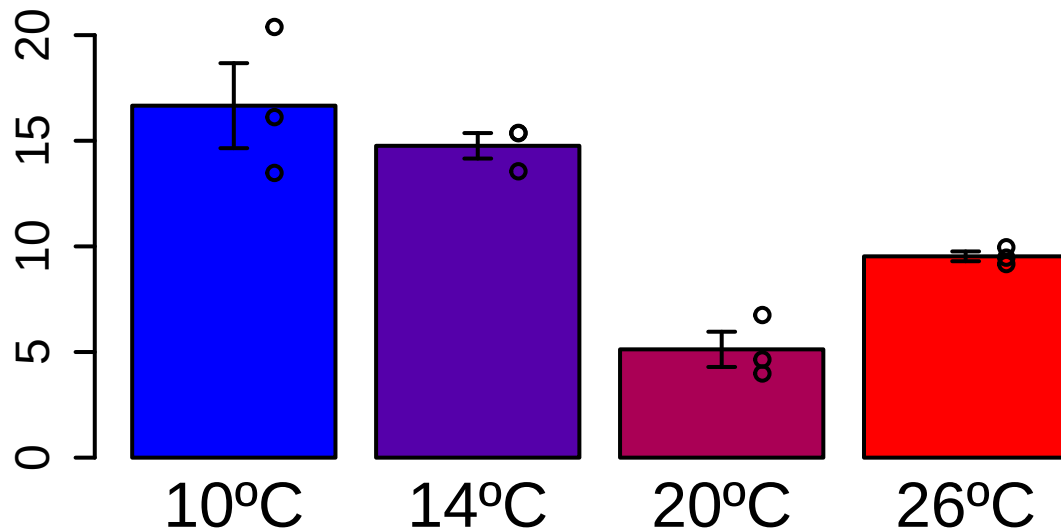
```
gene.profile.complete(gene.name = "ostta16g00910", #DGAT
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

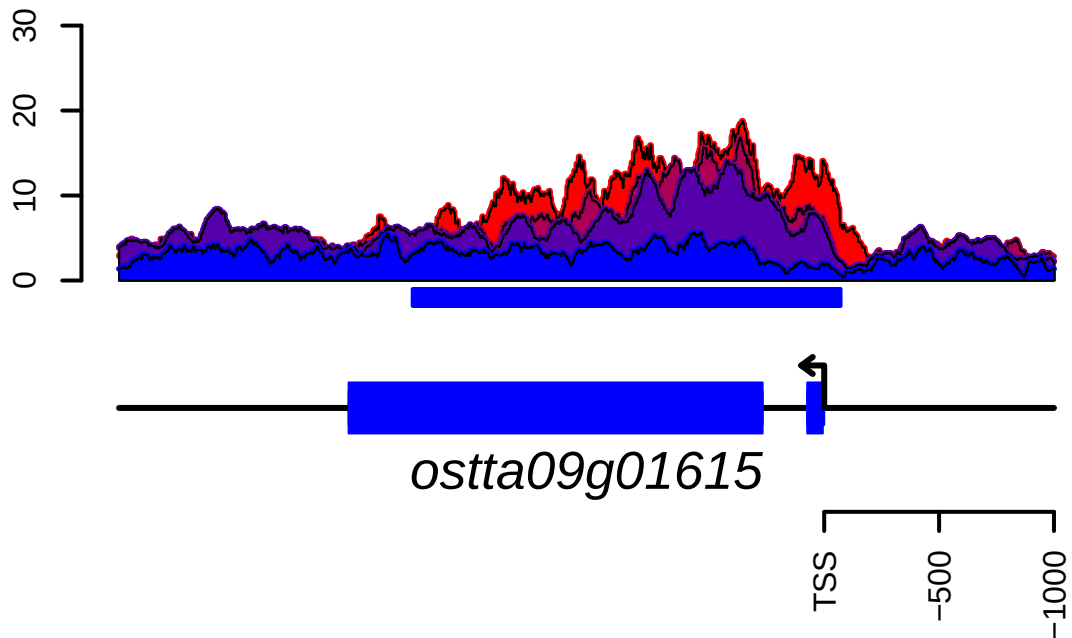
```
#jpeg(filename = "images/DGAT_ostta16g00910_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta16g00910", gene.expression=gene.expression)
```



```
#dev.off()
```

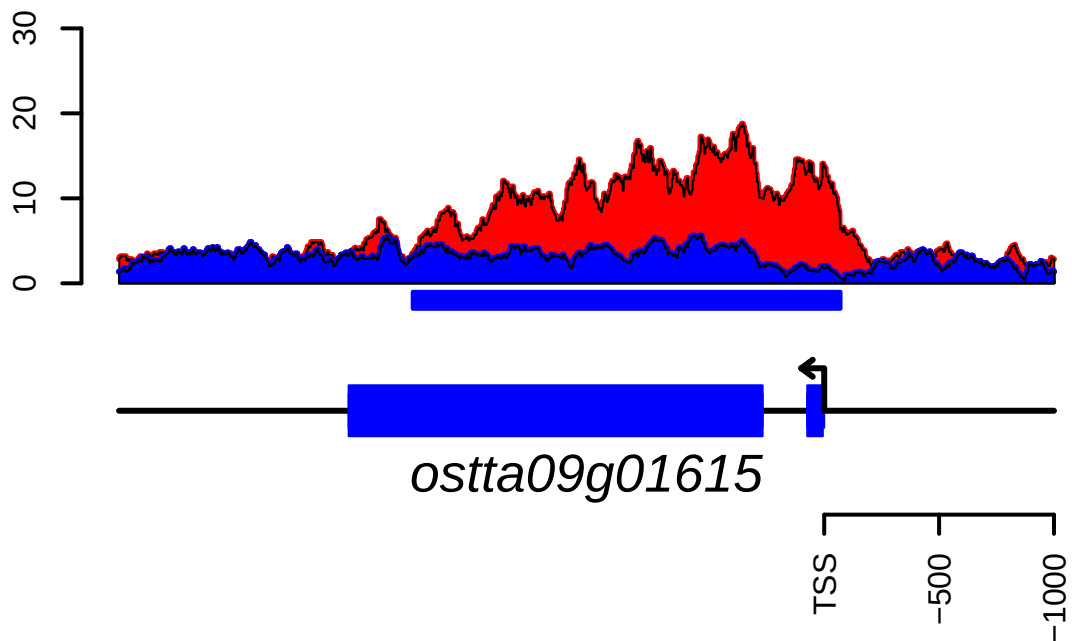
```
#jpeg(filename = "images/ABC_ostta09g01615.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta09g01615", #ABC
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_20_chip_2.bw",
    "h3k27me3/h3k27me3_ll_14_chip_1.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



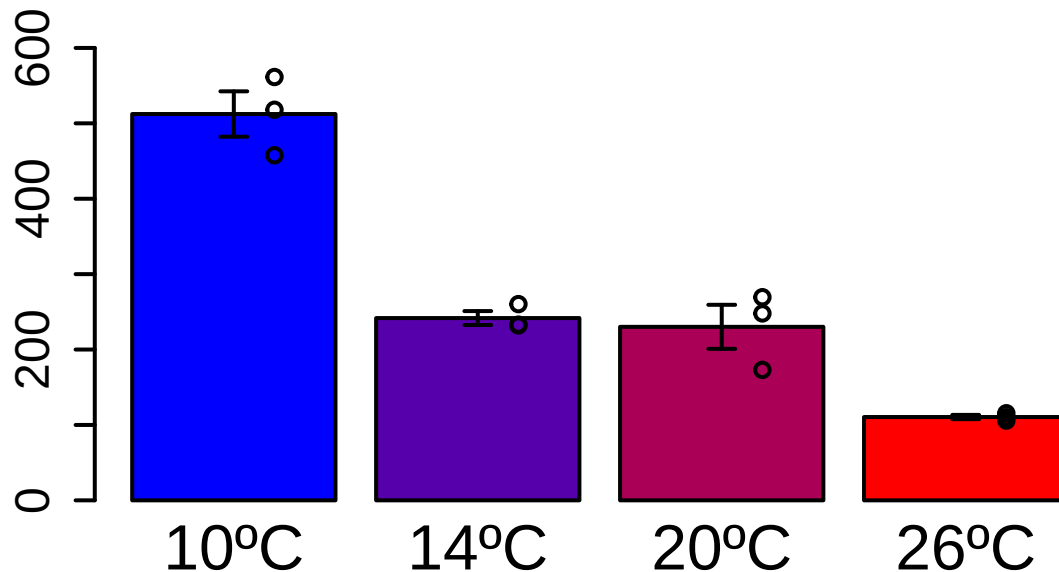
```
gene.profile.complete(gene.name = "ostta09g01615", #ABC
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb=TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

```
#jpeg(filename = "images/ABC_ostta09g01615_expression.jpg",res = 100,width = 600,height = 580)
gene.expression.barplot(gene.id = "ostta09g01615", gene.expression=gene.expression)
```



```
#dev.off()
```

Evolutionary comparison of H3K27me3 genes in *Ostreococcus tauri* and *Arabidopsis thaliana*

We compare according to orthology relationships the H3K27me3 marked genes found in *Ostreococcus tauri* in this study with those found in *Arabidopsis thaliana* in our previous study “H2A monoubiquitination in *Arabidopsis thaliana* is generally independent of LHP1 and PRC2 activity. Zhou Y, Romero-Campero FJ, Gómez-Zambrano Á, Turck F, Calonje M. *Genome Biol.* 2017 Apr 12;18(1):69. doi: 10.1186/s13059-017-1197-z.”.

First, we load the H3K27me3 marked genes in *Arabidopsis thaliana*:

```
atha.k27.genes <- unique(c(read.table(file="h3k27me3_h2aub_atha.txt",header = F,as.is=T)[[1]],
                           read.table(file="only_h3k27me3_atha.txt",header = F,as.is=T)[[1]]))
length(atha.k27.genes)
```

```
## [1] 6843
```

```
length(atha.k27.genes)/34100
```

```
## [1] 0.2006745
```

```
length(all.k27.genes)
```

```
## [1] 994
```

```
length(all.k27.genes)/7668
```

```
## [1] 0.1296296
```

```
at.transposons <- read.table(file="atha_data/transposable_elements.txt",as.is=T)[[1]]
length(intersect(atha.k27.genes,at.transposons))
```

```
## [1] 162
```

```
length(intersect(atha.k27.genes,at.transposons))/length(at.transposons)
```

```
## [1] 0.04150653
```

```
length(intersect(atha.k27.genes,at.transposons))/length(atha.k27.genes)
```

```
## [1] 0.02367383
```

H3k27me3 marks are found in 6843 genes in Arabidopsis accounting for 20% of its entire genome, compared to the 994 genes associated to H3K27me3 marks corresponding to 13% of the entire *Ostreococcus* genome. H3K27me3 has been shown to associate primarily with developmental genes rather than TE being H3K9me2/H3K29me3 the main repressive epigenetic mark for transposable elements. Nonetheless, some autonomous TE near genes have been found marked by H3K27me3 specially in euchromatic regions in Arabidopsis. In our study we have found a similar role for H3K27me3 in *Ostreococcus tauri*. In our pico alga this mark also associate primarily with non TE although some TEs are found to be H3K27me3 marked.

In order to determine in the H3K27me3 marked genes in Arabidopsis and *Ostreococcus* are evolutionarily related we load gene orthology relationships as orthogroups between all the genes in *Ostreococcus tauri* and Arabidopsis thaliana genomes. Then we determine the orthogroups from each species containing at least one H3K27me3 marked genes to determine H3K27me3 marked orthogroups that are compared using a Venn diagram.

```
ostta.atha.orthogroups <- read.table(file="ostta_atha_orthogroups.tsv",header=T,sep="\t")
ostta.atha.orthogroups <- subset(ostta.atha.orthogroups,
                                ostreococcus_tauri != "" | arabidopsis_thaliana != "")
nrow(ostta.atha.orthogroups)
```

```
## [1] 10710
```

```
number.orthogroups <- nrow(ostta.atha.orthogroups)
ostta.orthogroups <- c()
for(i in 1:length(all.k27.genes))
{
  # print(i)
  current.gene <- all.k27.genes[[i]]

  j <- 1
  while(j < number.orthogroups & !current.gene %in%
        strsplit(x = ostta.atha.orthogroups$ostreococcus_tauri[[j]],
                  split = ", ")[[1]])
  {
    j <- j + 1
  }

  if(current.gene %in%
      strsplit(x = ostta.atha.orthogroups$ostreococcus_tauri[[j]],
                split = ", ")[[1]])
  {
    ostta.orthogroups <- c(ostta.orthogroups,
                          ostta.atha.orthogroups$Orthogroup[j])
  }
}
ostta.orthogroups <- unique(ostta.orthogroups)
length(ostta.orthogroups)
```

```
## [1] 749
```

```

atha.orthogroups <- c()
for(i in 1:length(atha.k27.genes))
{
  # print(i)
  current.gene <- atha.k27.genes[[i]]

  j <- 1
  while(j < number.orthogroups & !current.gene %in%
        strsplit(x = ostta.atha.orthogroups$arabidopsis_thaliana[[j]],
                  split = ",", " ")[[1]])
  {
    j <- j + 1
  }

  if(current.gene %in%
      strsplit(x = ostta.atha.orthogroups$arabidopsis_thaliana[[j]],
                split = ",", " ")[[1]])
  {
    atha.orthogroups <- c(atha.orthogroups,
                          ostta.atha.orthogroups$Orthogroup[j])
  }
}
length(atha.orthogroups)

## [1] 5745

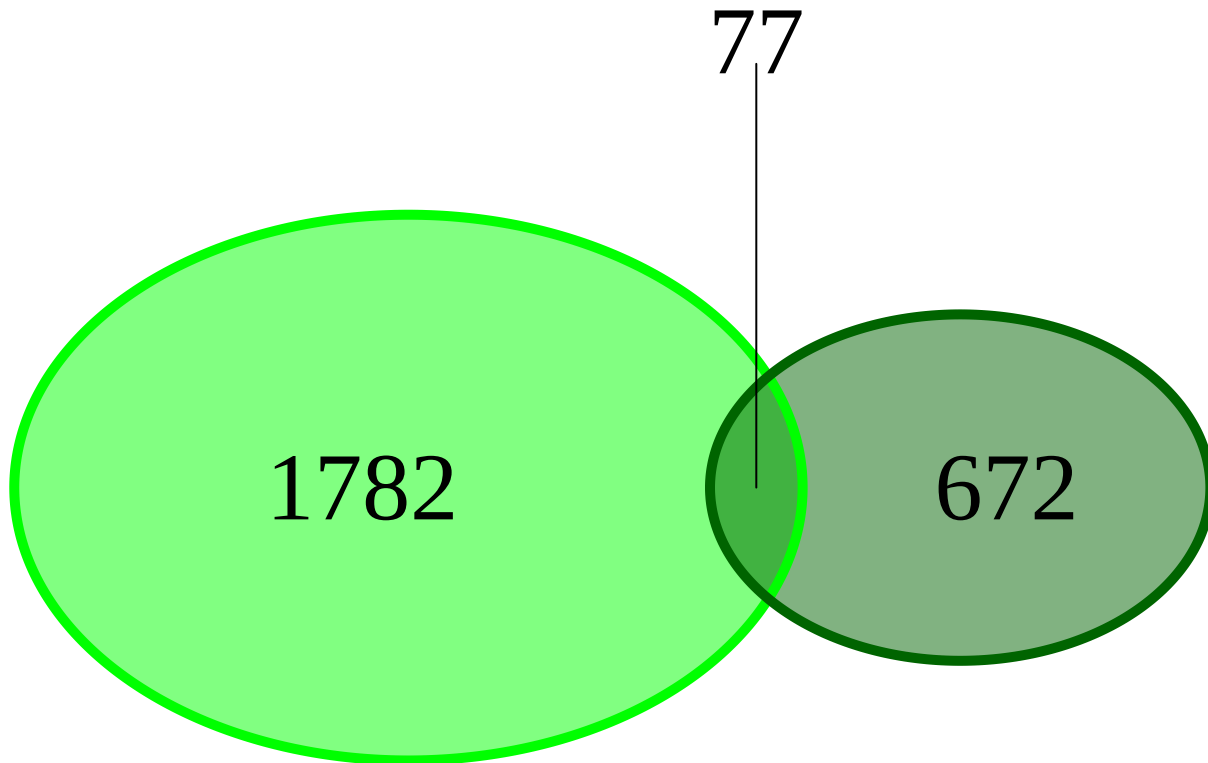
atha.orthogroups <- unique(atha.orthogroups)
length(atha.orthogroups)

## [1] 1859

all.k27.ostta.atha <- intersect(ostta.orthogroups,atha.orthogroups)

#jpeg(filename = "images/venn_diagram_k27_ostta_atha.jpg",res = 160,width = 1000,height = 1000)
grid.newpage()
draw.pairwise.venn(area1 = length(ostta.orthogroups),
                    area2 = length(atha.orthogroups),
                    cross.area = length(all.k27.ostta.atha),
                    lwd = 5,category = c("", ""),
                    euler.d = T,
                    col = c("darkgreen","green"),
                    fill = c("darkgreen","green"),
                    alpha = 0.5,cex = 3,cat.cex = 2)

```



```
## (polygon[GRID.polygon.1256], polygon[GRID.polygon.1257], polygon[GRID.polygon.1258], polygon[GRID.polygon.1259])
#dev.off()
overlap <- length(all.k27.ostta.atha)
size.1 <- length(ostta.orthogroups)
size.2 <- length(atha.orthogroups)
total.orthogroups <- nrow(ostta.atha.orthogroups)
phyper(overlap - 1, size.1, total.orthogroups - size.1, size.2, lower.tail = FALSE)

## [1] 1

ostta.specific.k27.orthogroups <- setdiff(ostta.orthogroups,atha.orthogroups)
length(ostta.specific.k27.orthogroups)

## [1] 672

sum(subset(ostta.atha.orthogroups, Orthogroup %in% ostta.specific.k27.orthogroups)$arabidopsis_thaliana)

## [1] 0.5223214

atha.specific.k27.orthogroups <- setdiff(atha.orthogroups,ostta.orthogroups)
sum(subset(ostta.atha.orthogroups, Orthogroup %in% atha.specific.k27.orthogroups)$ostreococcus_tauri != 0)

## [1] 0.2351291
```

We found 749 *Ostreococcus tauri* orthogroups and 1859 *Arabidopsis thaliana* orthogroups with at least one H3K27me3 marked gene in the corresponding species. These orthogroups will be referred to as H3K27me3 marked orthogroups. Only a non significant overlap consisting of 77 orthogroups was identified as commonly H3K27me3 marked in both species. 52% of the *Ostreococcus tauri* specifically H3K27me3 marked orthogroups posses *Arabidopsis thaliana* representatives. Viceversa

This indicates that although during evolution new gene families represented by orthogroups have emerged in *Arabidopsis* and have been

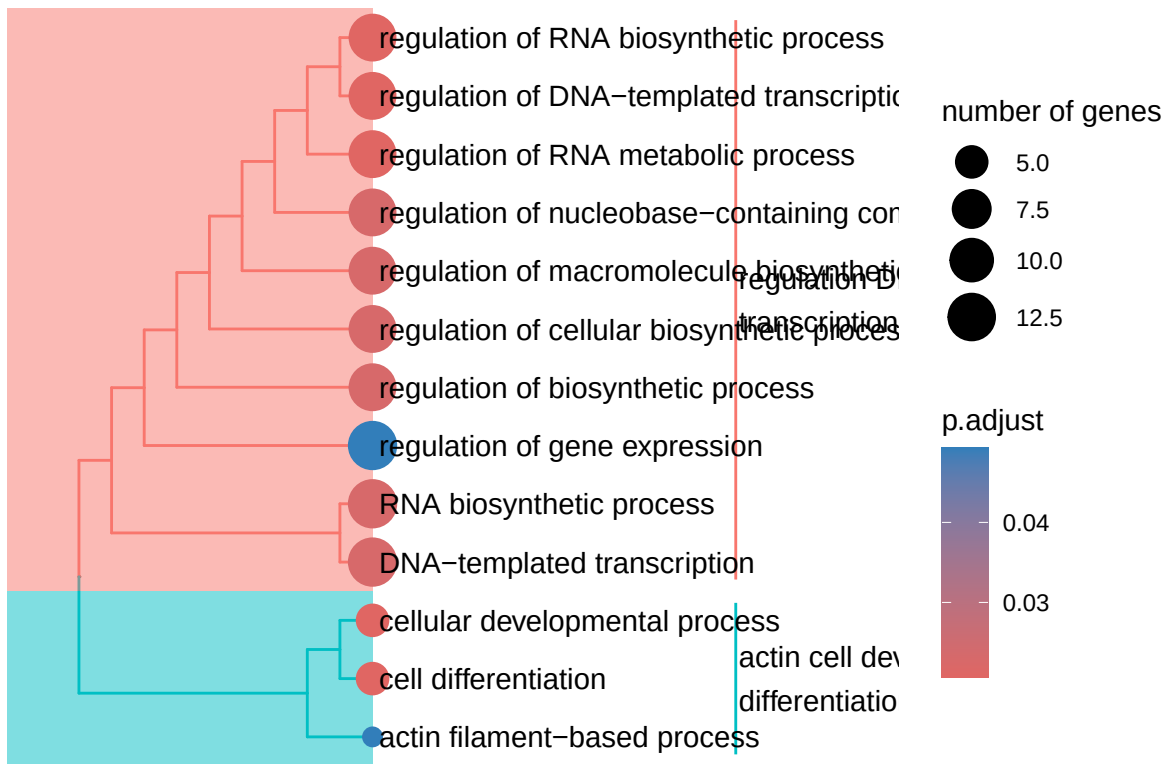
This indicates that H3K27me3 regulates very different biological processes in both species.

Next we explore the commonly H3K27me3 marked orthogroups in both species using GO and KEGG term enrichment analysis.

```
ostta.k27.ostta.atha <- intersect(unlist(strsplit(x = subset(ostta.atha.orthogroups, Orthogroup %in% all.

ostta.k27.ostta.atha.go.terms <- enrichGO(gene = ostta.k27.ostta.atha, pvalueCutoff = 0.05,
      qvalueCutoff = 0.05,
      OrgDb = org.Otauriv5.eg.db,
      ont = "BP",
      keyType = "GID")
write.table(x = as.data.frame(ostta.k27.ostta.atha.go.terms), file="tables/go_enrichment_ostta_common_ma

#jpeg(filename = "images/go_ostta_common_k27_orthogroups.jpg", res = 120, width = 1300, height = 500)
treemap(x = pairwise_termsim(ostta.k27.ostta.atha.go.terms), cluster.params = list(n = 2))
```



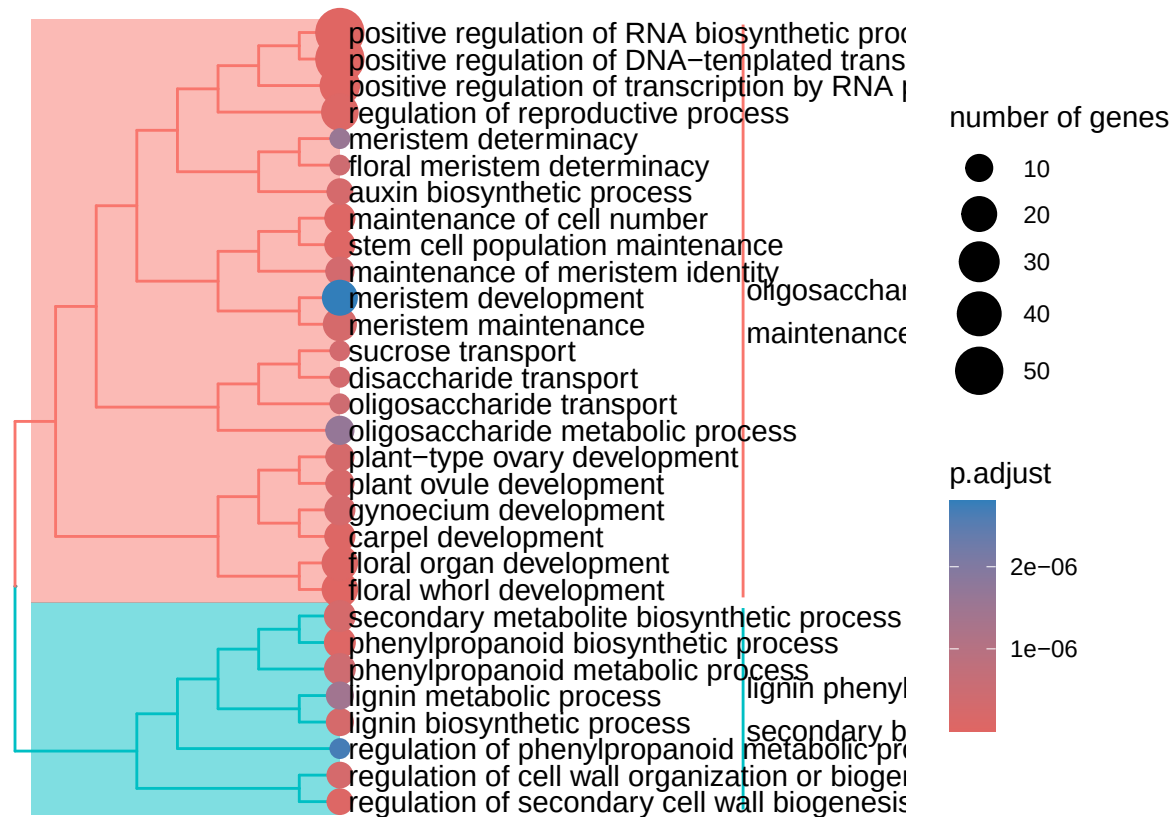
```
#dev.off()

library(org.At.tair.db)

##

atha.k27.ostta.atha <- intersect(unlist(strsplit(x = subset(ostta.atha.orthogroups, Orthogroup %in% all
atha.k27.ostta.atha.go.terms <- enrichGO(gene = atha.k27.ostta.atha, pvalueCutoff = 0.05,
      qvalueCutoff = 0.05,
      OrgDb = org.At.tair.db,
      ont = "BP",
      keyType = "TAIR")
write.table(x = as.data.frame(atha.k27.ostta.atha.go.terms), file="tables/go_enrichment_atha_common_mark
```

```
#jpeg(filename = "images/go_atha_common_k27_orthogroups.jpg",res = 120,width = 1300,height = 500)
treeplot(x = pairwise_termsim(atha.k27.ostta.atha.go.terms),cluster.params = list(n = 2))
```



```
#dev.off()
```

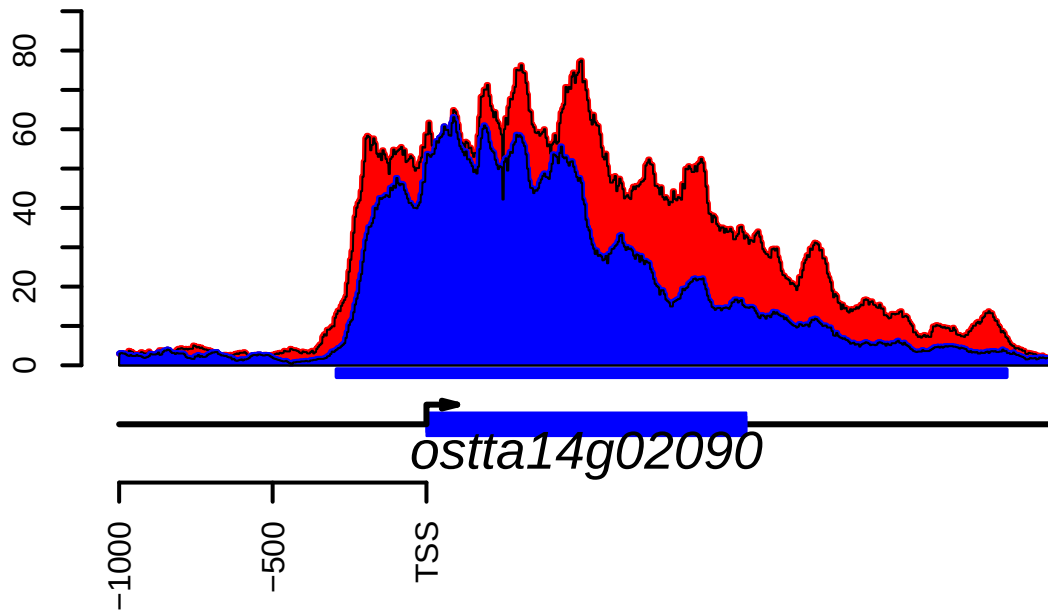
Gene expression control was the biological process most significant in the commonly H3K27me3 orthogroups in both species. 18 genes codifying for transcriptions factors in *Ostreococcus*, 10.7% of the entire set of transcription factors.

- ostta03g02220, ostta06g01110 and ostta14g02090 are three of the six AP2 transcription factors in *Ostreococcus tauri* that are marked by H3K27me3. These genes belong to the same orthogroup as the *Arabidopsis thaliana* H3K27me3 marked genes AINTEGUMENTA (AT4G37750), BABY BOOM (AT5G17430) and PLETHORA1 (AT3G20840) that are involved in the control of cell proliferation and stem cell establishment. Their role in *Ostreococcus tauri* remains to be characterized. Their H3K27me3 ChIP signal and expression level change significantly depending on the temperature, specially for ostta14g02090, suggesting a role in temperature acclimation.

```
#jpeg(filename = "images/AP2_ostta14g02090.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta14g02090",
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb = TxDb.Otauri.JGI)
```

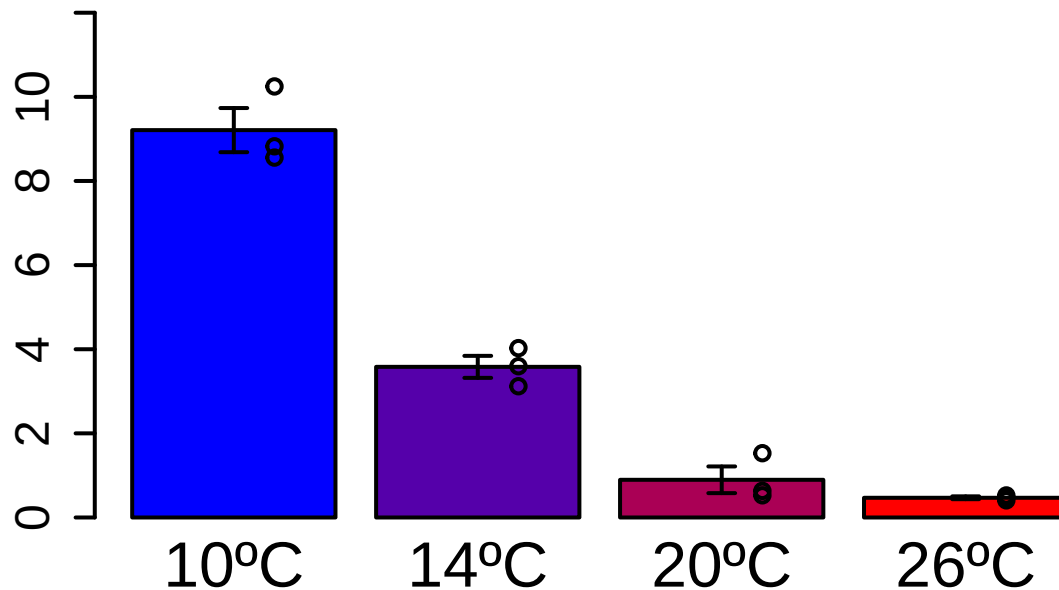
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
```

```
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



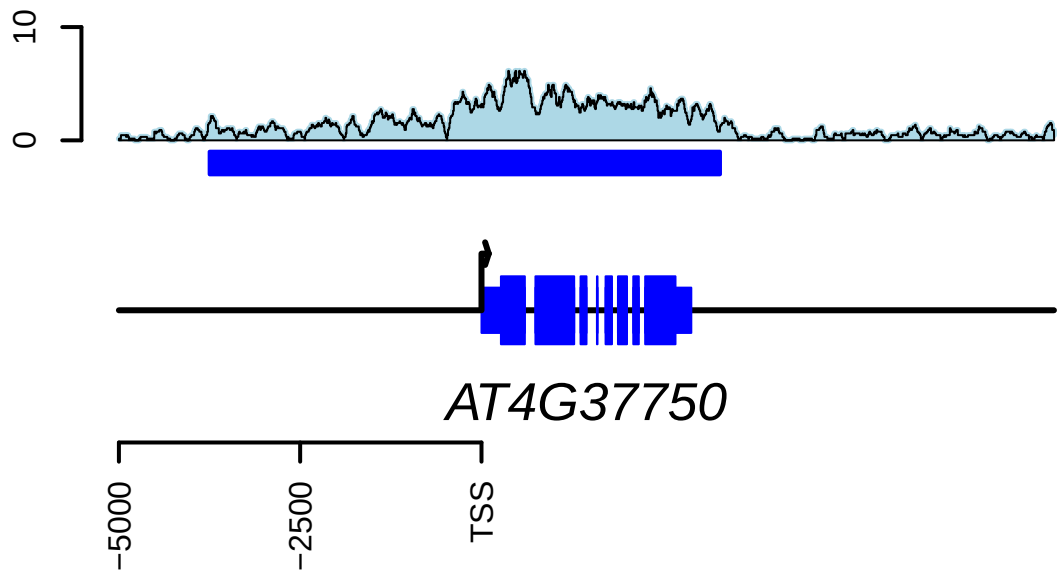
```
#dev.off()
```

```
gene.expression.barplot(gene.id = "ostta14g02090", gene.expression=gene.expression)
```



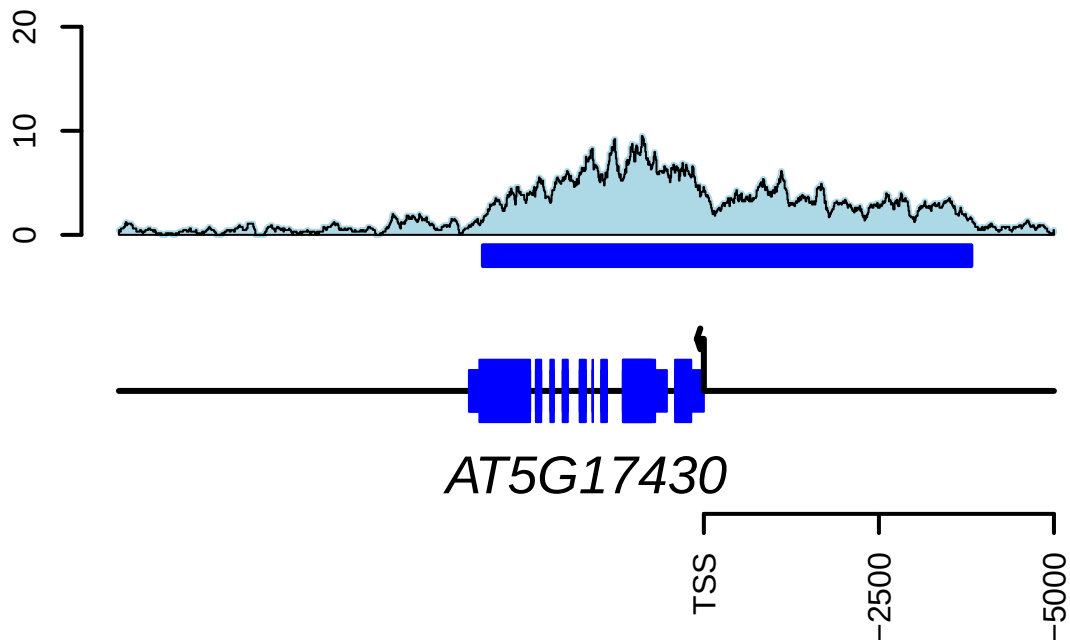
```
library("TxDb.Athaliana.BioMart.plantmart51")
gene.profile.complete(gene.name = "AT4G37750",
  promoter_length=5000,
  selected.bigwig.files = c("atha_data/h3k27me3_col0_7dag_original_1.bw"),
  cols = "lightblue",
  selected.bed.files = "atha_data/h3k27me3_col0.bed",
  txdb = TxDb.Athaliana.BioMart.plantmart51)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#jpeg(filename = "images/BBM_AT5G17430.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "AT5G17430",
  promoter_length=5000,
  selected.bigwig.files = c("atha_data/h3k27me3_col0_7dag_original_1.bw"),
  cols = "lightblue",
  selected.bed.files = "atha_data/h3k27me3_col0.bed",
  txdb = TxDb.Athaliana.BioMart.plantmart51)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

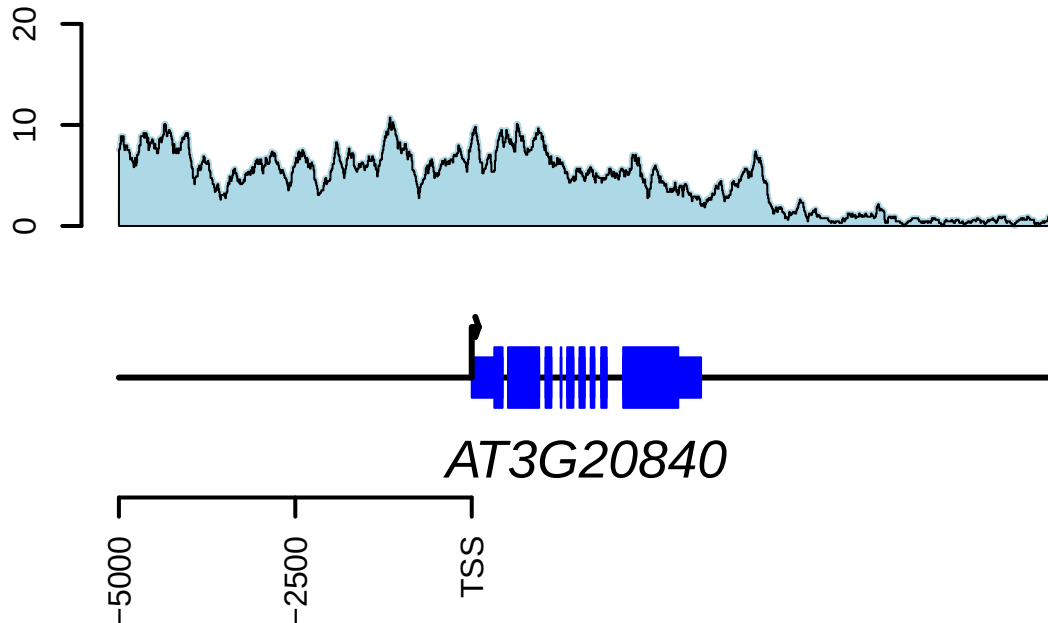


```
#dev.off()

gene.profile.complete(gene.name = "AT3G20840",
  promoter_length=5000,
  selected.bigwig.files = c("atha_data/h3k27me3_col0_7dag_original_1.bw"),
```

```
cols = "lightblue",
selected.bed.files = "atha_data/h3k27me3_col10.bed",
txdb = TxDb.Athaliana.BioMart.plantmart51)
```

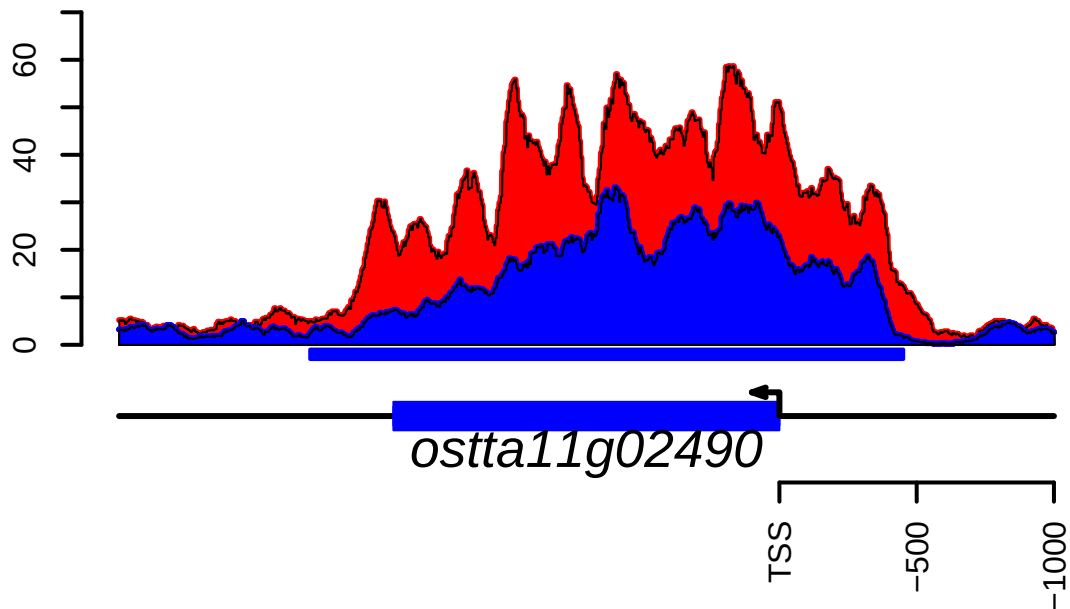
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



- ostta07g04340, ostta08g02190 and ostta11g02490 are the three WRKY transcription factors all of the them are H3K27me3 marked belonging to the orthogroup that contains the Group-II WRKY transcription factors that are also H3K27me3 marked in Arabidopsis thaliana being involved in biotic stress response such as WRKY40 (AT1G80840) and WRKY48 (AT5G49520) and cold stress response as WRY42 (AT4G04450). Their role in *Ostreococcus tauri* remains to be characterized. The H3K27me3 ChIP signal and expression level change significantly depending on the temperature for ostta11g02490, suggesting a role in temperature acclimation.

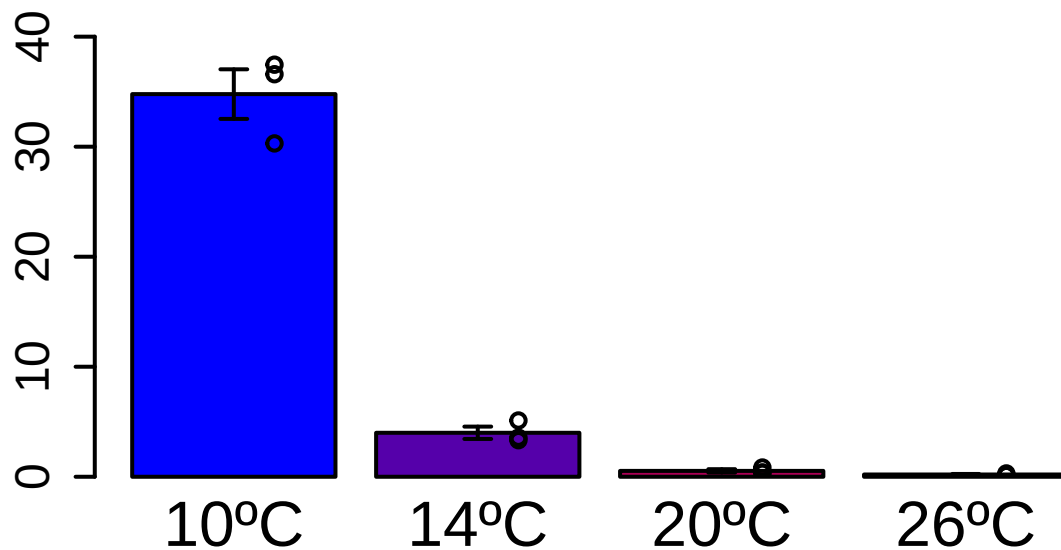
```
#jpeg(filename = "images/WRKY_ostta11g02490.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta11g02490",
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



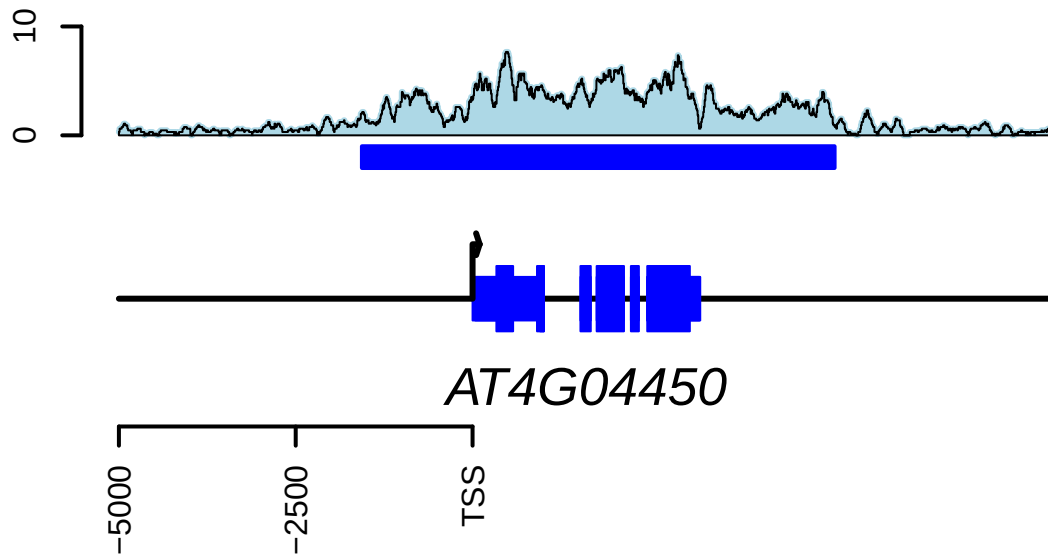
```
#dev.off()
```

```
gene.expression.barplot(gene.id = "ostta11g02490", gene.expression=gene.expression)
```



```
#jpeg(filename = "images/WRKY42_AT4G04450.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "AT4G04450",
  promoter_length=5000,
  selected.bigwig.files = c("atha_data/h3k27me3_col0_7dag_original_1.bw"),
  cols = "lightblue",
  selected.bed.files = "atha_data/h3k27me3_col0.bed",
  txdb = TxDb.Athaliana.BioMart.plantmart51)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

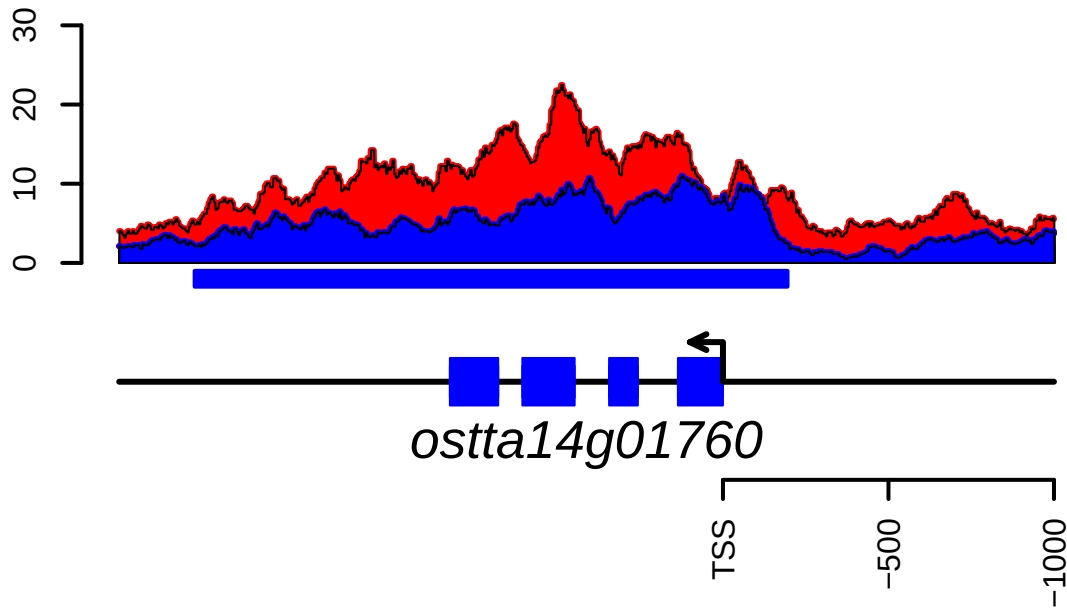


```
#dev.off()
```

- ostta14g01760 is a RWP-RK transcription factor that belongs to the same orthogroup as the Arabidopsis thaliana RWP-RK AT1G18790 and AT5G66990. Both its H3K27me3 ChIP signal and expression level change significantly depending on temperature suggesting a role in temperature acclimation in Ostreococcus.

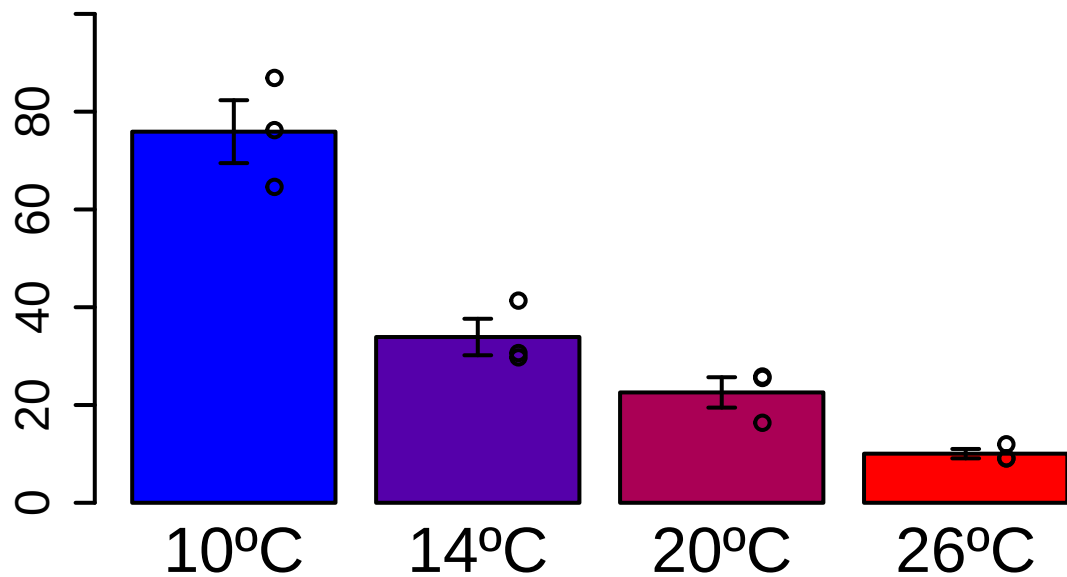
```
#jpeg(filename = "images/INT_ostta11g02575.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta14g01760",
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
    "h3k27me3/h3k27me3_11_10_chip_2.bw",
    "h3k27me3/h3k27me3_11_10_chip_2.bw",
    "h3k27me3/h3k27me3_11_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

```
gene.expression.barplot(gene.id = "ostta14g01760", gene.expression=gene.expression)
```



* The single MADS-box transcription factor in *Ostreococcus*, *ostta01g05160*, is H3K27me3 marked. It belongs to the same orthogroup as MADS-box H3K27me3 marked genes in *Arabidopsis* such as *FLOWERING LOCUS C* (AT5G10140), *SHORT VEGETATIVE PHASE* (AT2G22540) and *AGAMOUS* (AT4G18960). These *Arabidopsis* genes are involved in the floral transition and floral meristem identity. The role of this MADS-box transcription factor in *Ostreococcus* remains to be characterized. Although its H3K27me3 ChIP signal changes significantly depending on temperature its expression level is unaffected.

```
#jpeg(filename = "images/MADs_ostta01g05160.jpg", res = 100, width = 600, height = 580)
gene.profile.complete(gene.name = "ostta01g05160",
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",
```

```

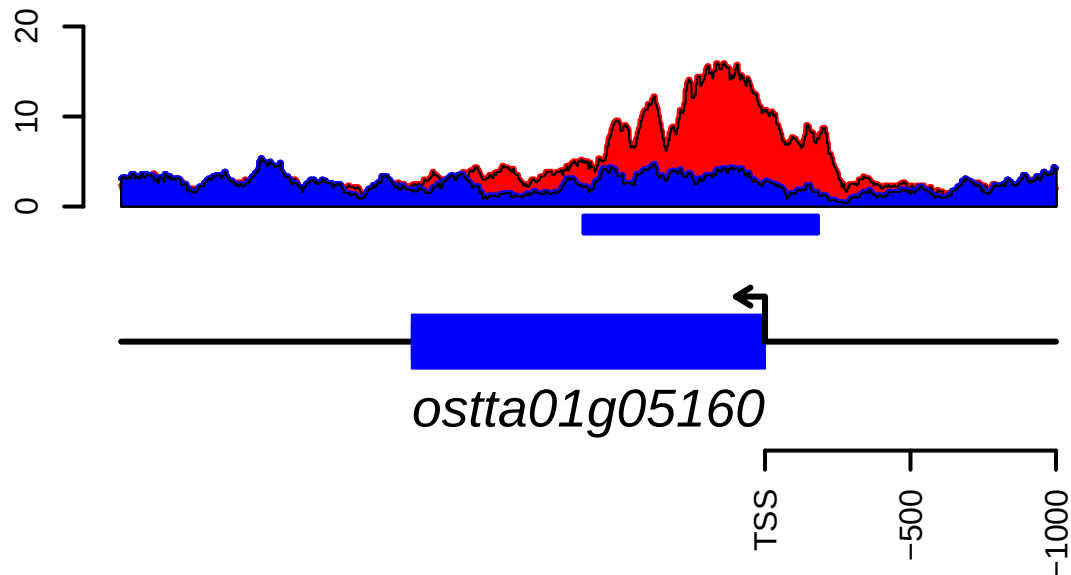
                                "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
cols = rev(colorRampPalette(c("blue", "red"))(4)),
selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
txdb=TxDb.Otauri.JGI)

```

```

## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

```

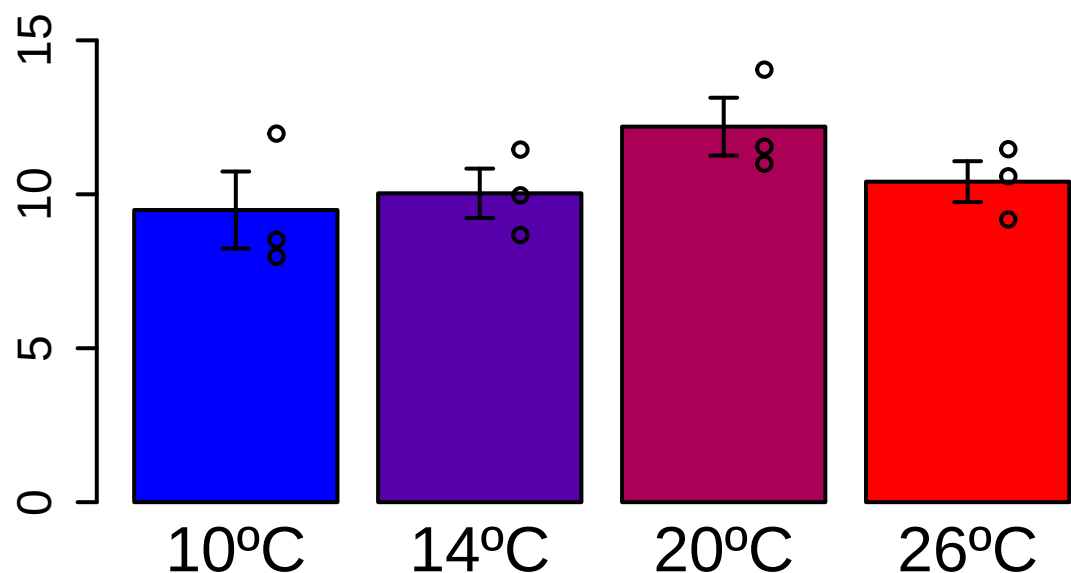


```

#dev.off()

gene.expression.barplot(gene.id = "ostta01g05160", gene.expression=gene.expression)

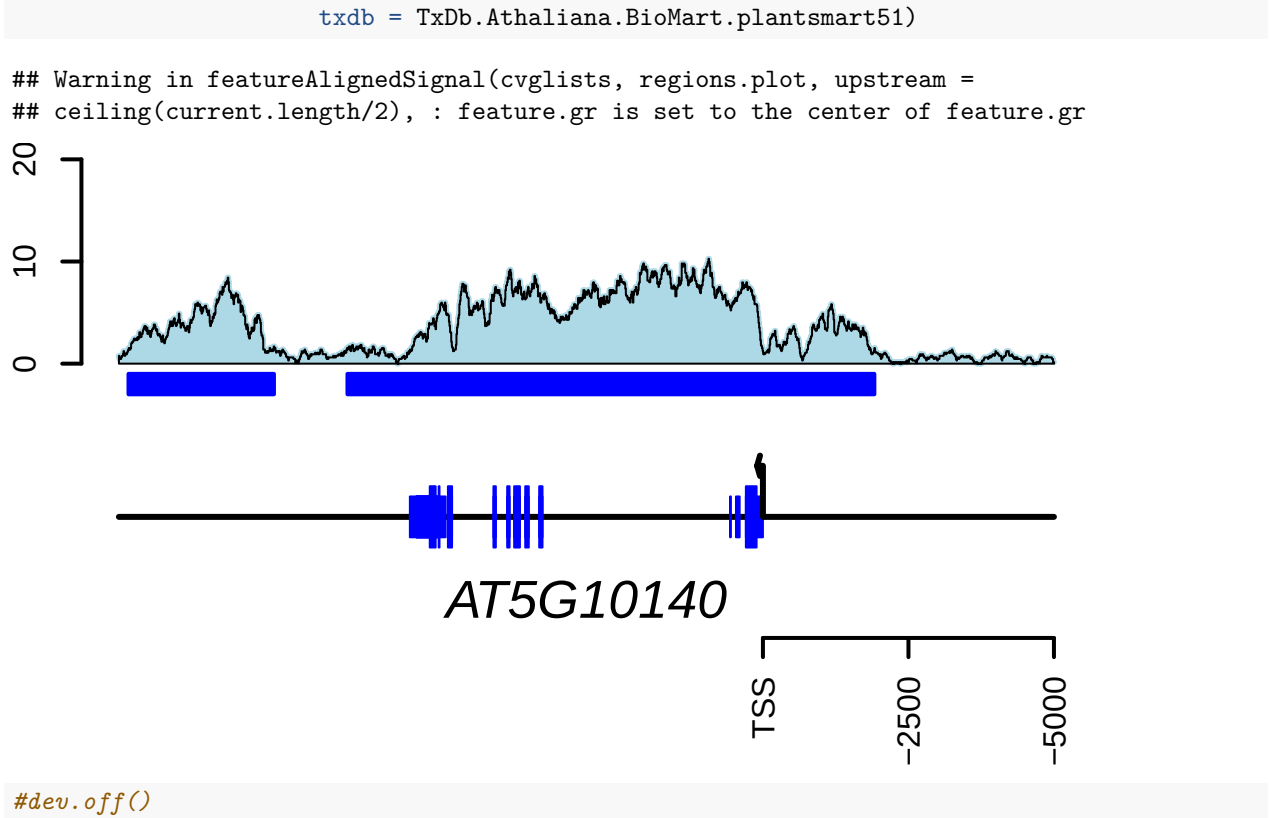
```



```

#jpeg(filename = "images/FLC_AT5G10140.jpg", res = 100, width = 600, height = 580)
gene.profile.complete(gene.name = "AT5G10140",
                      promoter_length=5000,
                      selected.bigwig.files = c("atha_data/h3k27me3_col0_7dag_original_1.bw"),
                      cols = "lightblue",
                      selected.bed.files = "atha_data/h3k27me3_col0.bed",

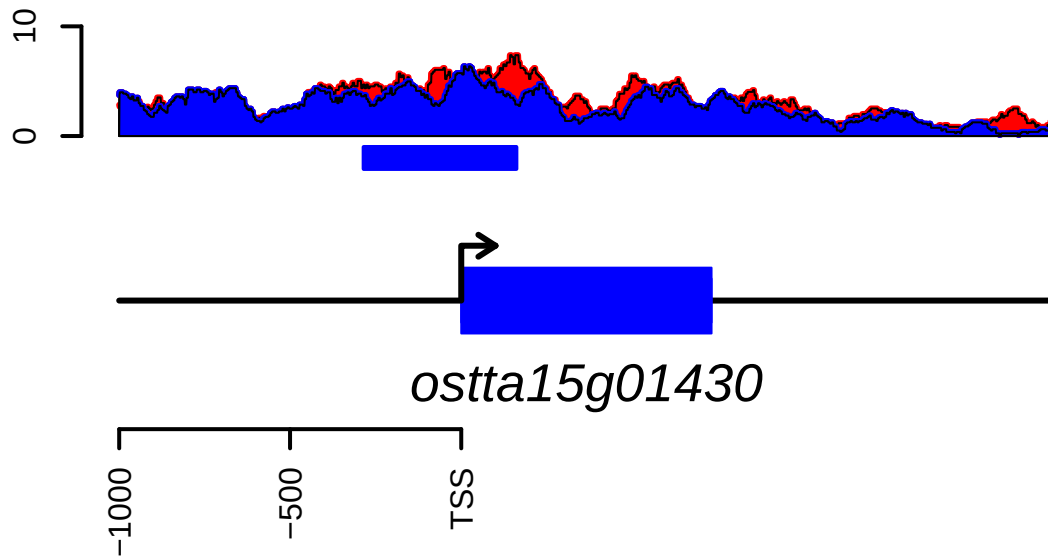
```



- ostta15g01430 is a H3K27me3 marked MYB gene belonging to the same orthogroup of H3K27me3 marked Arabidopsis MYBs as involved in meristem formation regulation REGULATOR OF AXILLARY MERISTEMS3 (AT3G49690) and LATERAL ORGAN FUSION 2 (AT1G69560).

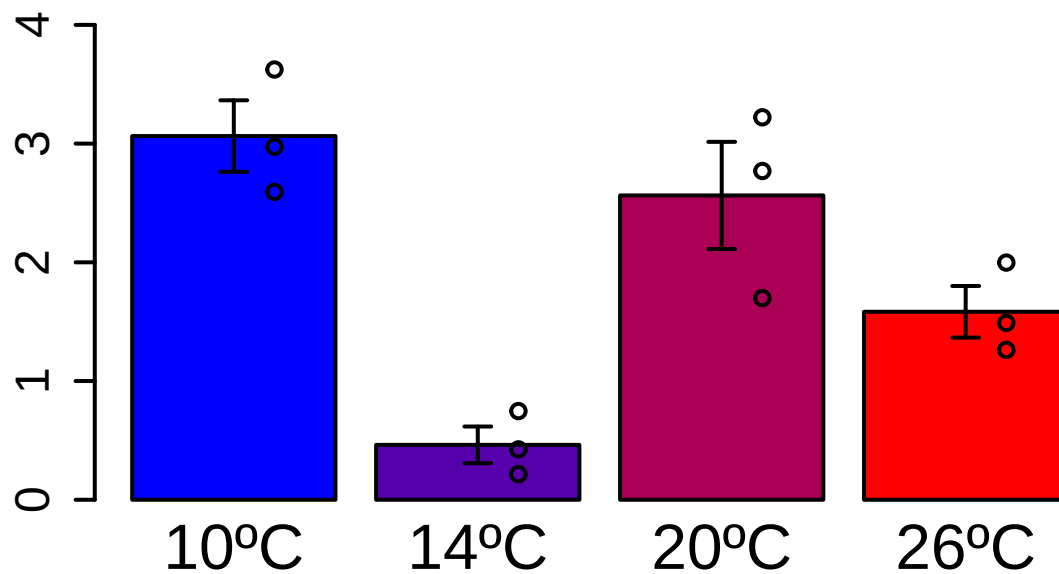
```
#jpeg(filename = "images/INT_ostta11g02575.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta15g01430",
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()

gene.expression.barplot(gene.id = "ostta15g01430", gene.expression=gene.expression)
```



Next we explore the *Ostreococcus tauri* specific H3K27me3 marked orthogroups using GO and KEGG term enrichment analysis.

```
ostta.specific.k27.orthogroups <- setdiff(ostta.orthogroups,atha.orthogroups)
length(ostta.specific.k27.orthogroups)

## [1] 672

sum(subset(ostta.atha.orthogroups, Orthogroup %in% ostta.specific.k27.orthogroups)$arabidopsis_thaliana)

## [1] 0.5223214

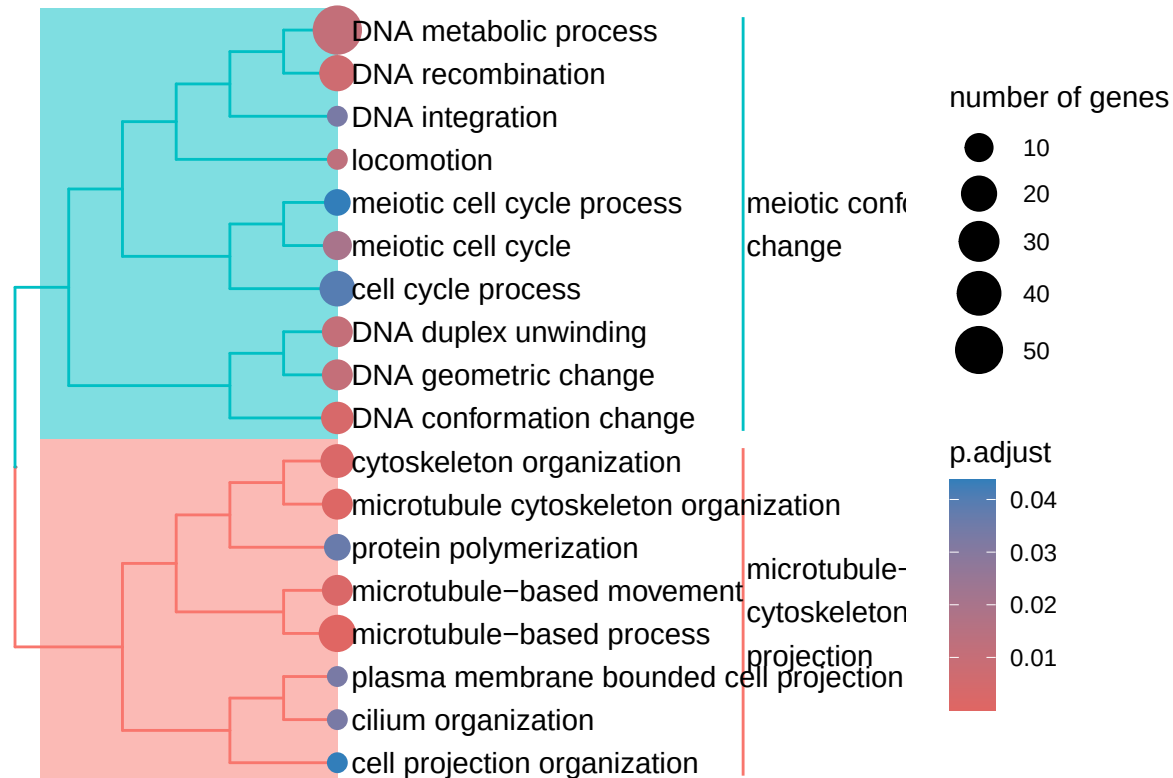
ostta.specific.k27 <- unlist(strsplit(x = subset(ostta.atha.orthogroups, Orthogroup %in% ostta.specific.k27.orthogroups),
                                     split = "\t"))

ostta.specific.k27.go.terms <- enrichGO(gene = intersect(all.k27.genes,ostta.specific.k27), pvalueCutoff = 0.05,
                                         qvalueCutoff = 0.05,
```

```

OrgDb = org.Otauriv5.eg.db,
ont = "BP",
keyType = "GID")
write.table(x = as.data.frame(ostta.specific.k27.go.terms),file="tables/go_enrichment_ostta_specific_ma
#jpeg(filename = "images/go_ostta_specific_k27_orthogroups.jpg",res = 120,width = 1300,height = 500)
treeplot(x = pairwise_termsim(ostta.specific.k27.go.terms),cluster.params = list(n = 2))

```



```
#dev.off()
```

We identified 672 orthogroups only marked in *Ostreococcus tauri*. 52.2% of these orthogroups also contained *Arabidopsis* genes. Therefore the reason why these orthogroups are specific to *Ostreococcus* is not related to the absence of *Arabidopsis* genes in them. This indicates that H3K27me3 regulates specific processes in *Ostreococcus* that are not regulated in *Arabidopsis*. Among these *Ostreococcus tauri* H3K27me3 specific biological processes we highlight:

- DNA integration with H3K27me3 marked genes like the integrase codified by ostta11g02575 and the reverse transcriptase ostta19g00515.

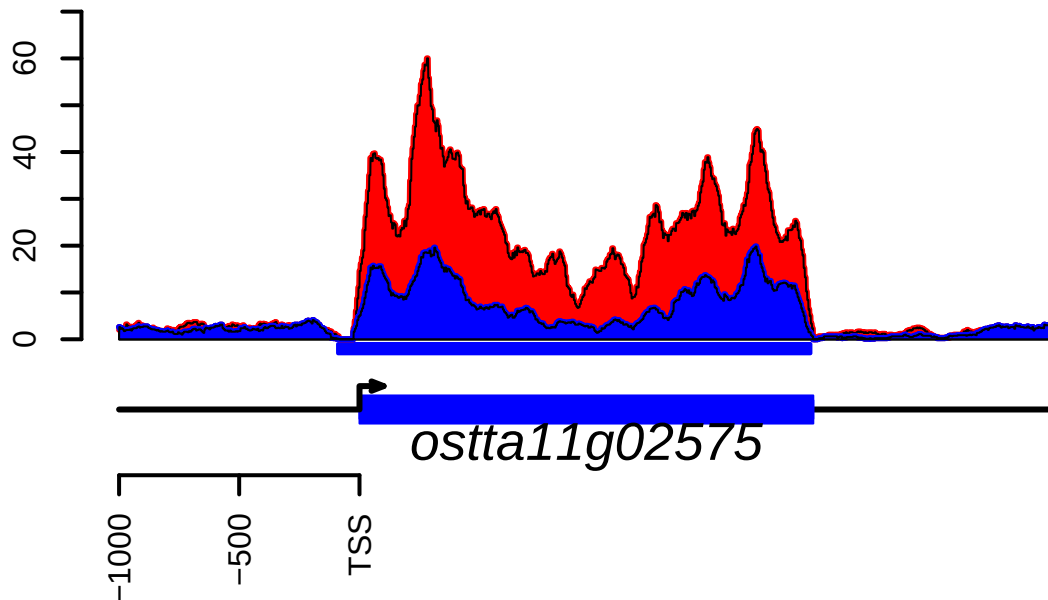
```

#jpeg(filename = "images/INT_ostta11g02575.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta11g02575",
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
                      selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
                      txdb = TxDb.Otauri.JGI)

```

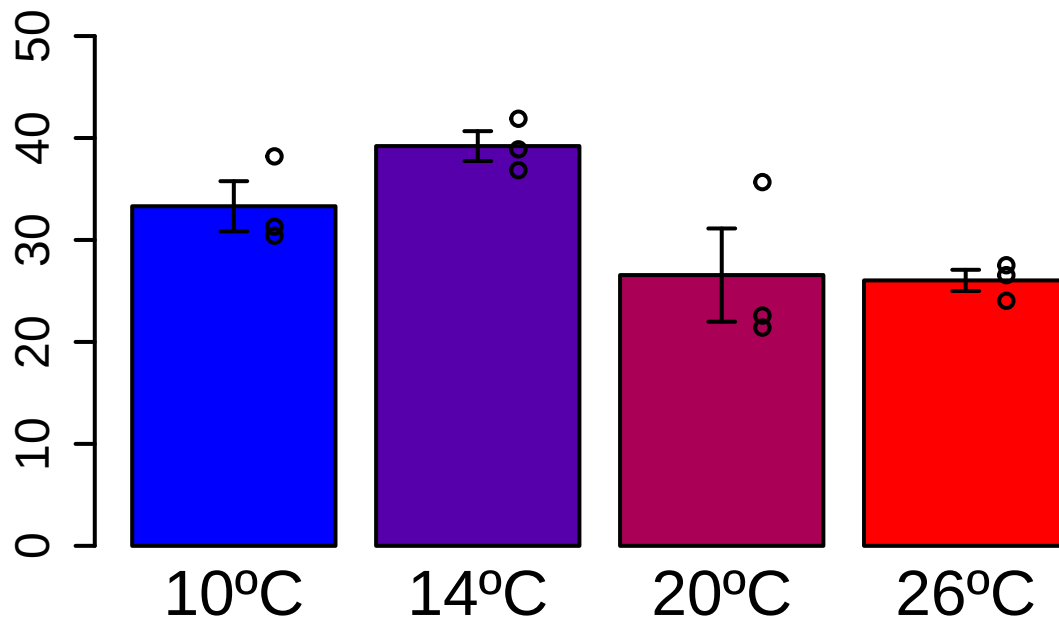
```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
```

```
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



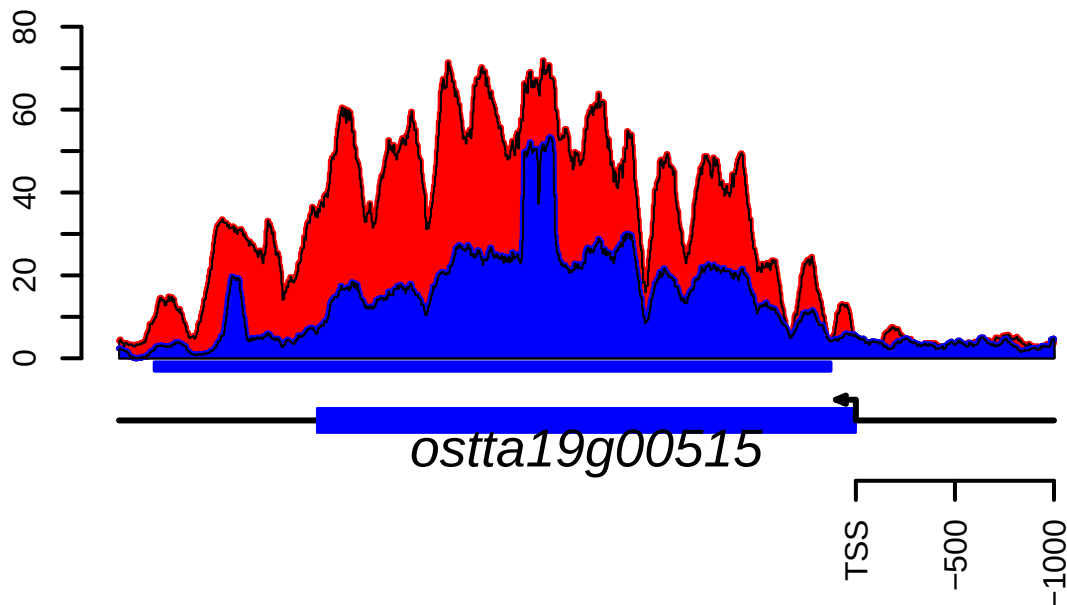
```
#dev.off()
```

```
gene.expression.barplot(gene.id = "ostta11g02575", gene.expression=gene.expression)
```

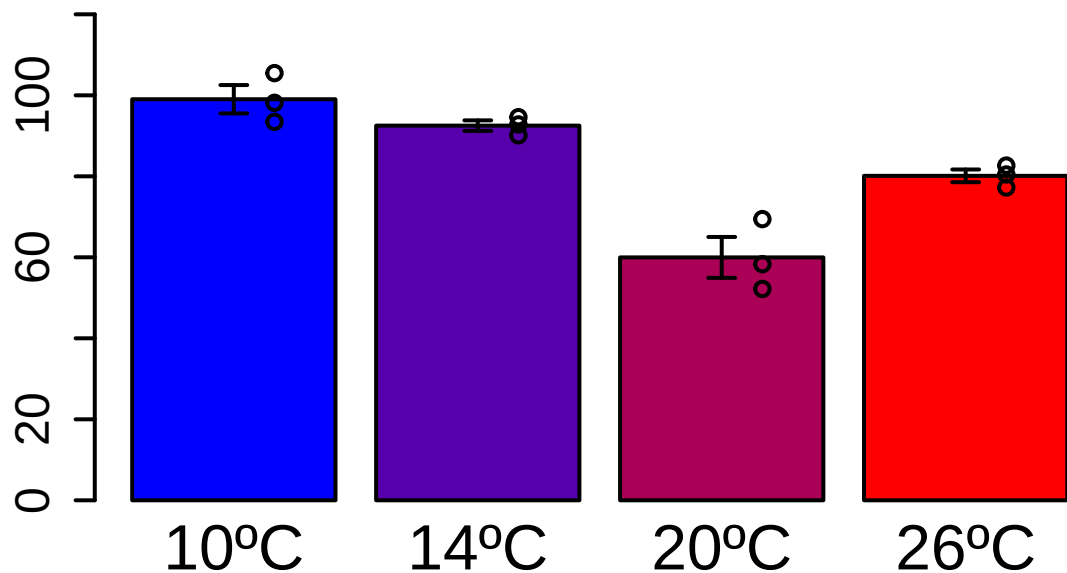


```
gene.profile.complete(gene.name = "ostta19g00515",
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
gene.expression.barplot(gene.id = "ostta19g00515", gene.expression=gene.expression)
```

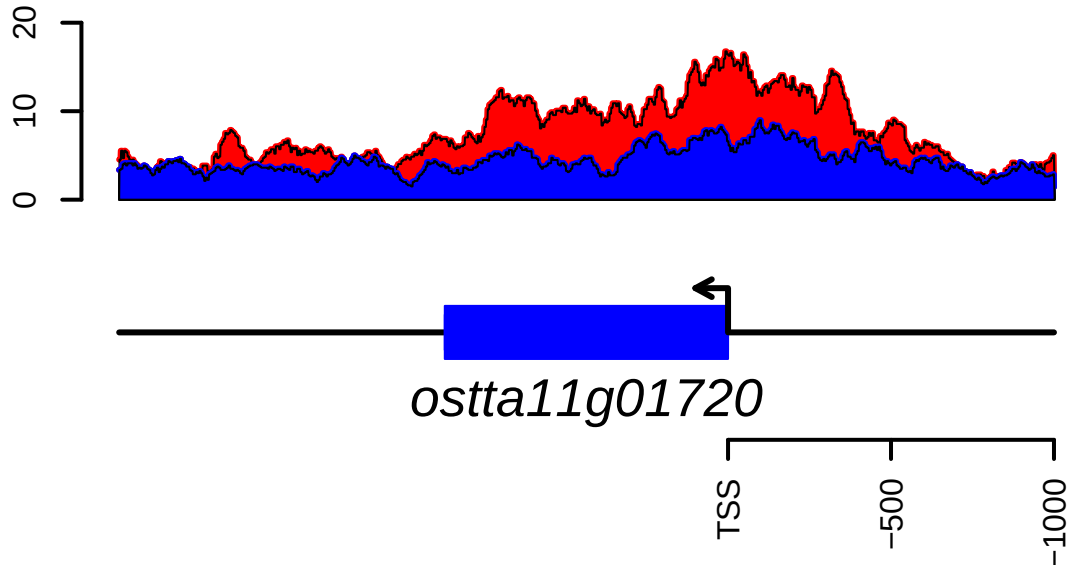


- Meiotic cell cycle with H3K27me3 marked genes specifically involved in the pachytene stage of meiosis particularly ostta11g01720 (Disrupted Meiotic cDNA 1, DMC1), ostta07g03600 (meiotic recombination protein SPO22) and ostta05g02370 (Pachytene Associated Protein 2, PA2). These proteins mediate chromosome remodeling and crossover by homologous recombination.

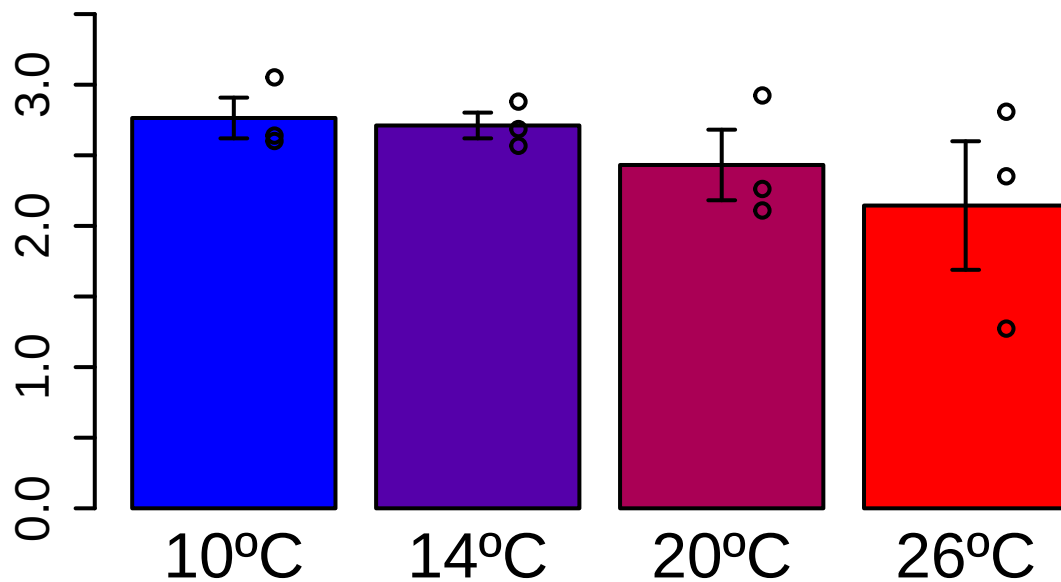
```
gene.profile.complete(gene.name = "ostta11g01720",
                      promoter_length=1000,
                      selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw",
                                                "h3k27me3/h3k27me3_11_10_chip_2.bw"),
                      cols = rev(colorRampPalette(c("blue", "red"))(4)),
```

```
selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



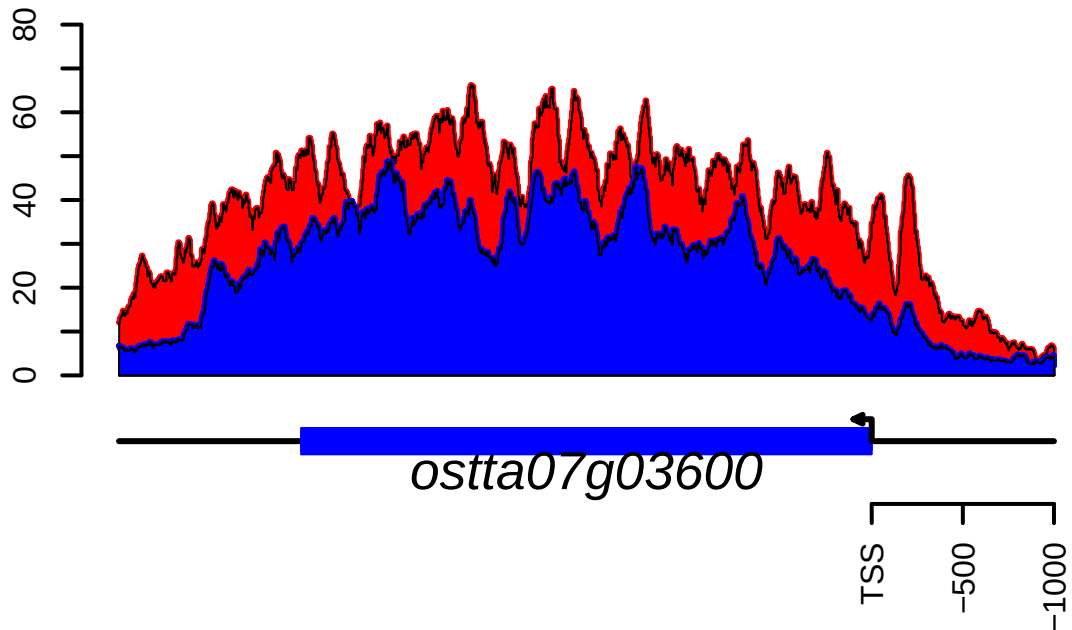
```
gene.expression.barplot(gene.id = "ostta11g01720", gene.expression=gene.expression)
```



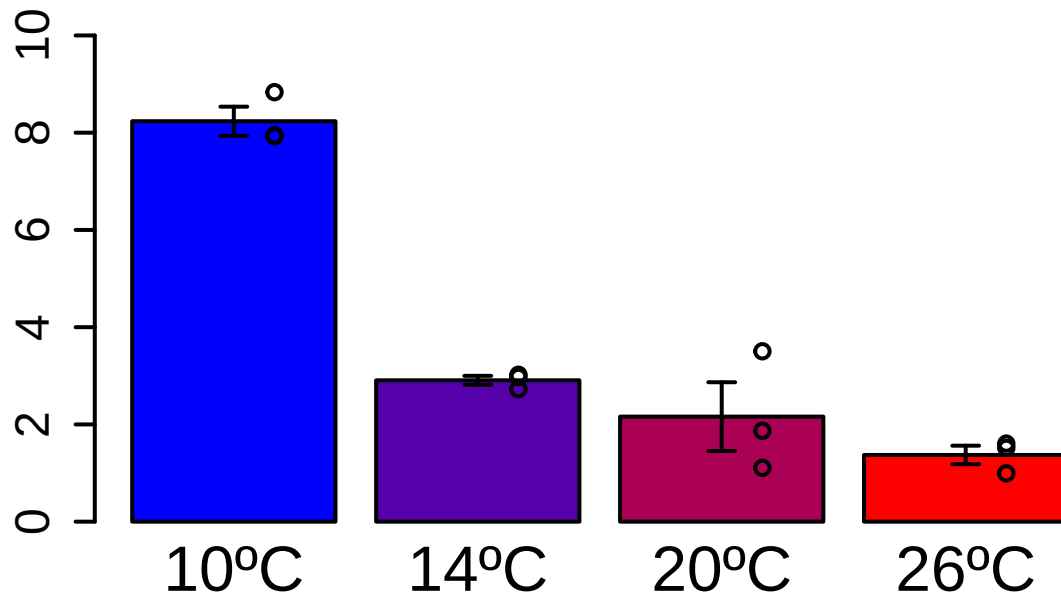
```
#jpeg(filename = "images/SP022_ostta07g03600.jpg", res = 100, width = 600, height = 580)
gene.profile.complete(gene.name = "ostta07g03600",
promoter_length=1000,
selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
"h3k27me3/h3k27me3_ll_10_chip_2.bw",
"h3k27me3/h3k27me3_ll_10_chip_2.bw",
"h3k27me3/h3k27me3_ll_10_chip_2.bw"),
cols = rev(colorRampPalette(c("blue", "red"))(4)),
selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
```

```
txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =  
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



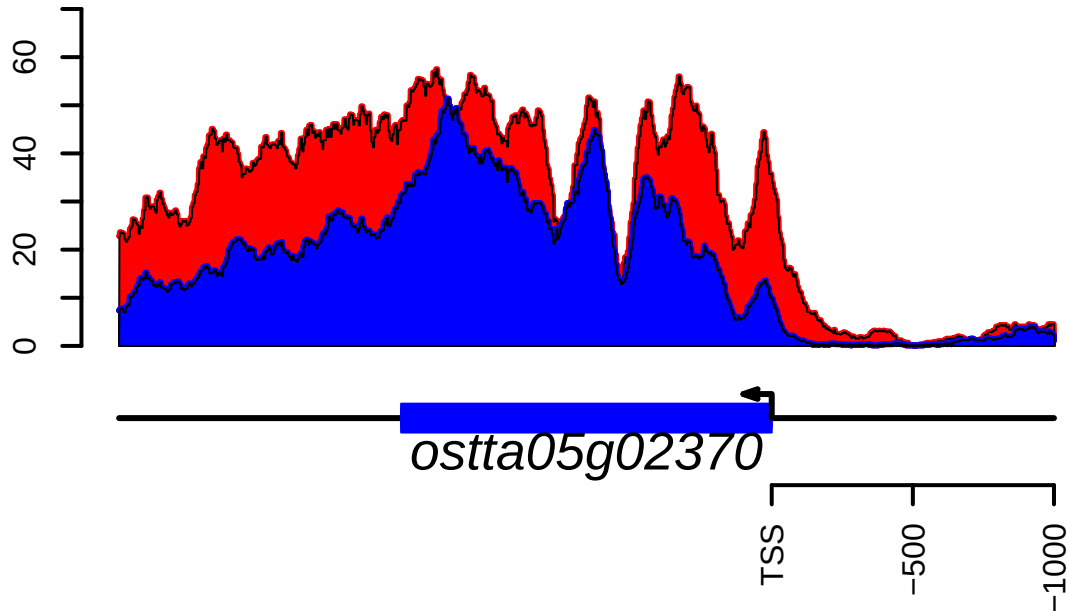
```
#dev.off()  
gene.expression.barplot(gene.id = "ostta07g03600", gene.expression=gene.expression)
```



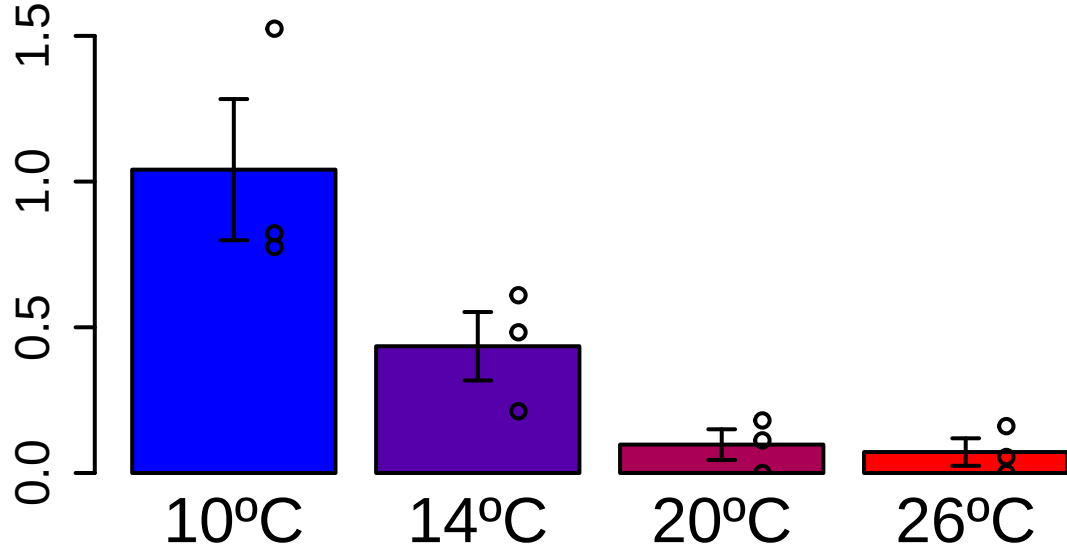
```
gene.profile.complete(gene.name = "ostta05g02370",  
  promoter_length=1000,  
  selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",  
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",  
    "h3k27me3/h3k27me3_ll_10_chip_2.bw",  
    "h3k27me3/h3k27me3_ll_10_chip_2.bw"),  
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
```

```
selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
gene.expression.barplot(gene.id = "ostta05g02370", gene.expression=gene.expression)
```

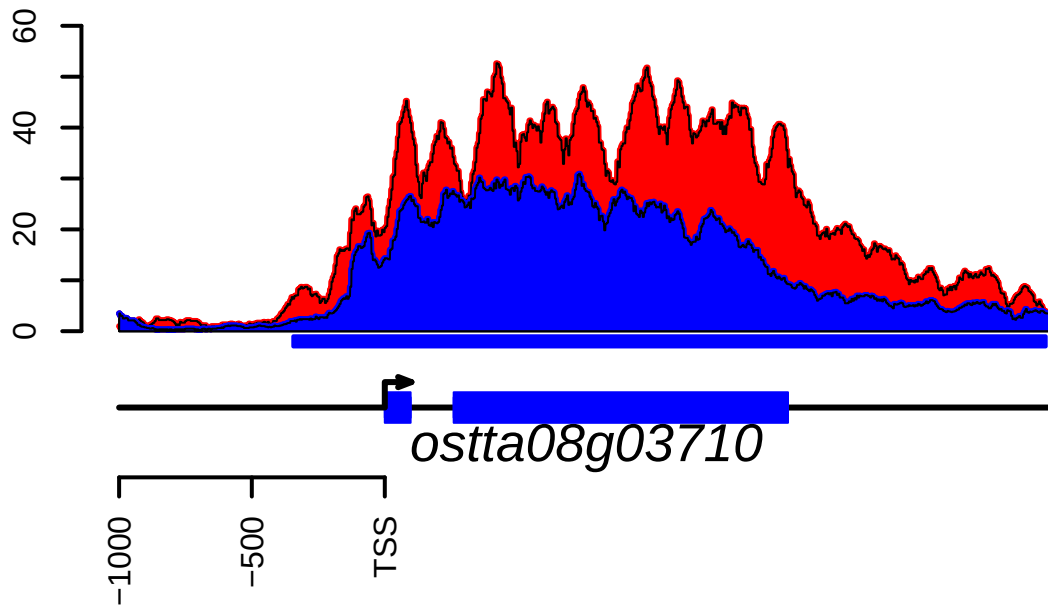


- Microtubule-based movement with H3K27me3 marked genes like tubulin alpha, ostta08g03710, dyneins, ostta14g00070 and ostta06g04460, and kinesins, ostta18g01040 and ostta07g03080.

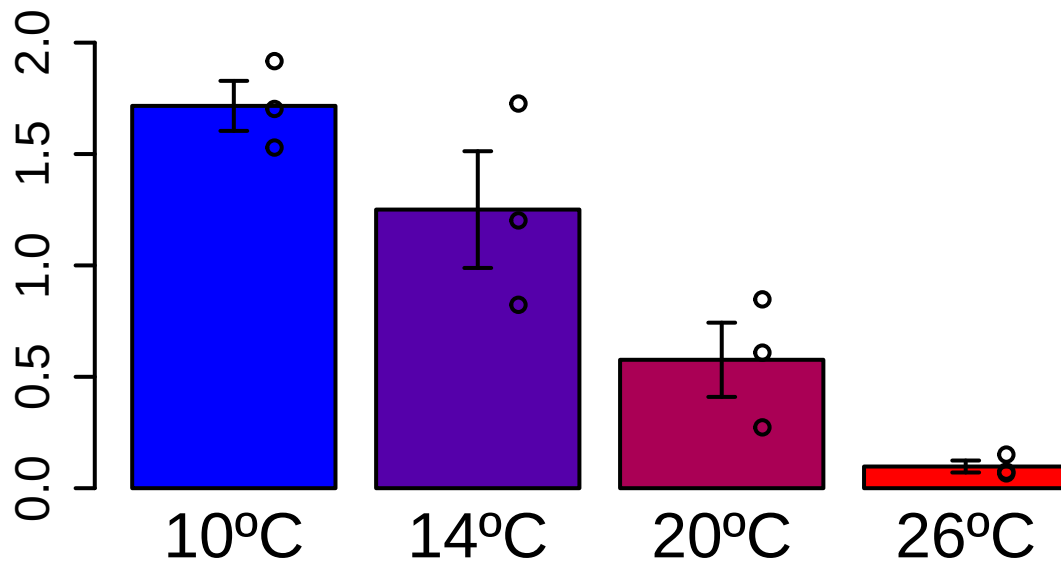
```
gene.profile.complete(gene.name = "ostta08g03710",
promoter_length=1000,
selected.bigwig.files = c("h3k27me3/h3k27me3_ll_26_chip_2.bw",
"h3k27me3/h3k27me3_ll_10_chip_2.bw",
"h3k27me3/h3k27me3_ll_10_chip_2.bw",
"h3k27me3/h3k27me3_ll_10_chip_2.bw"),
```

```
cols = rev(colorRampPalette(c("blue", "red"))(4)),
selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
txdb = TxDb.Otari.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



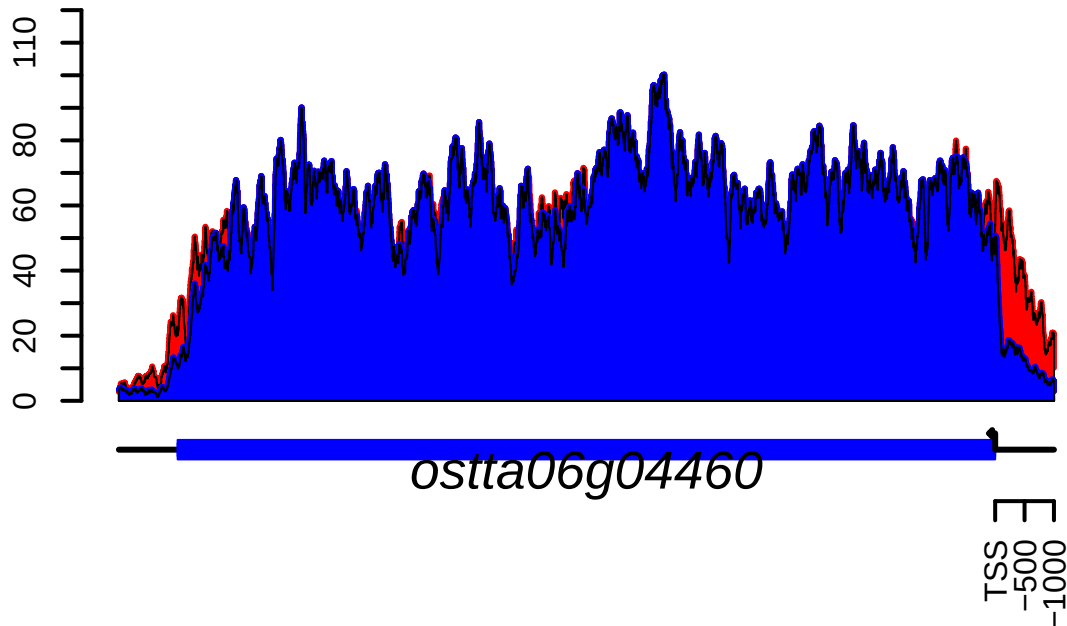
```
gene.expression.barplot(gene.id = "ostta08g03710", gene.expression=gene.expression)
```



```
#jpeg(filename = "images/DYN_ostta06g04460.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "ostta06g04460",
promoter_length=1000,
selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
"h3k27me3/h3k27me3_11_10_chip_2.bw",
"h3k27me3/h3k27me3_11_10_chip_2.bw",
"h3k27me3/h3k27me3_11_10_chip_2.bw"),
cols = rev(colorRampPalette(c("blue", "red"))(4)),
```

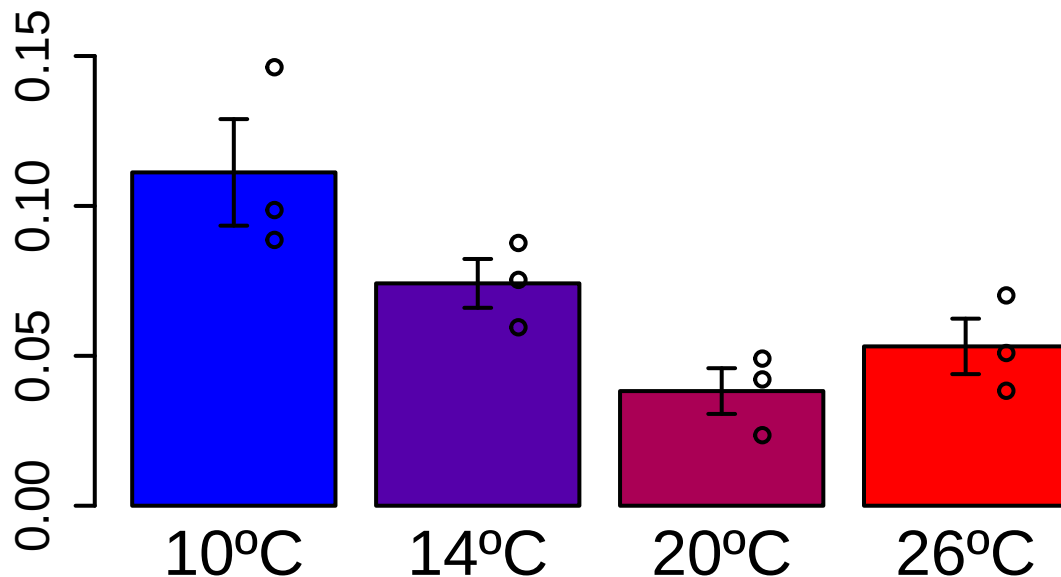
```
selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

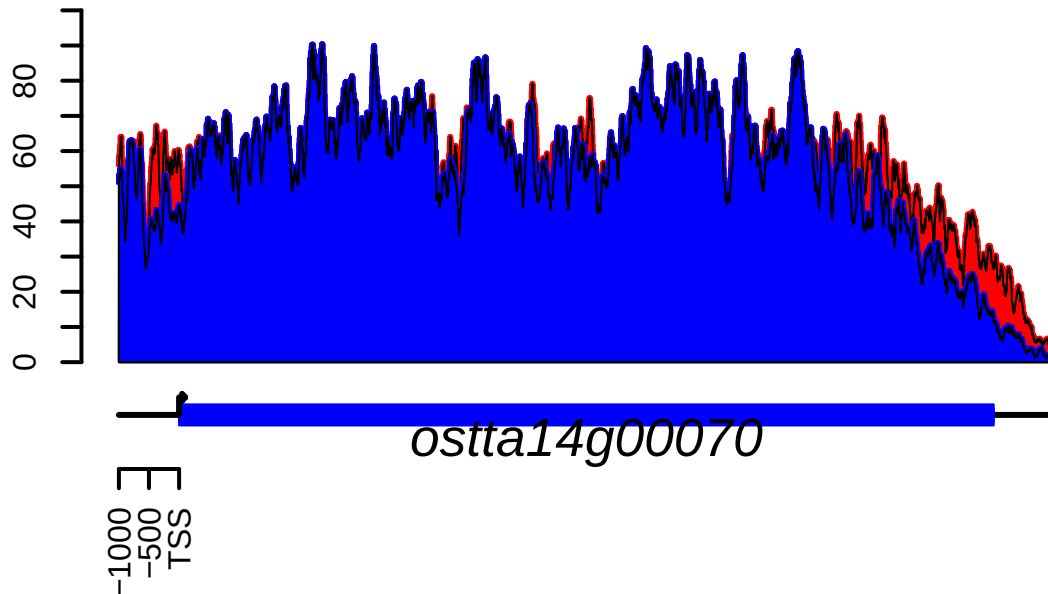
```
gene.expression.barplot(gene.id = "ostta06g04460", gene.expression=gene.expression)
```



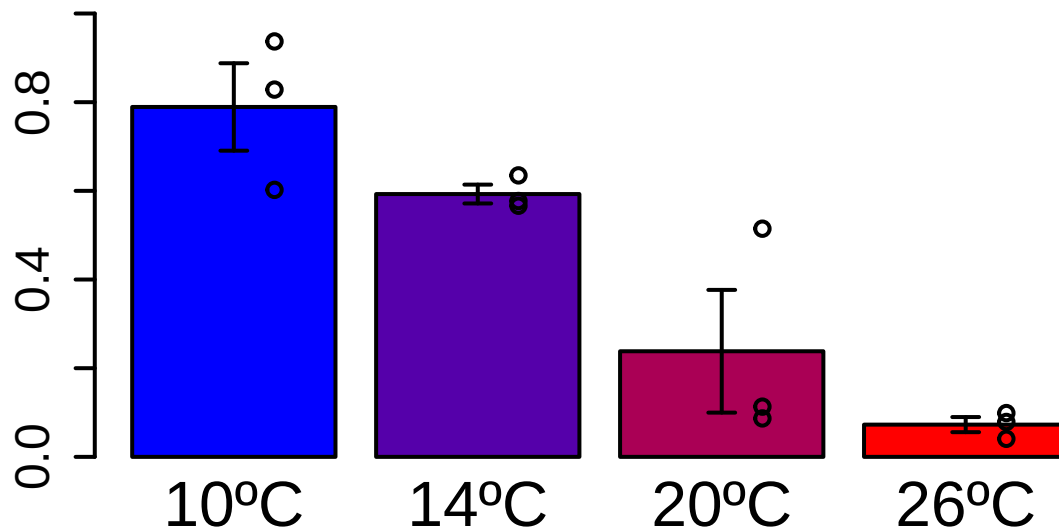
```
gene.profile.complete(gene.name = "ostta14g00070",
promoter_length=1000,
selected.bigwig.files = c("h3k27me3/h3k27me3_11_26_chip_2.bw",
"h3k27me3/h3k27me3_11_10_chip_2.bw",
"h3k27me3/h3k27me3_11_10_chip_2.bw",
"h3k27me3/h3k27me3_11_10_chip_2.bw"),
```

```
cols = rev(colorRampPalette(c("blue", "red"))(4)),
selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
gene.expression.barplot(gene.id = "ostta14g00070", gene.expression=gene.expression)
```



```
atha.specific.k27.orthogroups <- setdiff(atha.orthogroups,ostta.orthogroups)
length(atha.specific.k27.orthogroups)
```

```
## [1] 1782
```

```
sum(subset(ostta.atha.orthogroups, Orthogroup %in% atha.specific.k27.orthogroups)$ostreococcus_tauri !=
```

```
## [1] 0.2351291
```

```
atha.specific.k27 <- unlist(strsplit(x = subset(ostta.atha.orthogroups, Orthogroup %in% atha.specific.k
```

```

atha.specific.k27.go.terms <- enrichGO(gene = intersect(atha.k27.genes,atha.specific.k27), pvalueCutoff = 0.05,
                                     OrgDb = org.At.tair.db,
                                     ont = "BP",
                                     keyType = "TAIR")
write.table(x = as.data.frame(atha.specific.k27.go.terms),file="tables/go_enrichment_atha_specific_marks.txt",
           jpeg(filename = "images/go_atha_specific_k27_orthogroups.jpg",res = 120,width = 1300,height = 500)
treeplot(x = pairwise_termsim(atha.specific.k27.go.terms),cluster.params = list(n = 3),showCategory = 1,
dev.off()

## pdf
## 2

```

We found 1782 H3K27me3 orthogroups in Arabidopsis that are not marked in *Ostreococcus*, 419 orthogroups of this orthogroups corresponding to 23.51% have an *Ostreococcus* gene. This indicates that newly emerged orthogroups in the Arabidopsis lineage are recruited under H3K27me3 control. While some orthogroups, 419, that are not marked by H3K27me3 in *Ostreococcus* are regulated by this mark in Arabidopsis suggesting they have acquired new functions during evolution that need to be regulated by H3K27me3.

H3K27me3 Arabidopsis specific genes are involved in specialized secondary metabolism. For example:

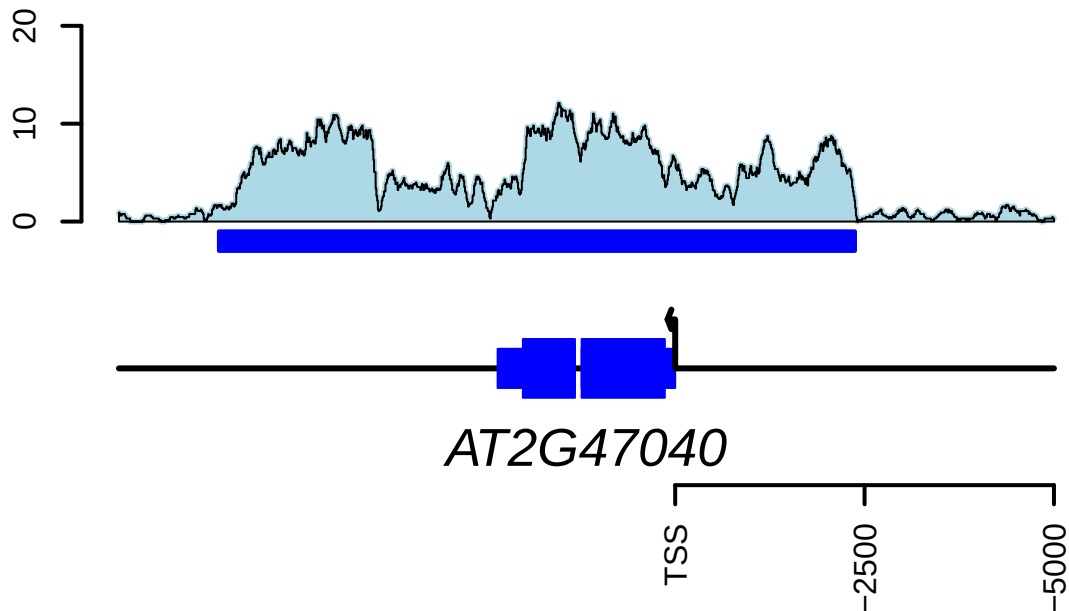
- VANGUARD1 (VGD1, AT2G47040) encodes a pectin methyltransferase (PME) that plays a crucial role in pollen tube growth by modifying the pectin composition of the cell wall, facilitating proper elongation and interaction with female reproductive tissues. Pectin is an important constituent of cell wall in Arabidopsis and it is absent in *Ostreococcus tauri* that possesses a glycoprotein-based layer surrounding the plasma membrane.

```

#jpeg(filename = "images/VGD1_AT2G47040.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "AT2G47040",
                      promoter_length=5000,
                      selected.bigwig.files = c("atha_data/h3k27me3_col0_7dag_original_1.bw"),
                      cols = "lightblue",
                      selected.bed.files = "atha_data/h3k27me3_col0.bed",
                      txdb = TxDb.Athaliana.BioMart.plantmart51)

## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr

```

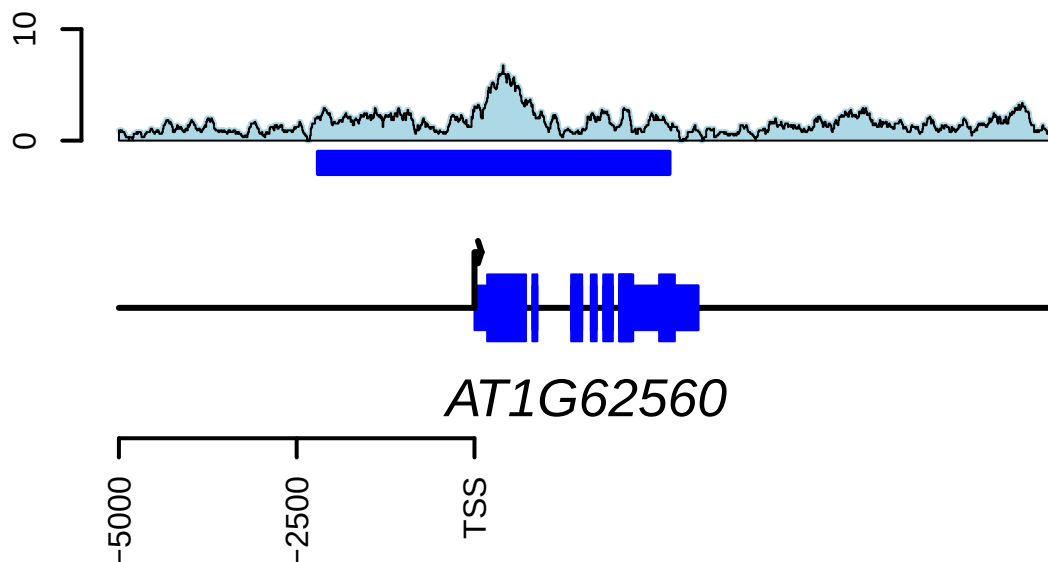


```
#dev.off()
```

- FLAVIN-MONOOXYGENASE GLUCOSINOLATE S-OXYGENASE 3 (AT1G62560, FMO GS-OX3) encodes a flavin-containing monooxygenases involved in biosynthesis of aliphatic glucosinolates. Glucosinolates are specific to Brassicaceae like Arabidopsis and that play a role in plant defense and sulfur metabolism.

```
#jpeg(filename = "images/FMO_AT1G62560.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "AT1G62560",
  promoter_length=5000,
  selected.bigwig.files = c("atha_data/h3k27me3_col0_7dag_original_1.bw"),
  cols = "lightblue",
  selected.bed.files = "atha_data/h3k27me3_col0.bed",
  txdb = TxDb.Athaliana.BioMart.plantmart51)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

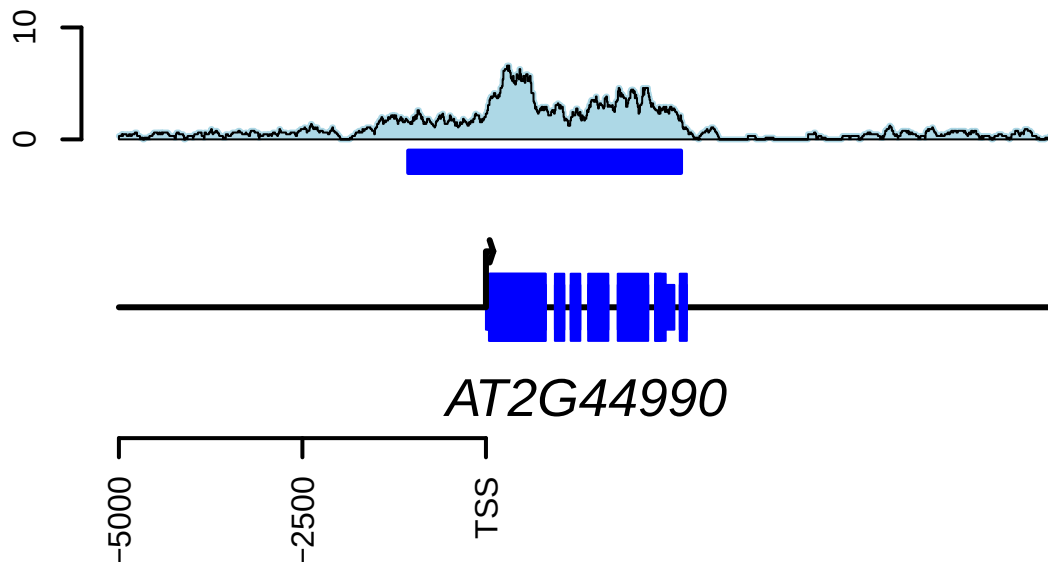


```
#dev.off()
```

In the specific H3K27me3 Arabidopsis orthogroups we also find orthogroups containing *Ostreococcus* genes that are not H3K27me3. For example, none of the genes in *Ostreococcus* encoding Carotenoid oxygenase, *ostta01g04480*, *ostta03g03900* or *ostta03g05630*, are H3K27me3 marked. While the single gene in Arabidopsis belonging to this orthogroup AT2G44990 (MAX3) is H3K27me3. This gene is involved in strigolactone biosynthesis which play a role in plant developmental processes such as flowering.

```
#jpeg(filename = "images/MAX3_AT2G44990.jpg",res = 100,width = 600,height = 580)
gene.profile.complete(gene.name = "AT2G44990",
  promoter_length=5000,
  selected.bigwig.files = c("atha_data/h3k27me3_col0_7dag_original_1.bw"),
  cols = "lightblue",
  selected.bed.files = "atha_data/h3k27me3_col0.bed",
  txdb = TxDb.Athaliana.BioMart.plantmart51)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```



```
#dev.off()
```

```
gene.profile.complete(gene.name = "ostta03g05630",
  promoter_length=1000,
  selected.bigwig.files = c("h3k27me3/h3k27me3_1l_26_chip_2.bw",
    "h3k27me3/h3k27me3_1l_10_chip_2.bw",
    "h3k27me3/h3k27me3_1l_10_chip_2.bw",
    "h3k27me3/h3k27me3_1l_10_chip_2.bw"),
  cols = rev(colorRampPalette(c("blue", "red"))(4)),
  selected.bed.files = "h3k27me3/h3k27me3_peaks.bed",
  txdb = TxDb.Otauri.JGI)
```

```
## Warning in featureAlignedSignal(cvglists, regions.plot, upstream =
## ceiling(current.length/2), : feature.gr is set to the center of feature.gr
```

